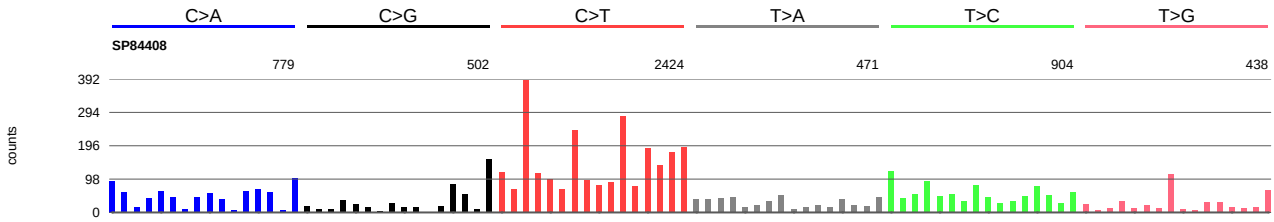
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471
904
438
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98
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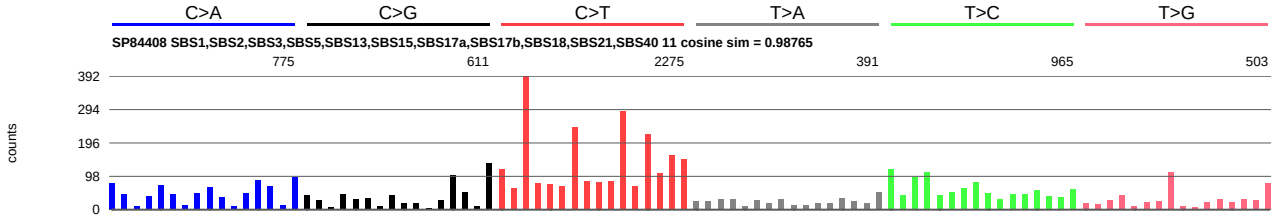
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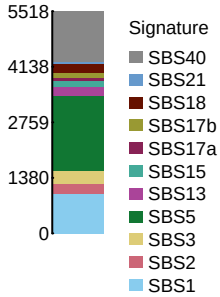
2275
391
965
503
counts
392
294
196
98
0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4138
2759
1380
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Signature
SBS40
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SBS18
SBS17b
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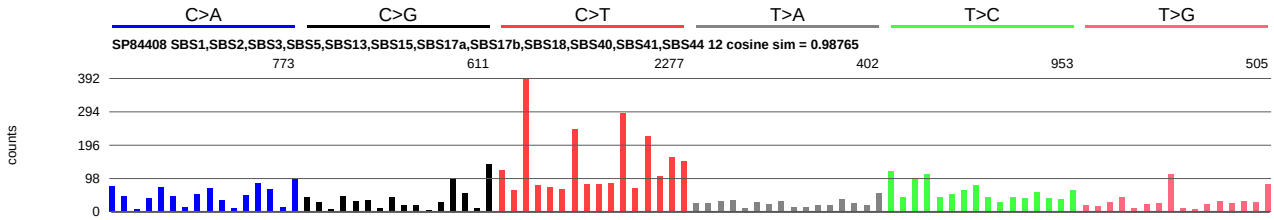
SBS13
SBS5
SBS3
SBS2
SBS1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SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 12 cosine sim = 0.98765
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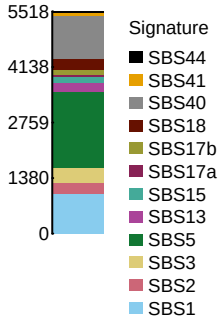
611
2277
402
953
505
counts
392
294
196
98
0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4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
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SBS15
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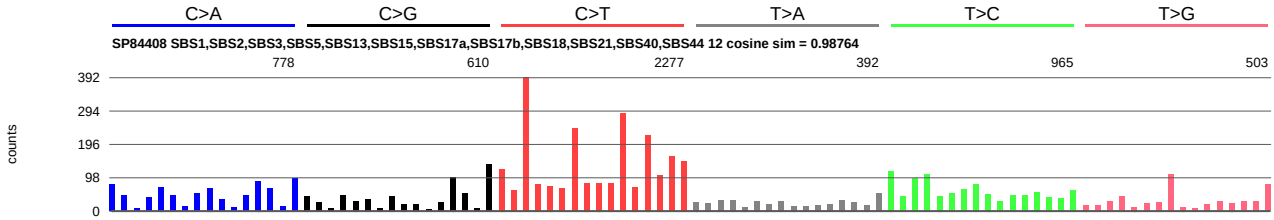
SBS2
SBS1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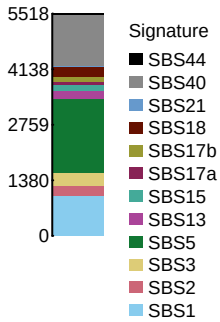
610
2277
392
965
503
counts
392
294
196
98
0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4138
2759
1380
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Signature
SBS44
SBS40
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SBS18
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SBS15
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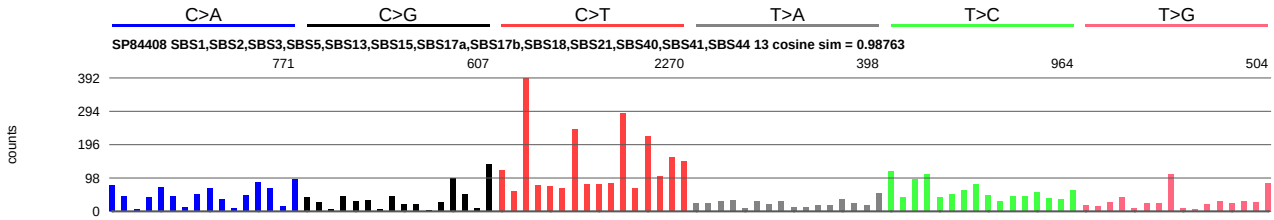
SBS2
SBS1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SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 13 cosine sim = 0.98763
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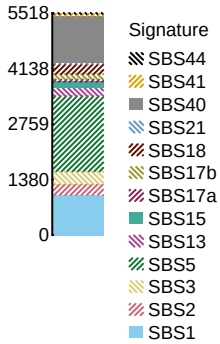
607
2270
398
964
504
counts
392
294
196
98
0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4138
2759
1380
0
Signature
SBS44
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SBS40
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SBS17b
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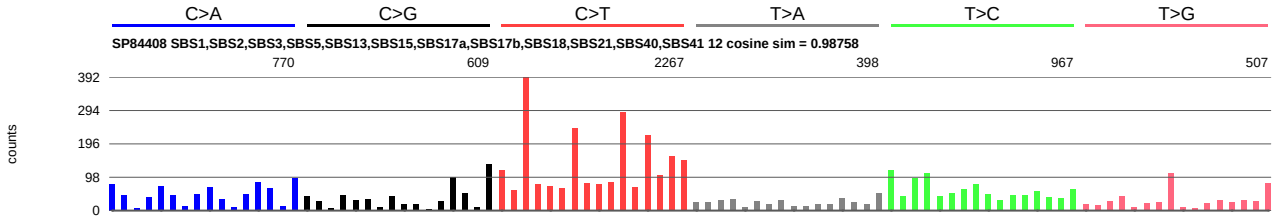
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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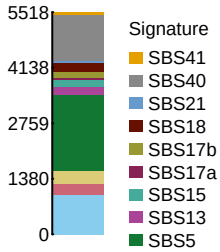
609
2267
398
967
507
counts
392
294
196
98
0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4138
2759
1380
0
Signature
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SBS40
SBS21
SBS18
SBS17b
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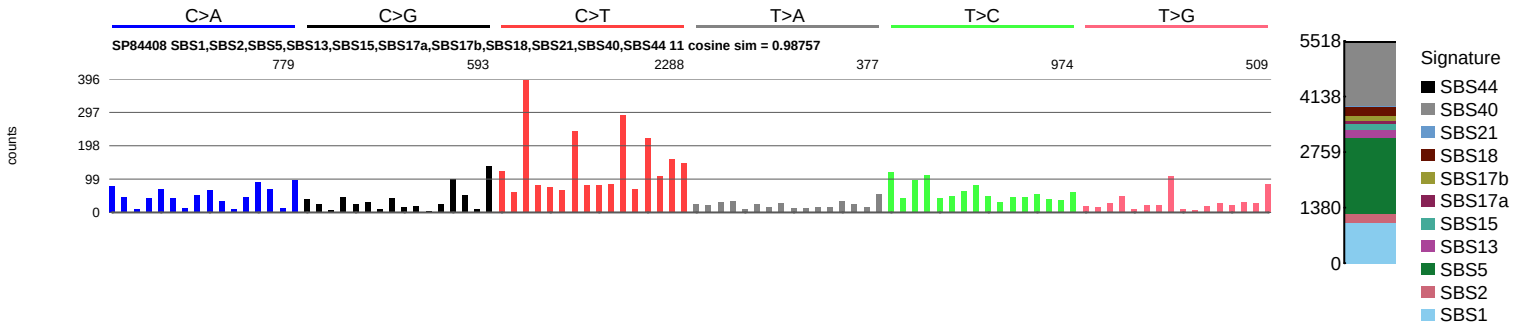
SBS15
SBS13
SBS5
SBS3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SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98757
779
593
2288
377
974
509
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
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SBS18
SBS17b
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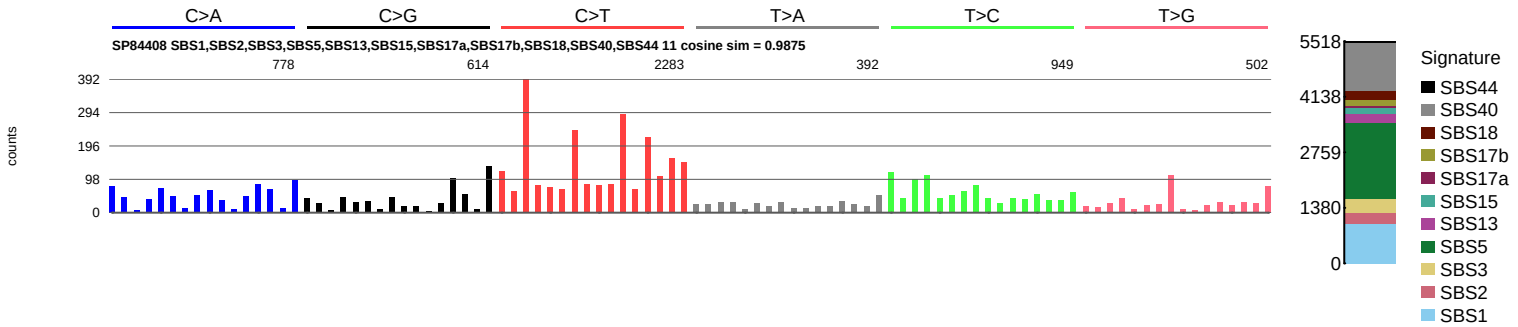
SBS2
SBS1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SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 11 cosine sim = 0.9875
778
614
2283
392
949
502
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
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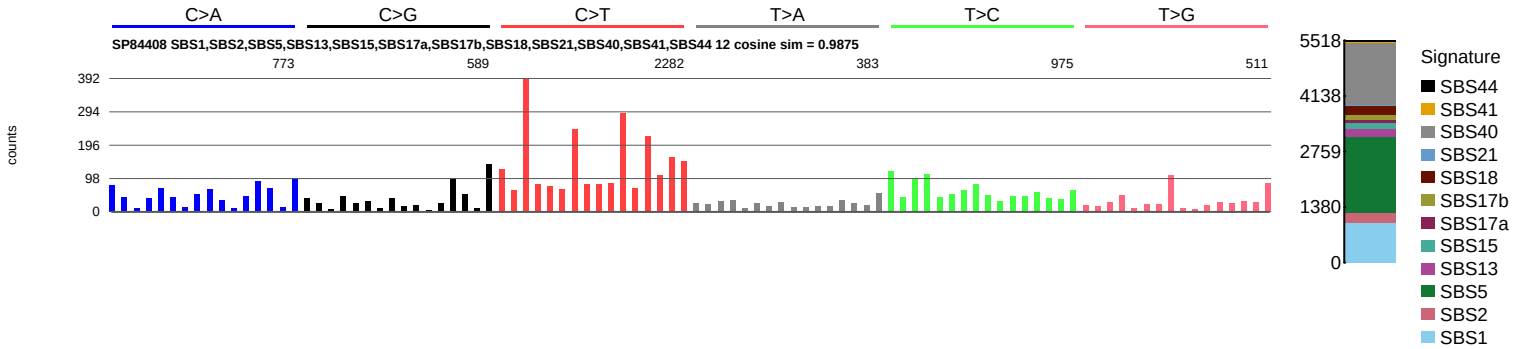
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.9875
773
589
2282
383
975
511
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
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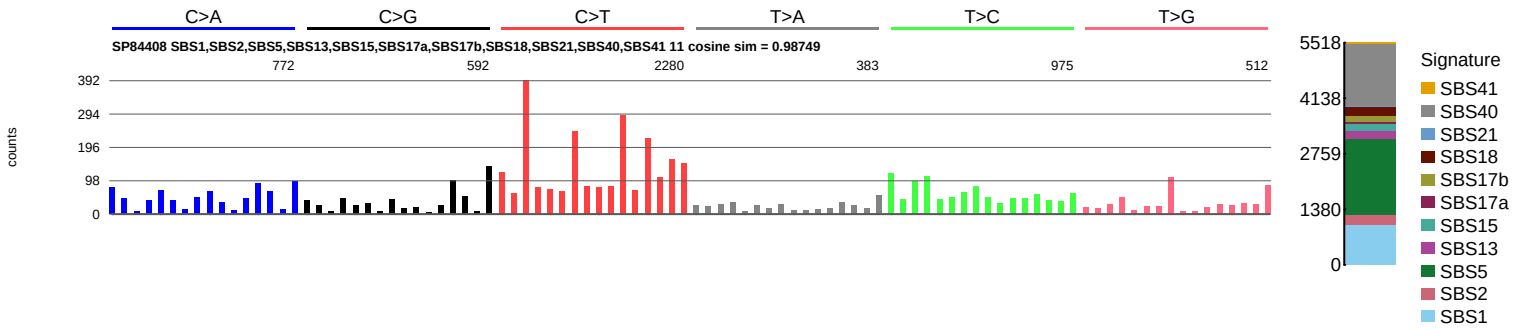
SBS2
SBS1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SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98749
772
592
2280
383
975
512
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
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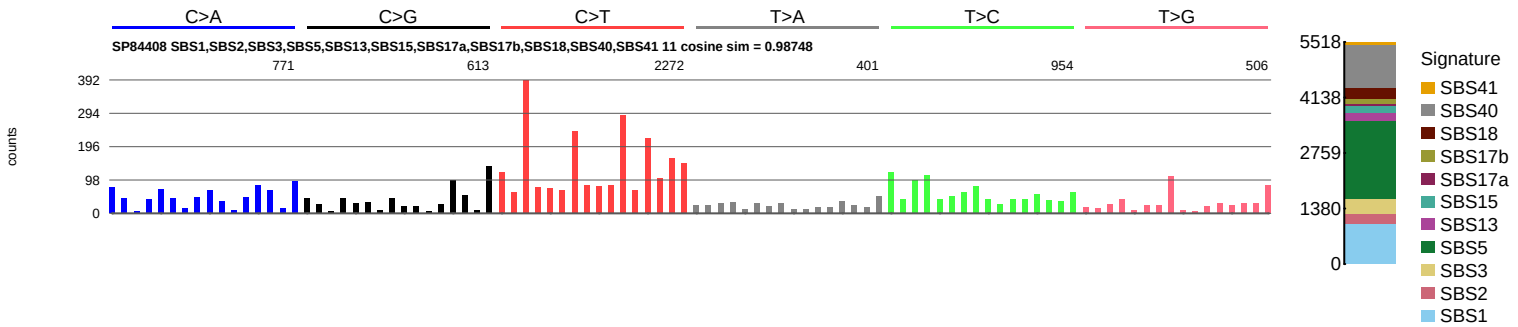
SBS2
SBS1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SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.98748
771
613
2272
401
954
506
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS41
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SBS18
SBS17b
SBS17a
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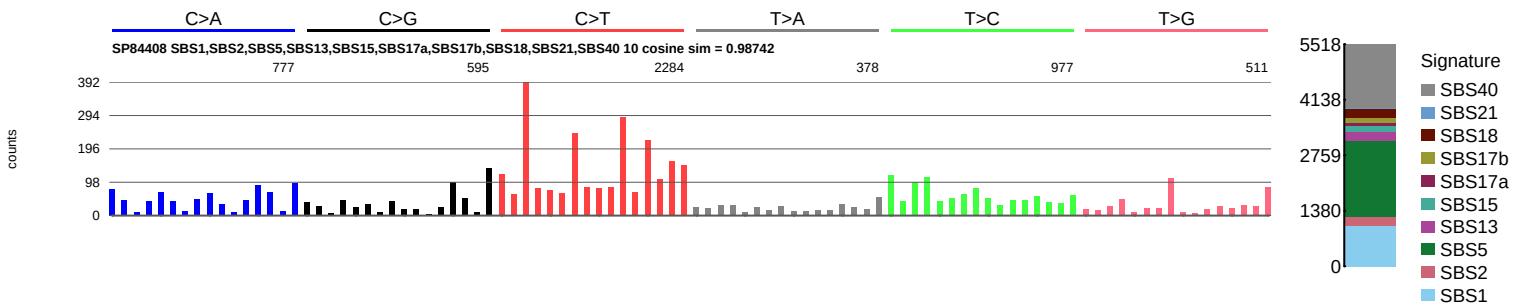
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98742
777
595
2284
378
977
511
counts
392
294
196
98
0
5518
4138
2759
1380
0
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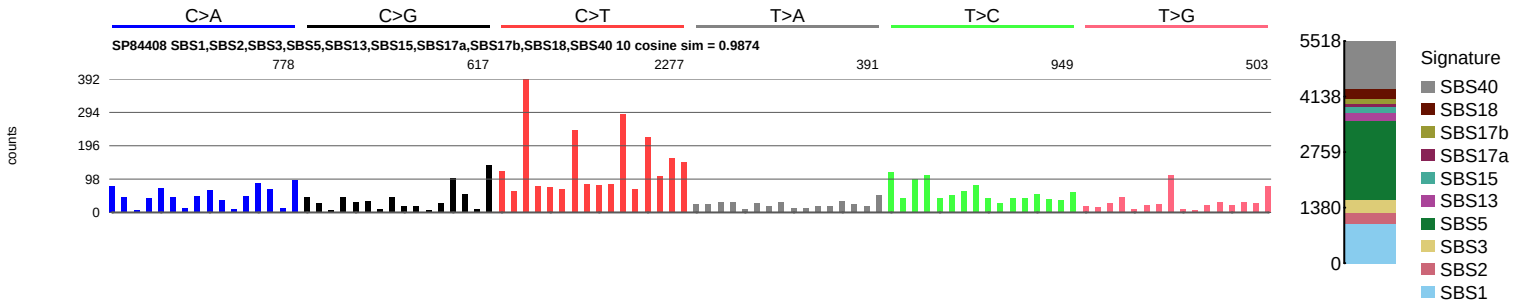
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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T>G
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778
617
2277
391
949
503
counts
392
294
196
98
0
5518
4138
2759
1380
0
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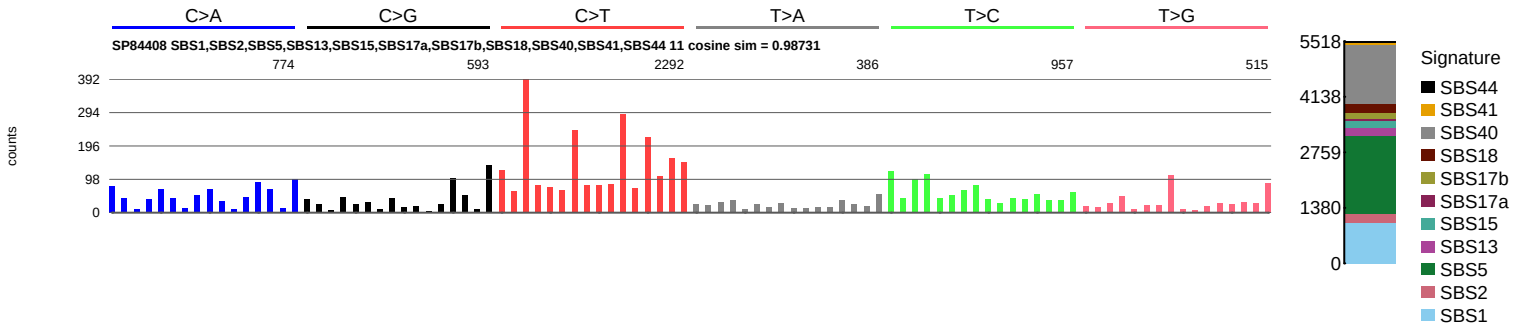
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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774
593
2292
386
957
515
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
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SBS18
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SBS15
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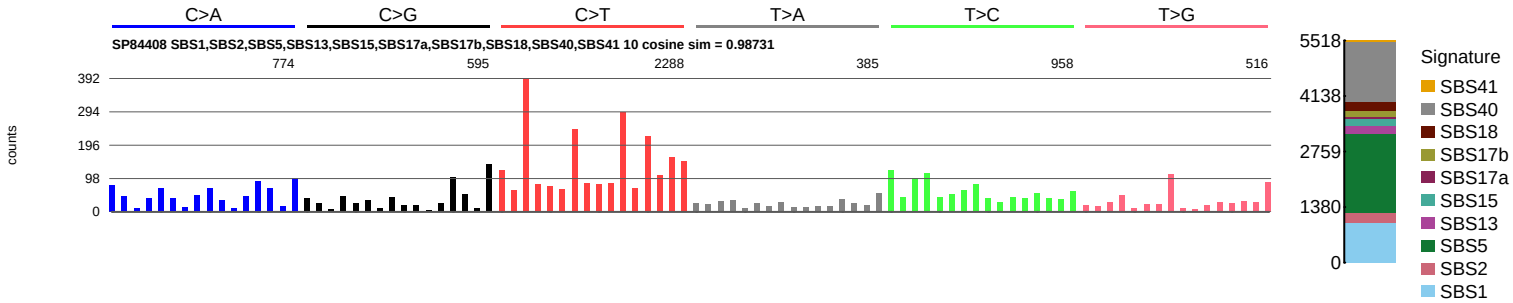
SBS5
SBS2
SBS1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T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98731
774
595
2288
385
958
516
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
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SBS40
SBS18
SBS17b
SBS17a
SBS15
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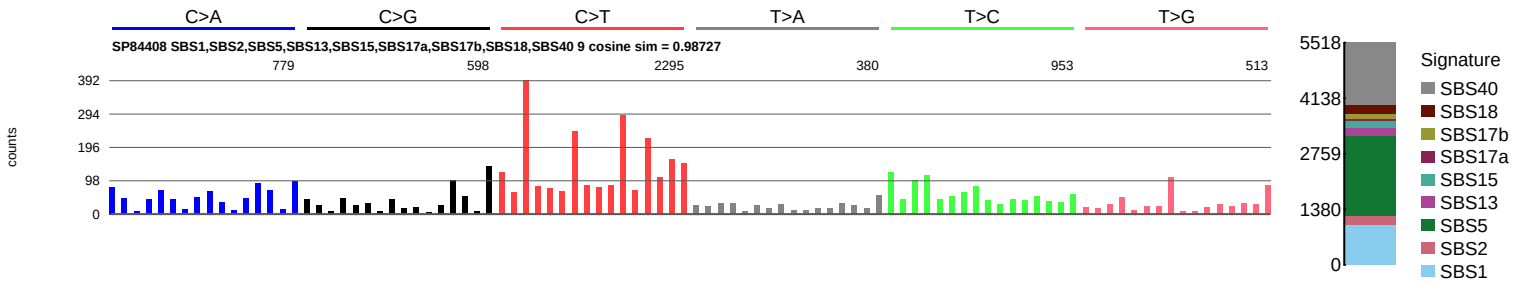
SBS5
SBS2
SBS1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T>C
T>G
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779
598
2295
380
953
513
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS40
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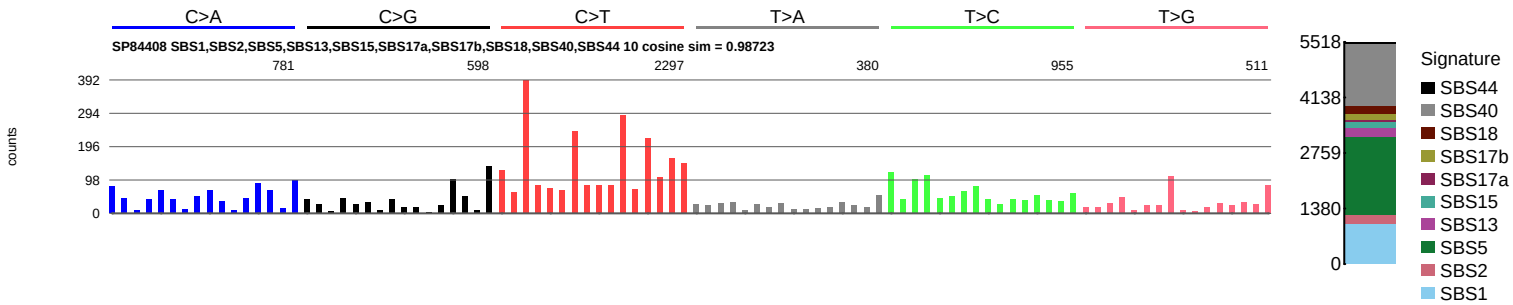
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98723
781
598
2297
380
955
511
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
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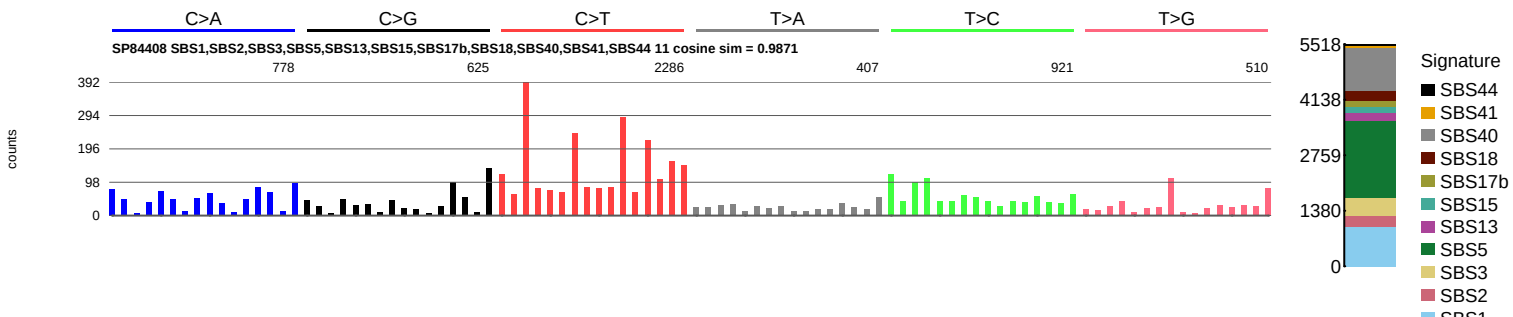
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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T>C
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778
625
2286
407
921
510
counts
392
294
196
98
0
5518
4138
2759
1380
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SBS18
SBS17b
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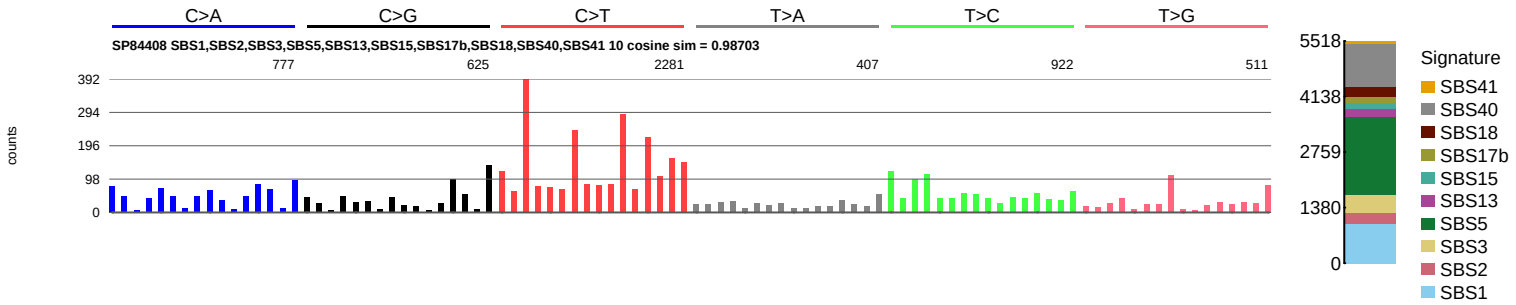
SBS3
SBS2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C>T
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T>G
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777
625
2281
407
922
511
counts
392
294
196
98
0
5518
4138
2759
1380
0
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SBS40
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SBS17b
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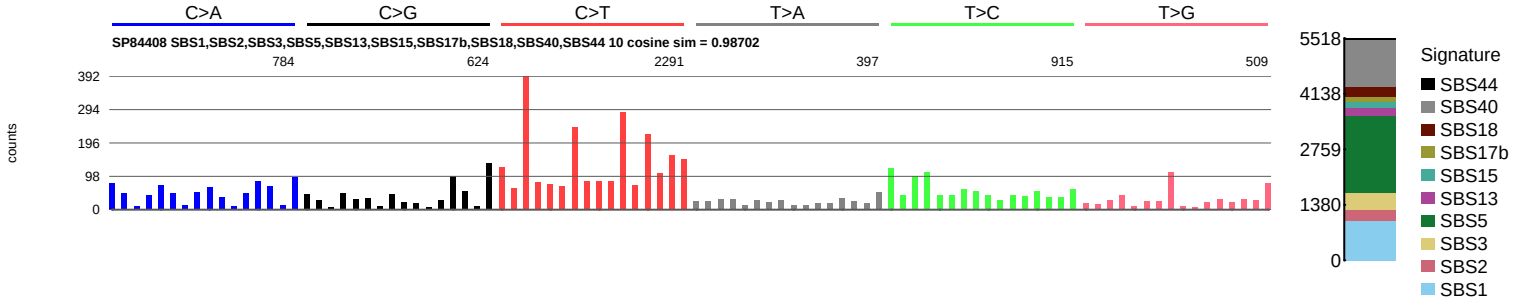
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98702
784
624
2291
397
915
509
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
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SBS40
SBS18
SBS17b
SBS15
SBS13
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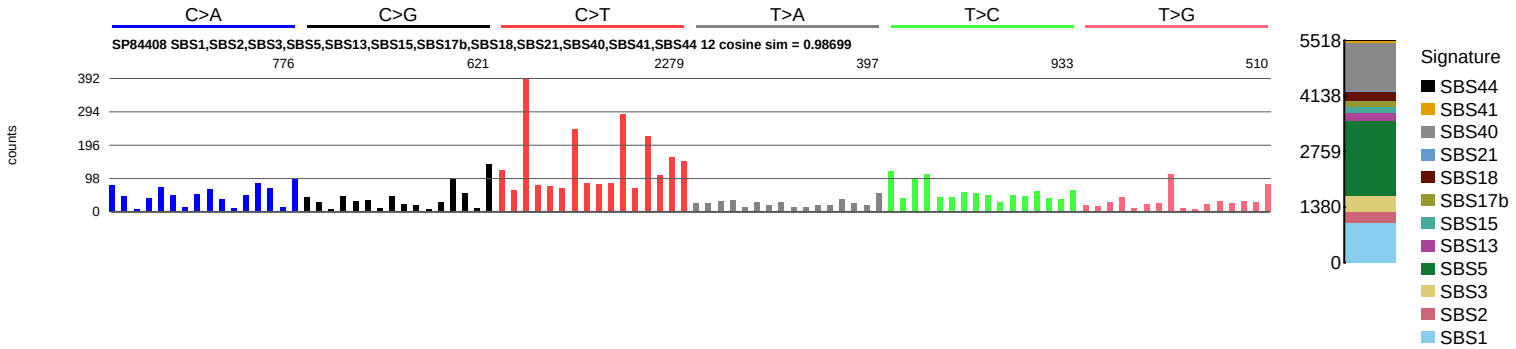
SBS2
SBS1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C>G
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T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98699
776
621
2279
397
933
510
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
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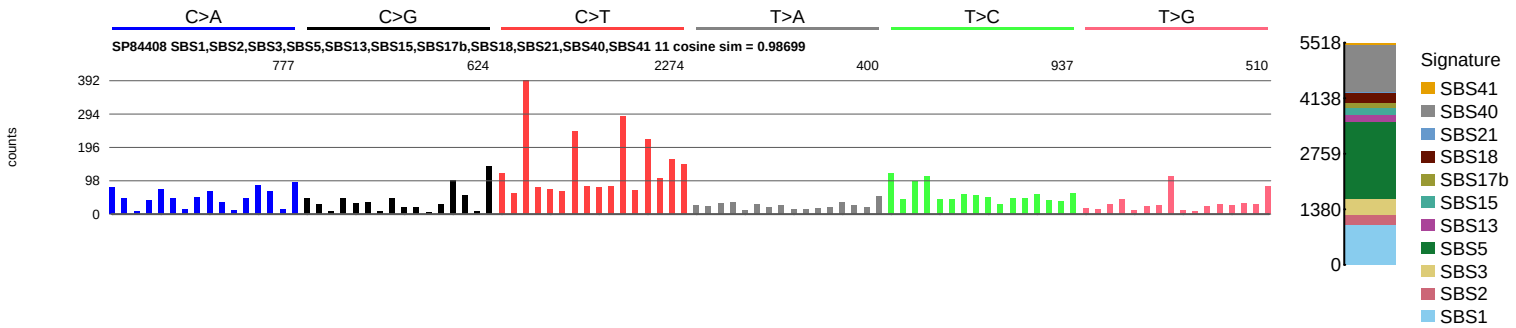
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98699
777
624
2274
400
937
510
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
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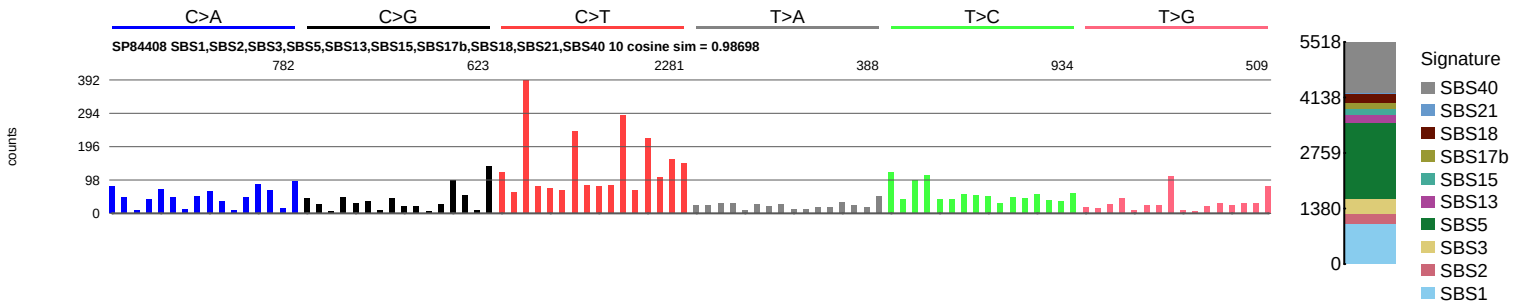
SBS1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C>G
C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98698
782
623
2281
388
934
509
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS40
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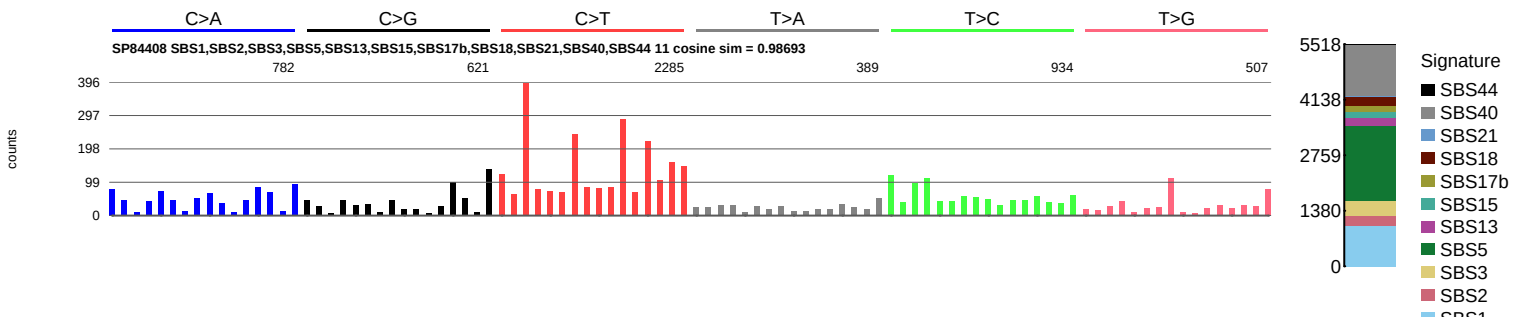
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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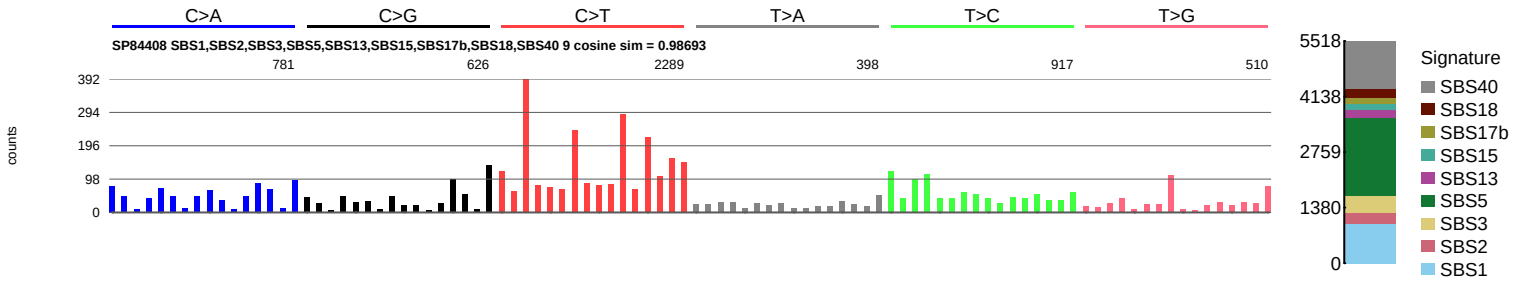
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98693
782
621
2285
389
934
507
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2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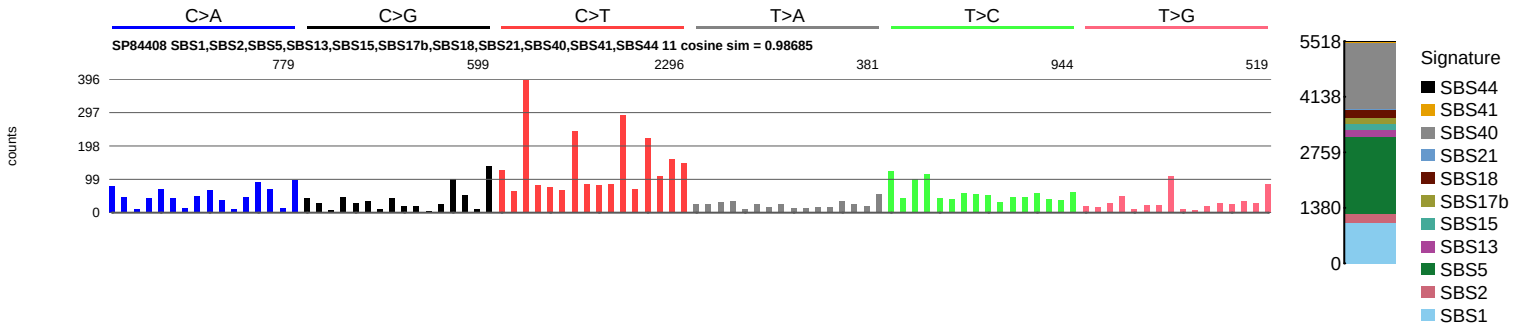
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 9 cosine sim = 0.98693
781
626
2289
398
917
510
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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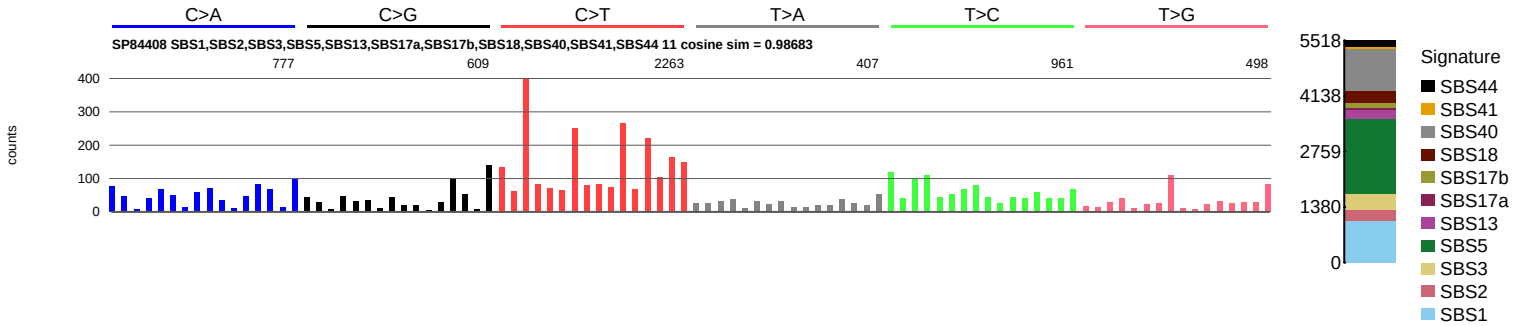
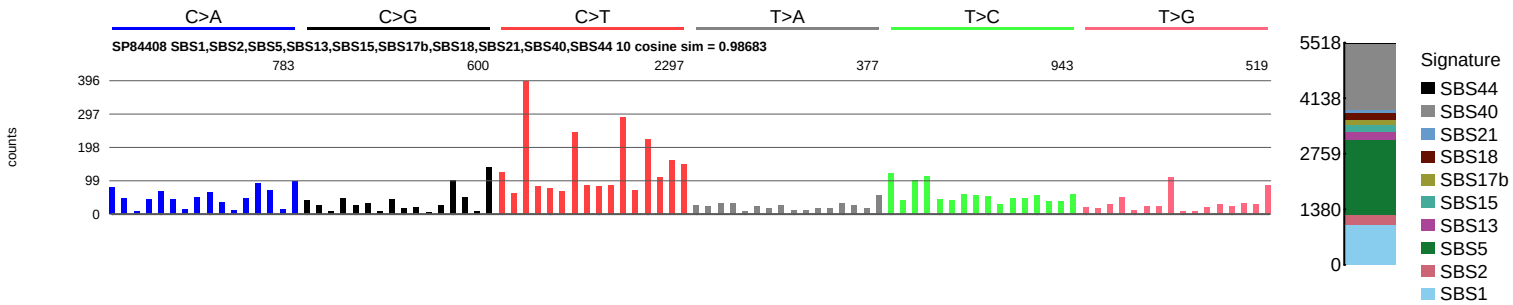
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98685
779
599
2296
381
944
519
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98683
777
609
2263
407
961
498
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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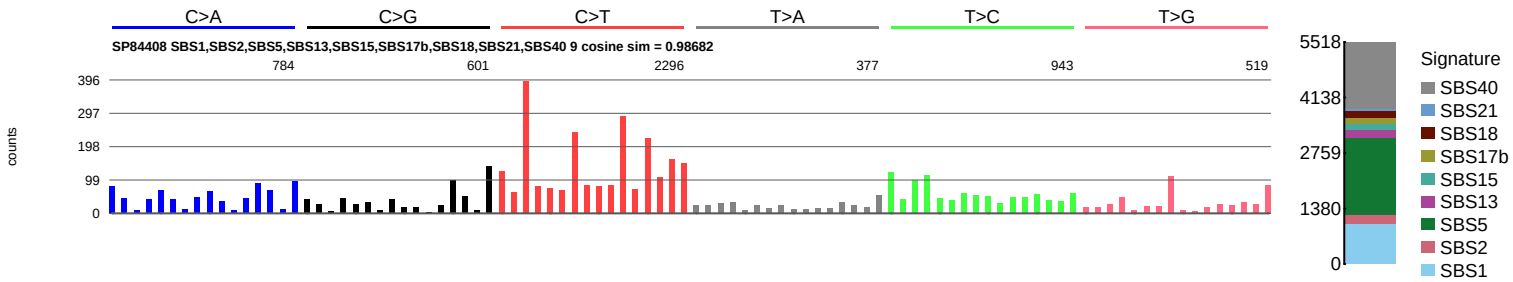
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98683
783
600
2297
377
943
519
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98682
784
601
2296
377
943
519
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
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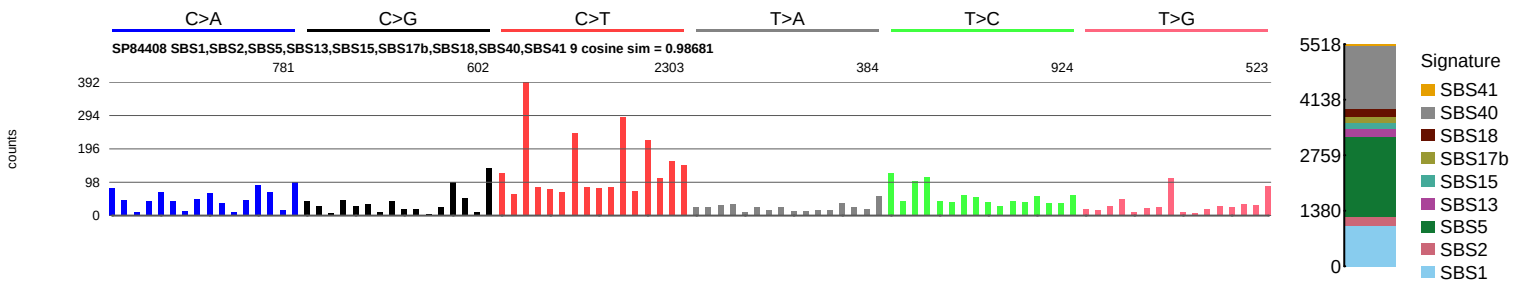
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98681
781
602
2303
384
924
523
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
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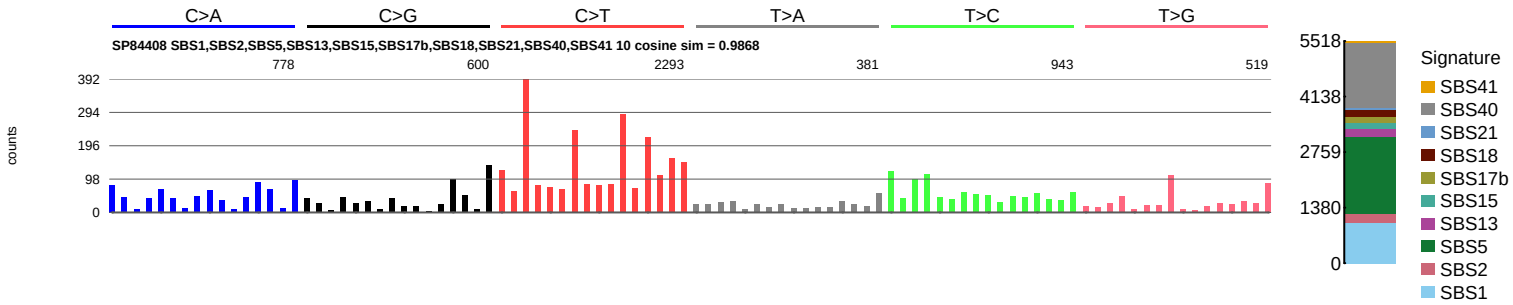
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.9868
778
600
2293
381
943
519
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
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SBS18
SBS17b
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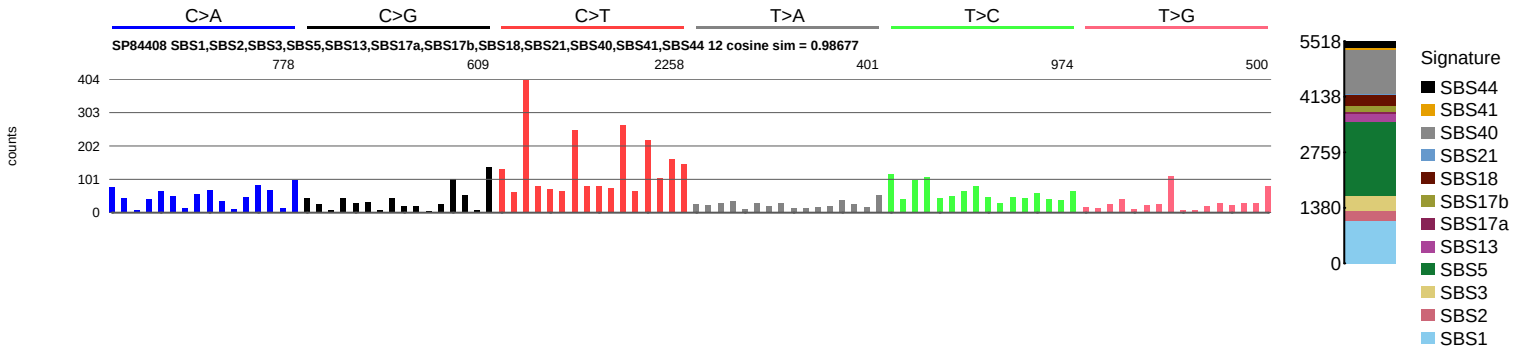
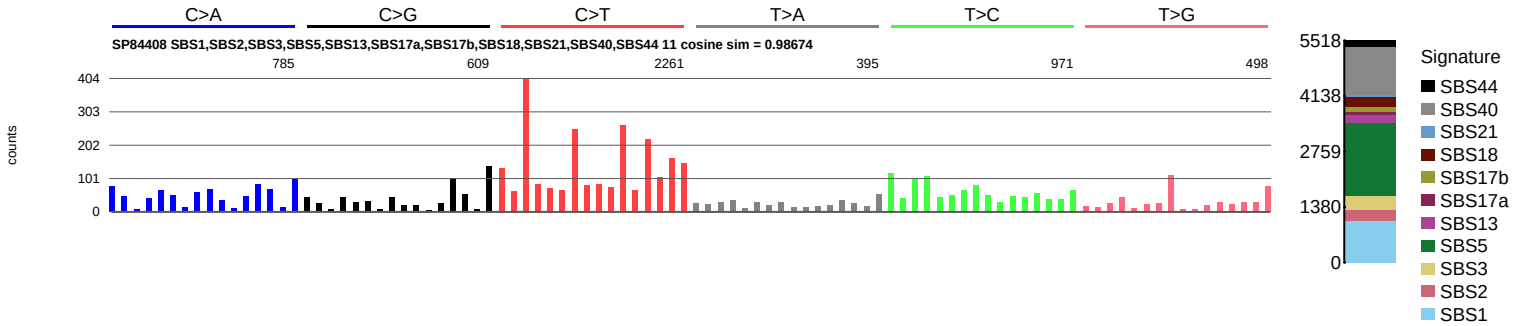
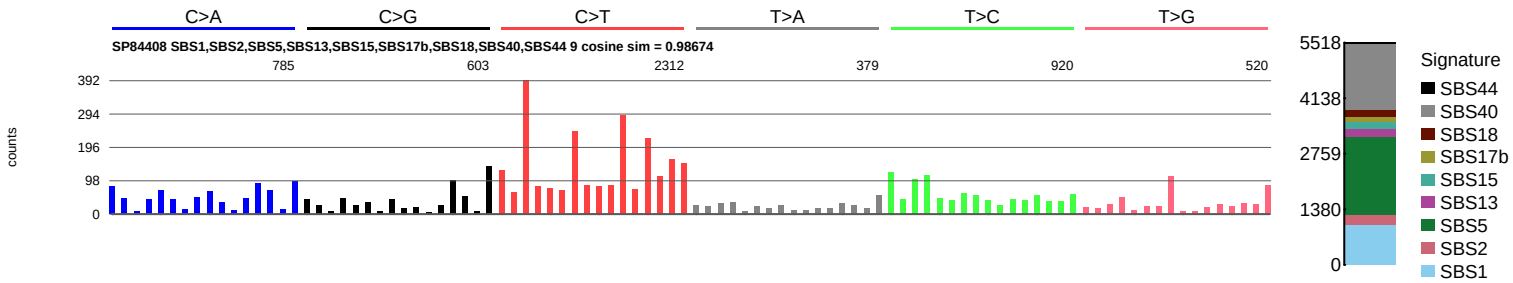
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98677
778
609
2258
401
974
500
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98674
785
609
2261
395
971
498
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98674
785
603
2312
379
920
520
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
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SBS17b
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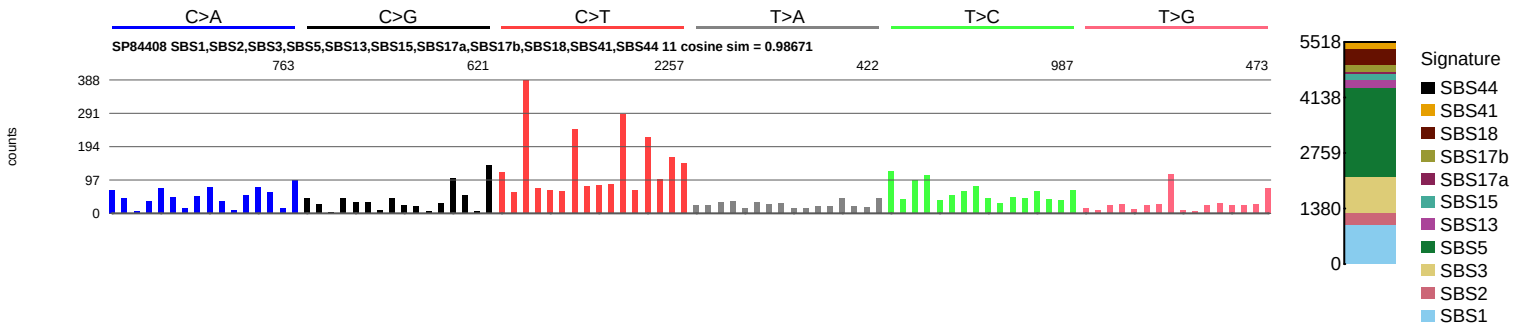
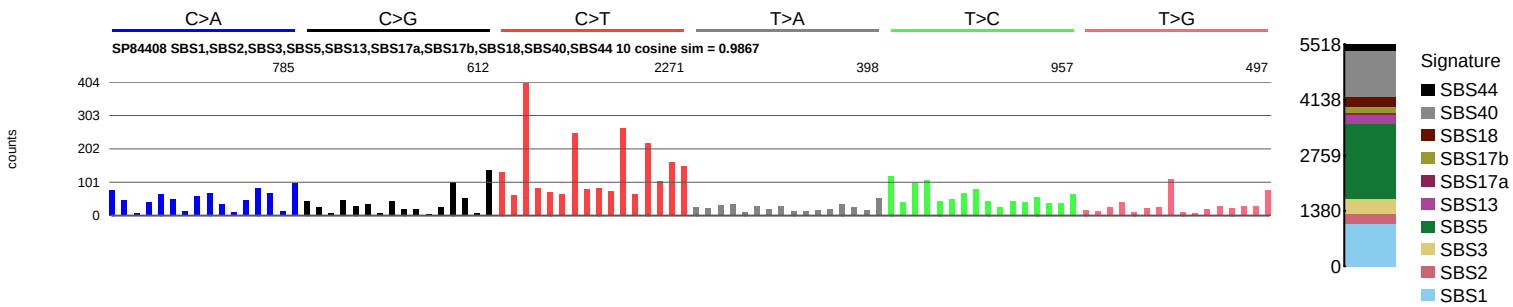
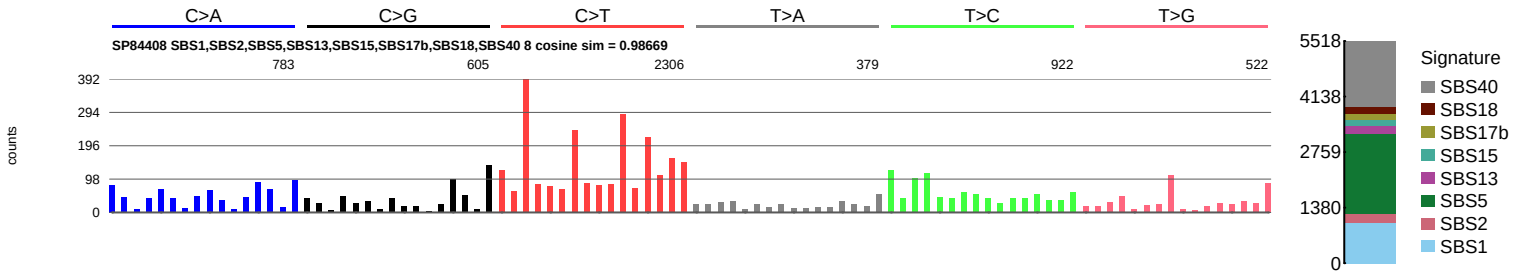
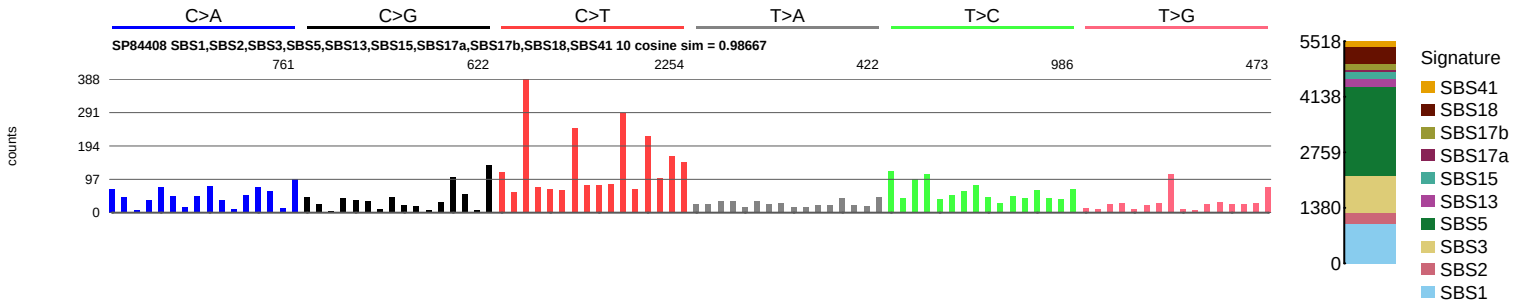
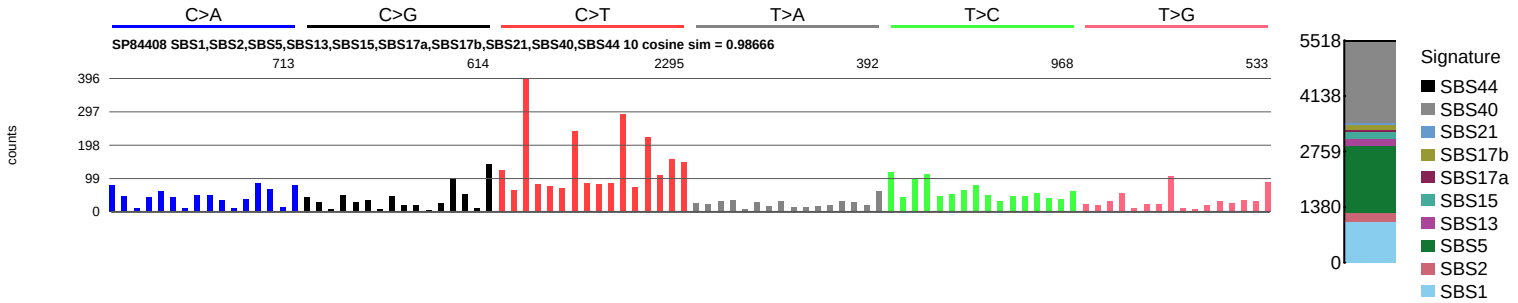
SBS5
SBS2
SBS1

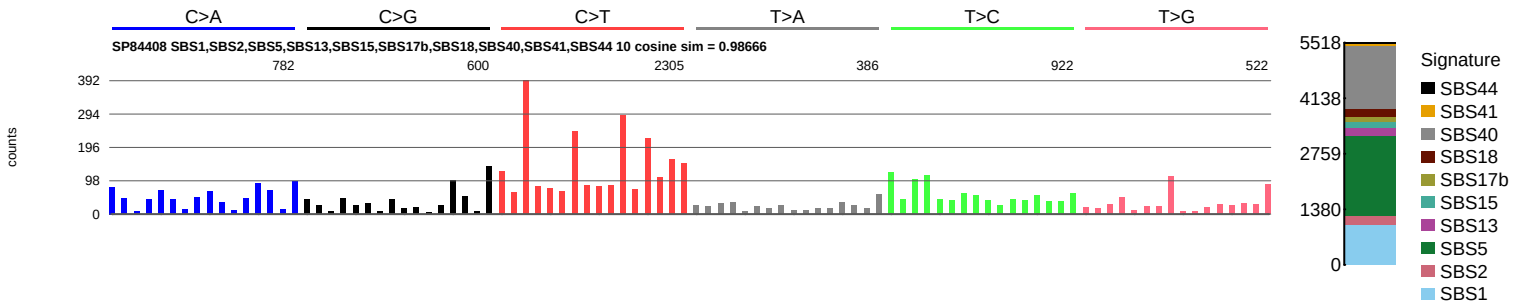

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.98671
763
621
2257
422
987
473
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

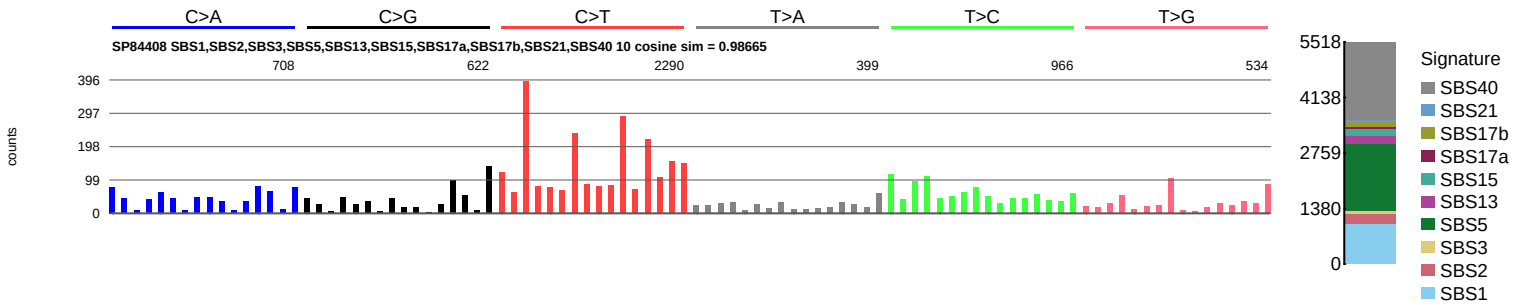

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.9867
785
612
2271
398
957
497
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

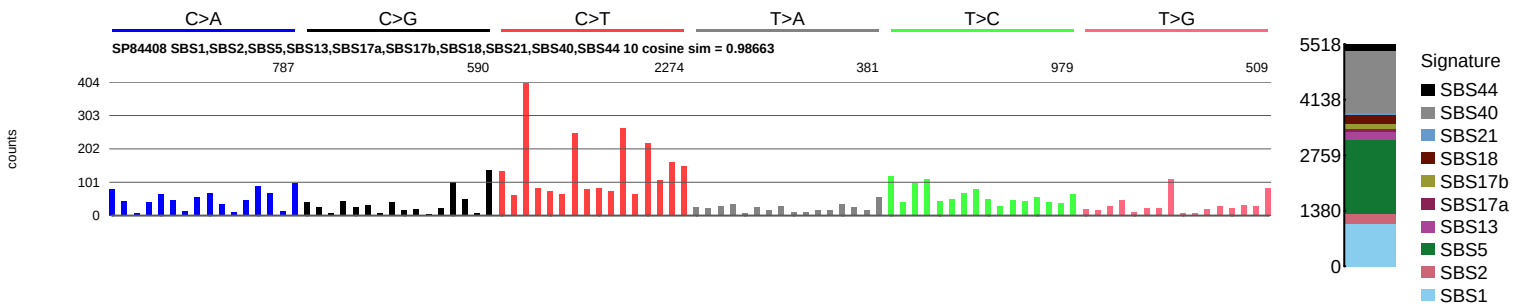

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98669
783
605
2306
379
922
522
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98667
761
622
2254
422
986
473
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98666
713
614
2295
392
968
533
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98666
782
600
2305
386
922
522
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 10 cosine sim = 0.98665
708
622
2290
399
966
534
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98663
787
590
2274
381
979
509
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
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SBS17b
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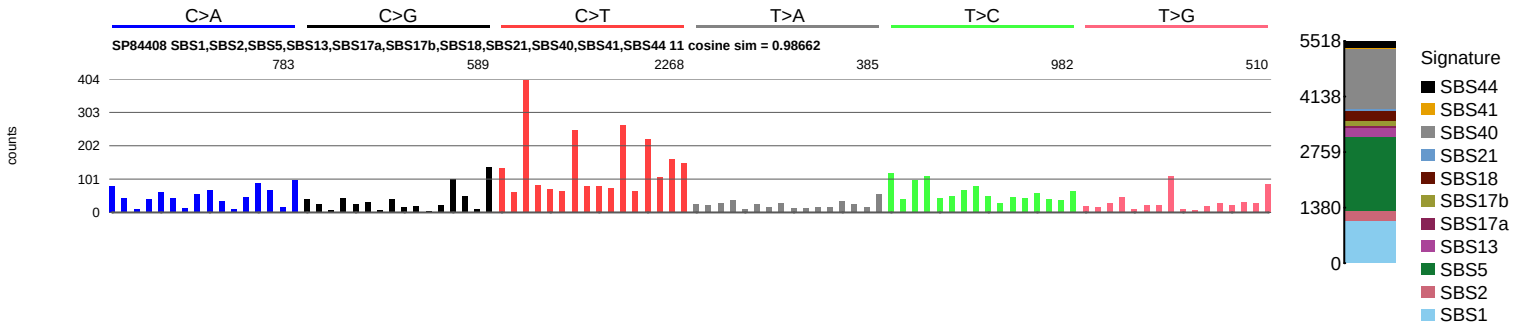
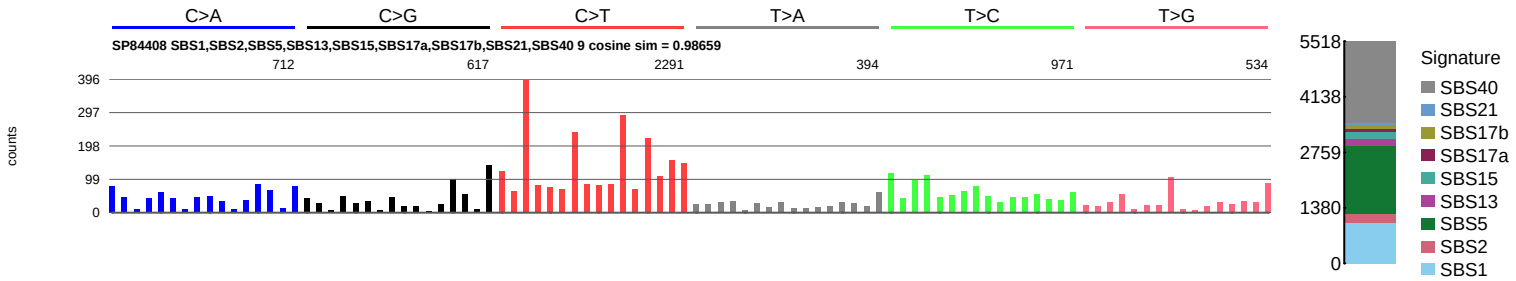
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98662
783
589
2268
385
982
510
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98659
712
617
2291
394
971
534
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS17a
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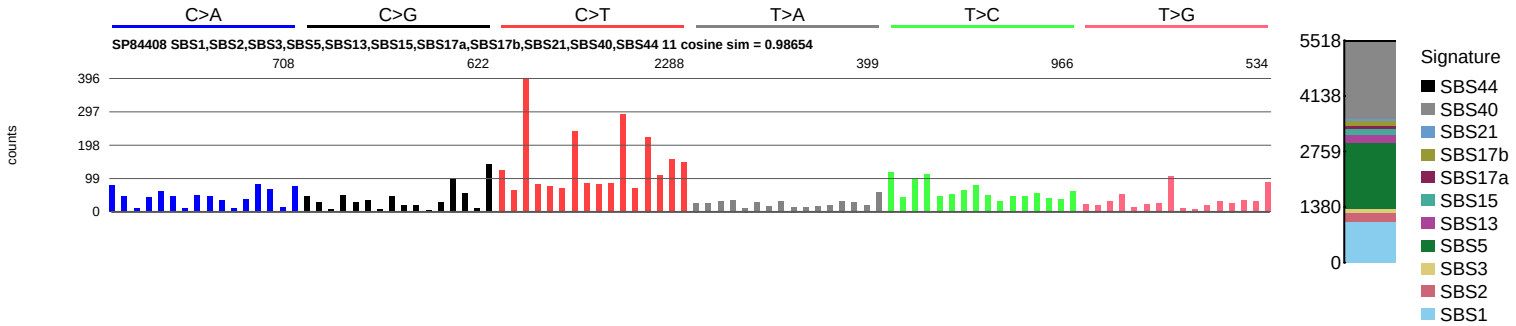
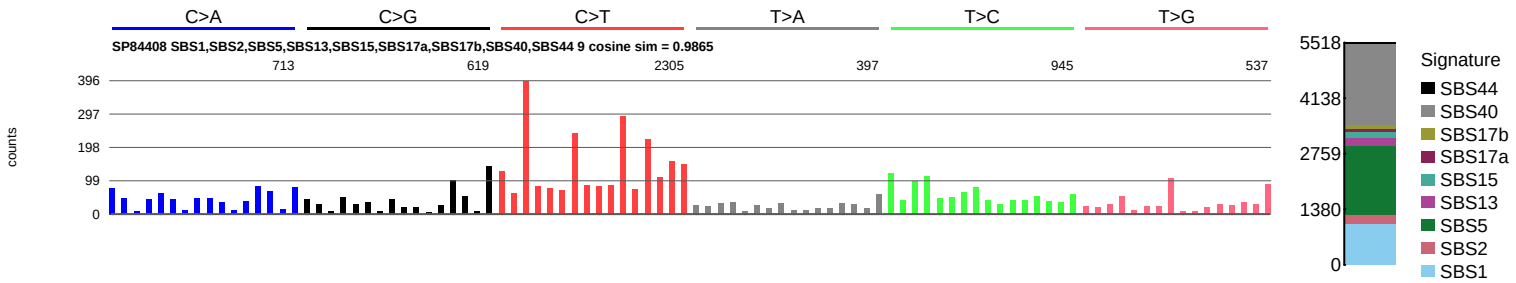
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 11 cosine sim = 0.98654
708
622
2288
399
966
534
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.9865
713
619
2305
397
945
537
counts
396
297
198
99
0
5518
4138
2759
1380
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Signature
SBS44
SBS40
SBS17b
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SBS15
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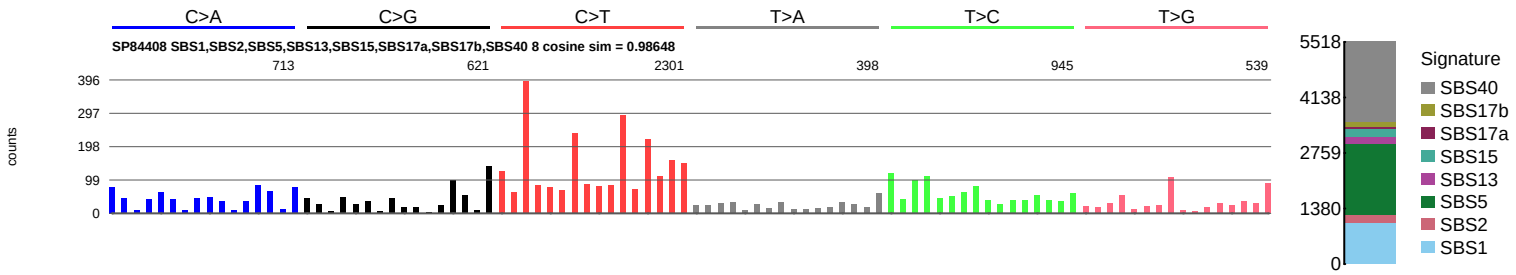
SBS1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C>G
C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98648
713
621
2301
398
945
539
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
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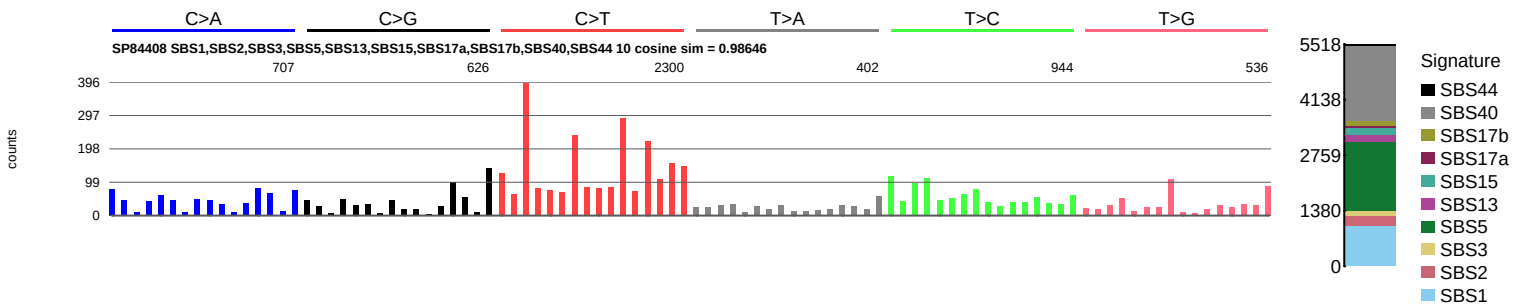
SBS2
SBS1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C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 10 cosine sim = 0.98646
707
626
2300
402
944
536
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
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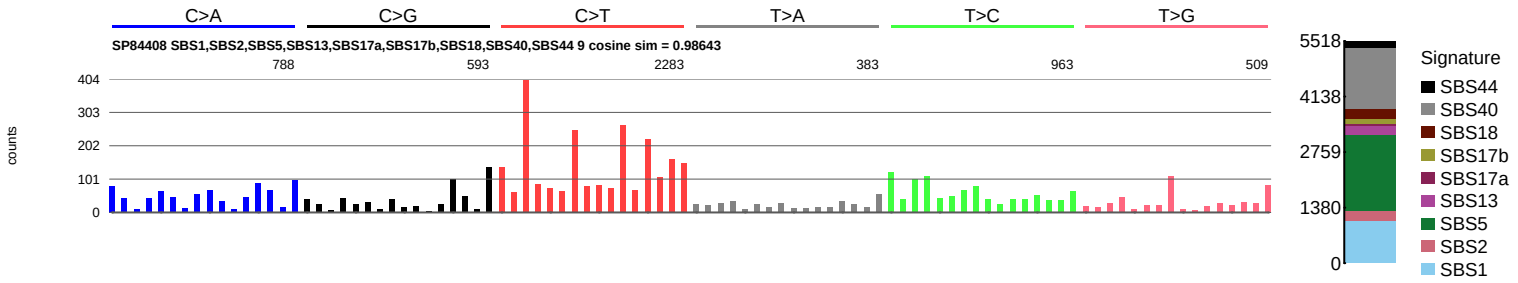
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98643
788
593
2283
383
963
509
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
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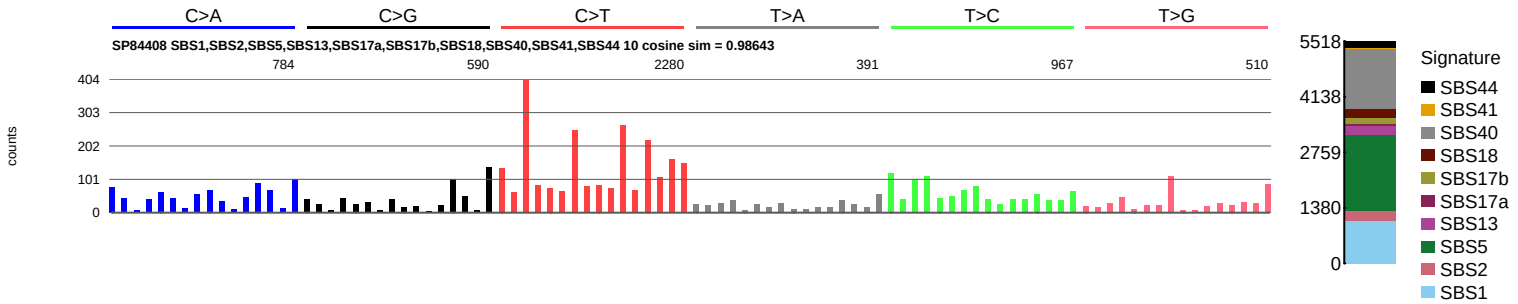
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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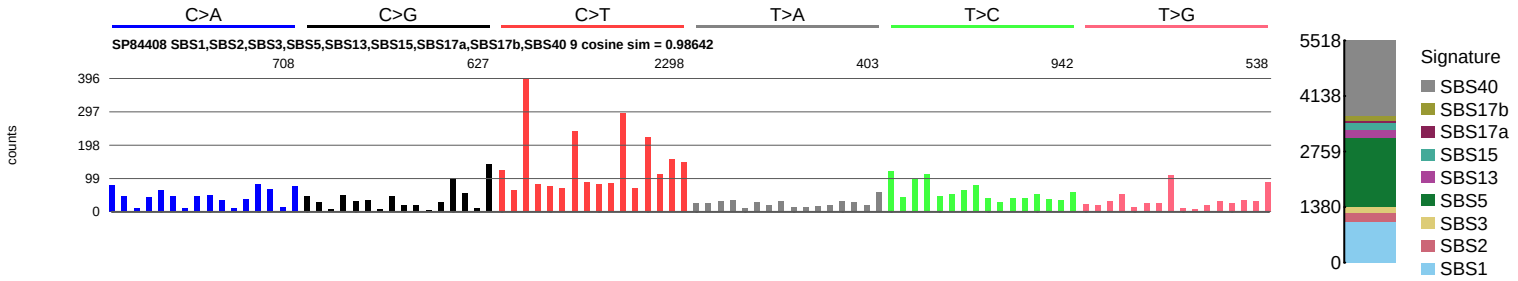
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98643
784
590
2280
391
967
510
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 9 cosine sim = 0.98642
708
627
2298
403
942
538
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
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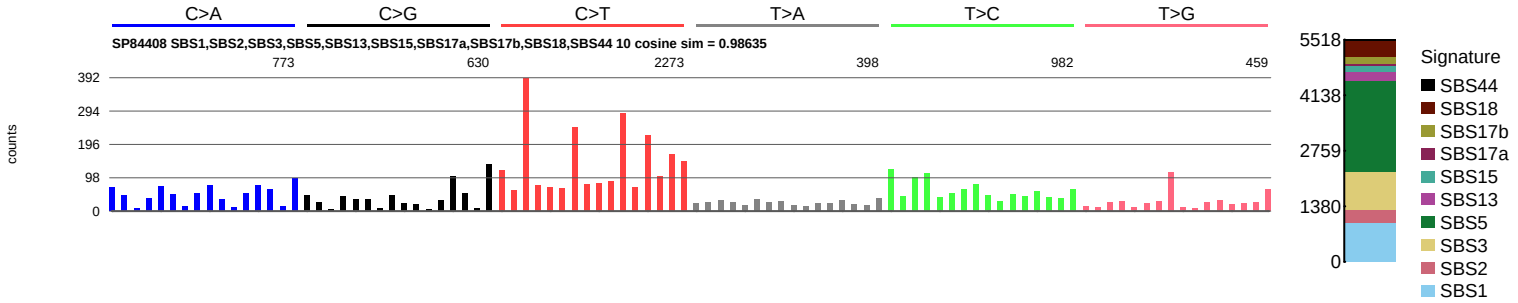
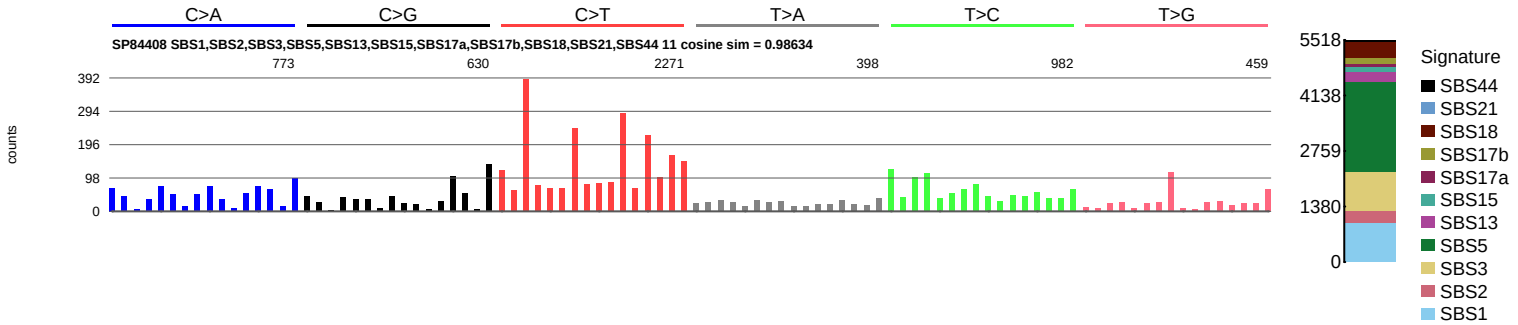
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98635
773
630
2273
398
982
459
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98634
773
630
2271
398
982
459
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
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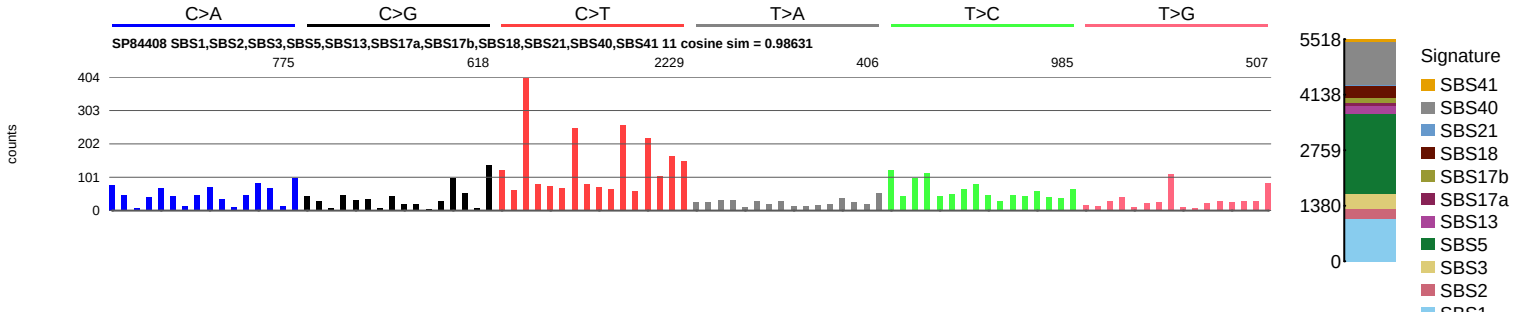
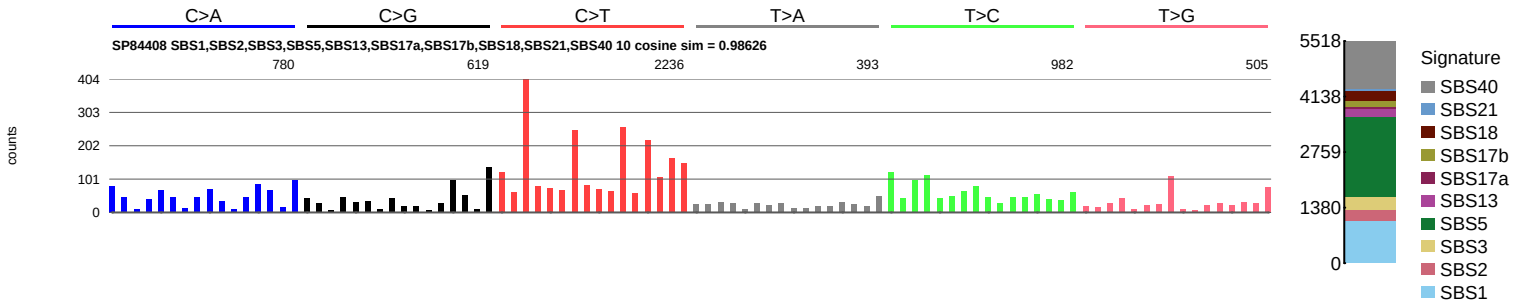
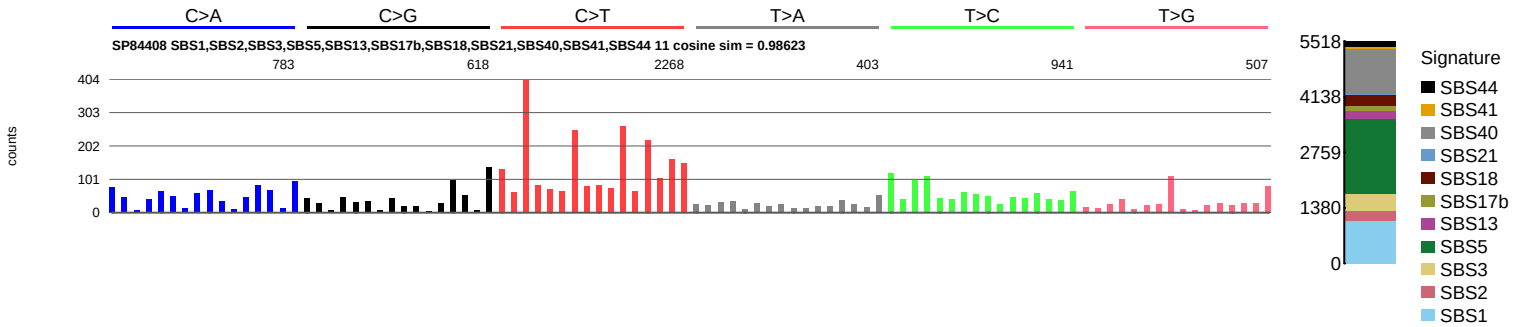
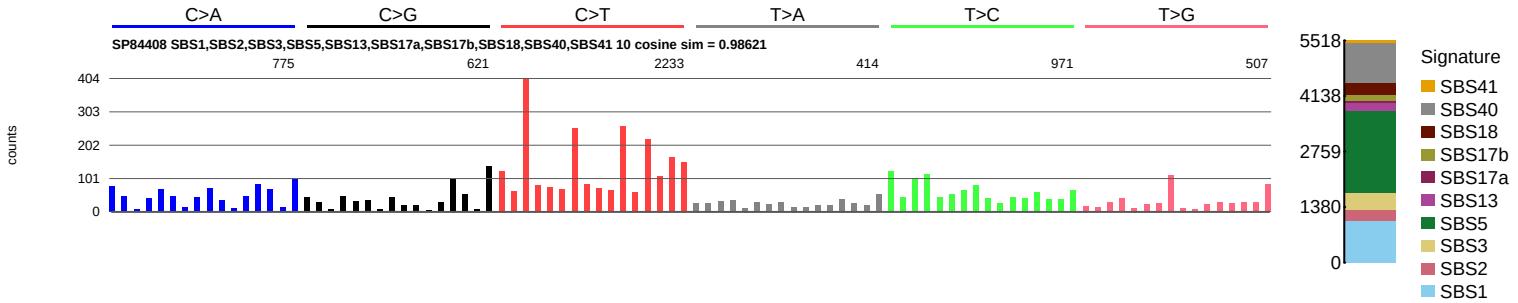
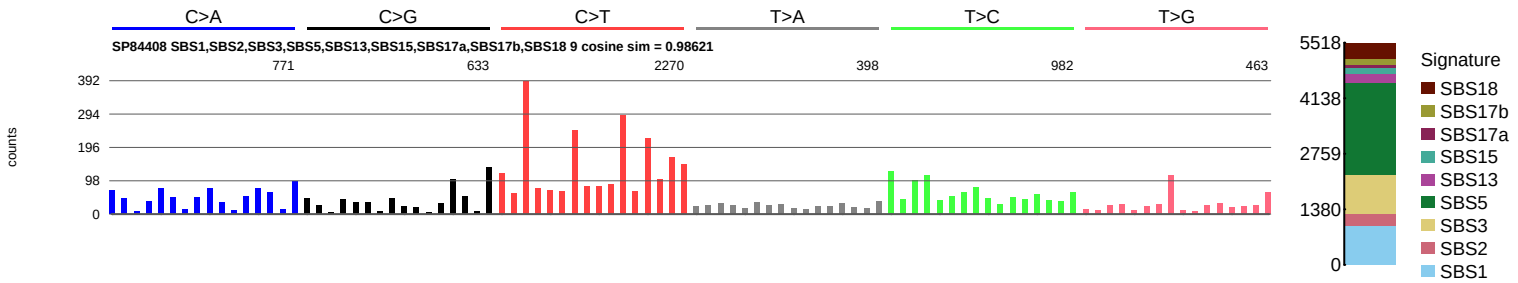
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98631
775
618
2229
406
985
507
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98626
780
619
2236
393
982
505
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98623
783
618
2268
403
941
507
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98621
775
621
2233
414
971
507
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98621
771
633
2270
398
982
463
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS18
SBS17b
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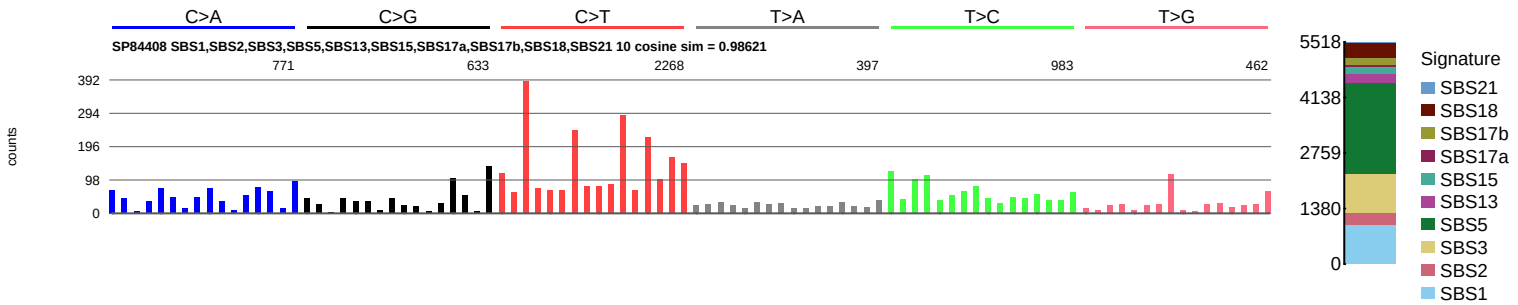
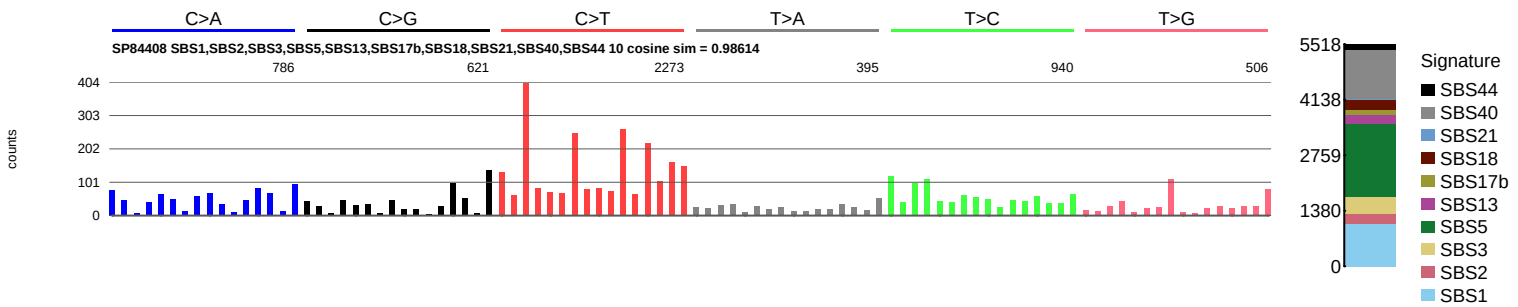
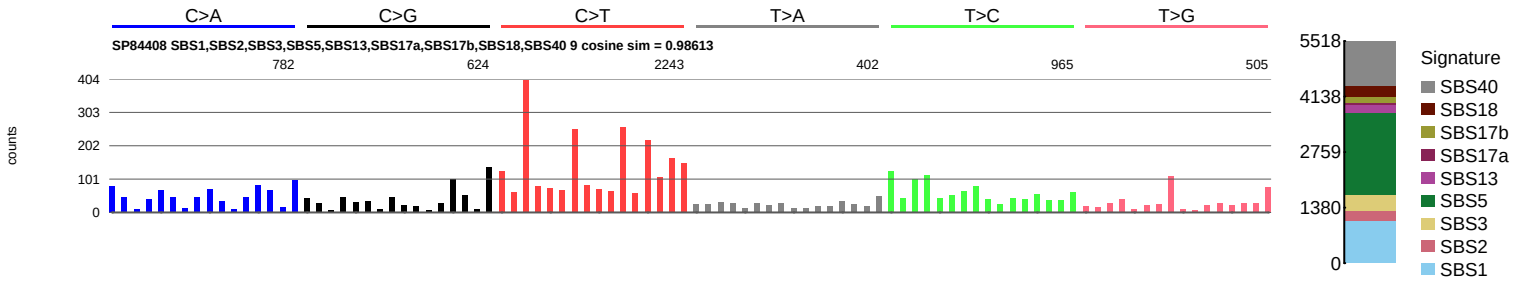
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98621
771
633
2268
397
983
462
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98614
786
621
2273
395
940
506
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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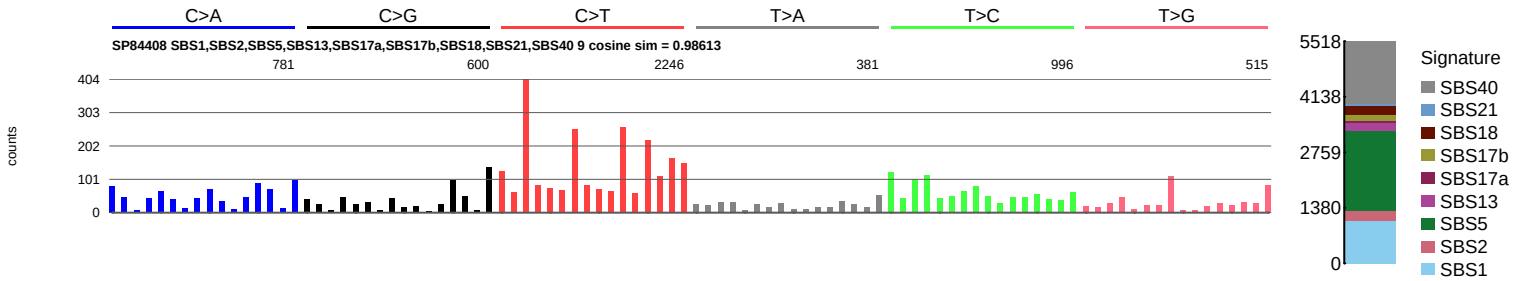
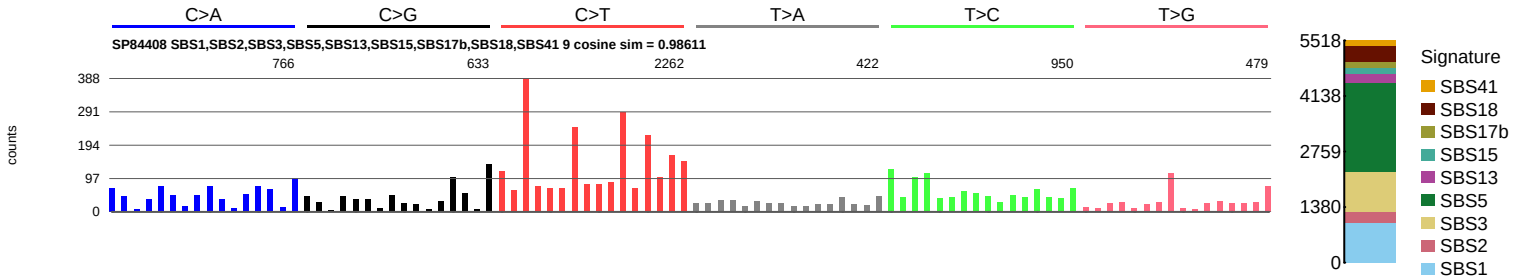
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98613
782
624
2243
402
965
505
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98613
781
600
2246
381
996
515
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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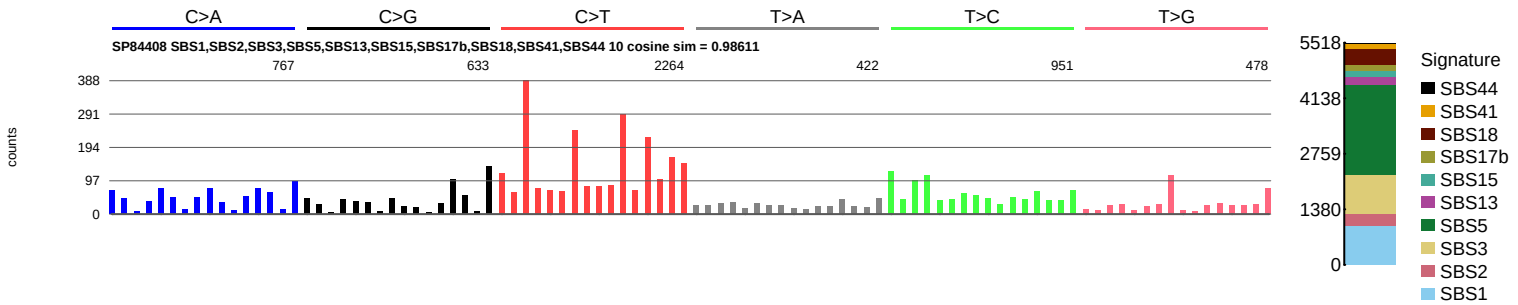
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98611
766
633
2262
422
950
479
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98611
767
633
2264
422
951
478
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
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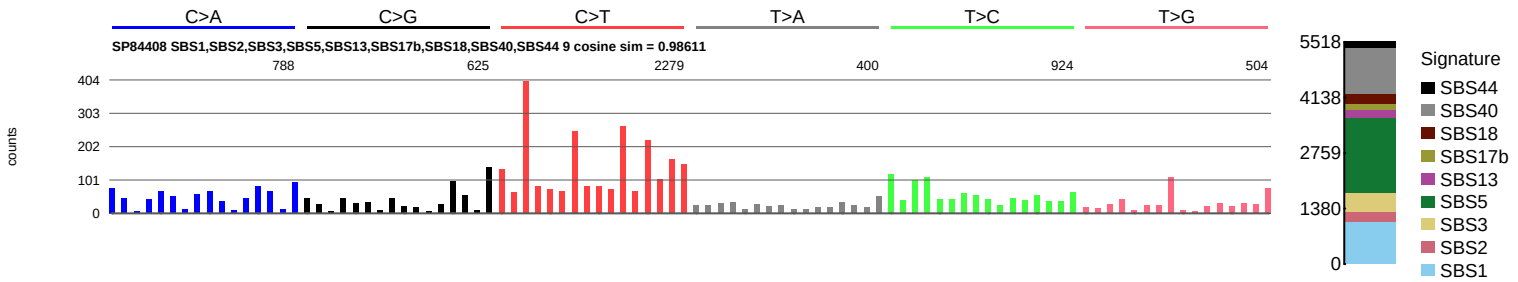
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98611
788
625
2279
400
924
504
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
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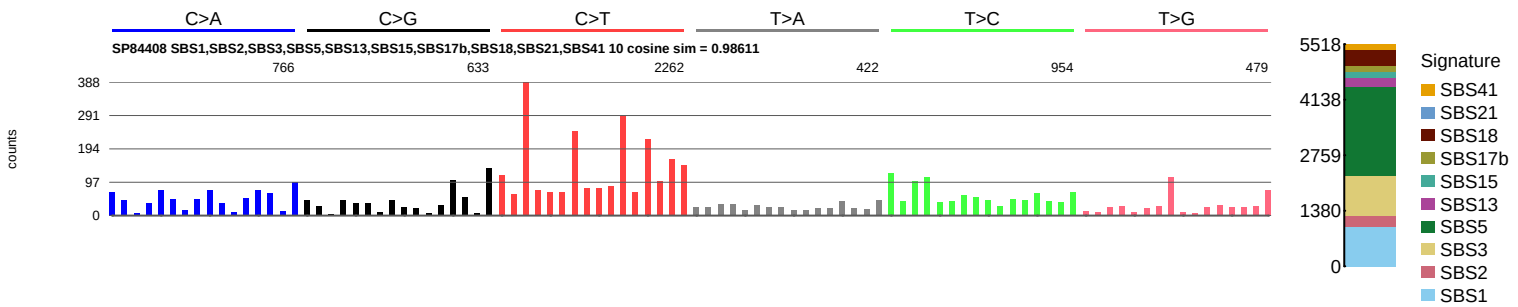
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98611
766
633
2262
422
954
479
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
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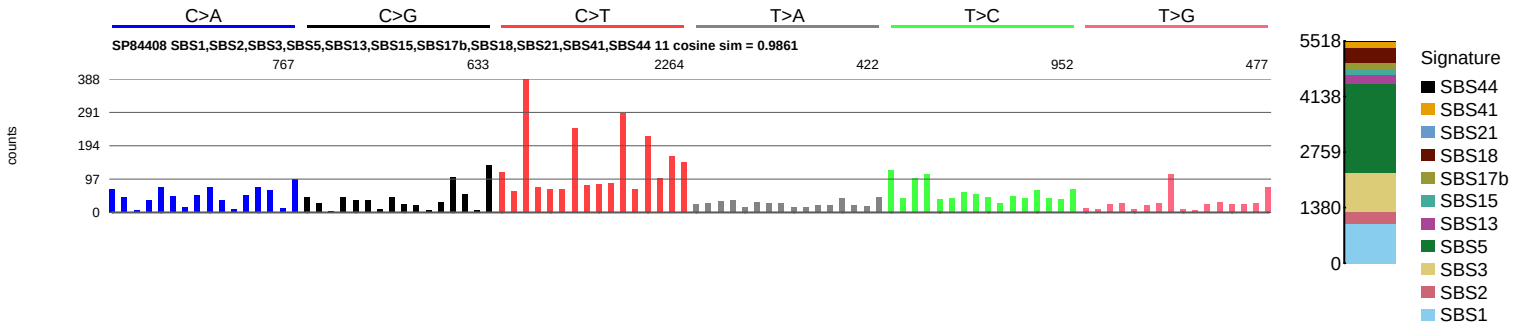
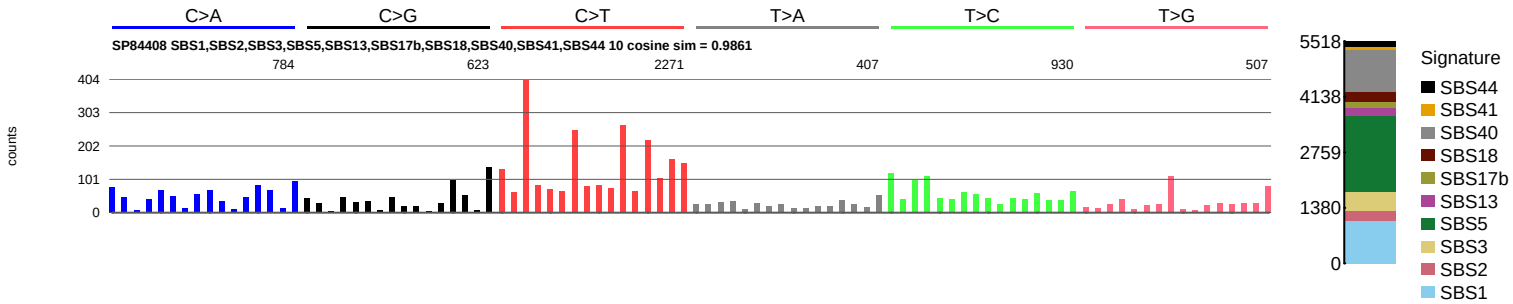
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.9861
767
633
2264
422
952
477
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.9861
784
623
2271
407
930
507
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
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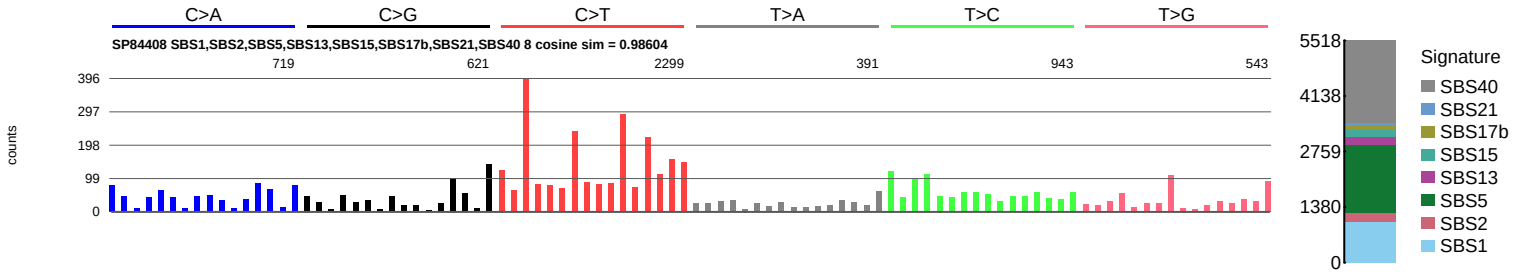
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40 8 cosine sim = 0.98604
719
621
2299
391
943
543
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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SBS21
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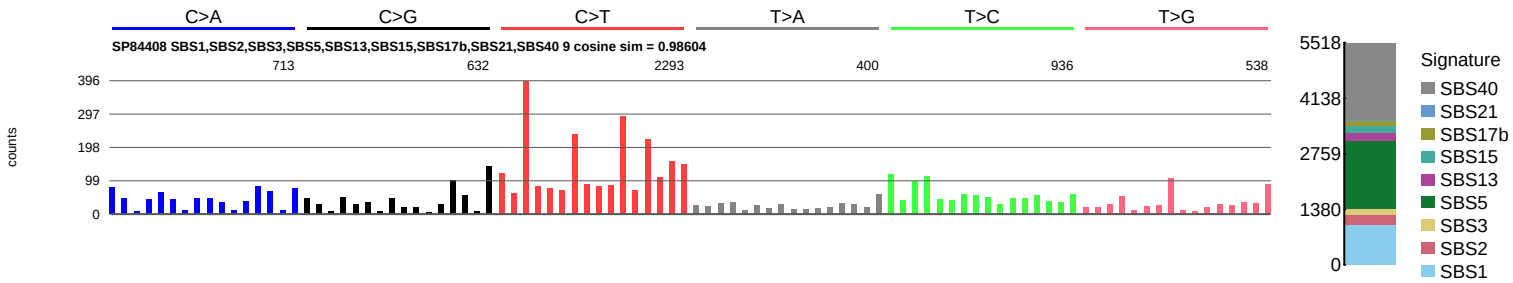
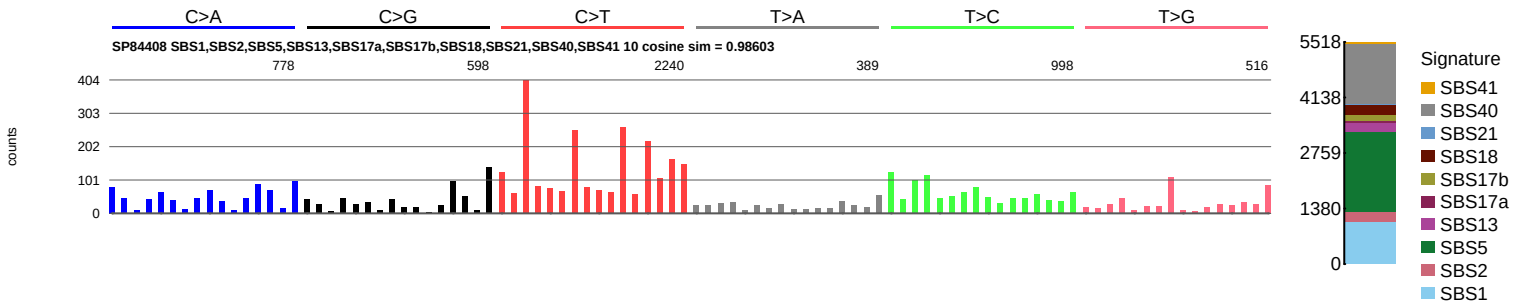
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40 9 cosine sim = 0.98604
713
632
2293
400
936
538
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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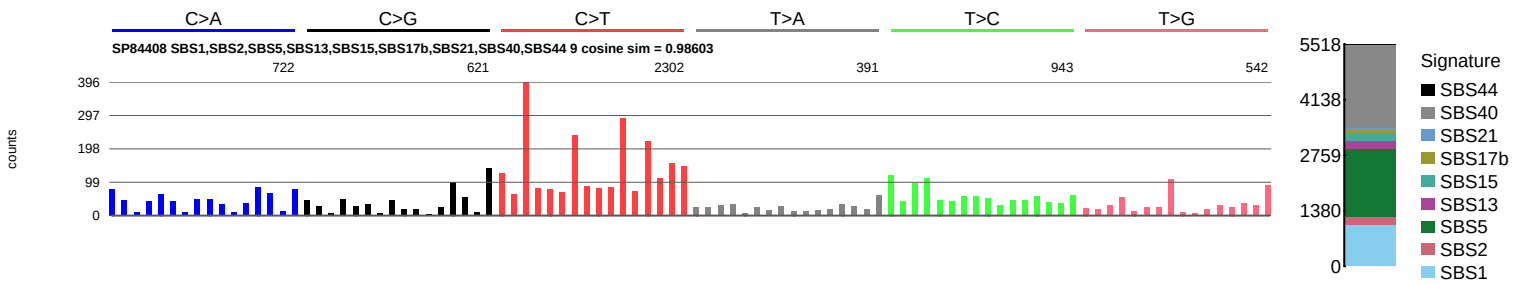
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98603
778
598
2240
389
998
516
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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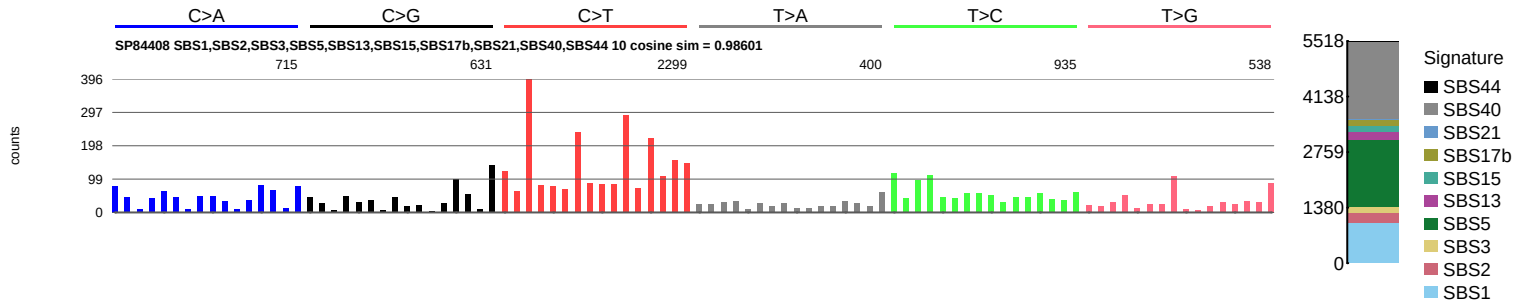
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98603
722
621
2302
391
943
542
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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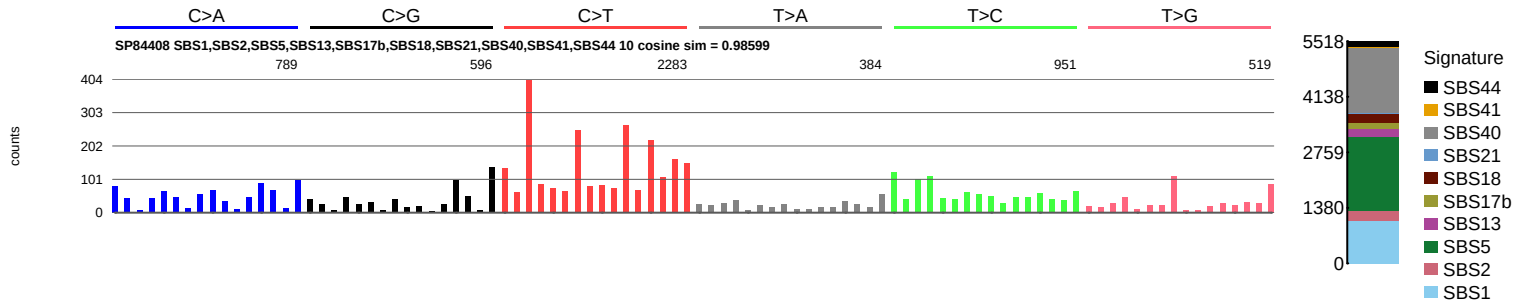
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98601
715
631
2299
400
935
538
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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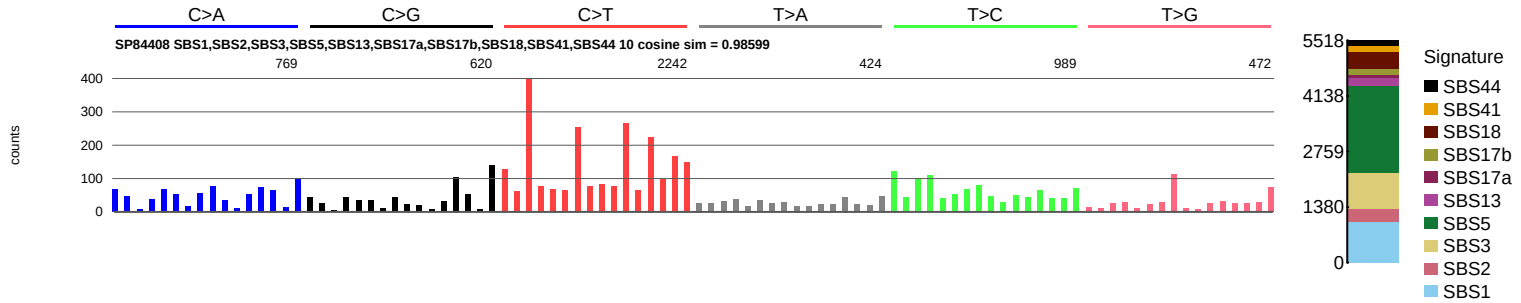
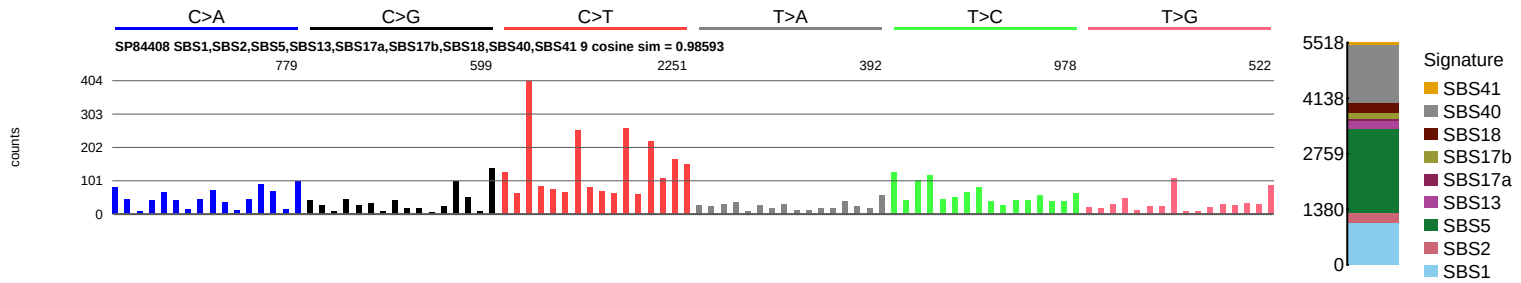
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98599
789
596
2283
384
951
519
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98599
769
620
2242
424
989
472
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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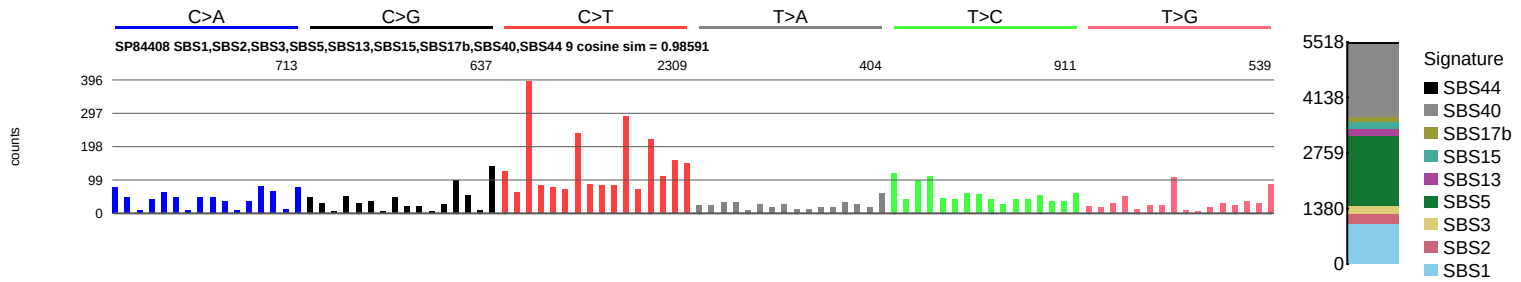
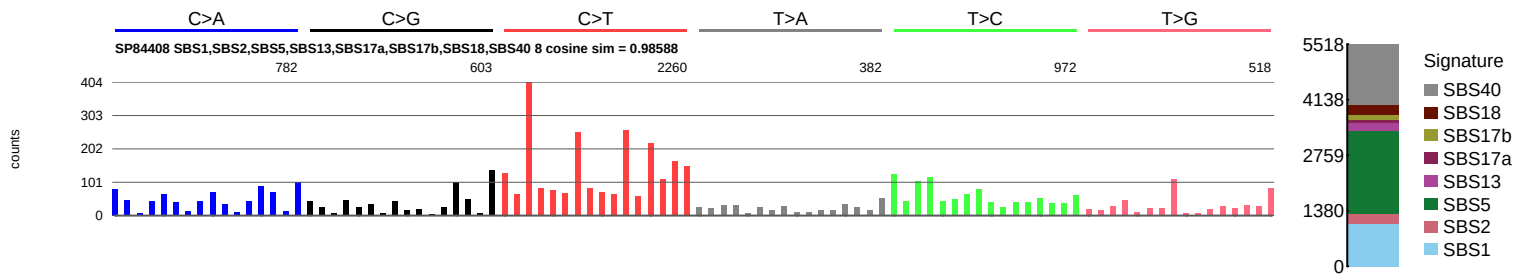
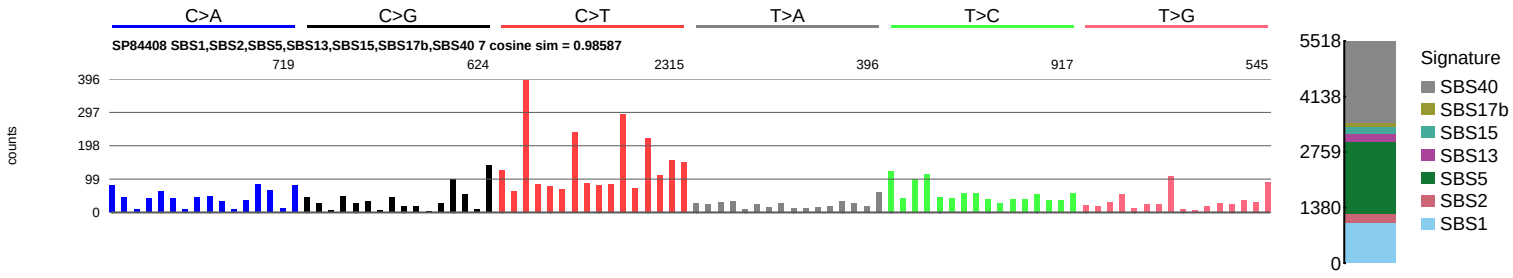
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98593
779
599
2251
392
978
522
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS44 9 cosine sim = 0.98591
713
637
2309
404
911
539
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98588
782
603
2260
382
972
518
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40 7 cosine sim = 0.98587
719
624
2315
396
917
545
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
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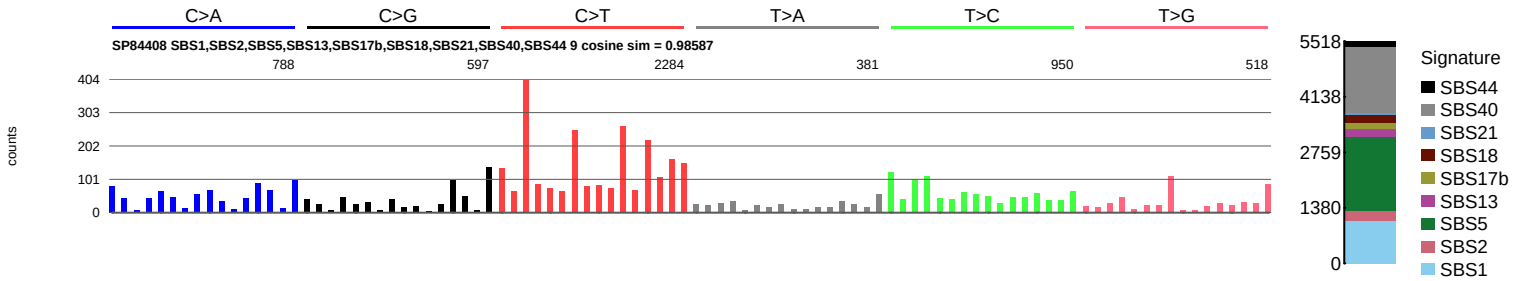
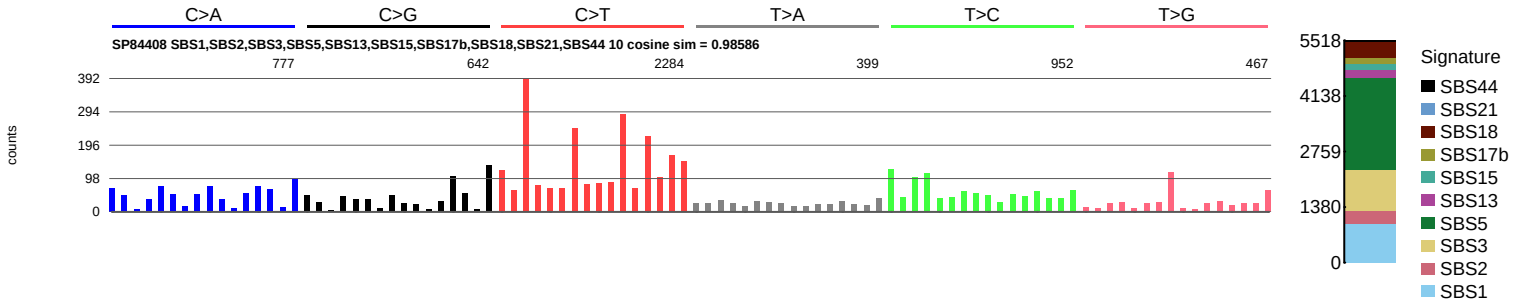
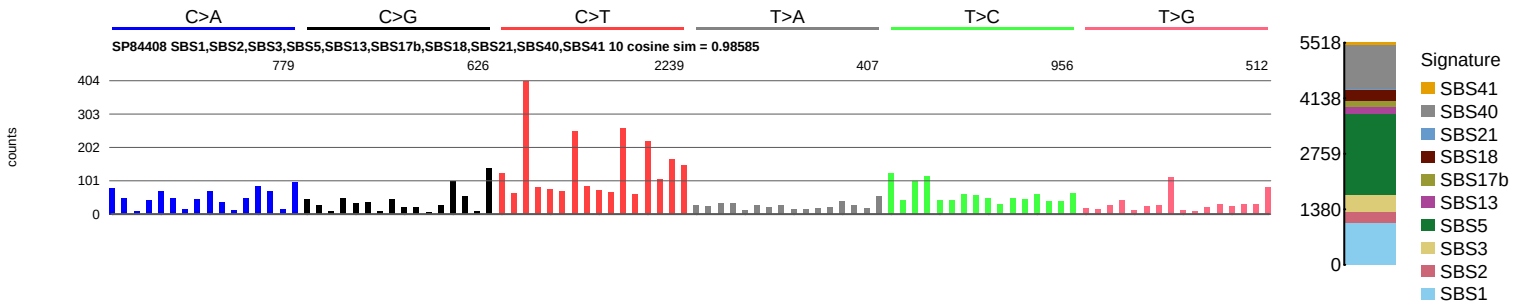
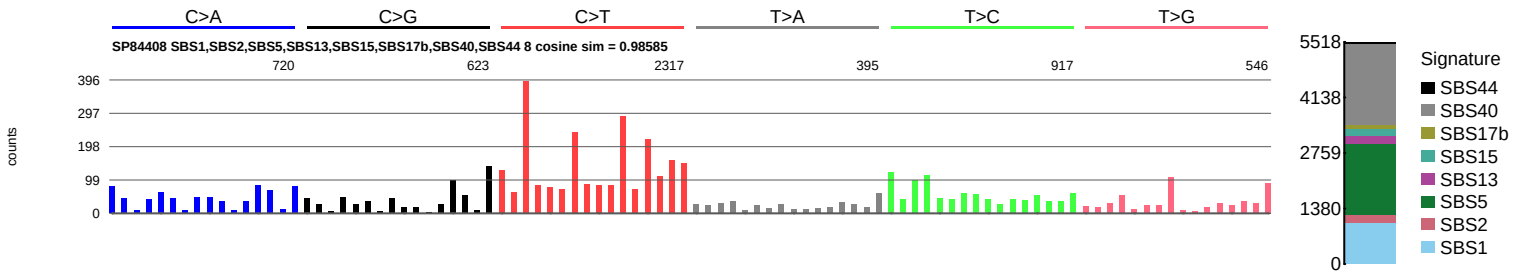
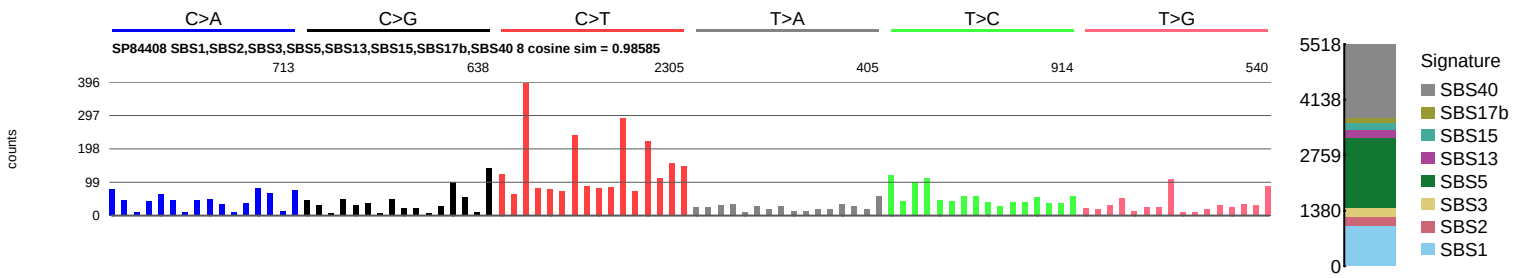
SBS5
SBS2
SBS1

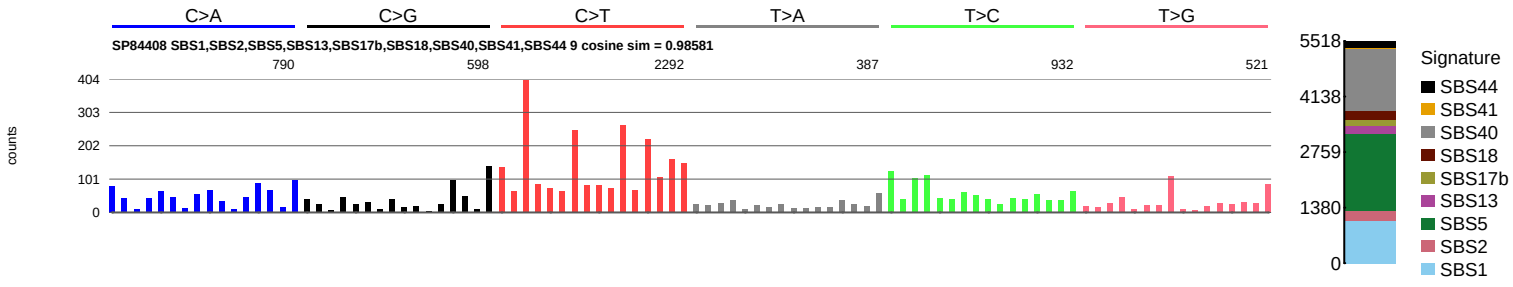

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98587
788
597
2284
381
950
518
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98586
777
642
2284
399
952
467
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98585
779
626
2239
407
956
512
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS44 8 cosine sim = 0.98585
720
623
2317
395
917
546
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40 8 cosine sim = 0.98585
713
638
2305
405
914
540
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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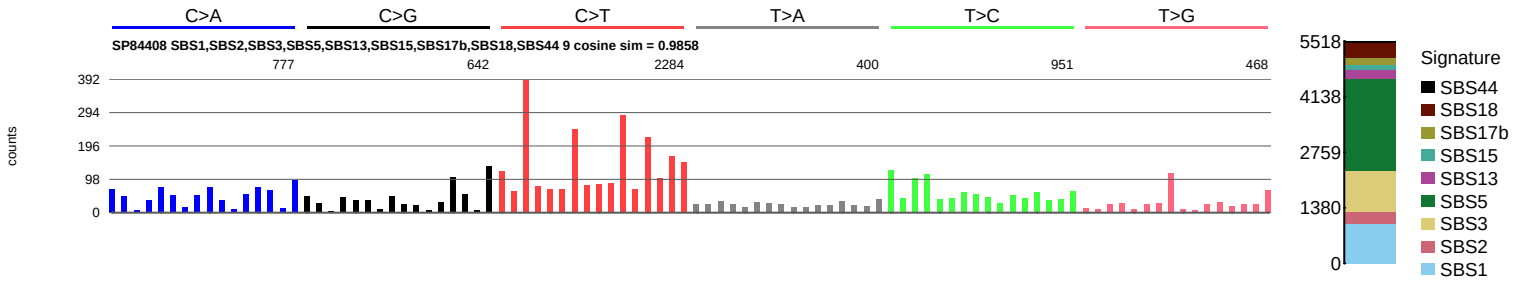
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98581
790
598
2292
387
932
521
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.9858
777
642
2284
400
951
468
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS18
SBS17b
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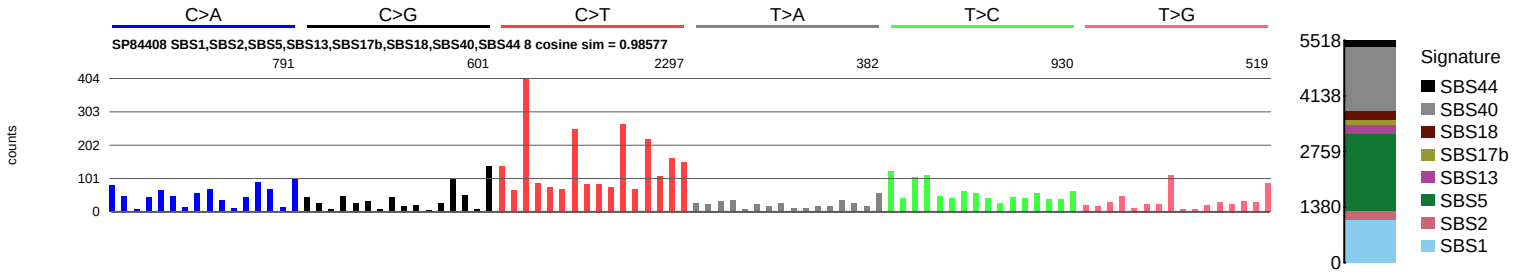
SBS3
SBS2
SBS1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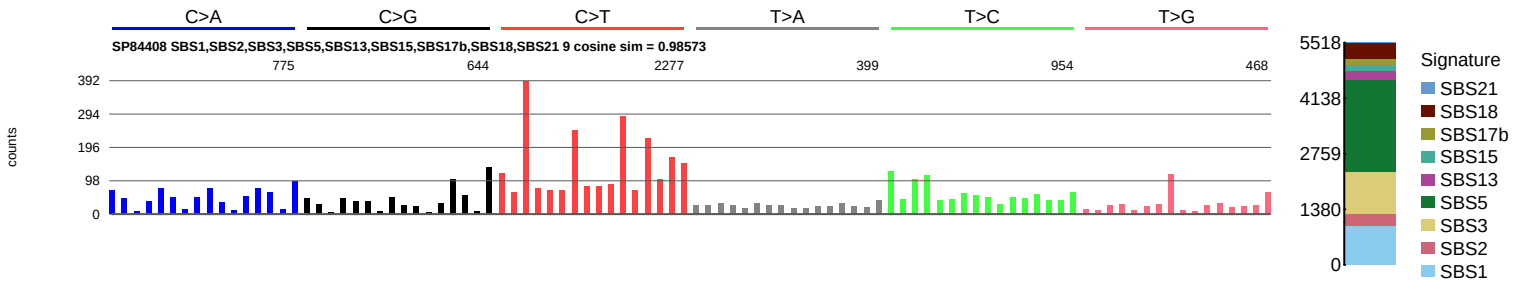
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98577
791
601
2297
382
930
519
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98573
775
644
2277
399
954
468
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS21
SBS18
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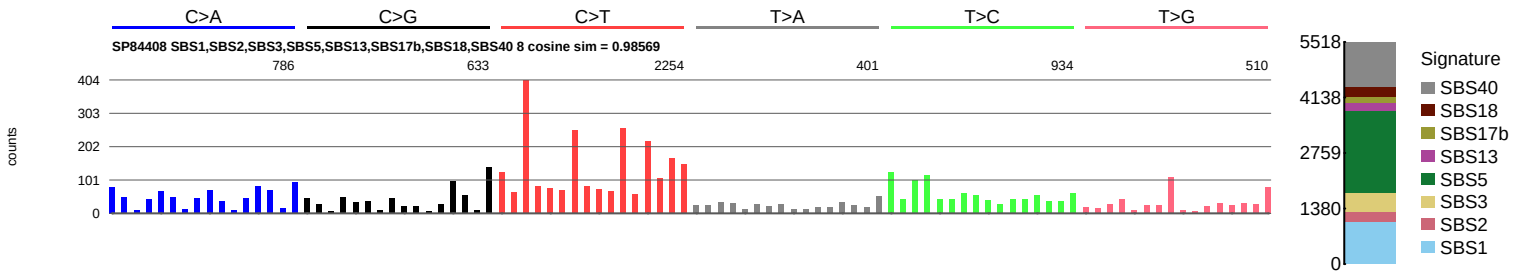
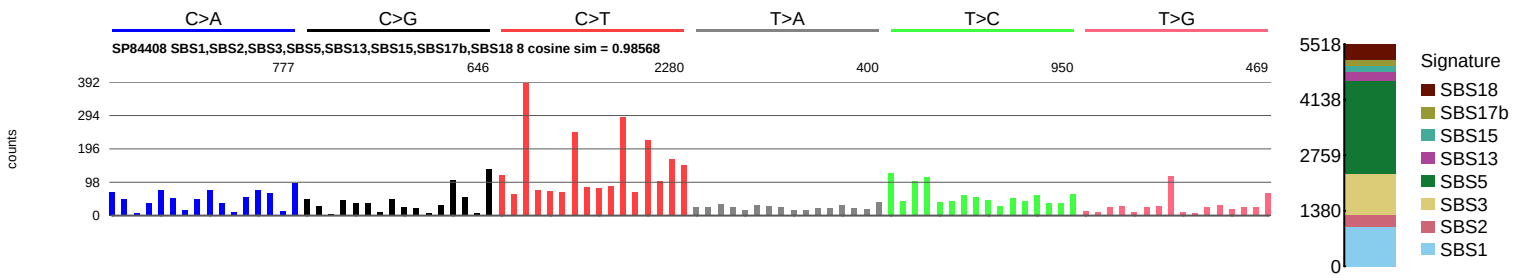
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40 8 cosine sim = 0.98569
786
633
2254
401
934
510
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98568
777
646
2280
400
950
469
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS18
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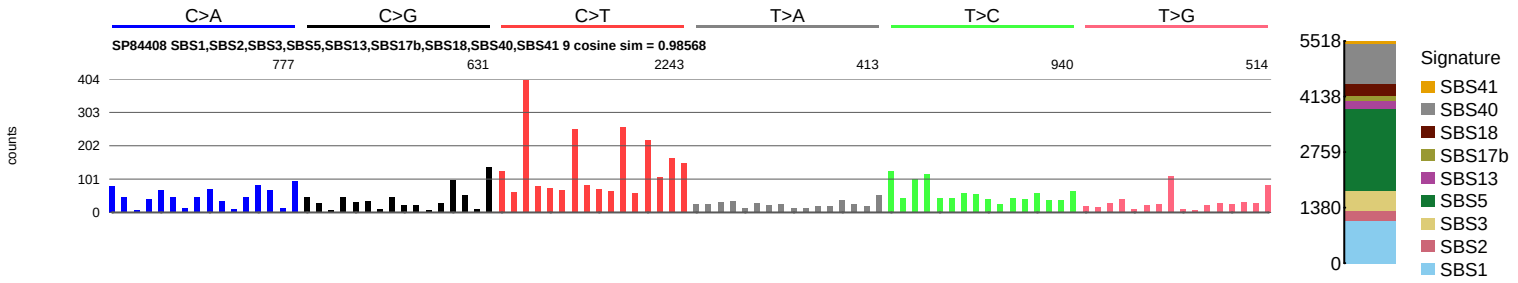
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98568
777
631
2243
413
940
514
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
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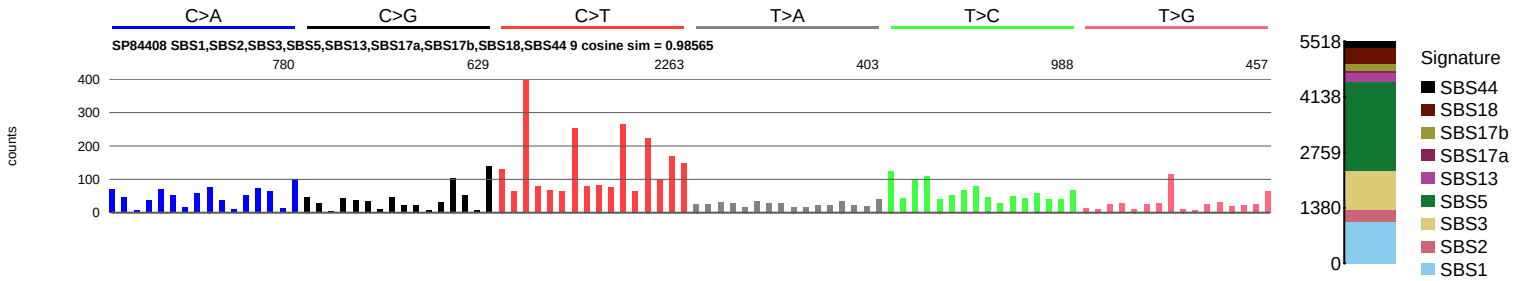
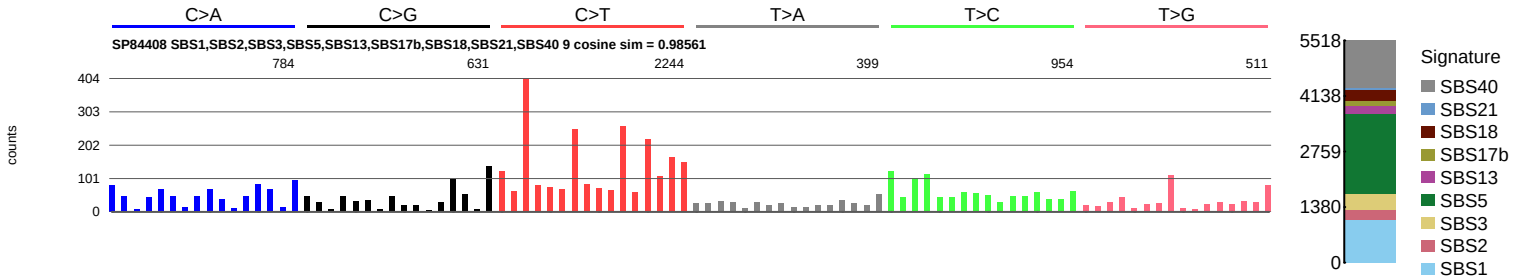
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98565
780
629
2263
403
988
457
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98561
784
631
2244
399
954
511
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
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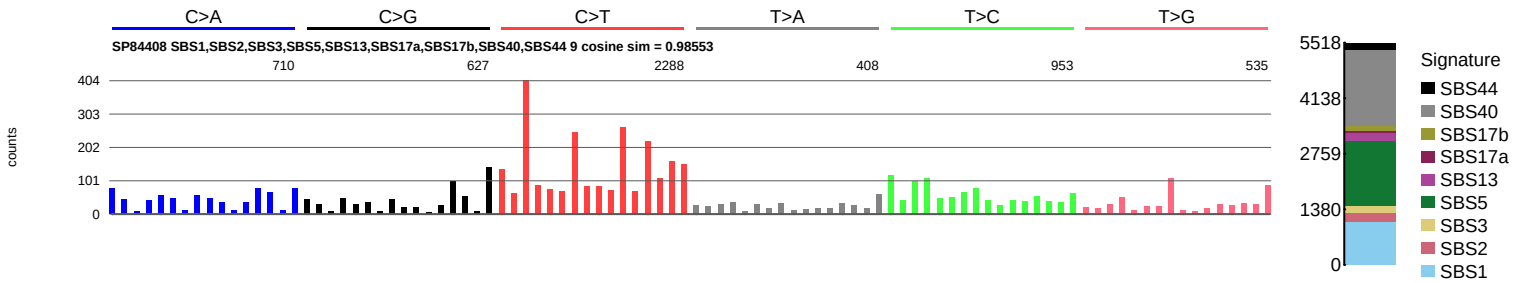
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98553
710
627
2288
408
953
535
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
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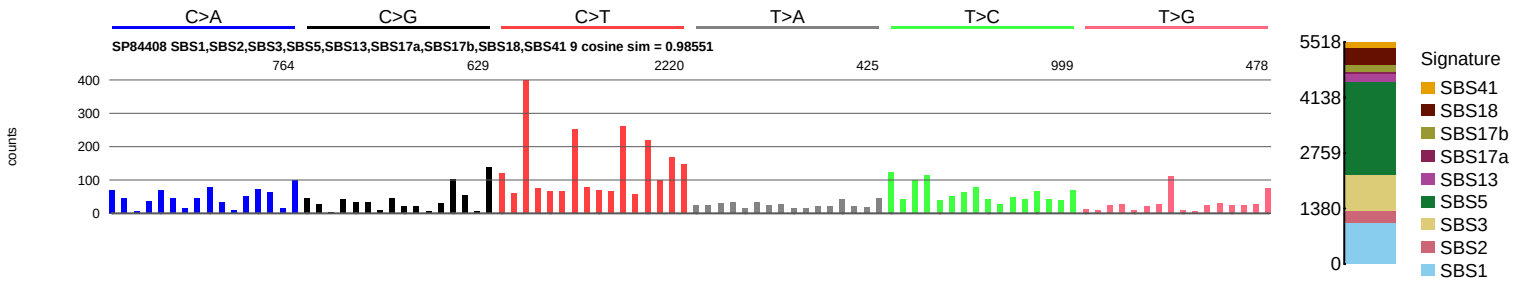
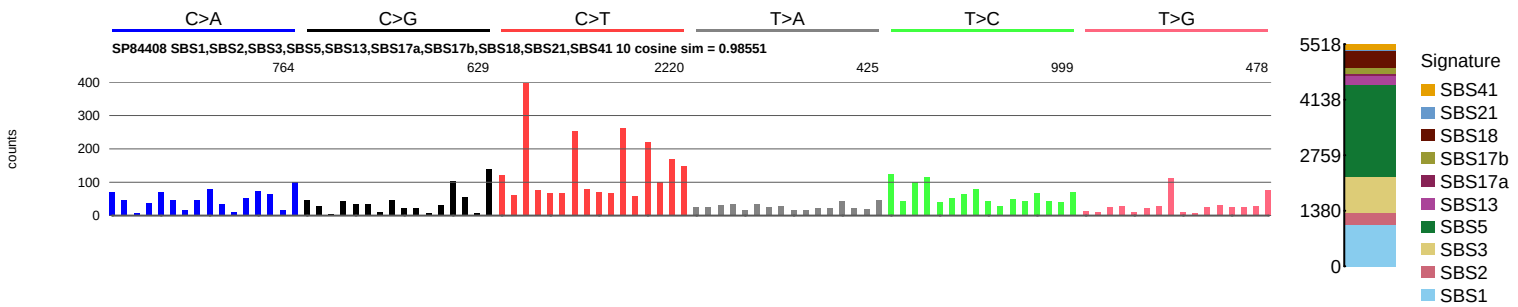
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98551
764
629
2220
425
999
478
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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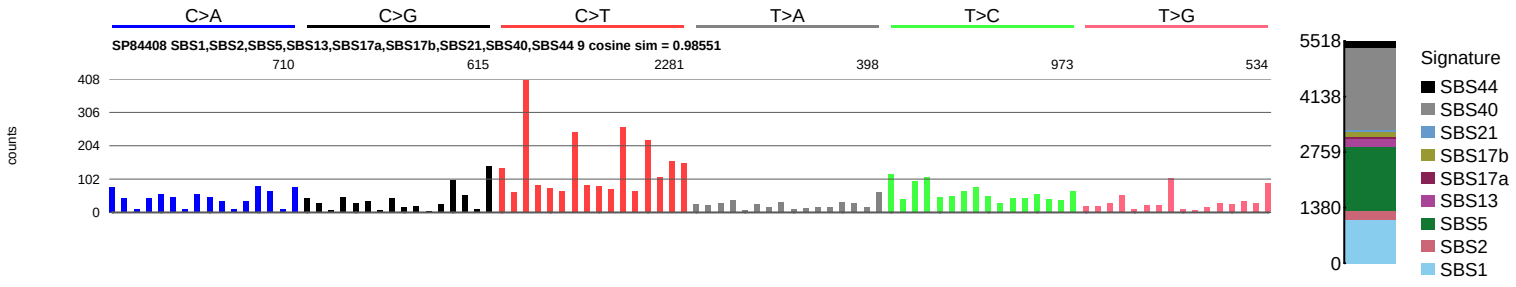
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98551
764
629
2220
425
999
478
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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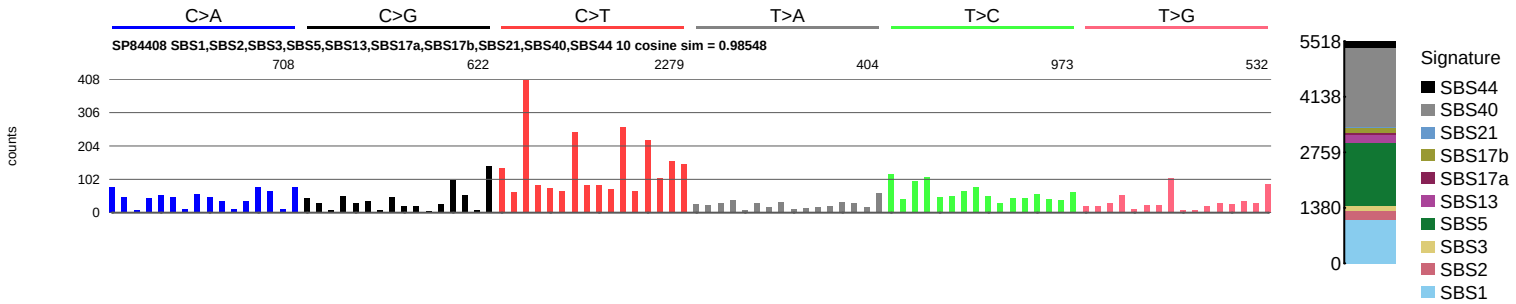
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98551
710
615
2281
398
973
534
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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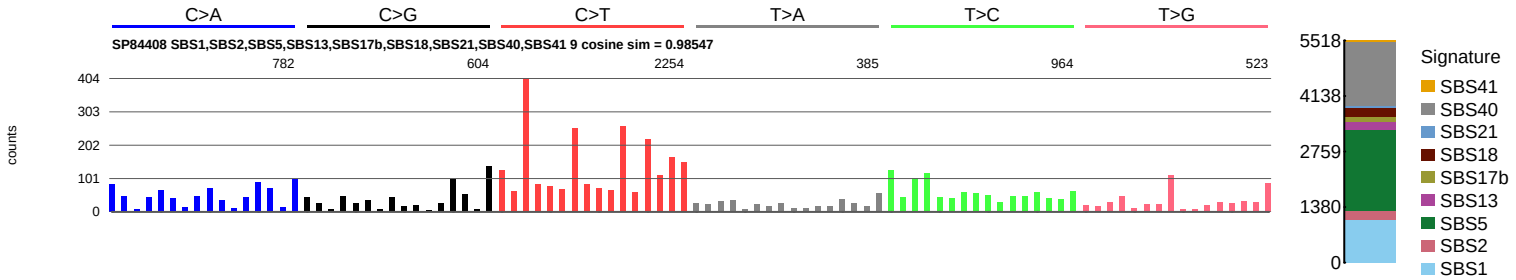
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98548
708
622
2279
404
973
532
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98547
782
604
2254
385
964
523
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
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SBS18
SBS17b
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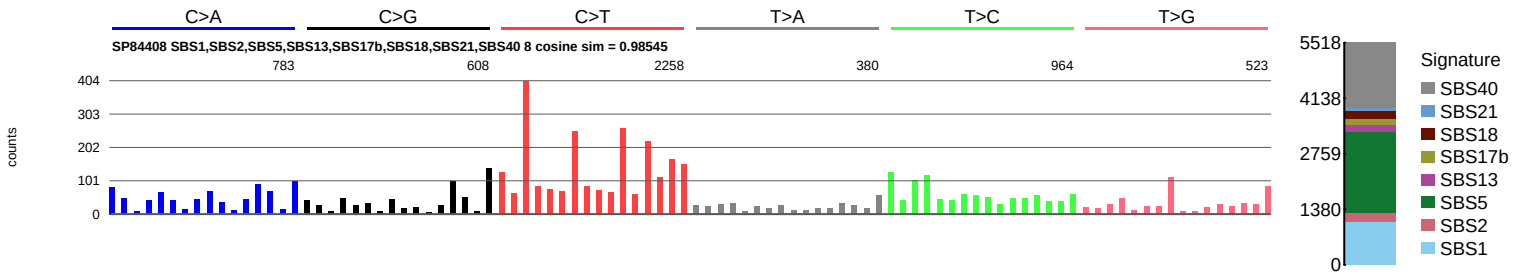
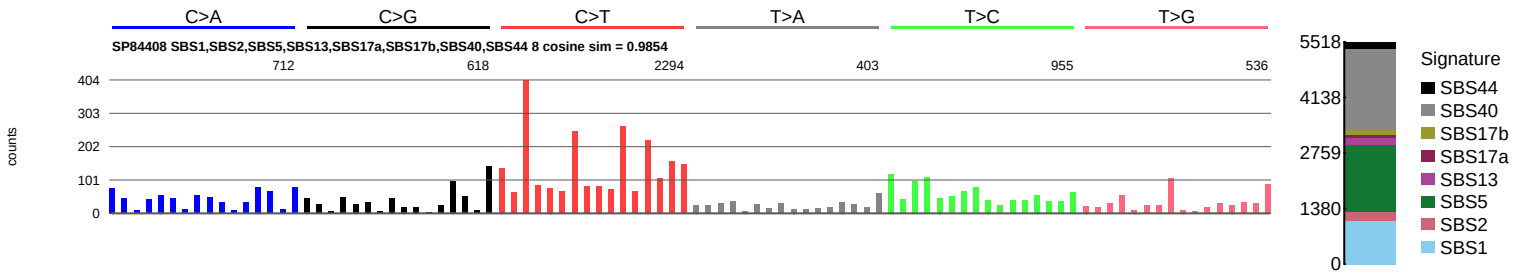
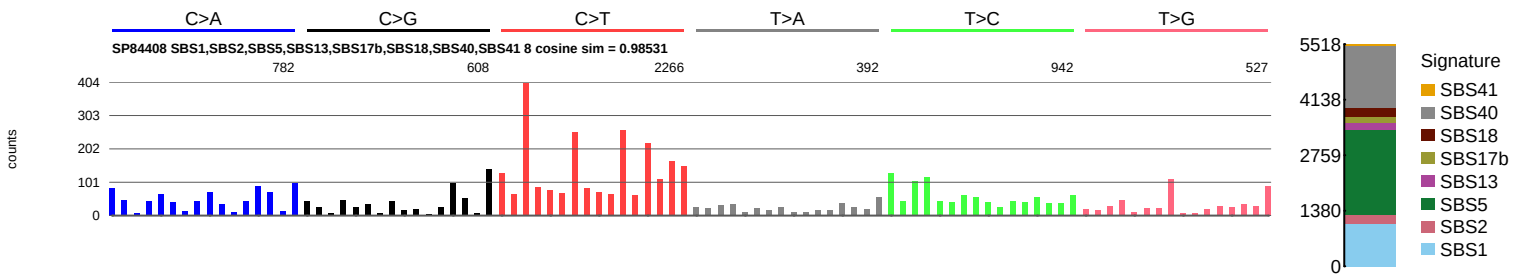
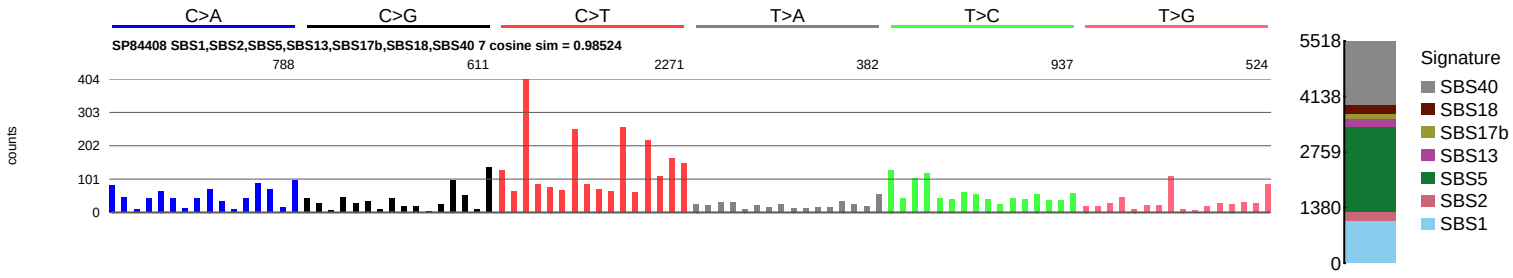
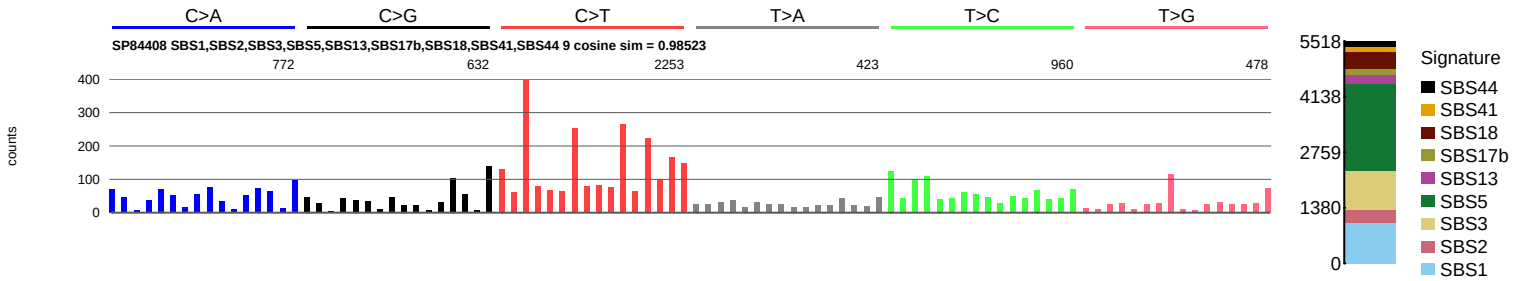
SBS5
SBS2
SBS1

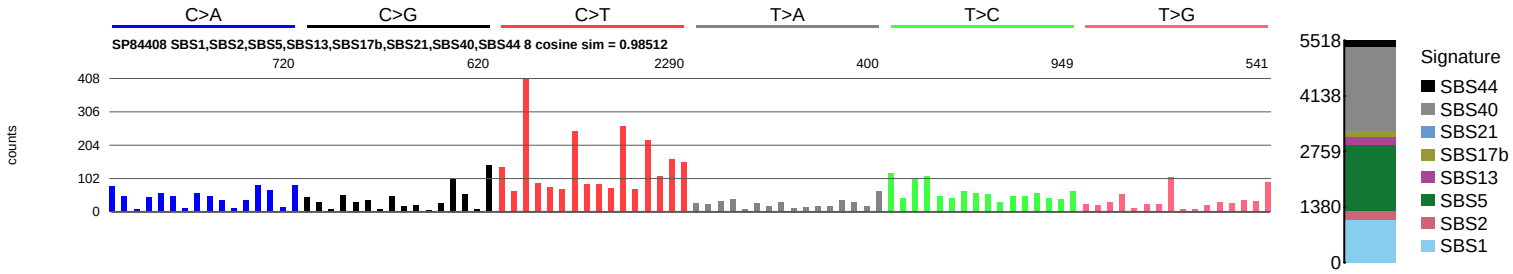

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98545
783
608
2258
380
964
523
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

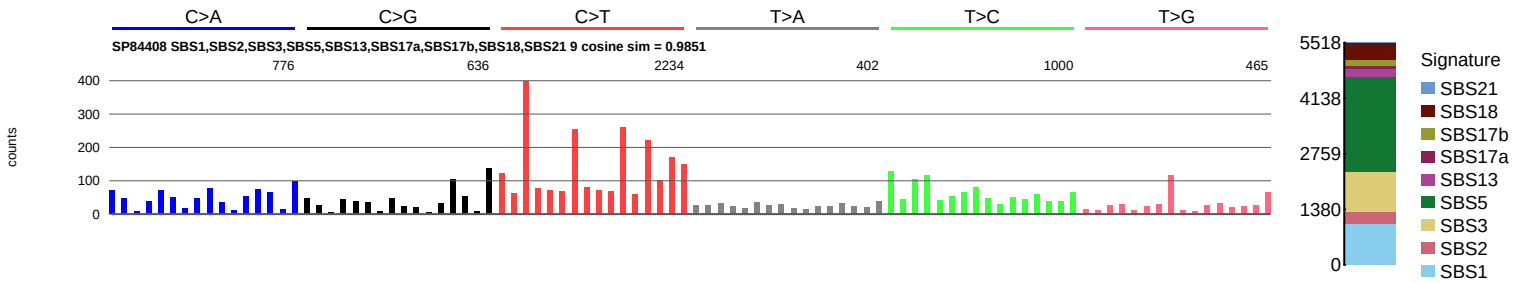

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.9854
712
618
2294
403
955
536
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

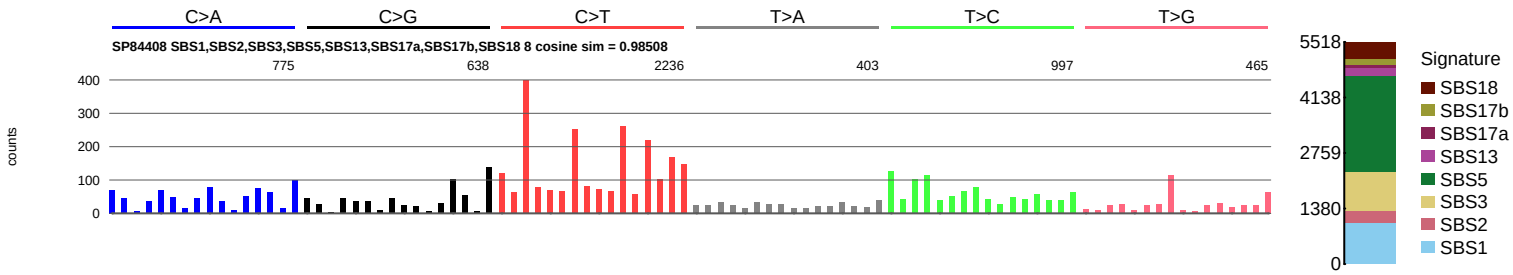

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98531
782
608
2266
392
942
527
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

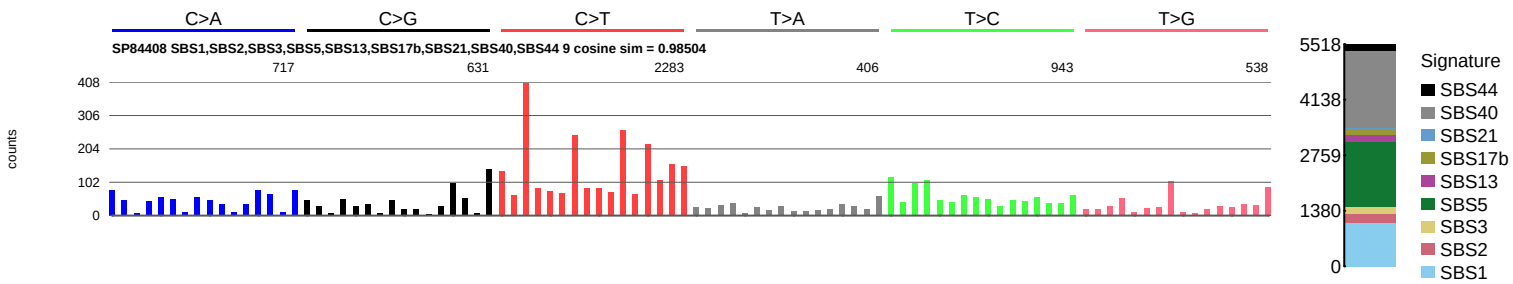

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98524
788
611
2271
382
937
524
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

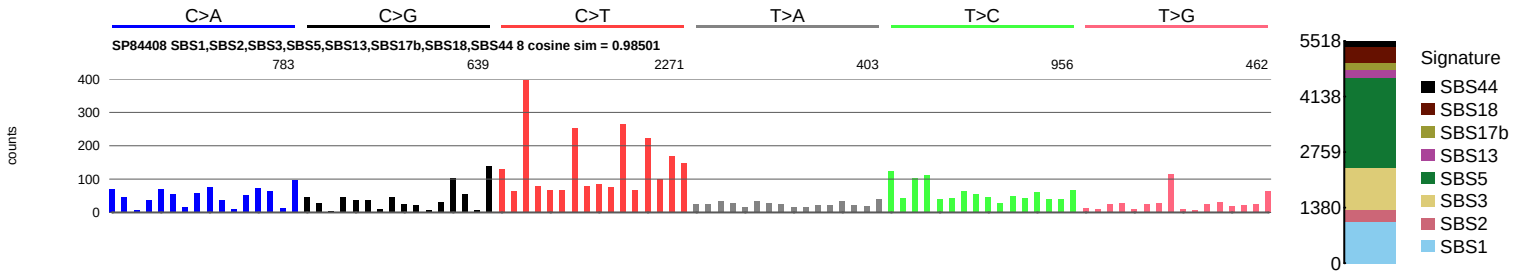

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98523
772
632
2253
423
960
478
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

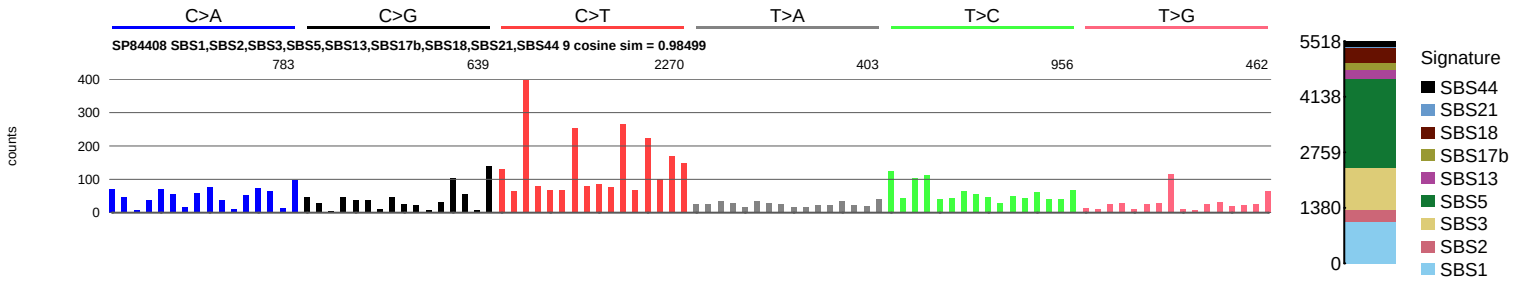

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98512
720
620
2290
400
949
541
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS2
SBS1

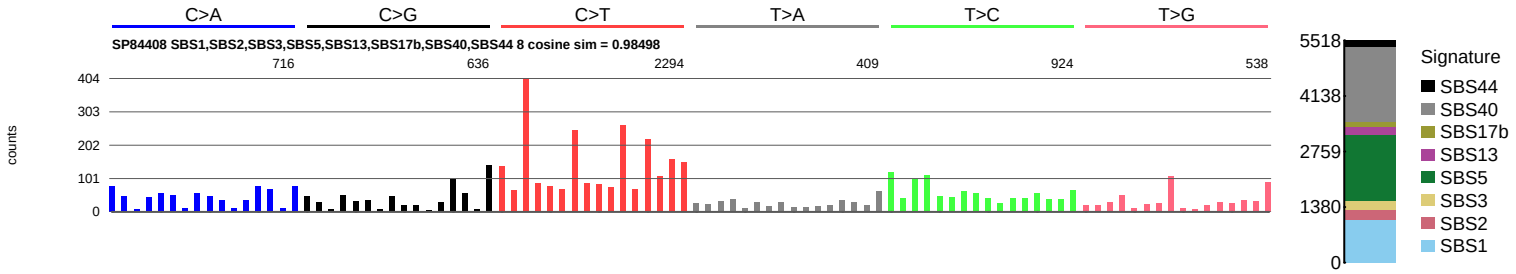

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.9851
776
636
2234
402
1000
465
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

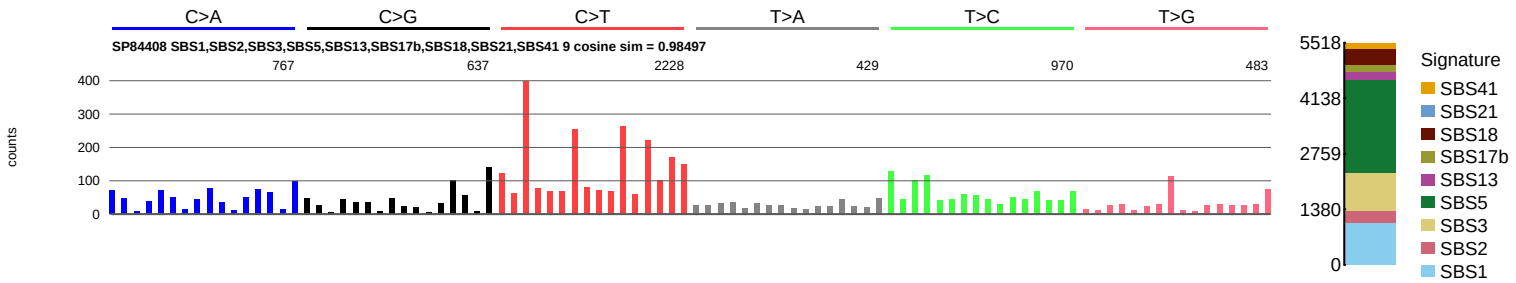

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98508
775
638
2236
403
997
465
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

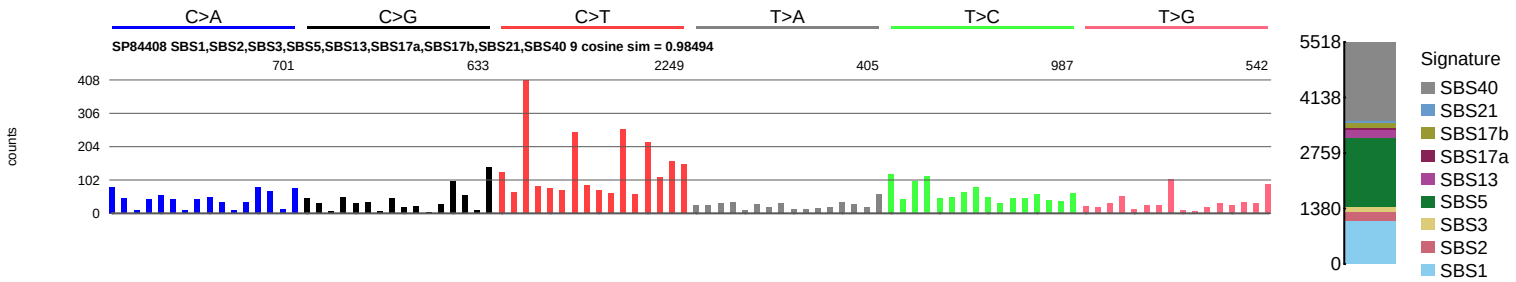

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98504
717
631
2283
406
943
538
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98501
783
639
2271
403
956
462
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98499
783
639
2270
403
956
462
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS40,SBS44 8 cosine sim = 0.98498
716
636
2294
409
924
538
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98497
767
637
2228
429
970
483
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98494
701
633
2249
405
987
542
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
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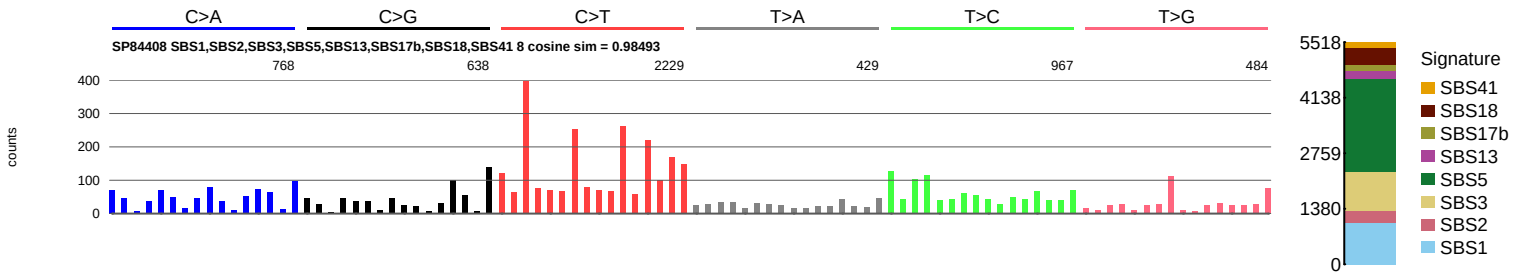
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.98493
768
638
2229
429
967
484
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS41
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SBS17b
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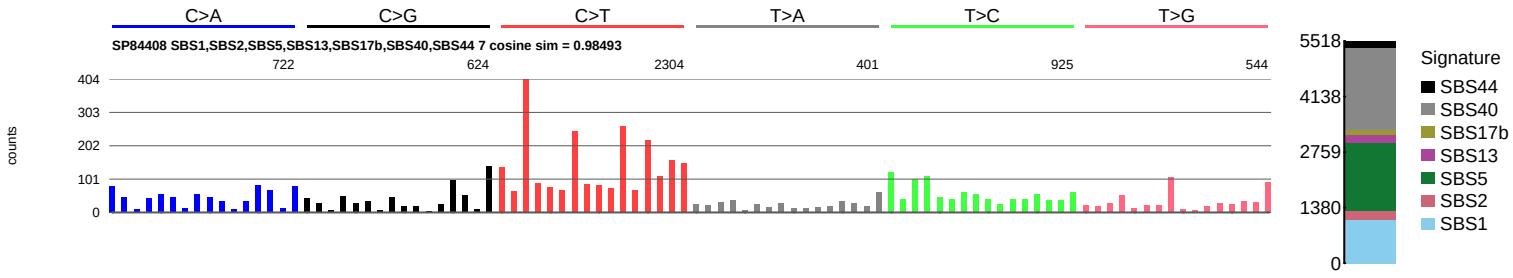
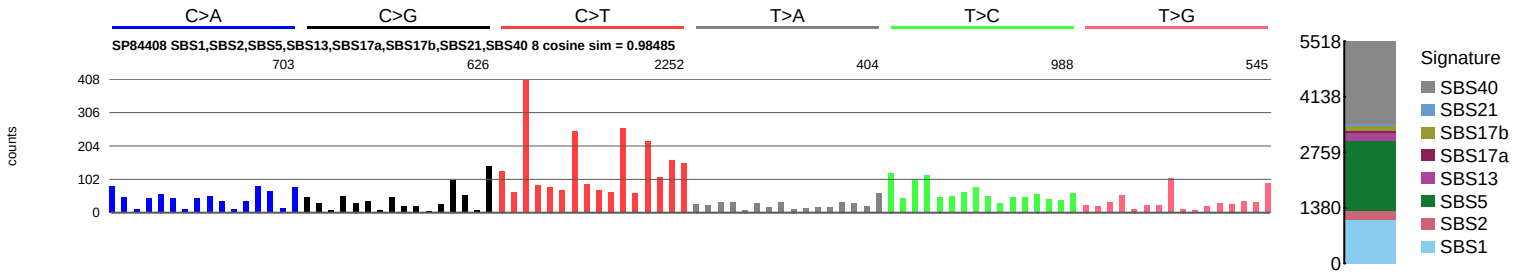
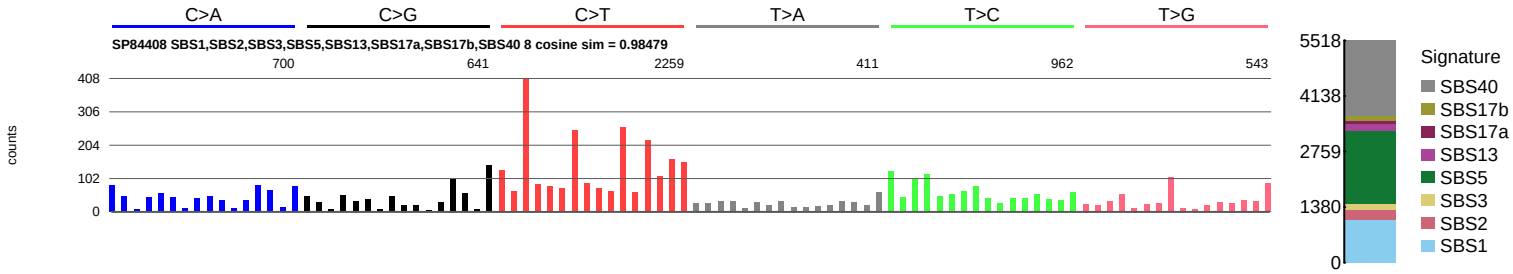
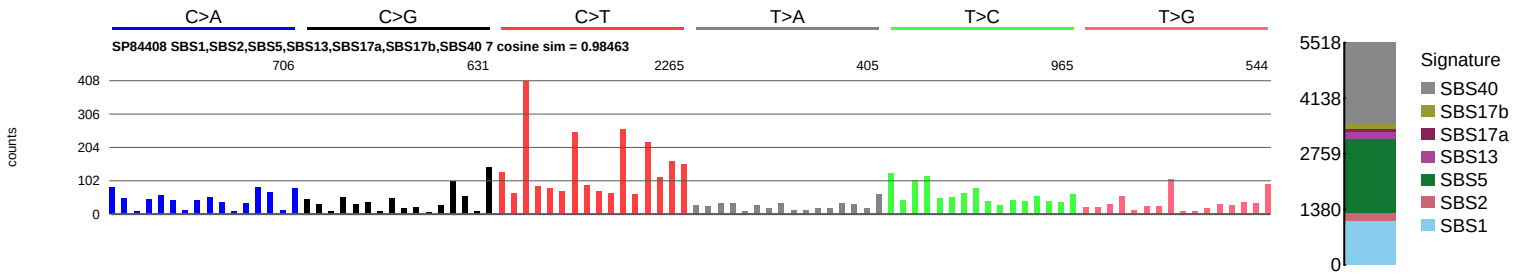
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS40,SBS44 7 cosine sim = 0.98493
722
624
2304
401
925
544
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98485
703
626
2252
404
988
545
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98479
700
641
2259
411
962
543
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98463
706
631
2265
405
965
544
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17b
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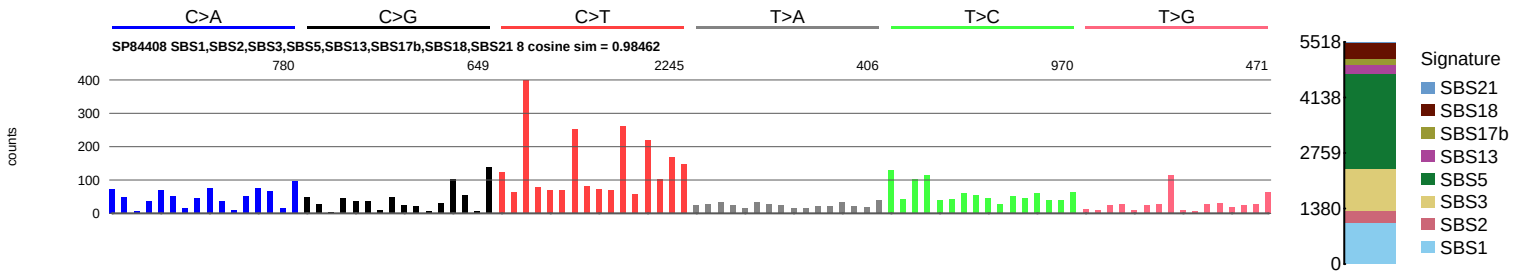
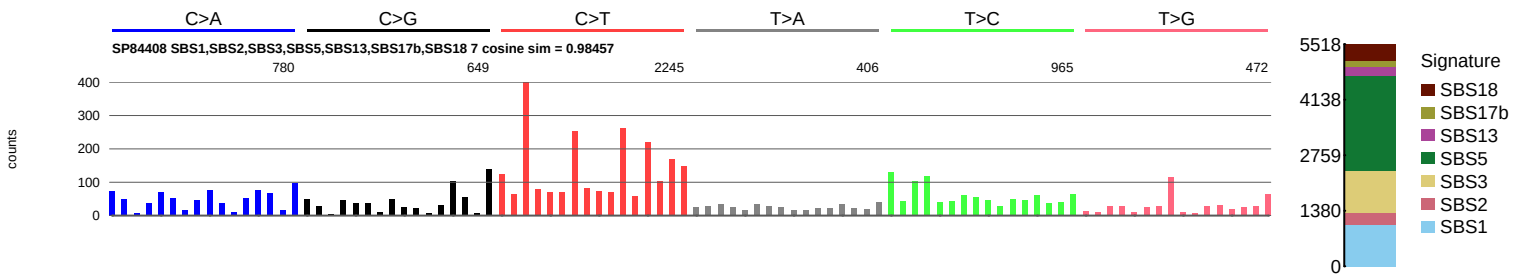
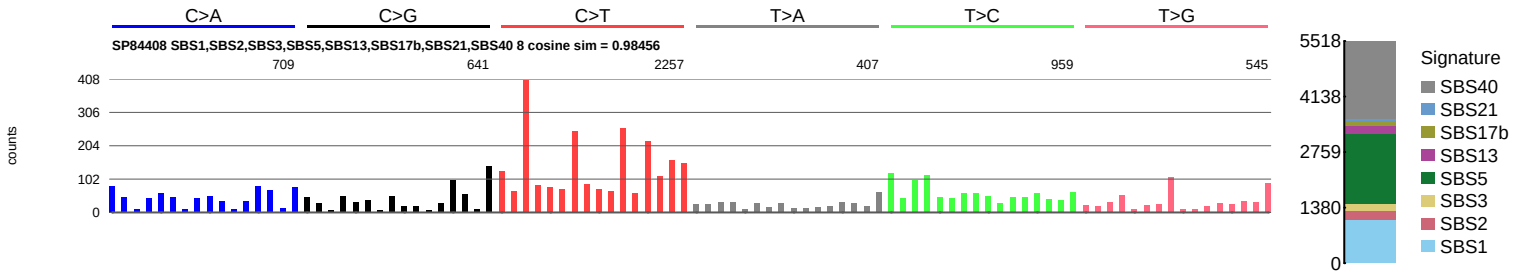
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98462
780
649
2245
406
970
471
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.98457
780
649
2245
406
965
472
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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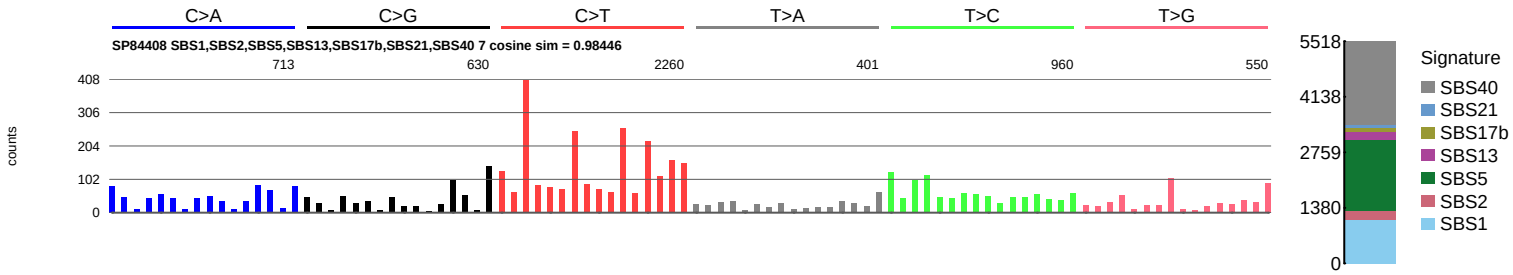
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS21,SBS40 8 cosine sim = 0.98456
709
641
2257
407
959
545
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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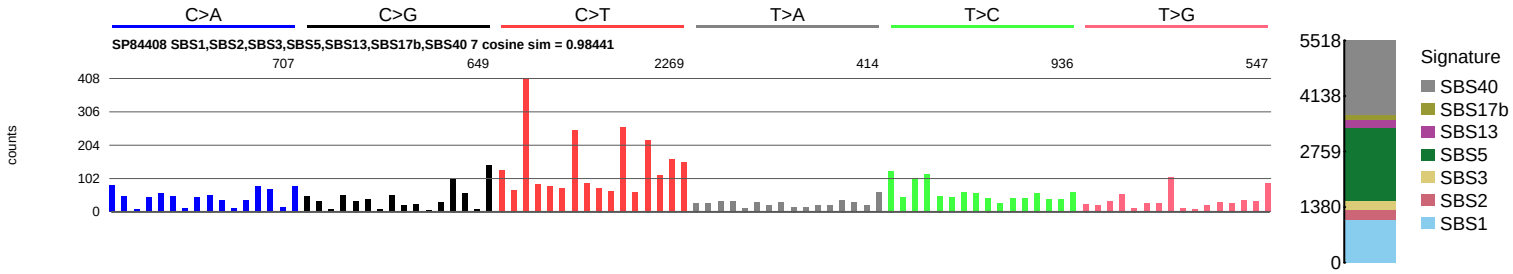
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS21,SBS40 7 cosine sim = 0.98446
713
630
2260
401
960
550
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS40 7 cosine sim = 0.98441
707
649
2269
414
936
547
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
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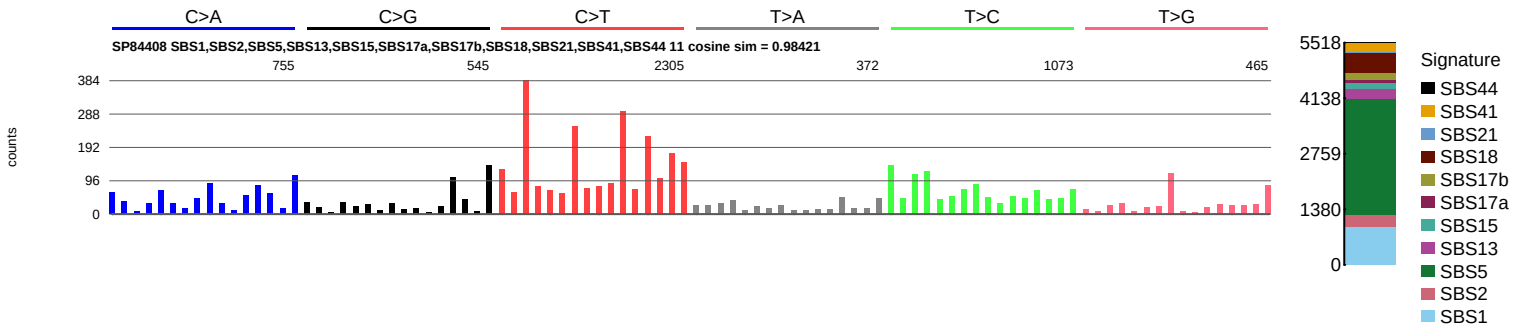
SBS2
SBS1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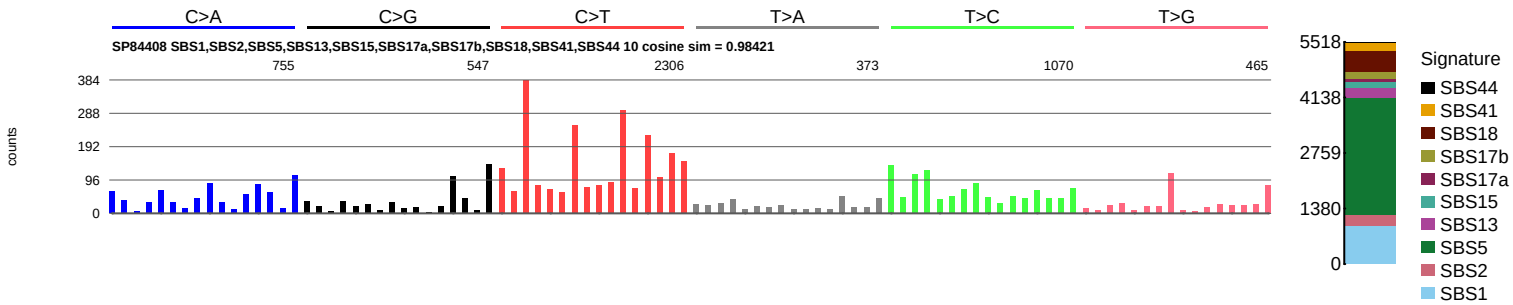
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98421
755
545
2305
372
1073
465
counts
384
288
192
96
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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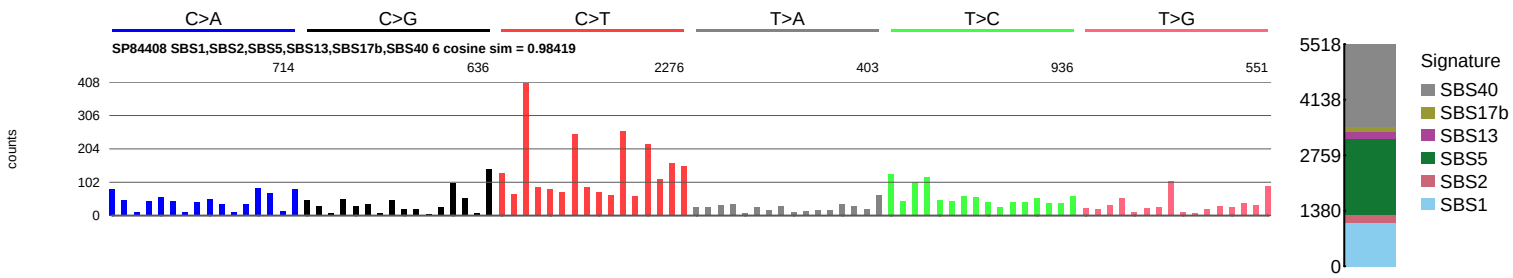
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98421
755
547
2306
373
1070
465
counts
384
288
192
96
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS40 6 cosine sim = 0.98419
714
636
2276
403
936
551
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
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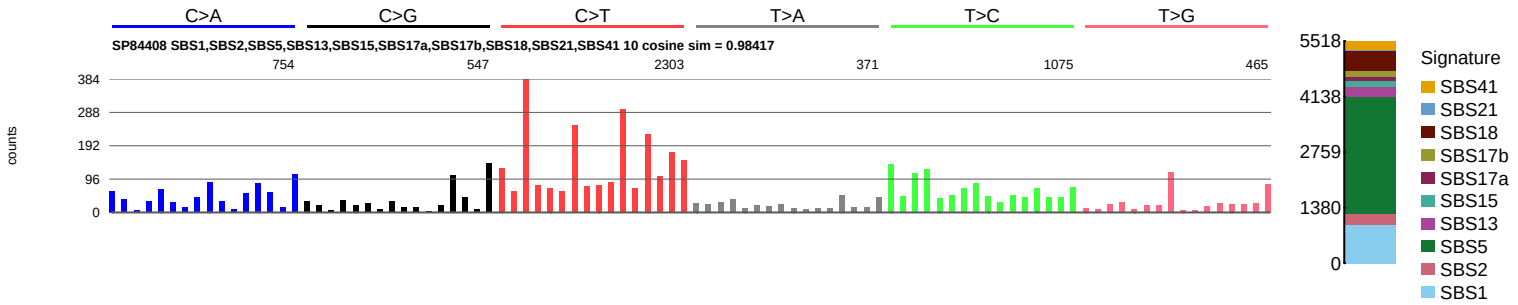
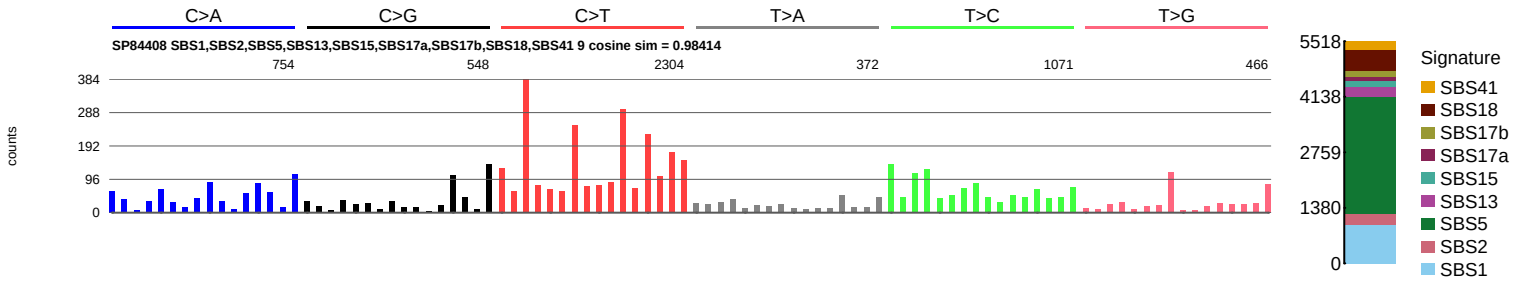
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98417
754
547
2303
371
1075
465
counts
384
288
192
96
0
5518
4138
2759
1380
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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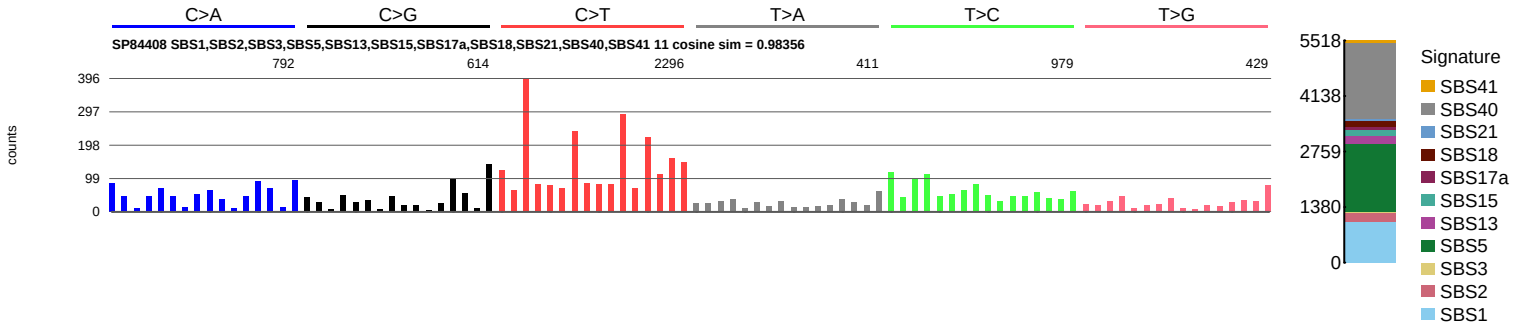
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98414
754
548
2304
372
1071
466
counts
384
288
192
96
0
5518
4138
2759
1380
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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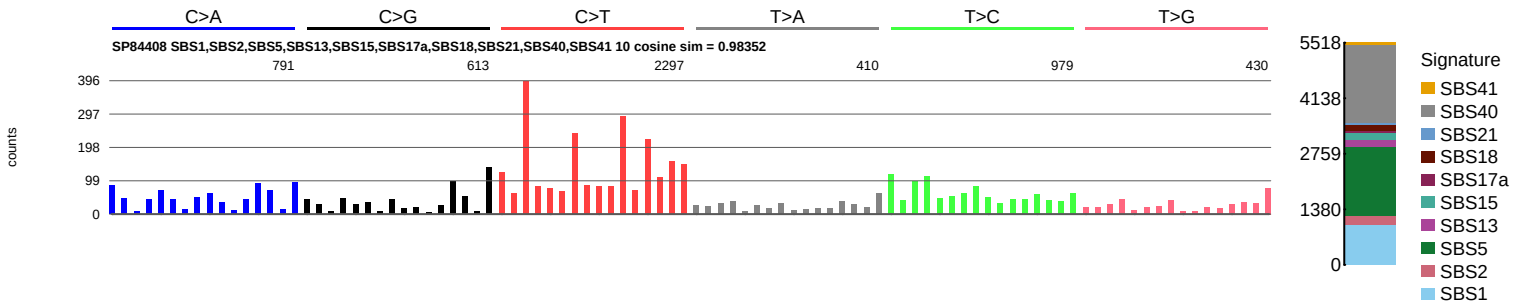
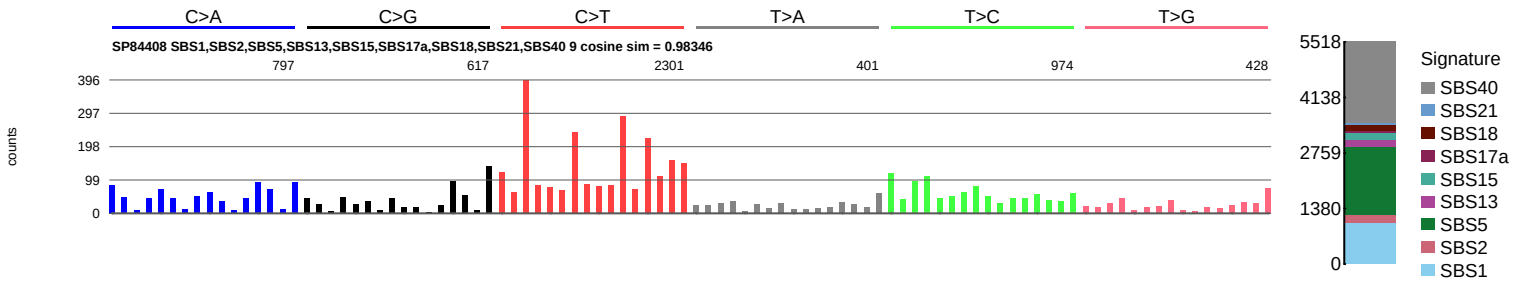
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98356
792
614
2296
411
979
429
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98352
791
613
2297
410
979
430
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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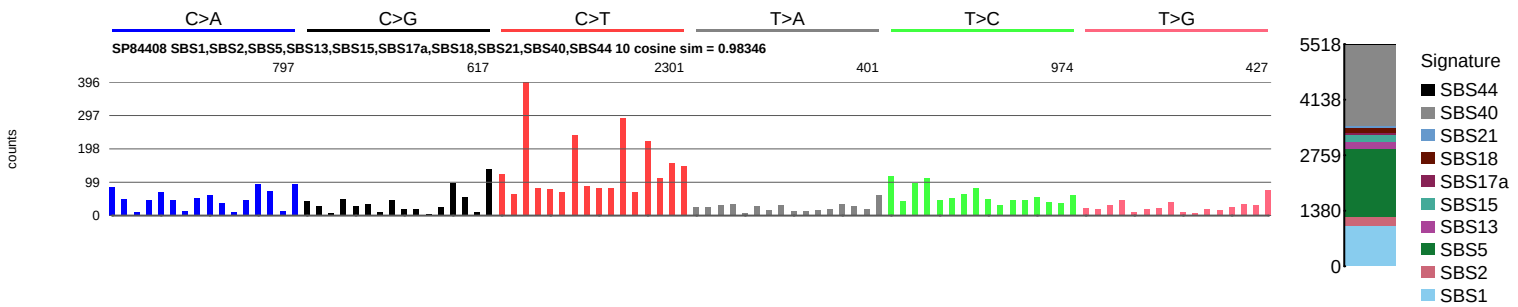
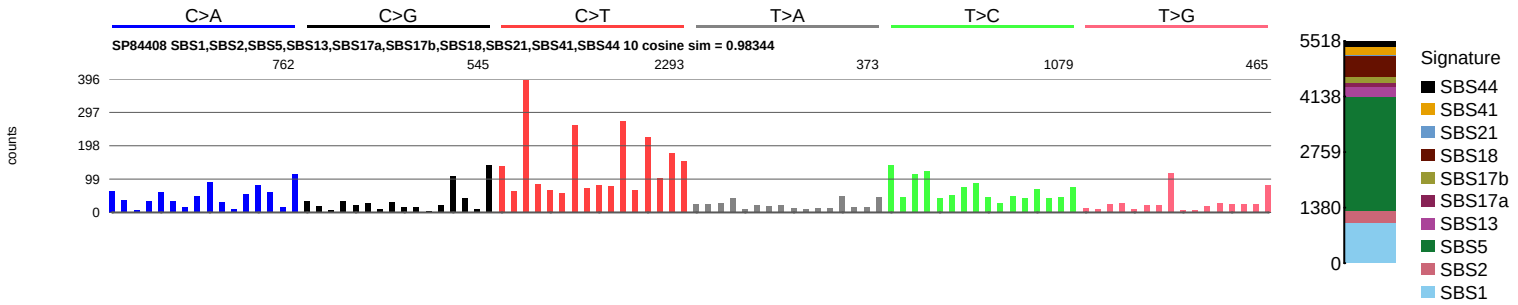
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98346
797
617
2301
401
974
428
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98346
797
617
2301
401
974
427
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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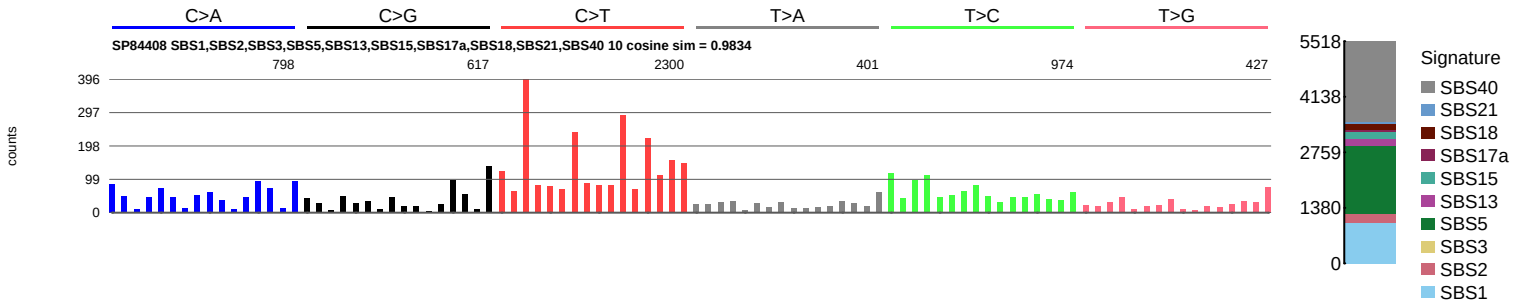
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98344
762
545
2293
373
1079
465
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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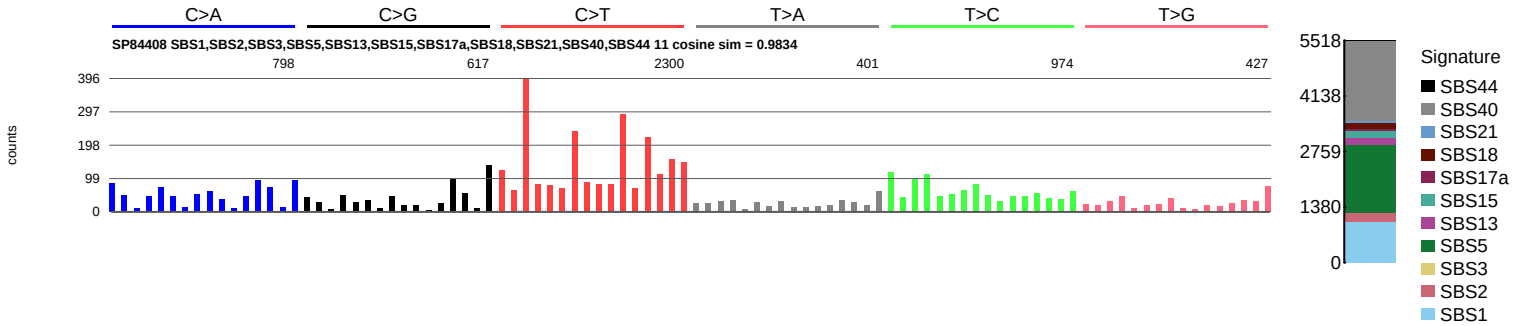
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 10 cosine sim = 0.9834
798
617
2300
401
974
427
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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T>C
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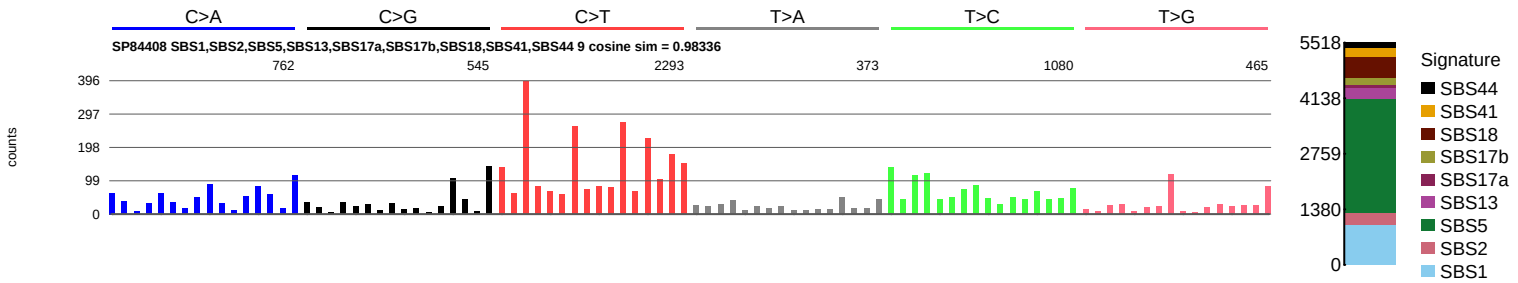
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.9834
798
617
2300
401
974
427
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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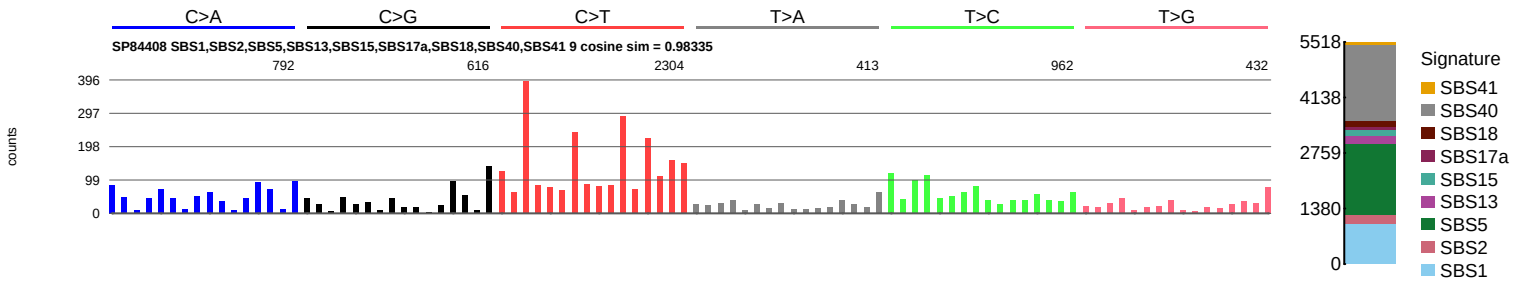
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98336
762
545
2293
373
1080
465
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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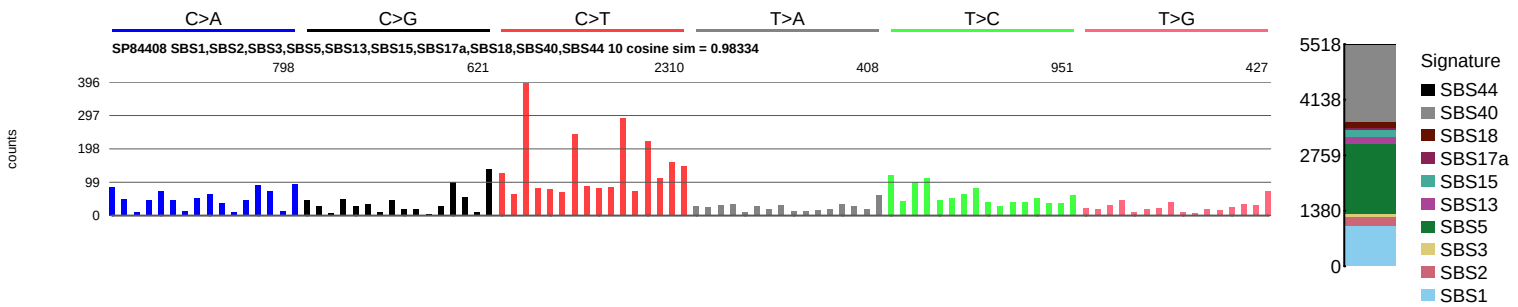
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98335
792
616
2304
413
962
432
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 10 cosine sim = 0.98334
798
621
2310
408
951
427
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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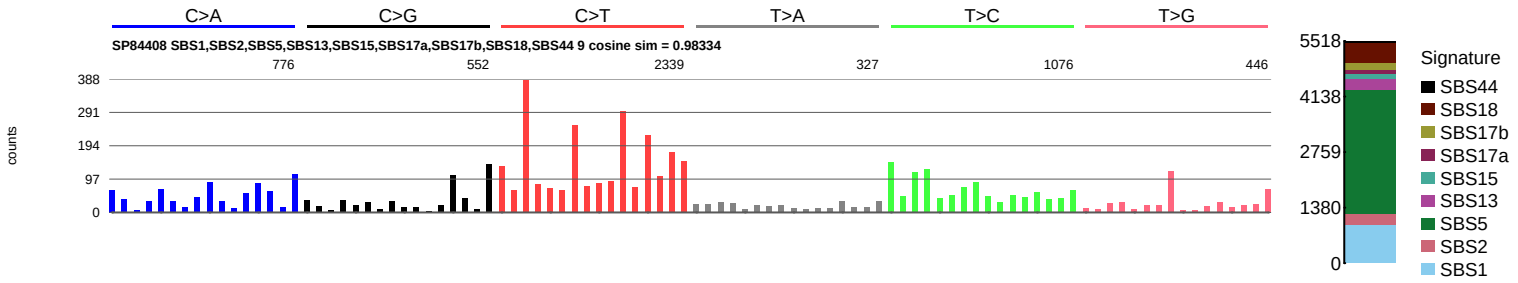
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98334
776
552
2339
327
1076
446
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS44
SBS18
SBS17b
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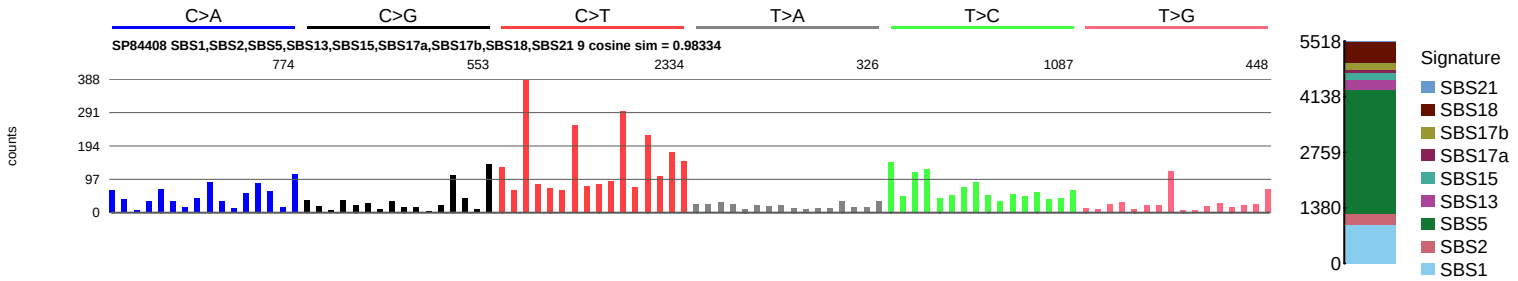
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98334
774
553
2334
326
1087
448
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS21
SBS18
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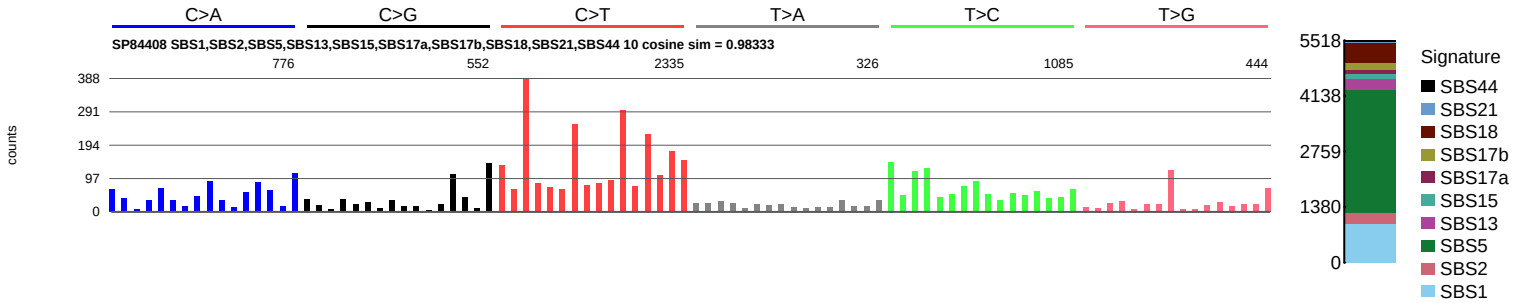
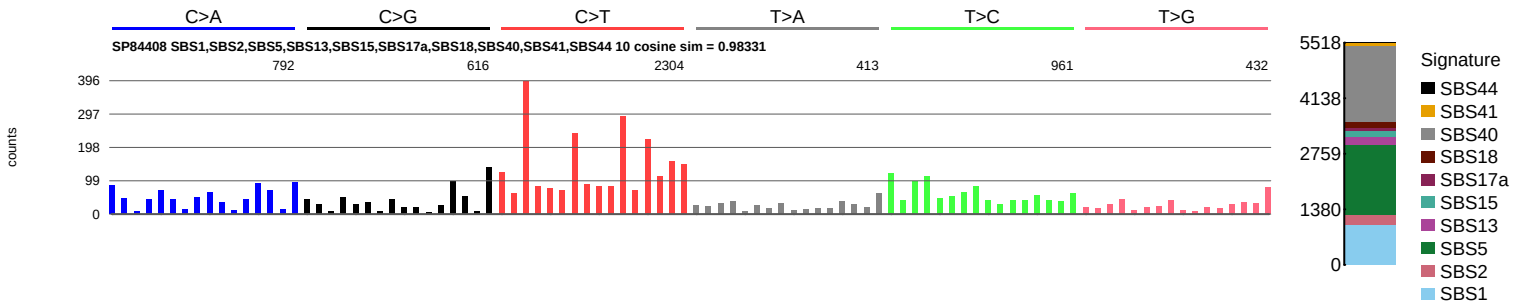
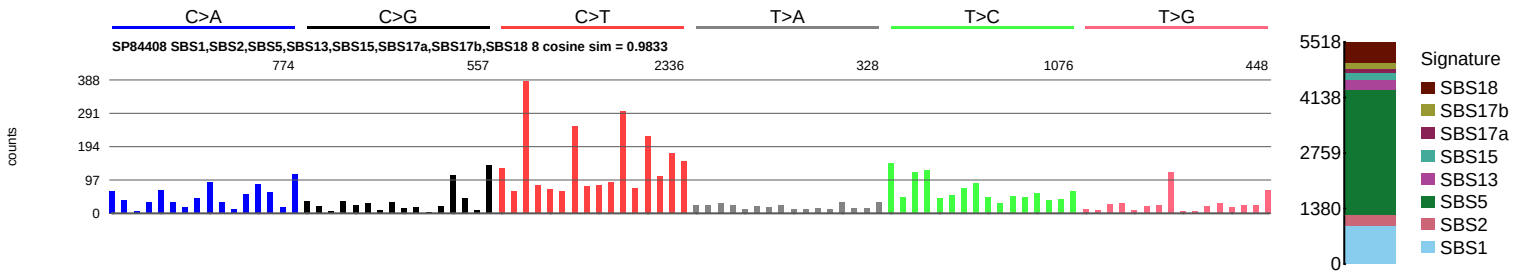
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98333
776
552
2335
326
1085
444
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98331
792
616
2304
413
961
432
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9833
774
557
2336
328
1076
448
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS18
SBS17b
SBS17a
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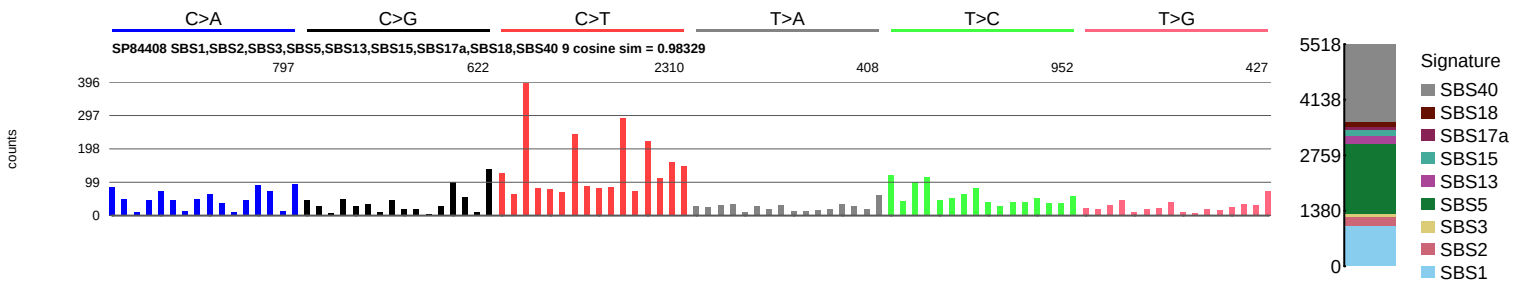
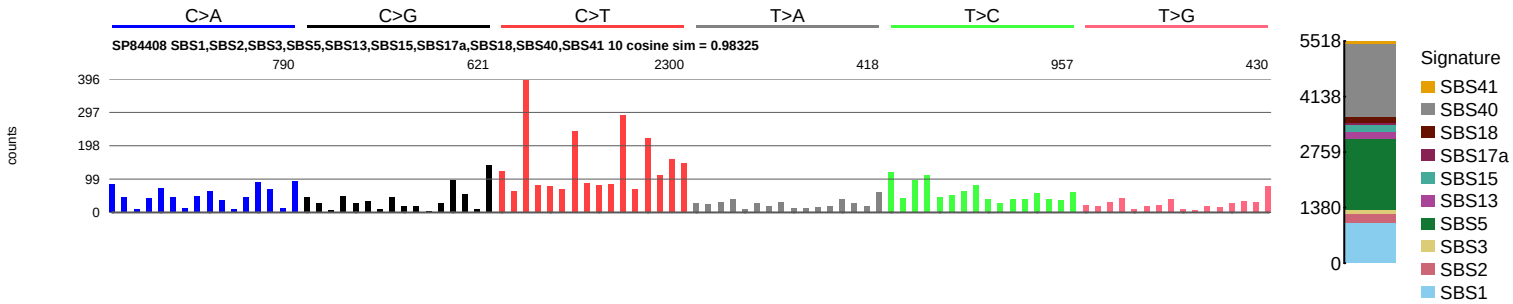
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 9 cosine sim = 0.98329
797
622
2310
408
952
427
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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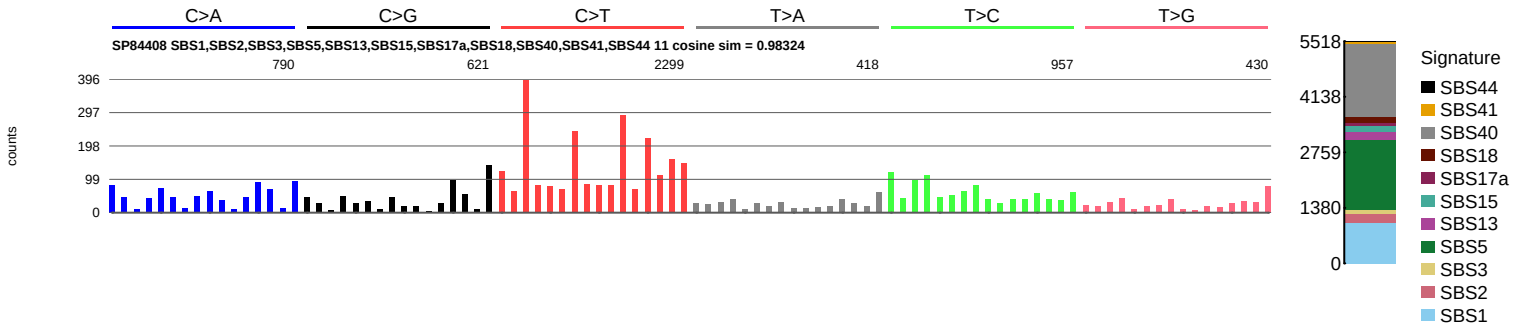
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 10 cosine sim = 0.98325
790
621
2300
418
957
430
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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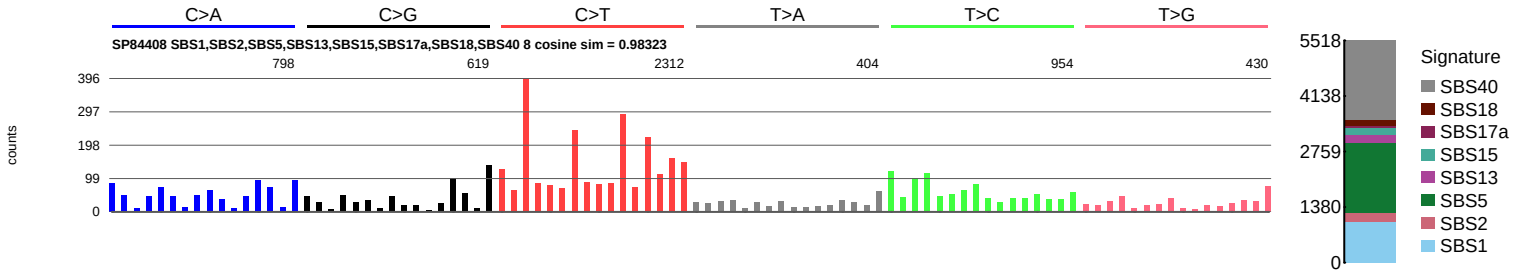
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98324
790
621
2299
418
957
430
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98323
798
619
2312
404
954
430
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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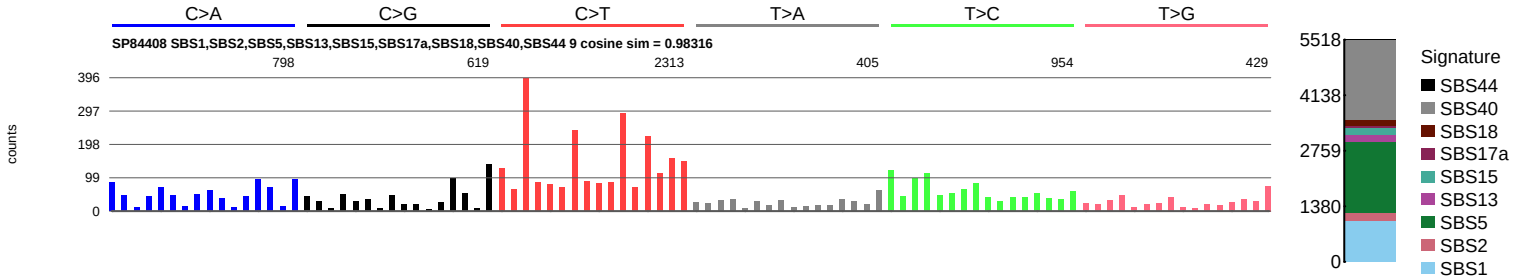
SBS2
SBS1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SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98316
798
619
2313
405
954
429
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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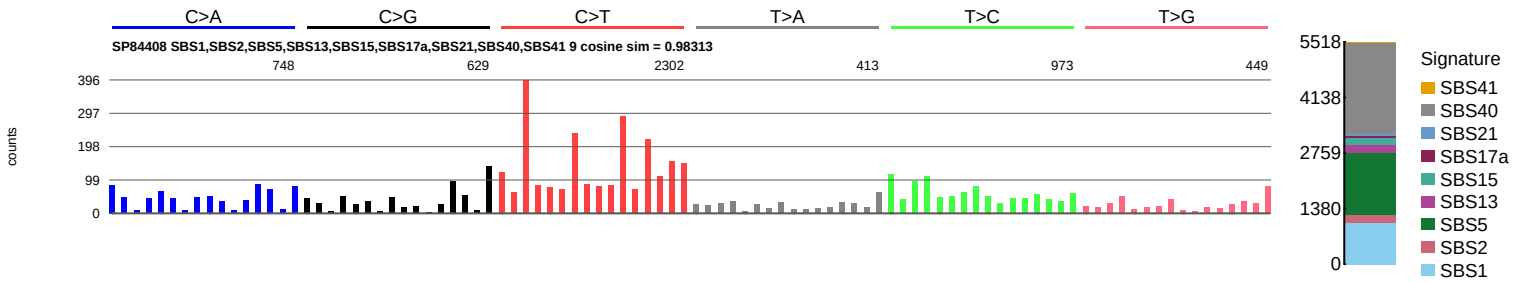
SBS2
SBS1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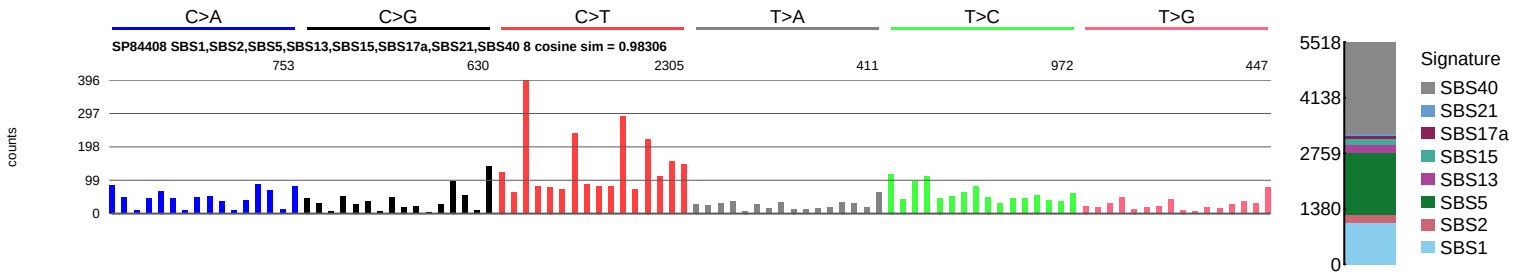
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41 9 cosine sim = 0.98313
748
629
2302
413
973
449
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40 8 cosine sim = 0.98306
753
630
2305
411
972
447
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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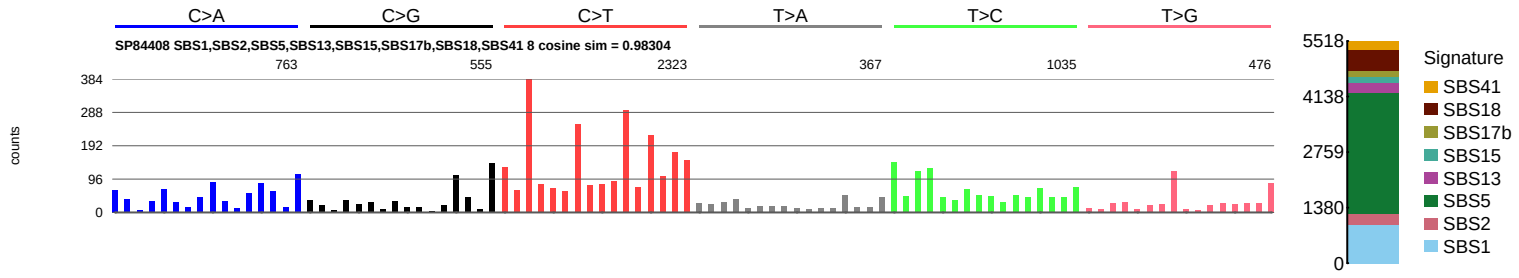
SBS2
SBS1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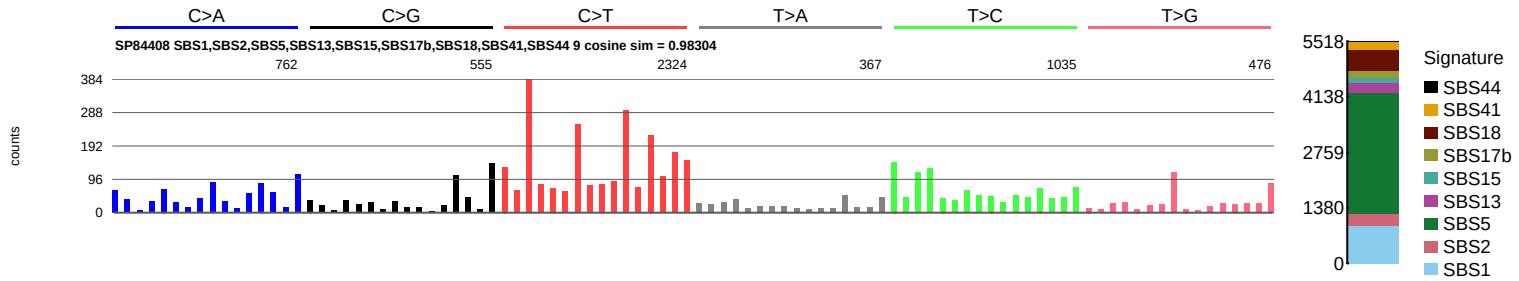
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98304
763
555
2323
367
1035
476
counts
384
288
192
96
0
5518
4138
2759
1380
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98304
762
555
2324
367
1035
476
counts
384
288
192
96
0
5518
4138
2759
1380
0
Signature
SBS44
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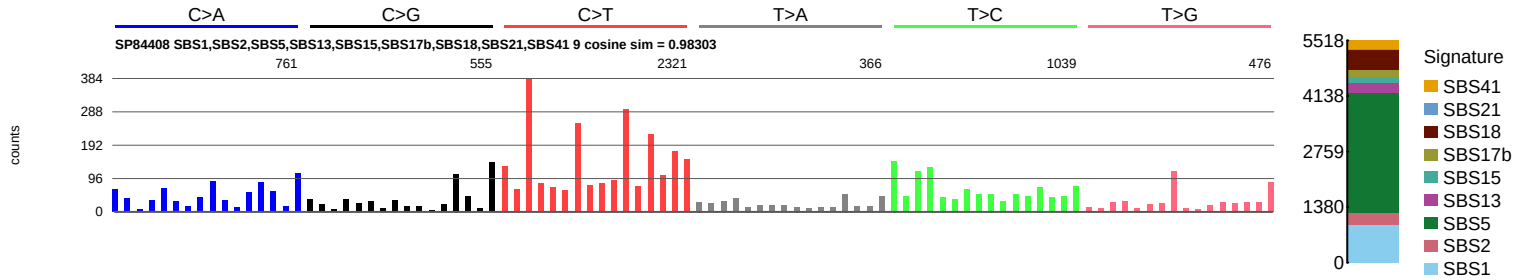
SBS2
SBS1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C>G
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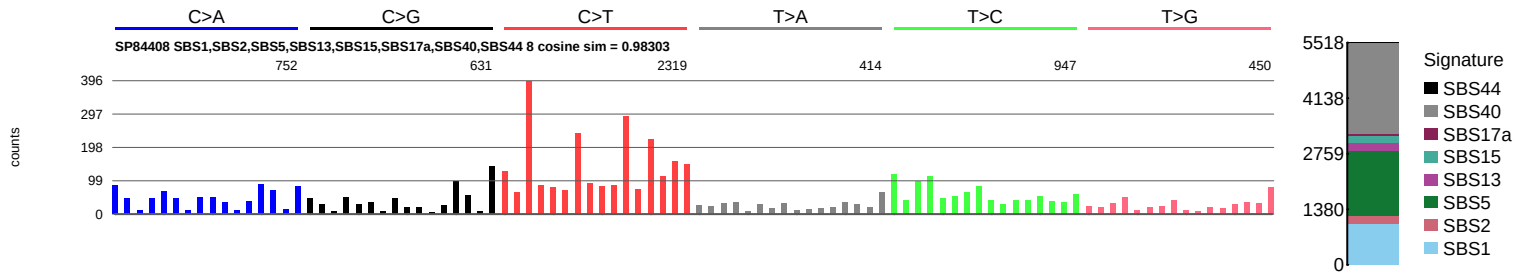
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98303
761
555
2321
366
1039
476
counts
384
288
192
96
0
5518
4138
2759
1380
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS44 8 cosine sim = 0.98303
752
631
2319
414
947
450
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
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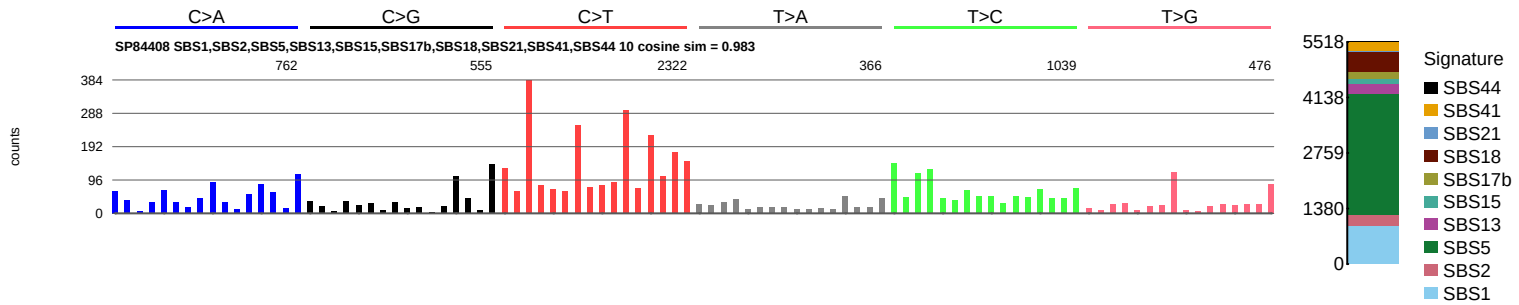
SBS2
SBS1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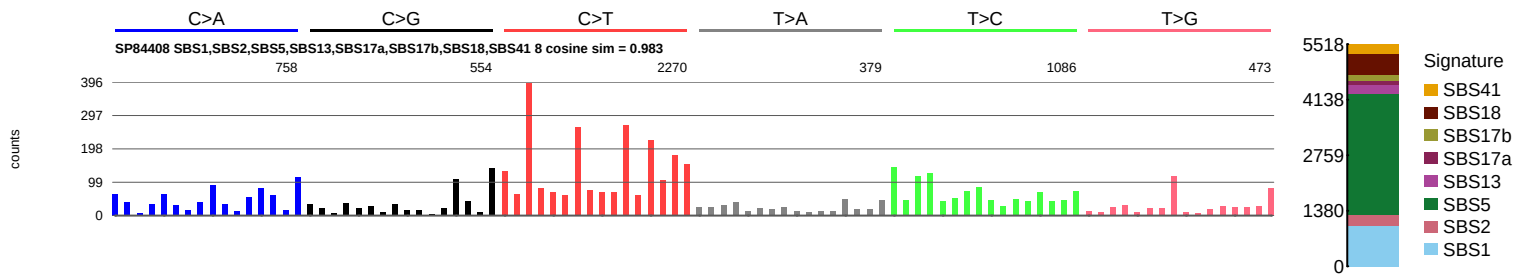
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.983
762
555
2322
366
1039
476
counts
384
288
192
96
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.983
758
554
2270
379
1086
473
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
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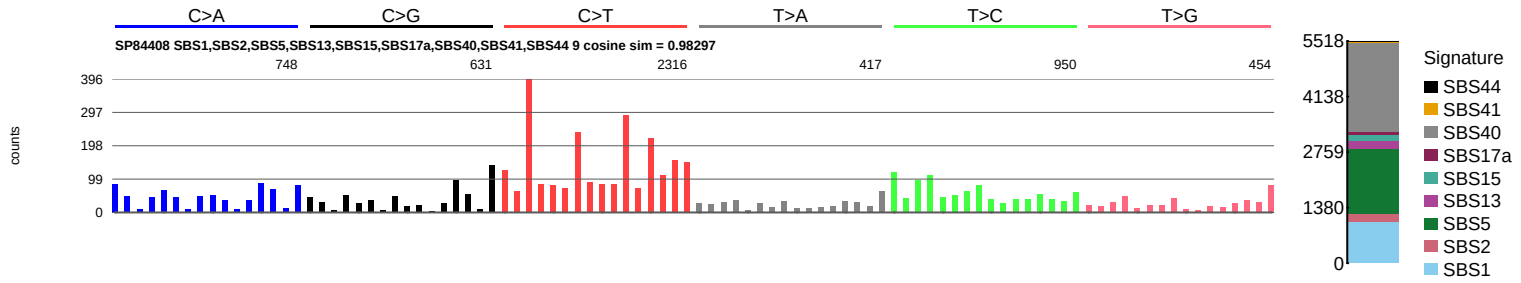
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS41,SBS44 9 cosine sim = 0.98297
748
631
2316
417
950
454
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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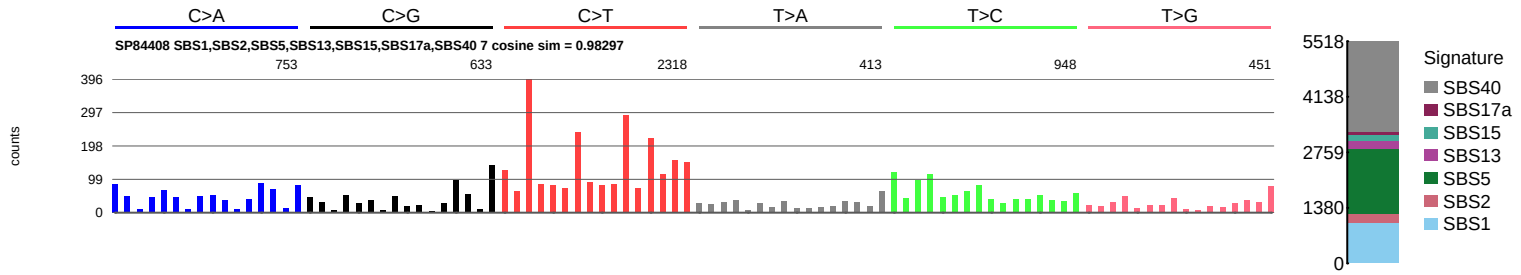
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40 7 cosine sim = 0.98297
753
633
2318
413
948
451
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17a
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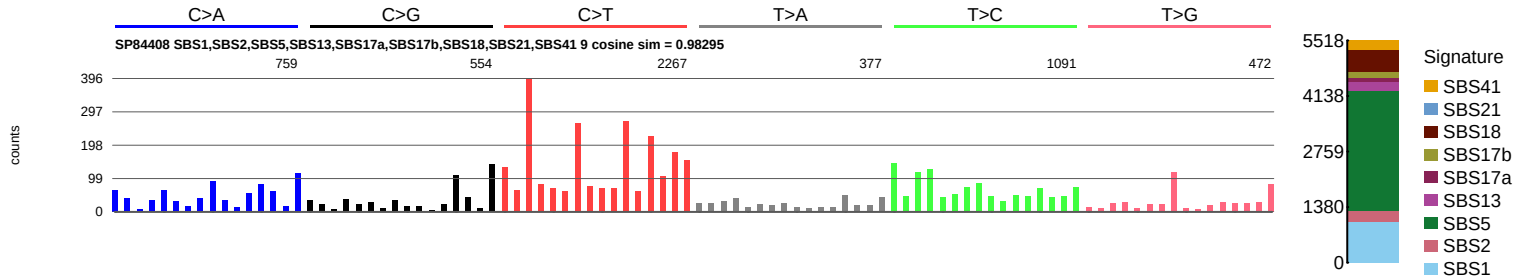
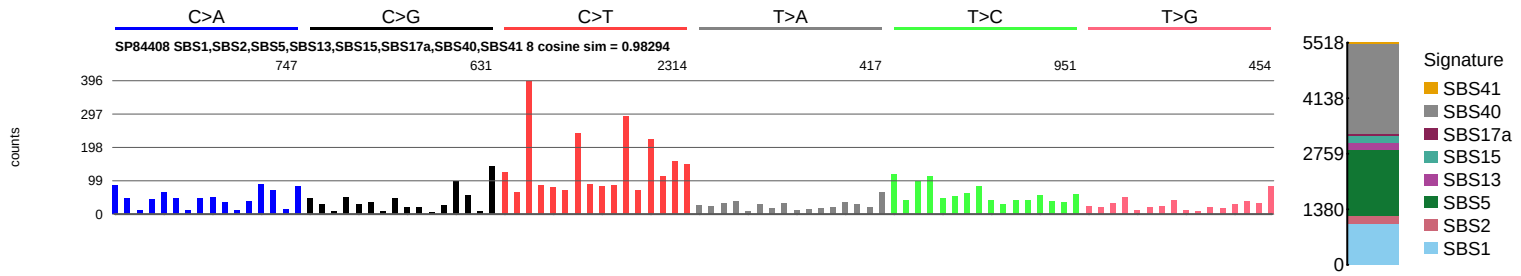
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98295
759
554
2267
377
1091
472
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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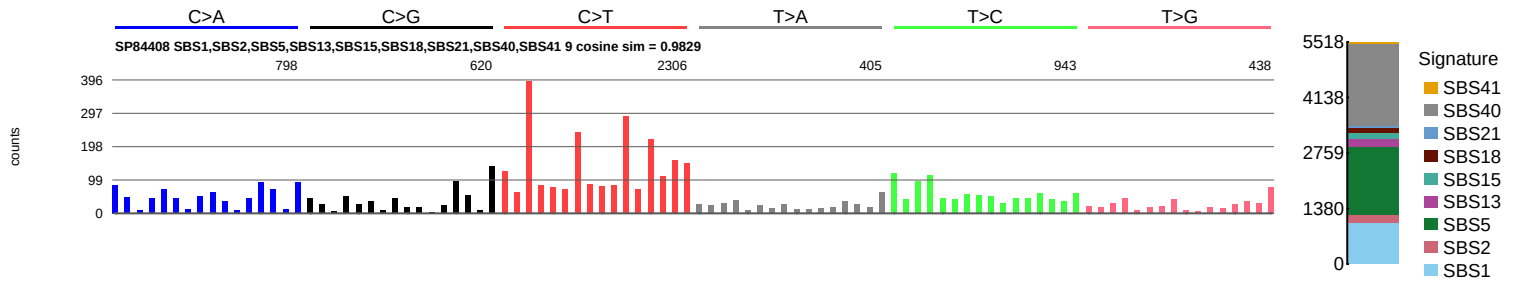
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS41 8 cosine sim = 0.98294
747
631
2314
417
951
454
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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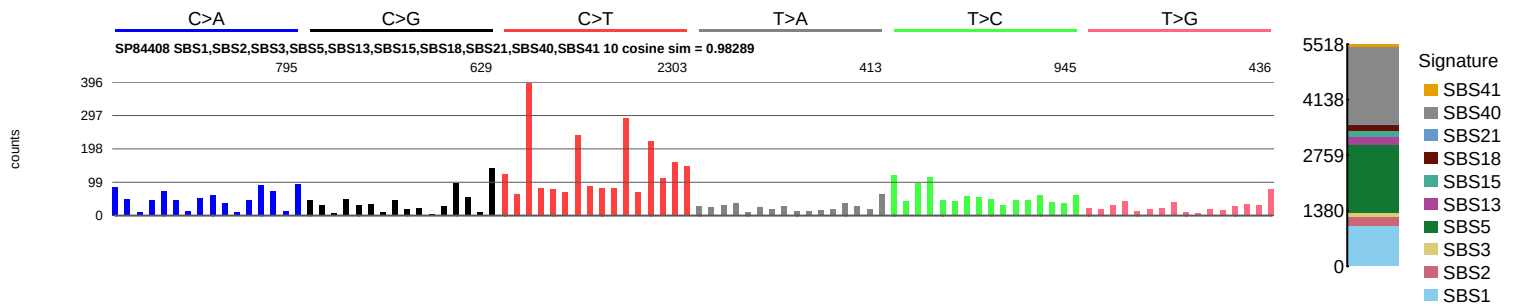
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9829
798
620
2306
405
943
438
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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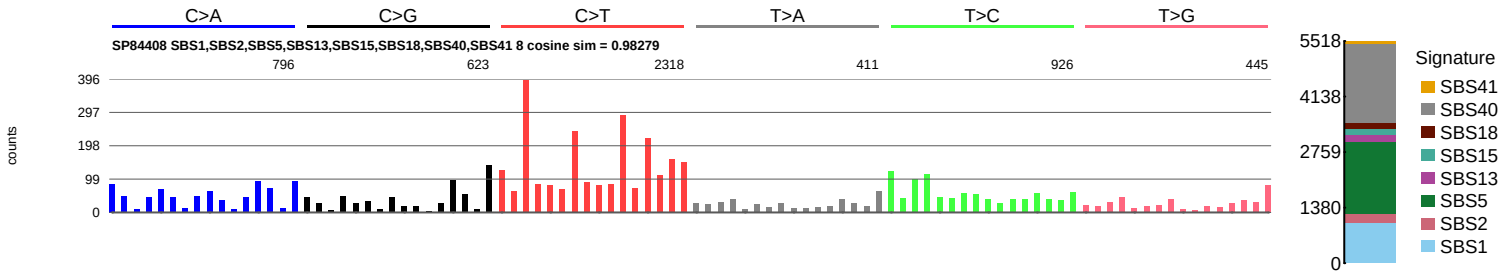
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98289
795
629
2303
413
945
436
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98279
796
623
2318
411
926
445
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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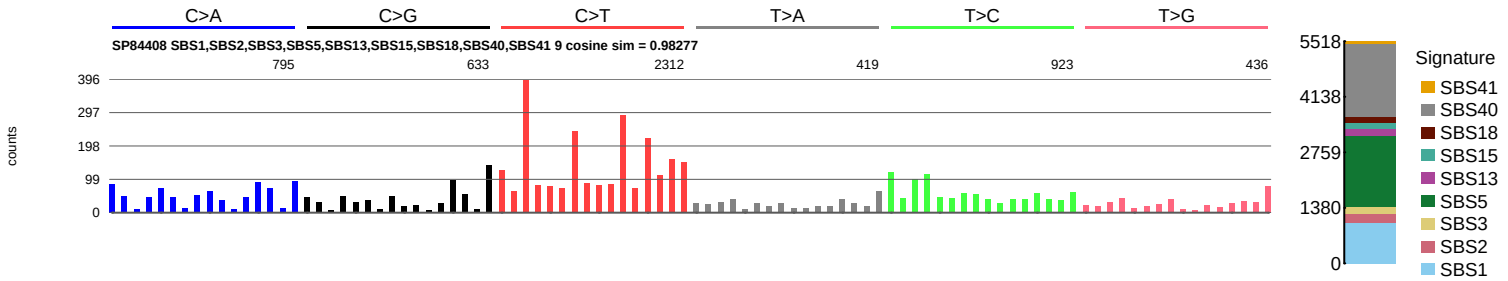
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 9 cosine sim = 0.98277
795
633
2312
419
923
436
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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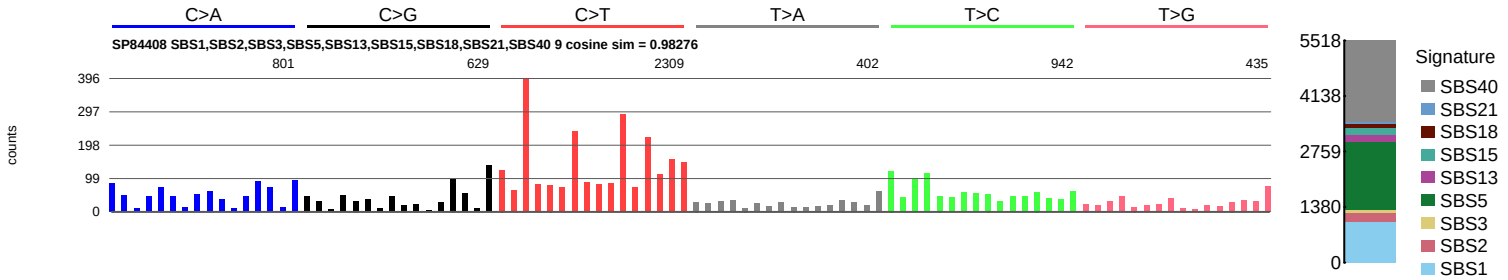
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 9 cosine sim = 0.98276
801
629
2309
402
942
435
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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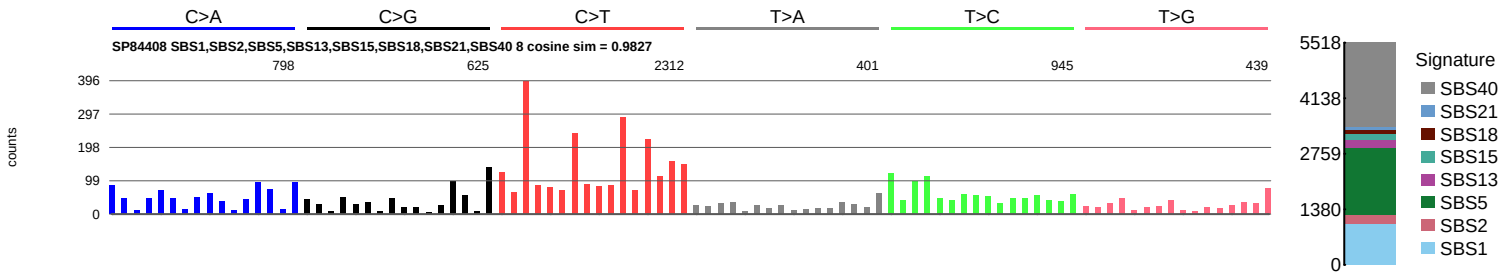
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.9827
798
625
2312
401
945
439
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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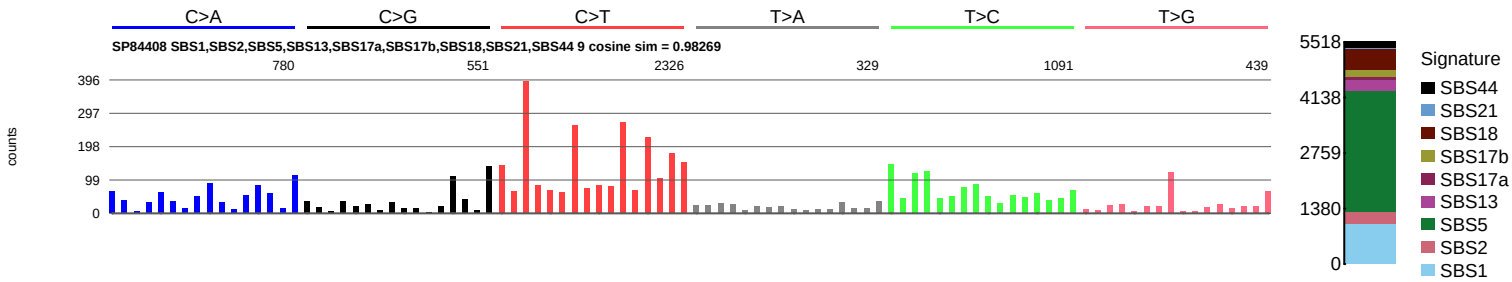
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98269
780
551
2326
329
1091
439
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
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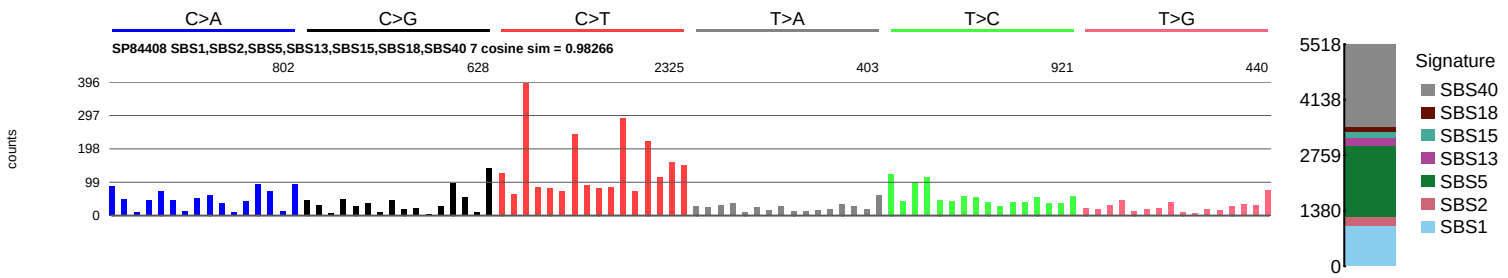
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.98266
802
628
2325
403
921
440
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
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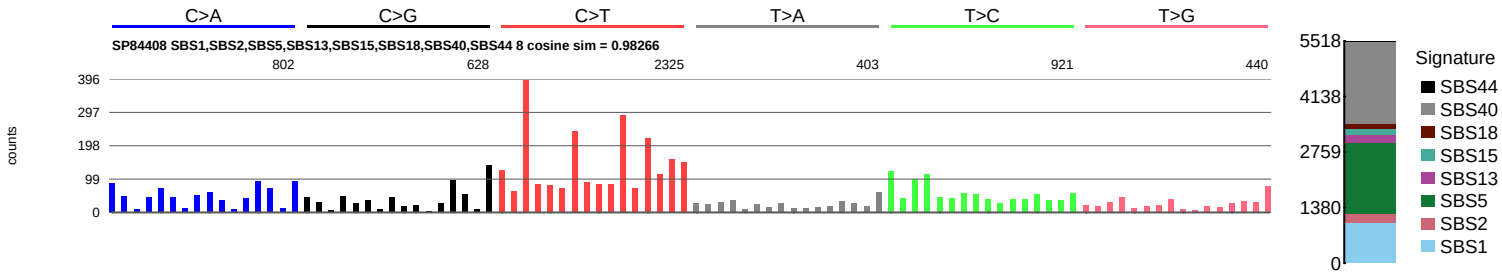
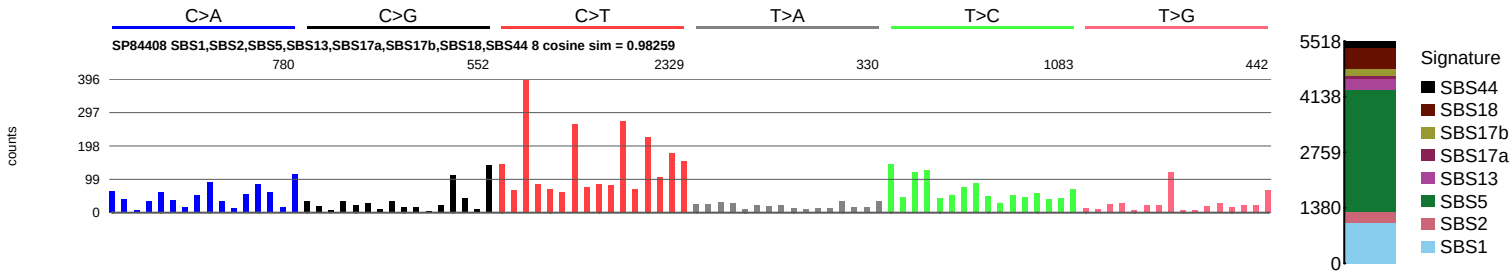
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98266
802
628
2325
403
921
440
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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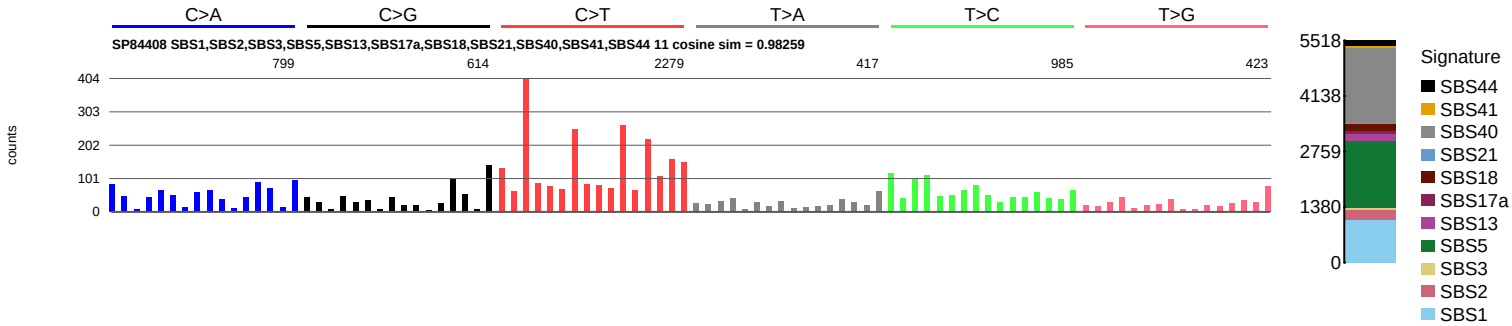
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98259
780
552
2329
330
1083
442
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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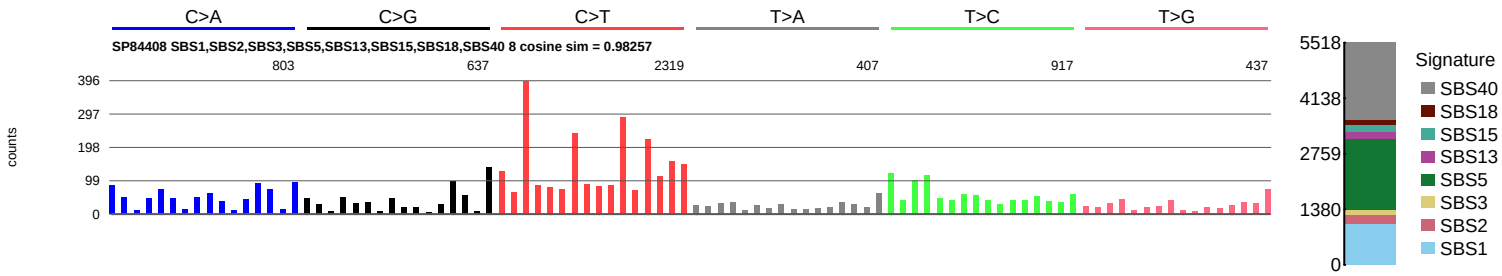
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98259
799
614
2279
417
985
423
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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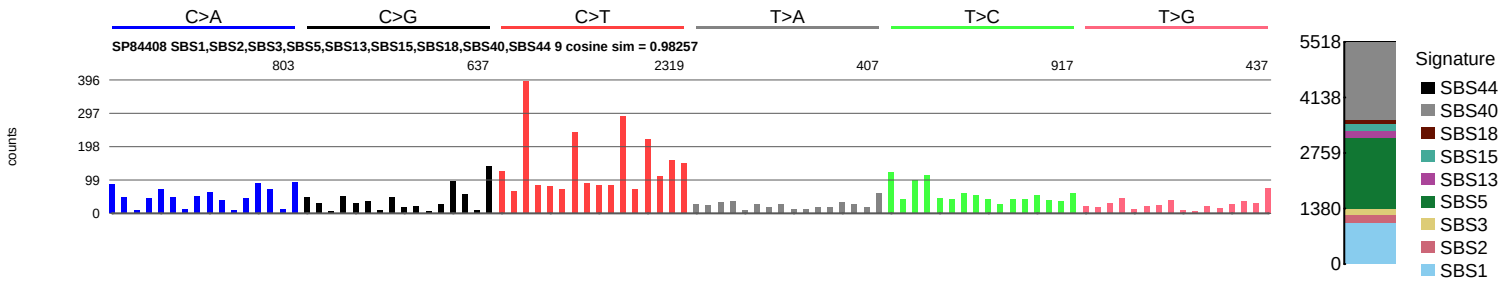
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40 8 cosine sim = 0.98257
803
637
2319
407
917
437
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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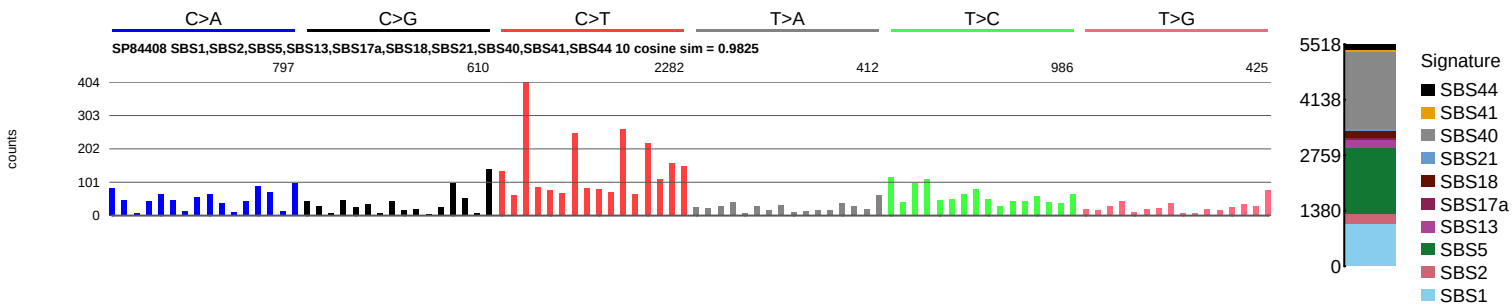
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS44 9 cosine sim = 0.98257
803
637
2319
407
917
437
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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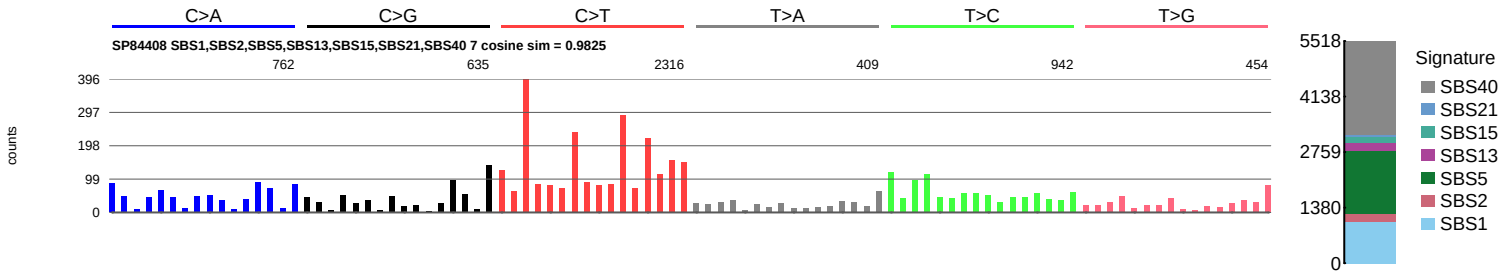
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9825
797
610
2282
412
986
425
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS21,SBS40 7 cosine sim = 0.9825
762
635
2316
409
942
454
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS15
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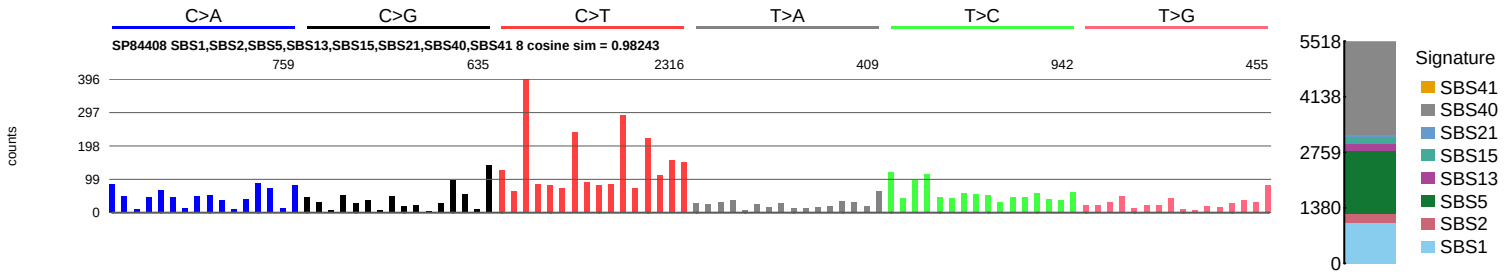
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS21,SBS40,SBS41 8 cosine sim = 0.98243
759
635
2316
409
942
455
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS15
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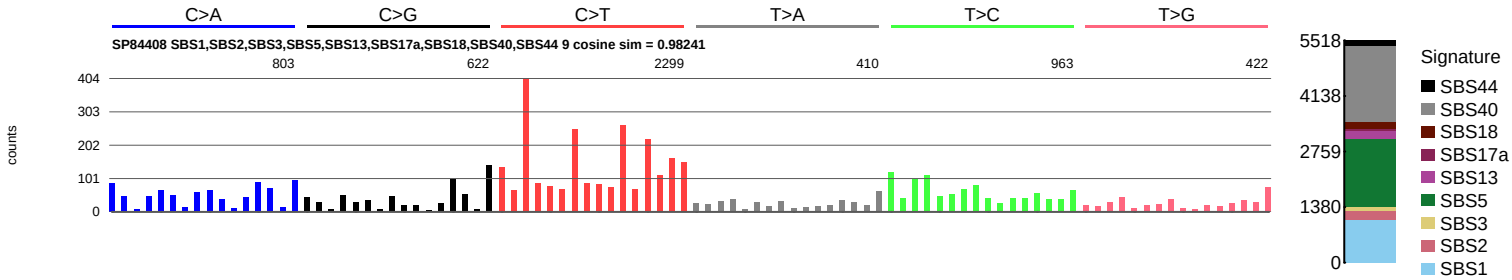
SBS5
SBS2
SBS1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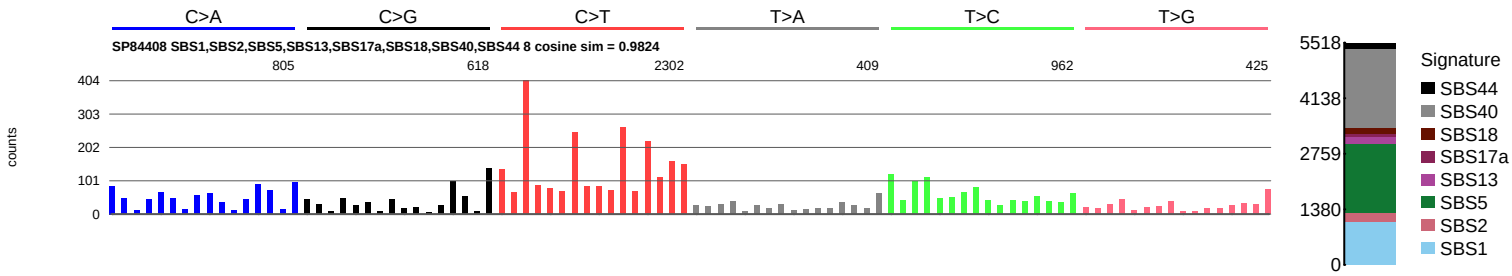
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98241
803
622
2299
410
963
422
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.9824
805
618
2302
409
962
425
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
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SBS17a
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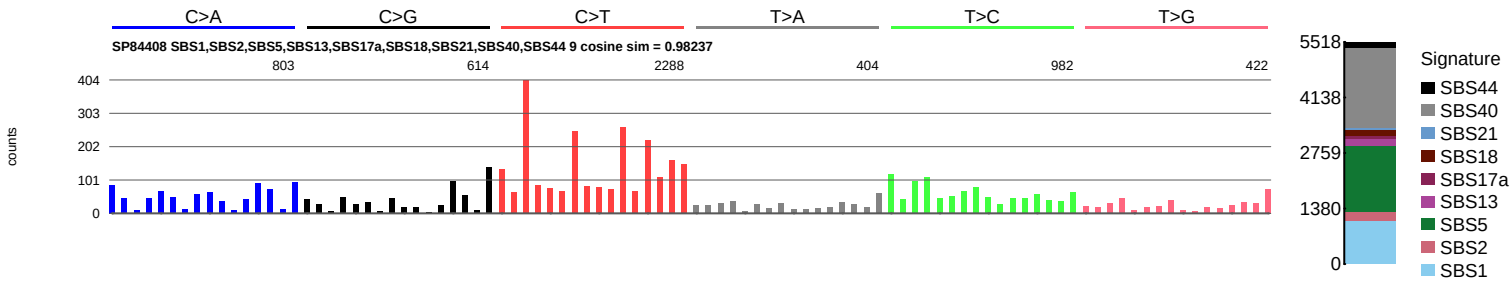
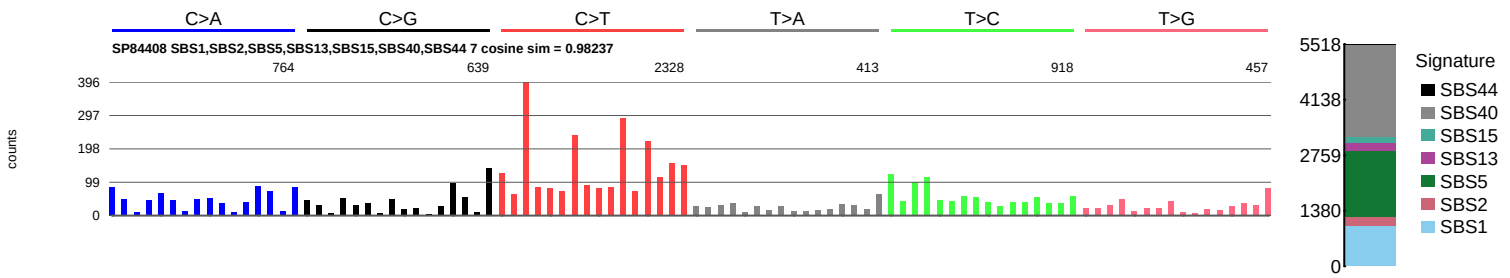
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98237
803
614
2288
404
982
422
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40,SBS44 7 cosine sim = 0.98237
764
639
2328
413
918
457
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
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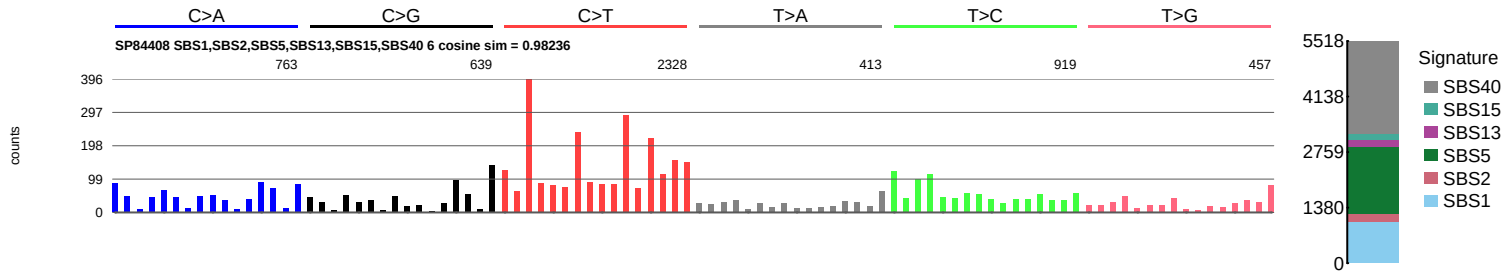
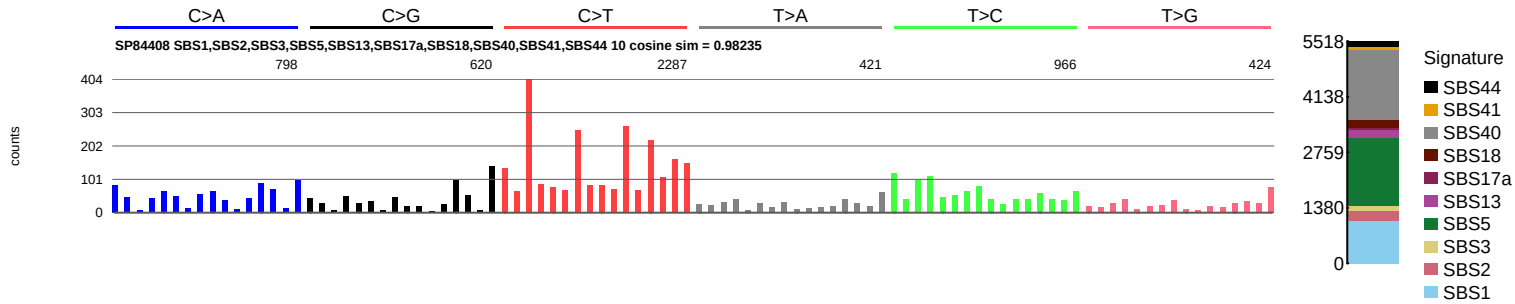
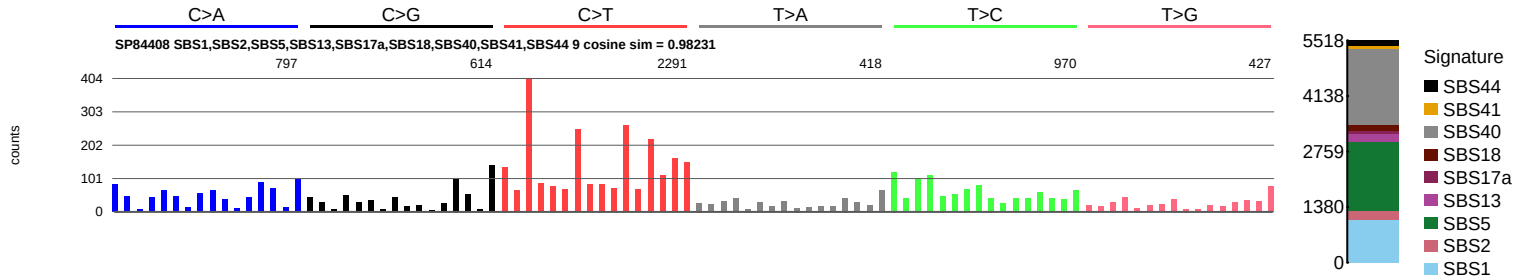
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40 6 cosine sim = 0.98236
763
639
2328
413
919
457
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98235
798
620
2287
421
966
424
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98231
797
614
2291
418
970
427
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
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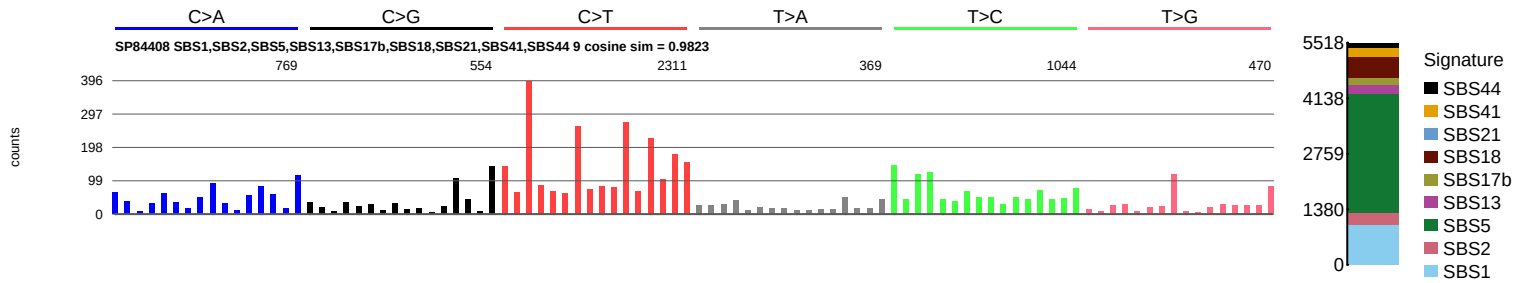
SBS5
SBS2
SBS1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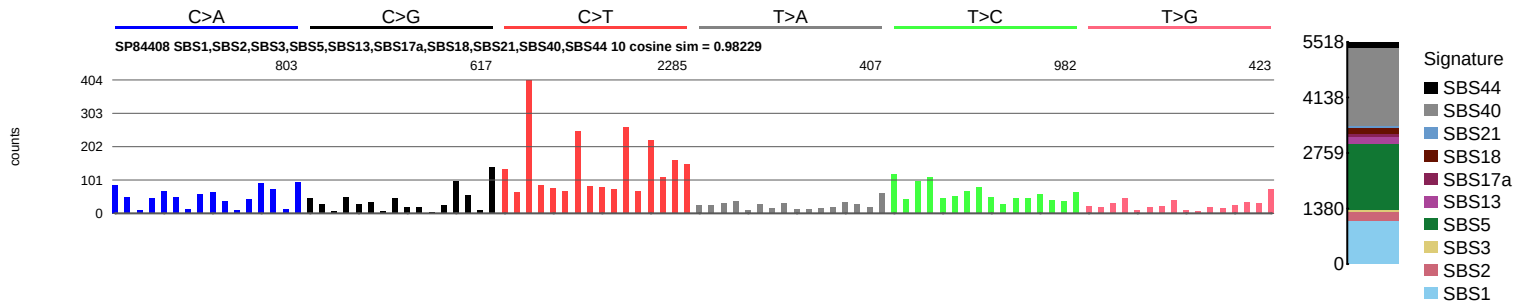
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9823
769
554
2311
369
1044
470
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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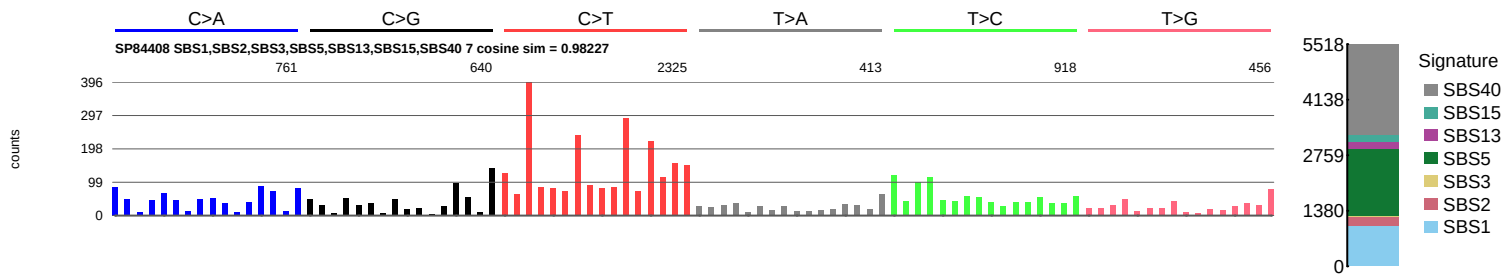
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98229
803
617
2285
407
982
423
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40 7 cosine sim = 0.98227
761
640
2325
413
918
456
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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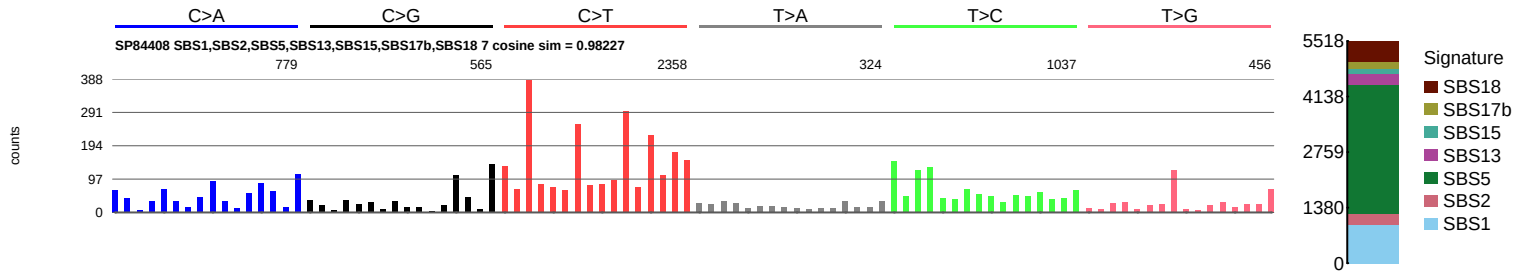
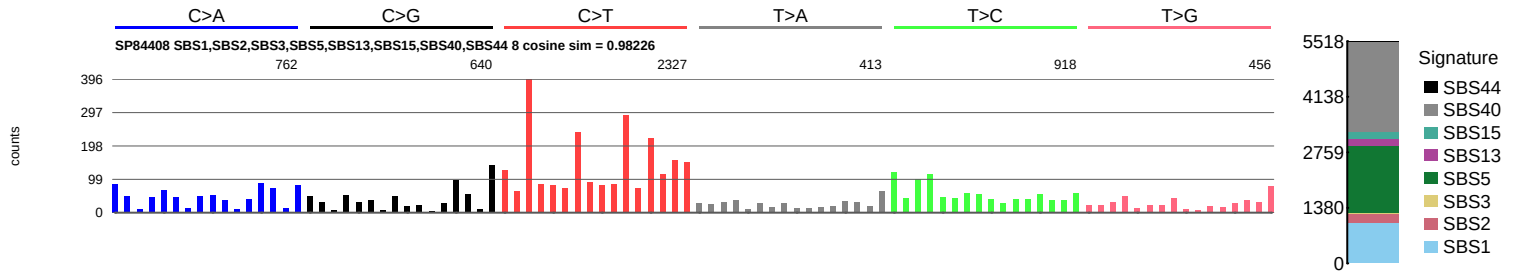
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18 7 cosine sim = 0.98227
779
565
2358
324
1037
456
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40,SBS44 8 cosine sim = 0.98226
762
640
2327
413
918
456
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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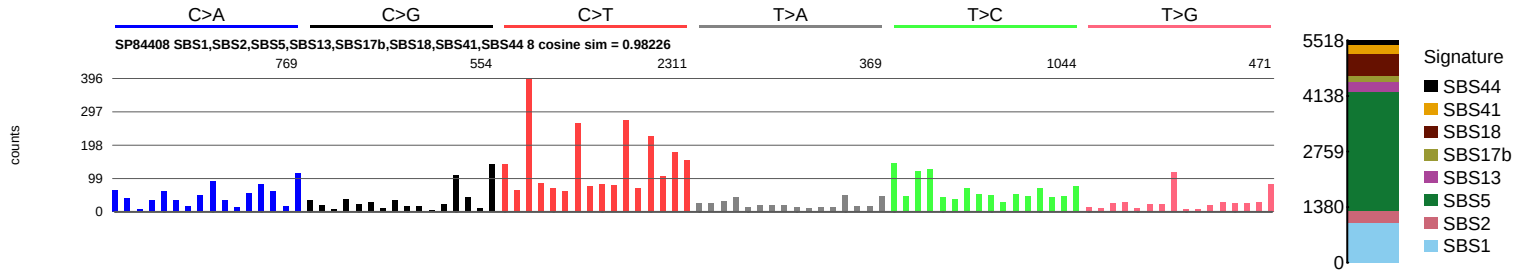
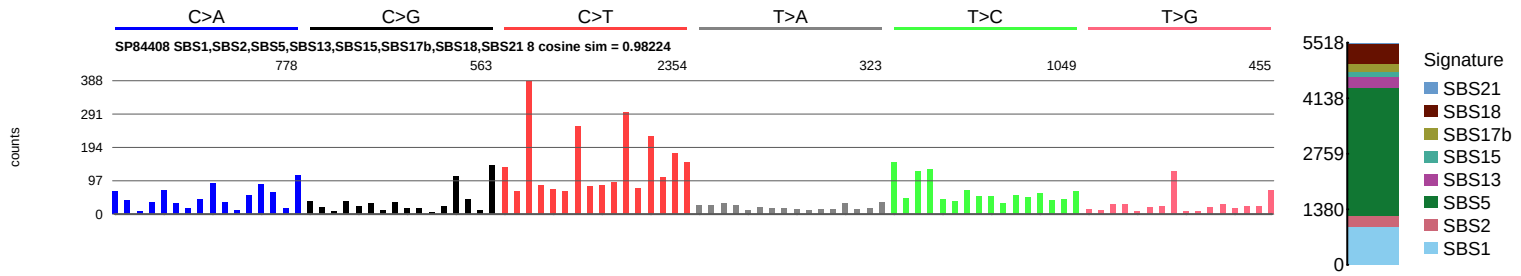
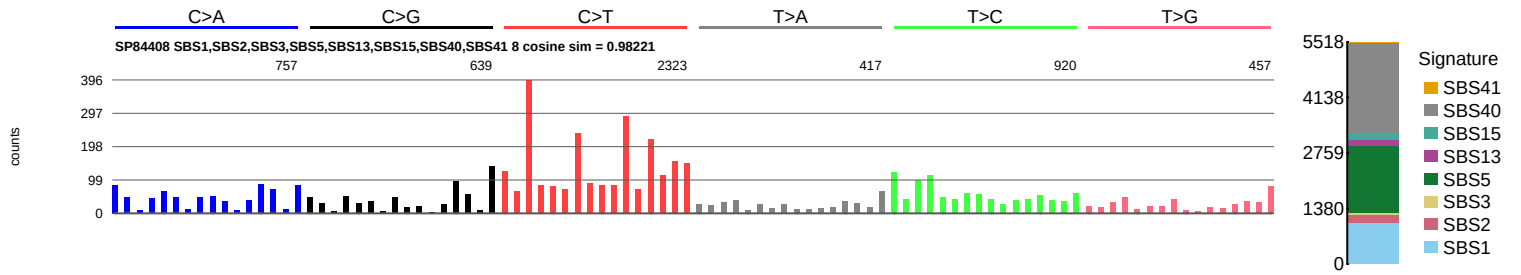
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98226
769
554
2311
369
1044
471
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98224
778
563
2354
323
1049
455
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40,SBS41 8 cosine sim = 0.98221
757
639
2323
417
920
457
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS15
SBS13
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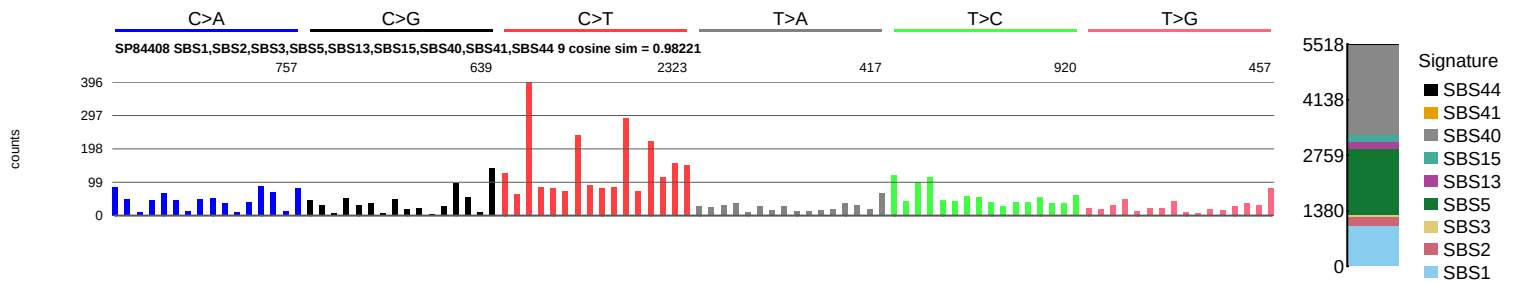
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40,SBS41,SBS44 9 cosine sim = 0.98221
757
639
2323
417
920
457
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS15
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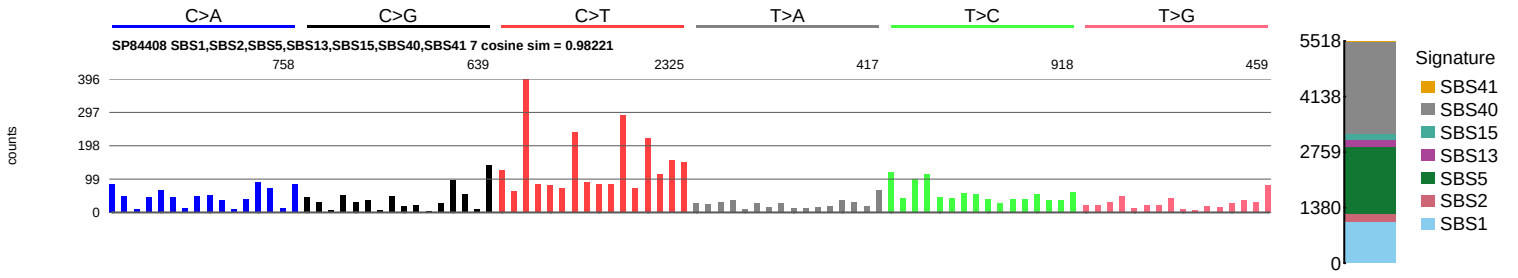
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40,SBS41 7 cosine sim = 0.98221
758
639
2325
417
918
459
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
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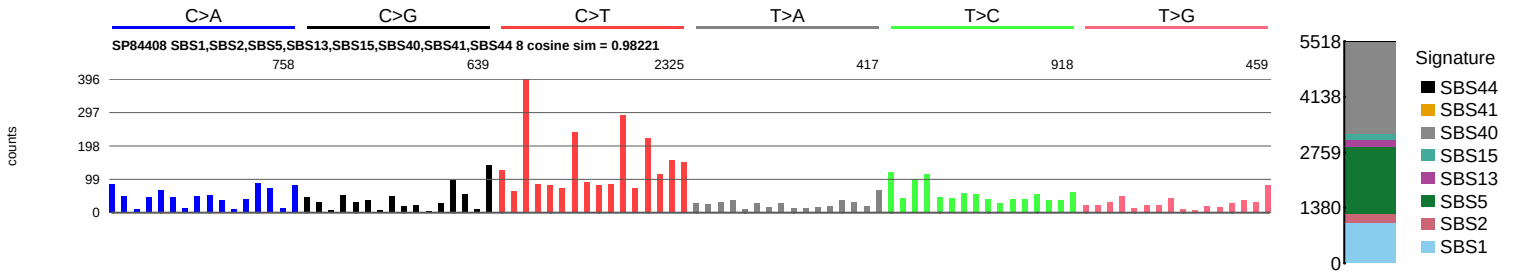
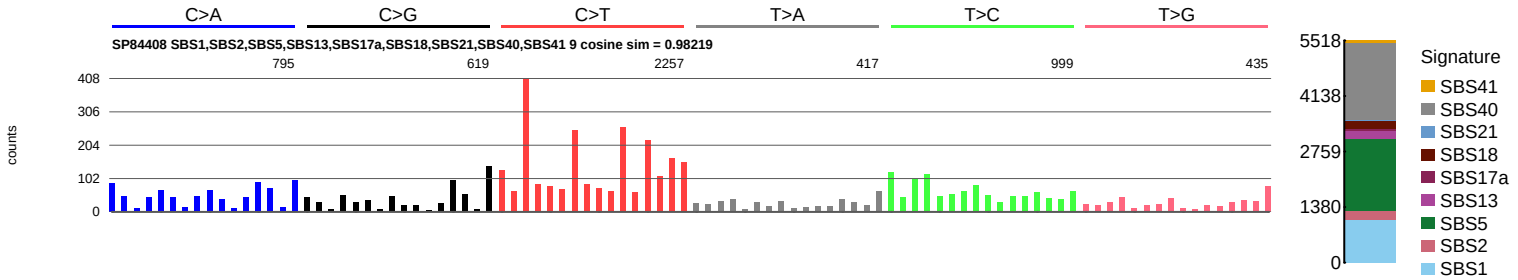
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40,SBS41,SBS44 8 cosine sim = 0.98221
758
639
2325
417
918
459
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS15
SBS13
SBS5
SBS2
SBS1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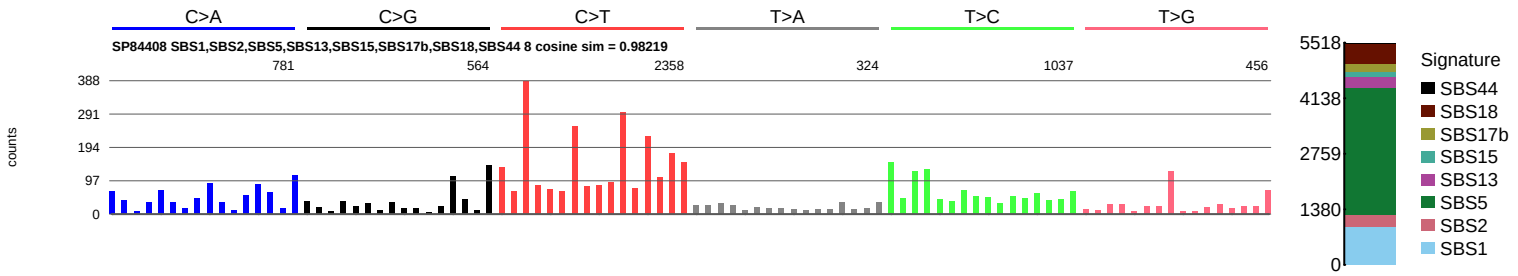
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98219
795
619
2257
417
999
435
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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C>T
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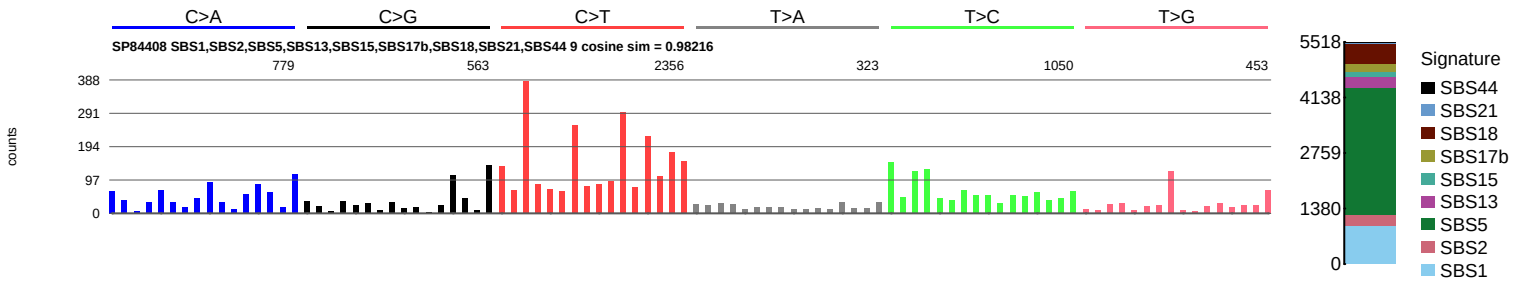
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98219
781
564
2358
324
1037
456
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>T
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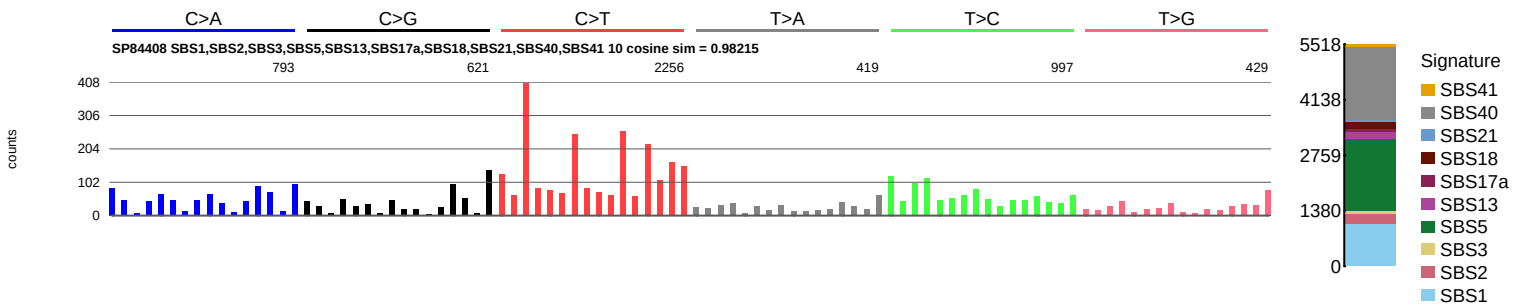
SP84408 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98216
779
563
2356
323
1050
453
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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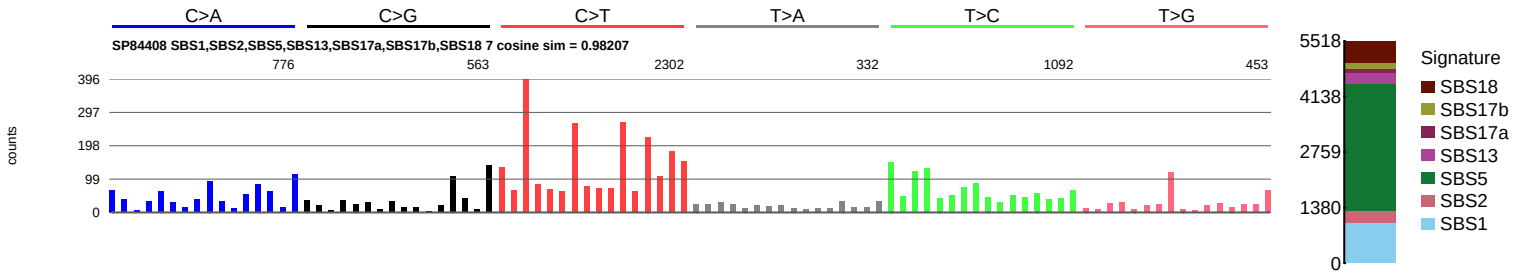
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98215
793
621
2256
419
997
429
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98207
776
563
2302
332
1092
453
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS18
SBS17b
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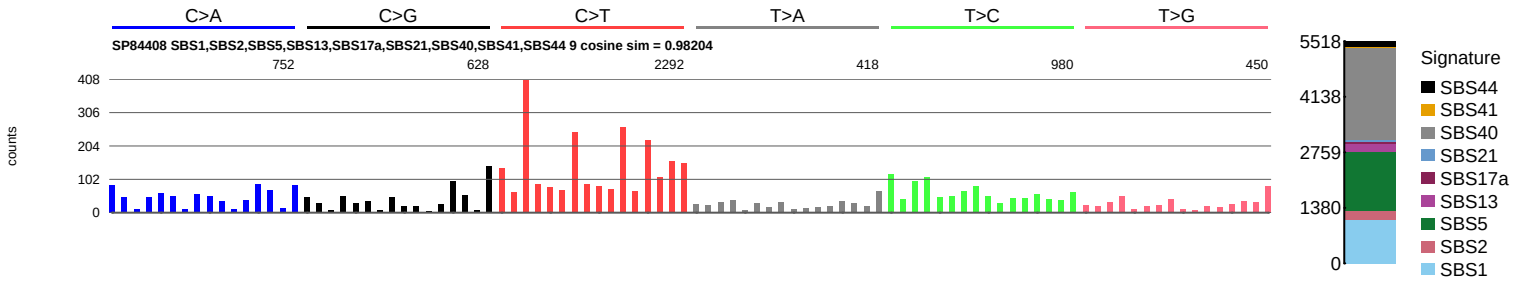
SBS2
SBS1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C>G
C>T
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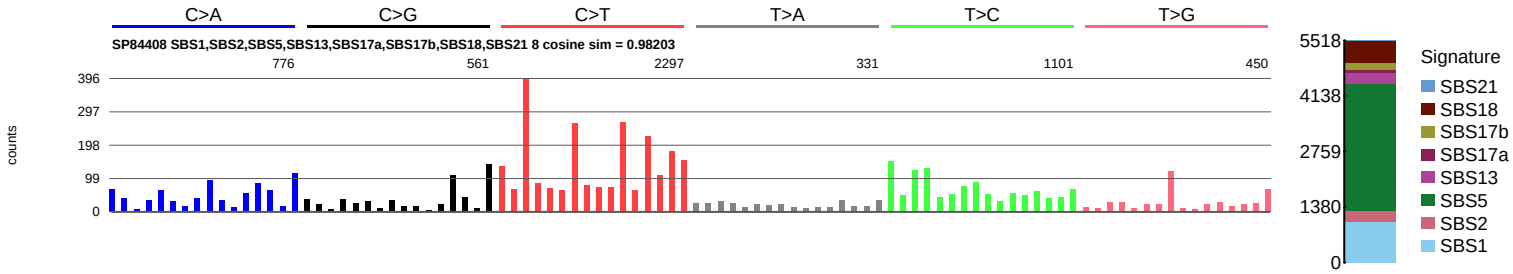
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98204
752
628
2292
418
980
450
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS2
SBS1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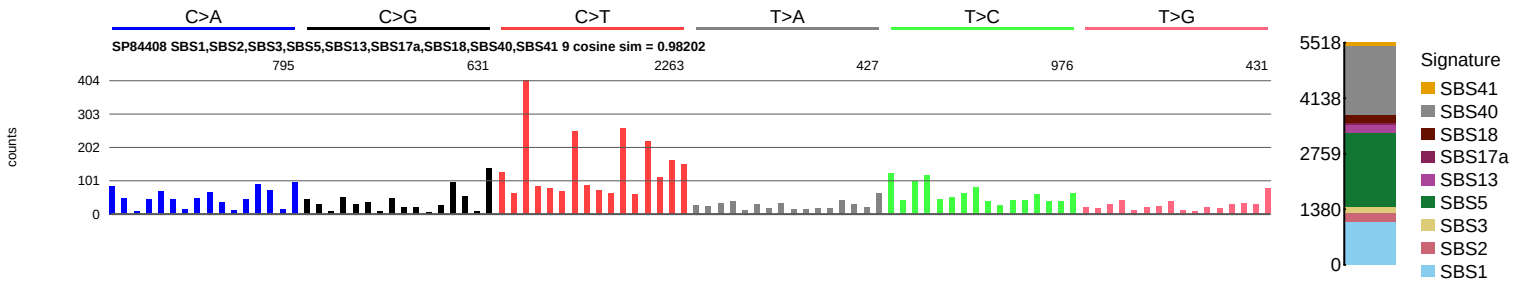
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98203
776
561
2297
331
1101
450
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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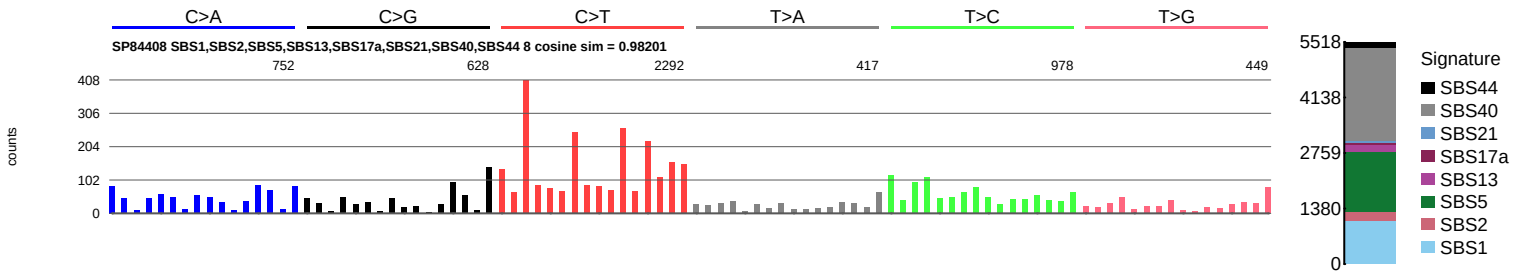
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98202
795
631
2263
427
976
431
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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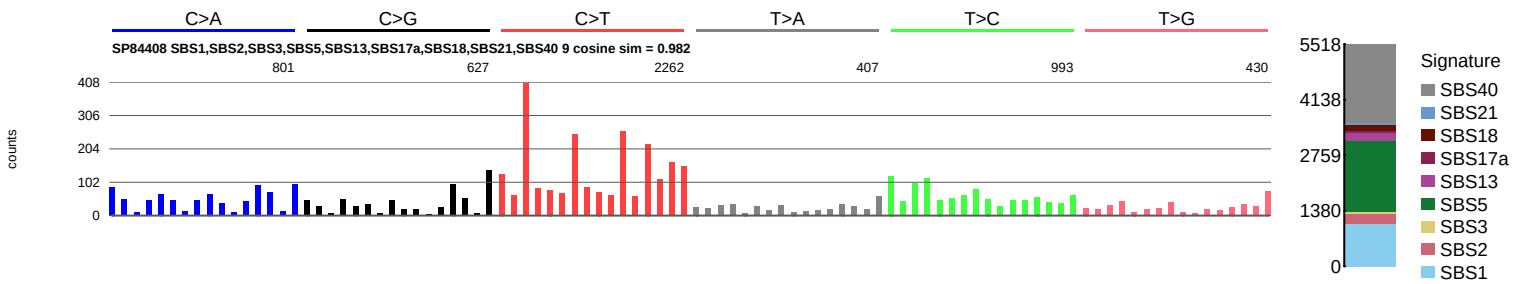
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS44 8 cosine sim = 0.98201
752
628
2292
417
978
449
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS2
SBS1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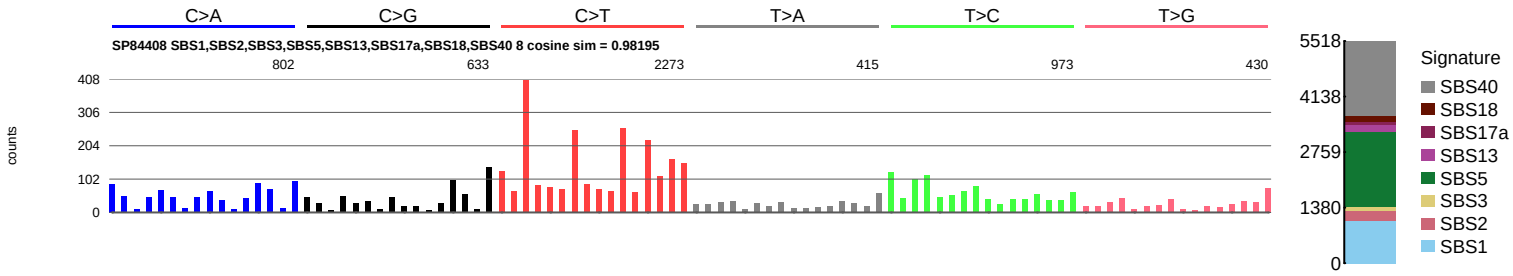
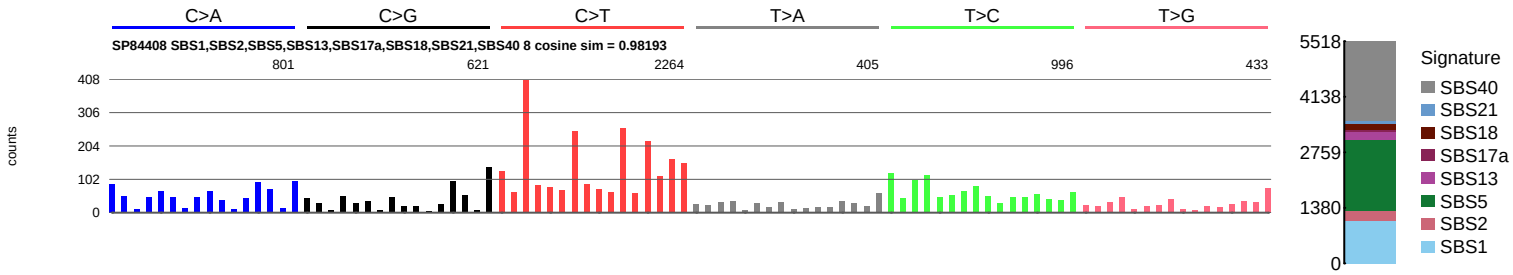
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.982
801
627
2262
407
993
430
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40 8 cosine sim = 0.98195
802
633
2273
415
973
430
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98193
801
621
2264
405
996
433
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
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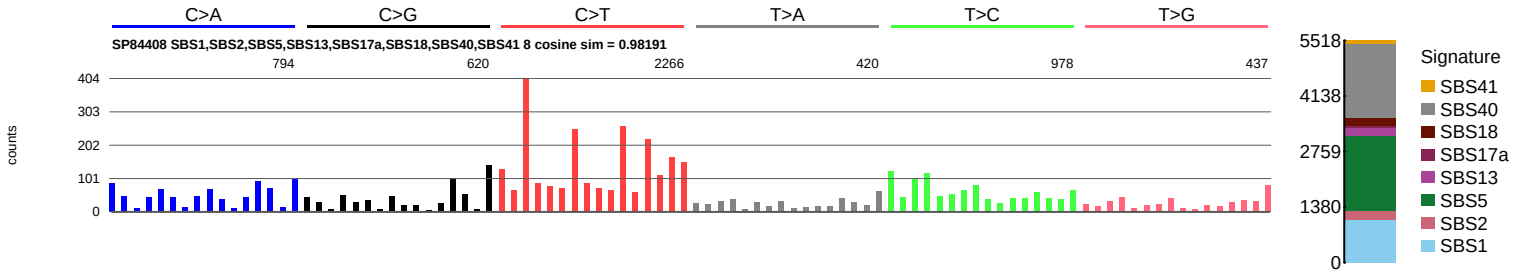
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98191
794
620
2266
420
978
437
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
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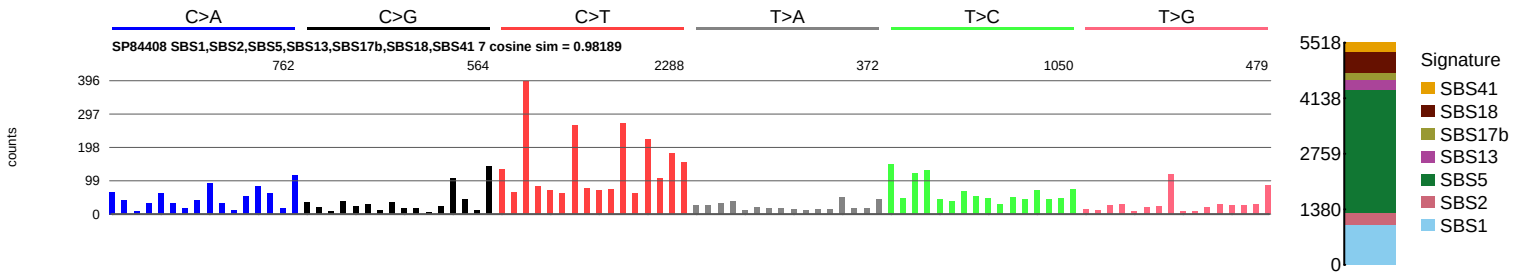
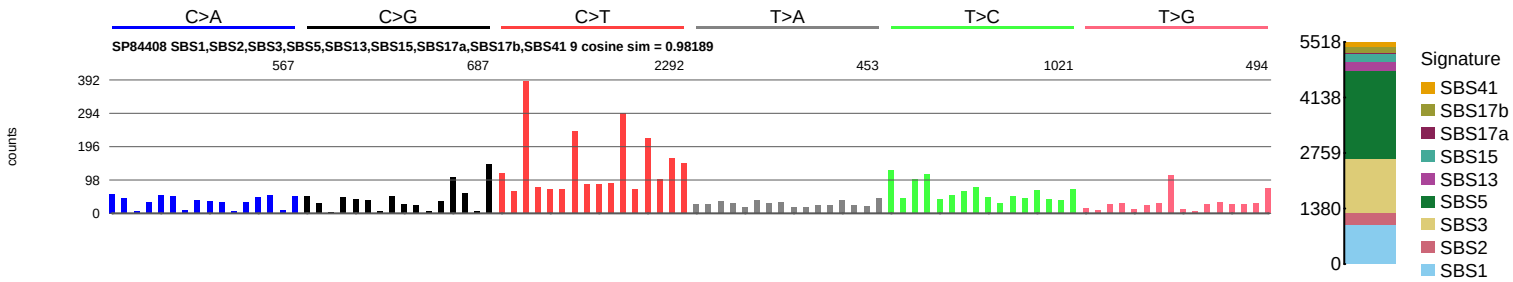
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98189
762
564
2288
372
1050
479
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 9 cosine sim = 0.98189
567
687
2292
453
1021
494
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS41
SBS17b
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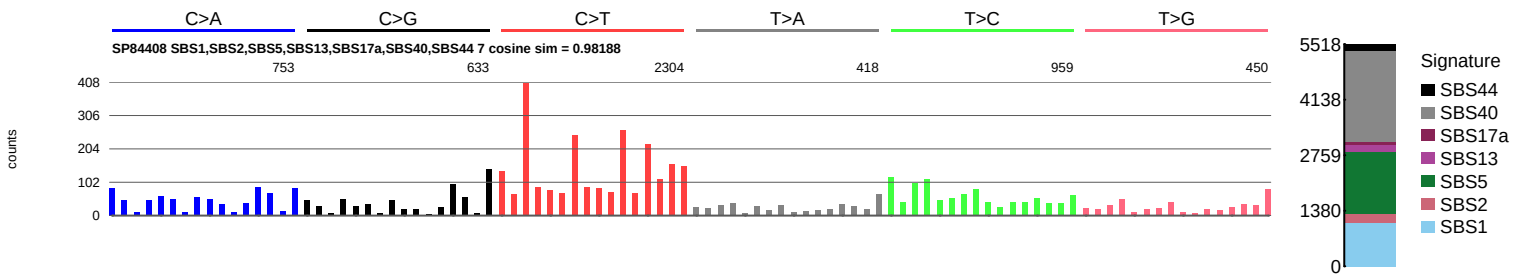
SBS2
SBS1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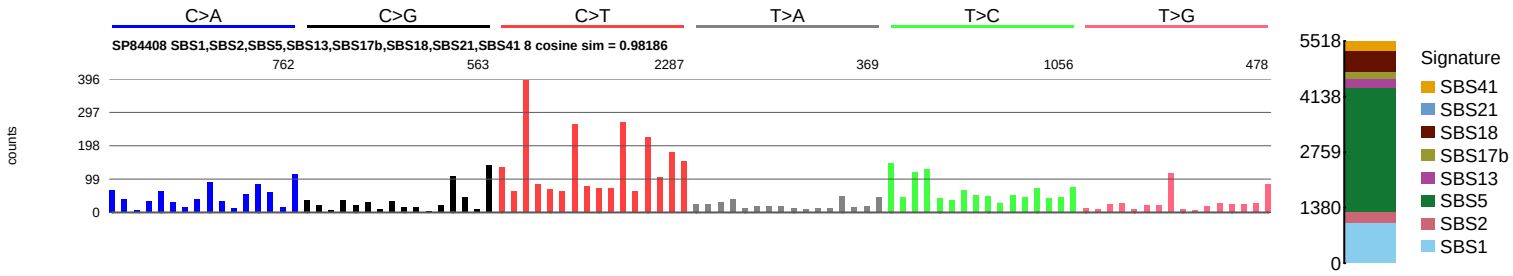
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40,SBS44 7 cosine sim = 0.98188
753
633
2304
418
959
450
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98186
762
563
2287
369
1056
478
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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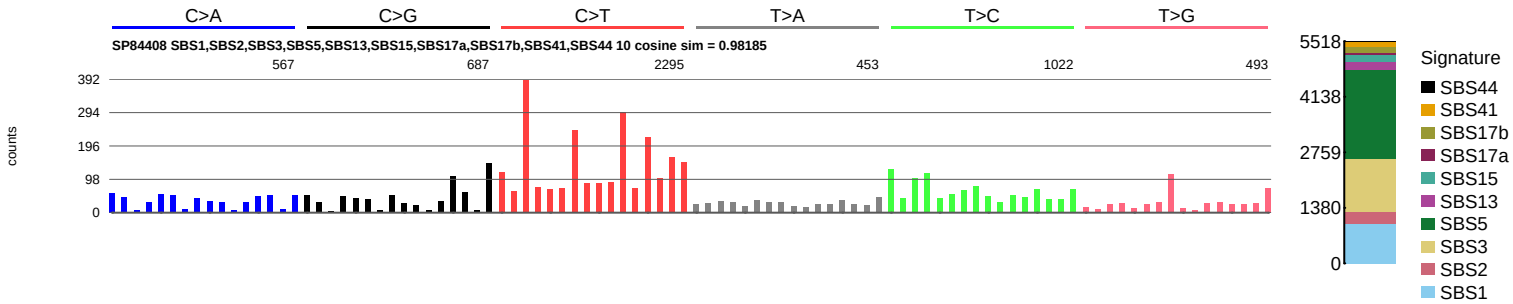
SBS2
SBS1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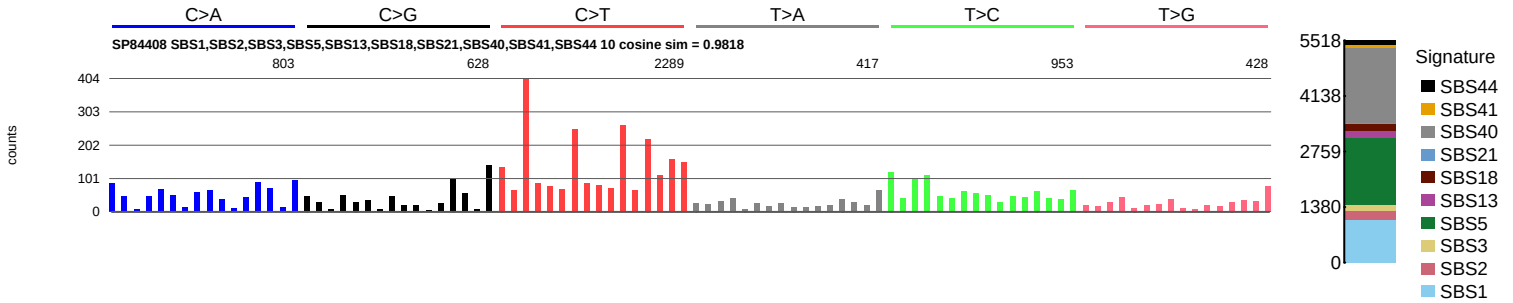
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 10 cosine sim = 0.98185
567
687
2295
453
1022
493
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9818
803
628
2289
417
953
428
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
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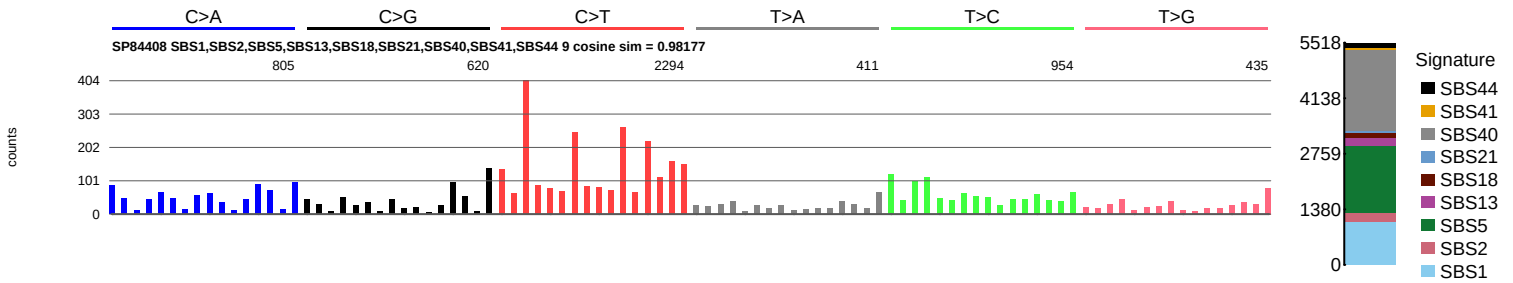
SBS2
SBS1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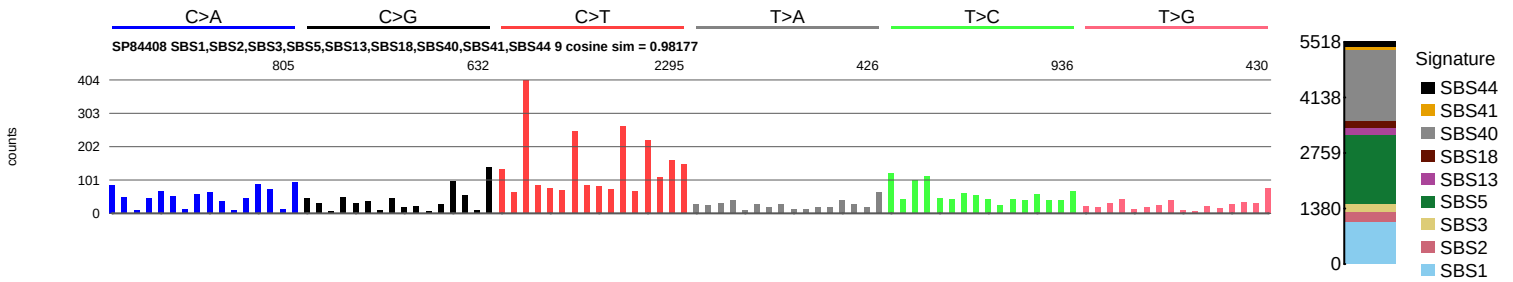
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98177
805
620
2294
411
954
435
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1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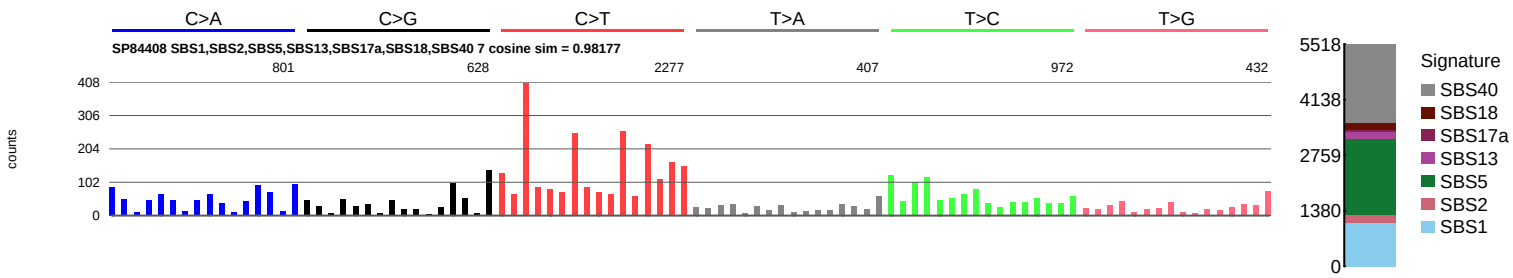
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98177
805
632
2295
426
936
430
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.98177
801
628
2277
407
972
432
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
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SBS17a
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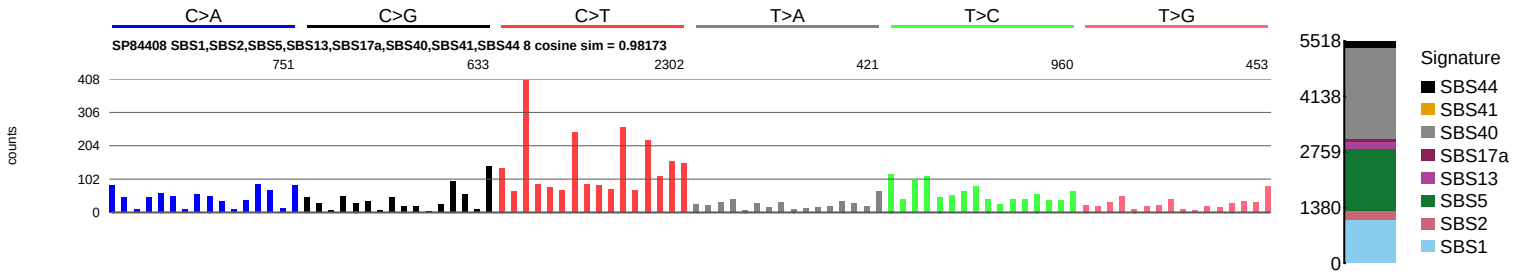
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40,SBS41,SBS44 8 cosine sim = 0.98173
751
633
2302
421
960
453
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
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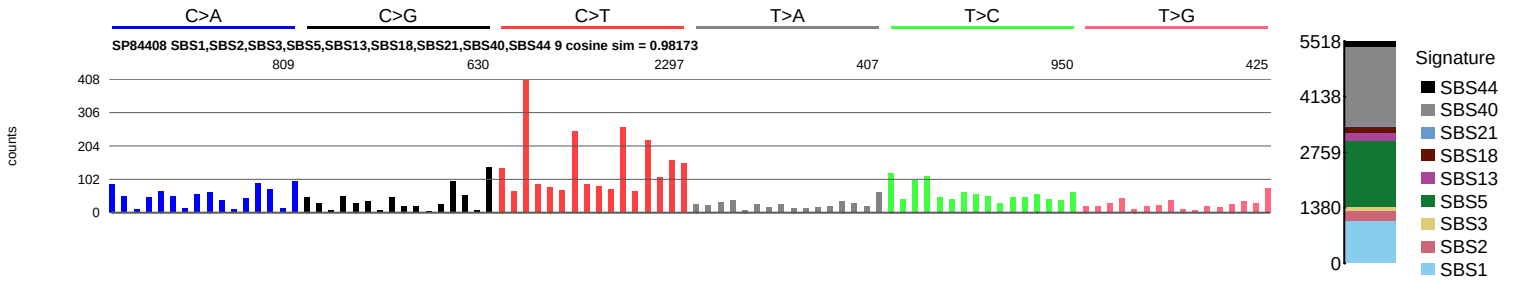
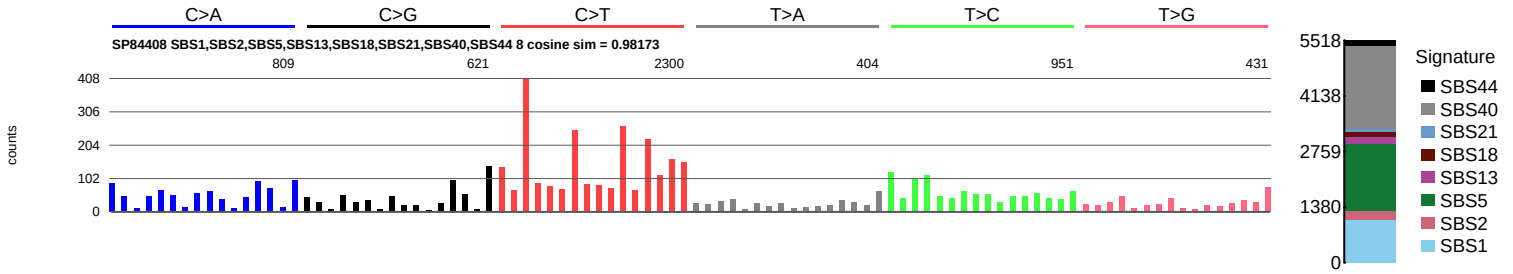
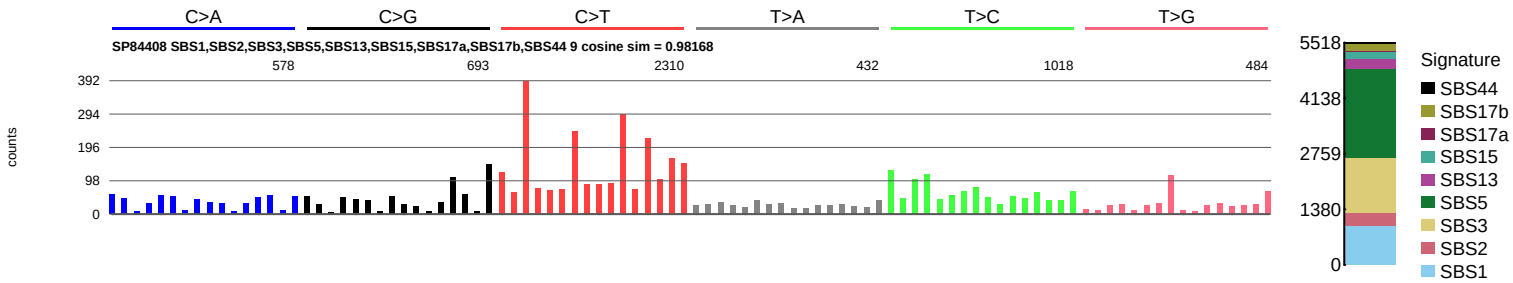
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98173
809
630
2297
407
950
425
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98173
809
621
2300
404
951
431
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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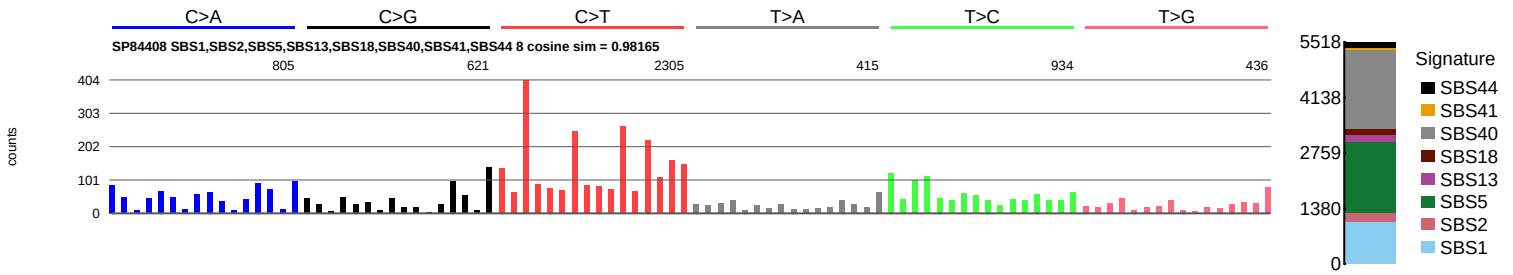
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 9 cosine sim = 0.98168
578
693
2310
432
1018
484
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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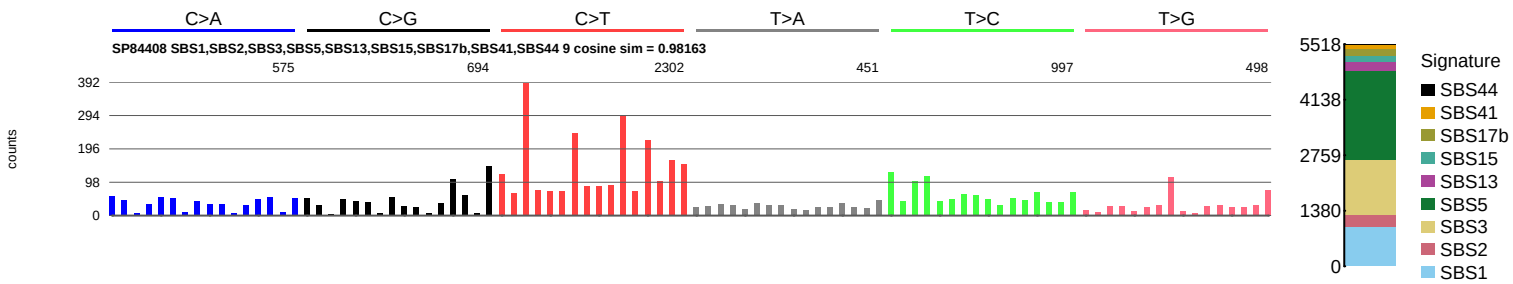
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98165
805
621
2305
415
934
436
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1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C>T
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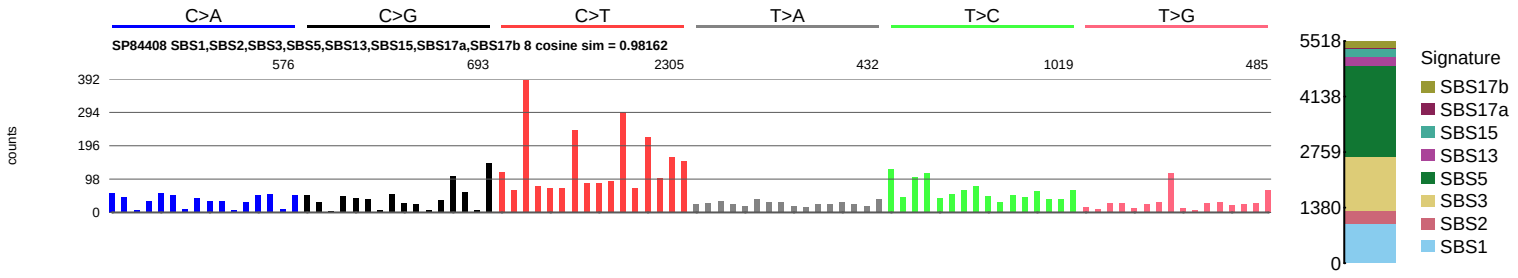
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS41,SBS44 9 cosine sim = 0.98163
575
694
2302
451
997
498
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 8 cosine sim = 0.98162
576
693
2305
432
1019
485
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS17b
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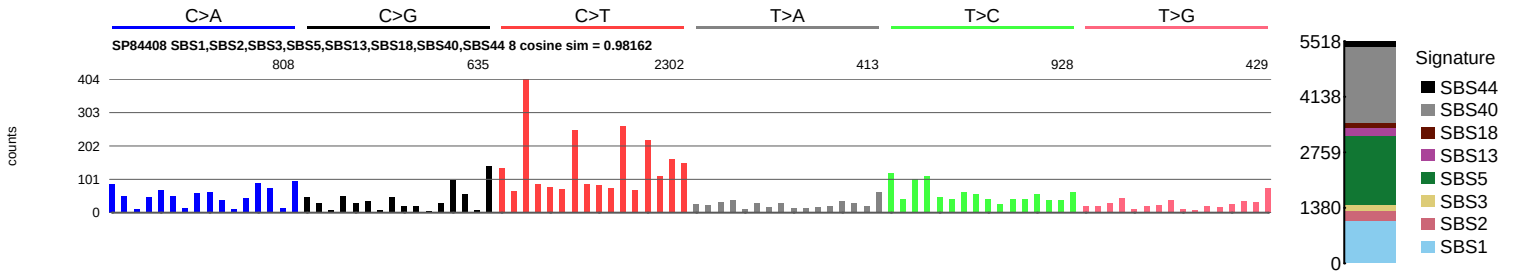
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS44 8 cosine sim = 0.98162
808
635
2302
413
928
429
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
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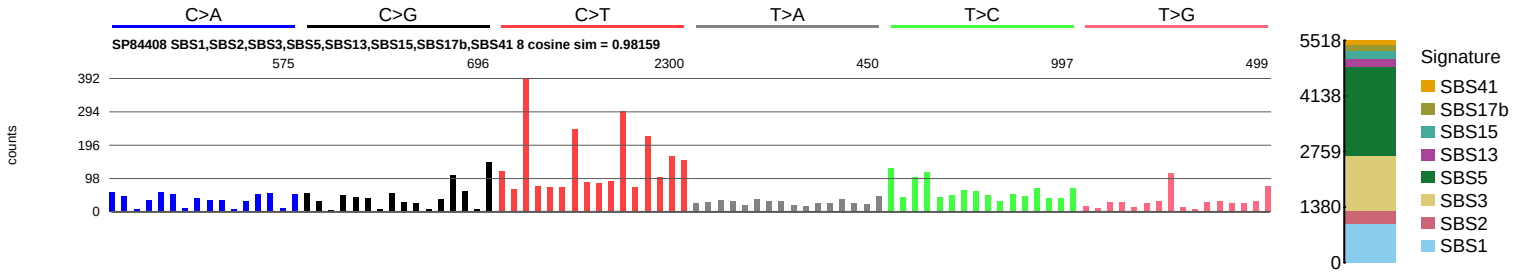
SBS2
SBS1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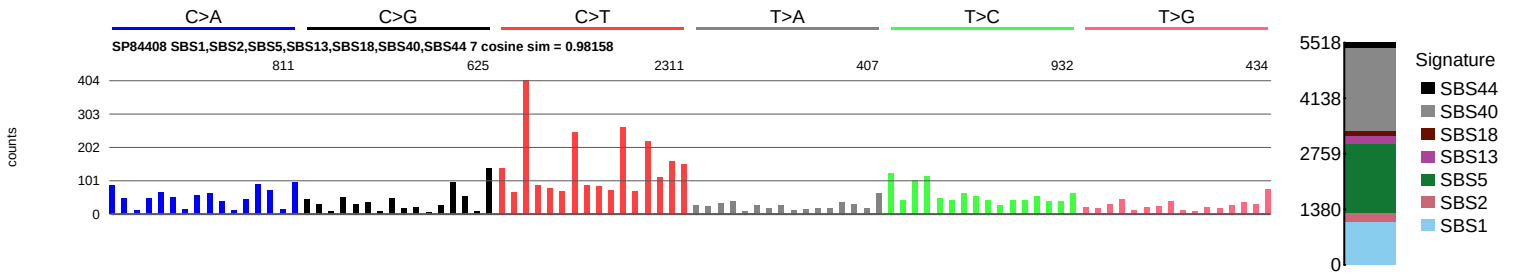
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS41 8 cosine sim = 0.98159
575
696
2300
450
997
499
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS41
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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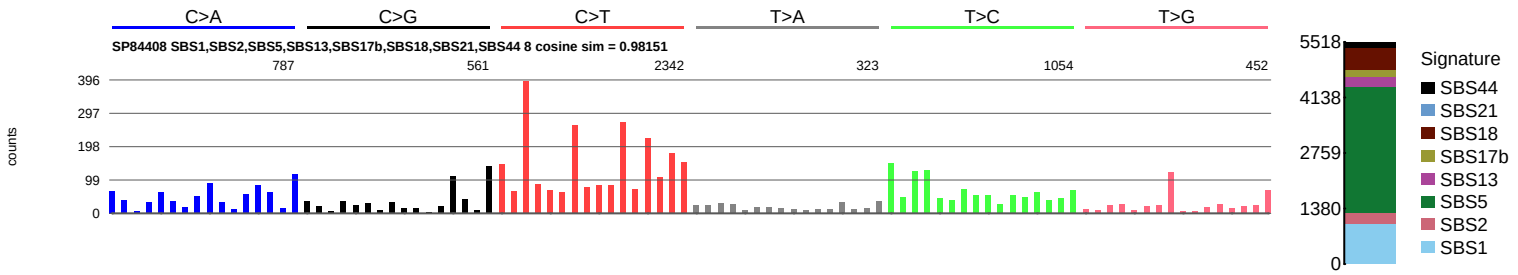
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.98158
811
625
2311
407
932
434
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98151
787
561
2342
323
1054
452
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS21
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SBS17b
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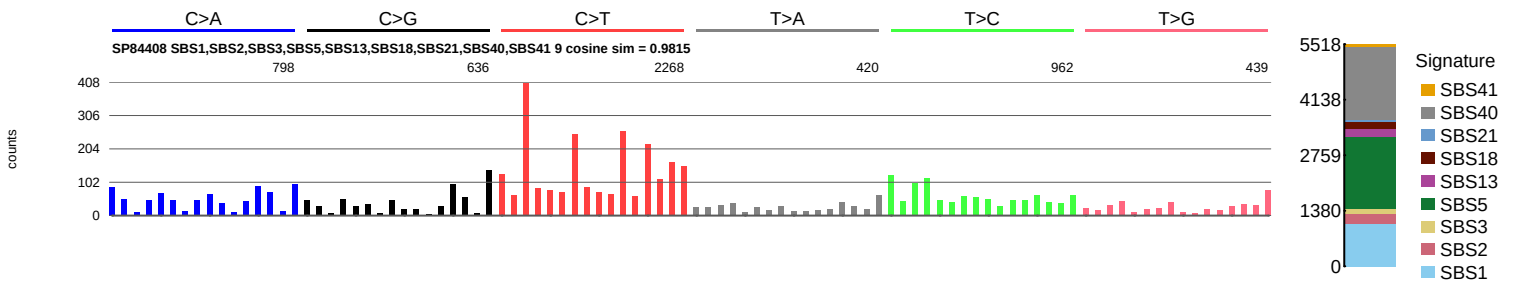
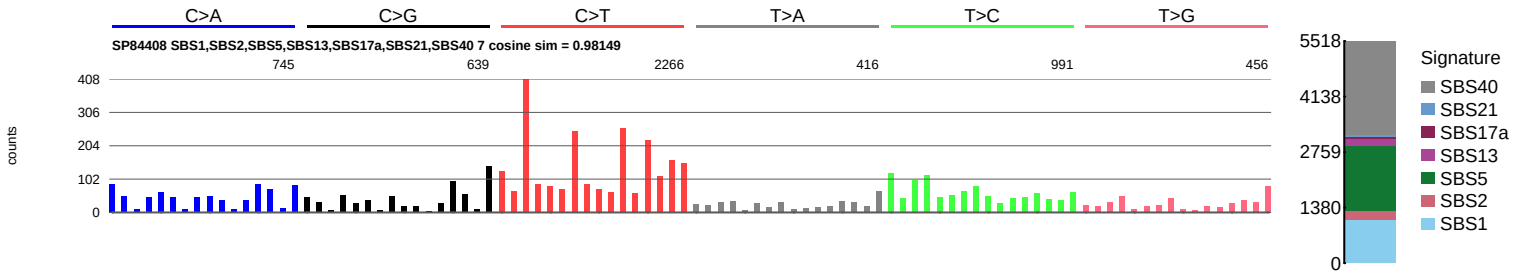
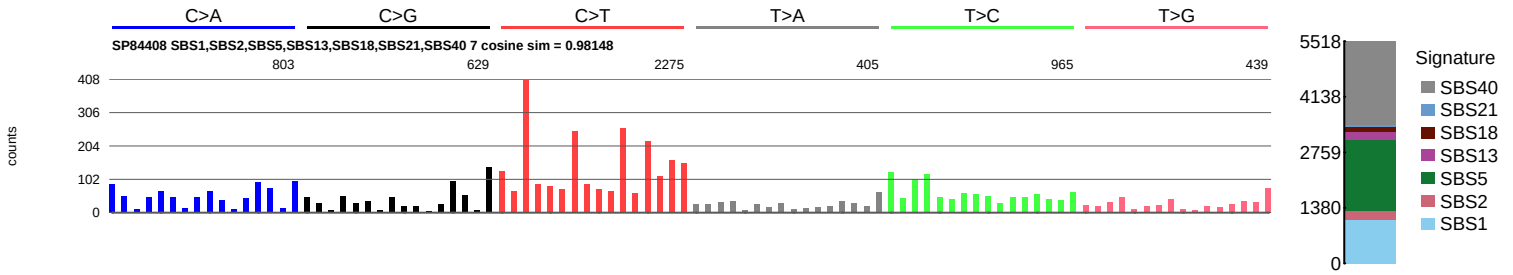
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9815
798
636
2268
420
962
439
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40 7 cosine sim = 0.98149
745
639
2266
416
991
456
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40 7 cosine sim = 0.98148
803
629
2275
405
965
439
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
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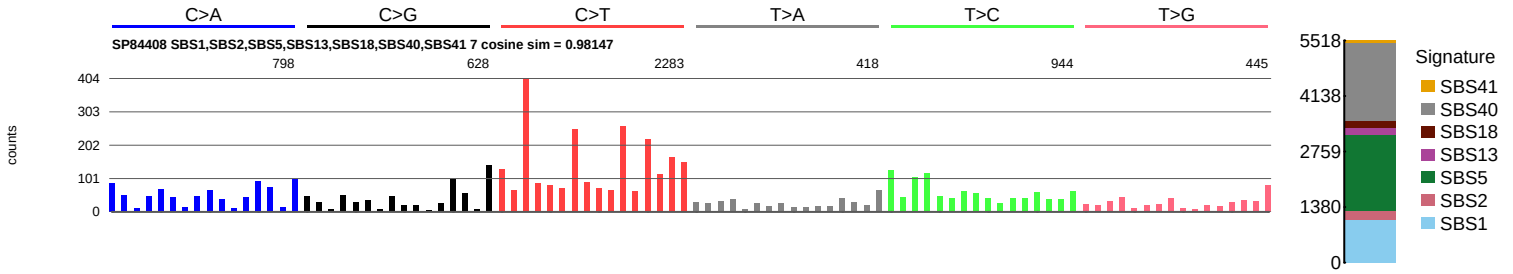
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.98147
798
628
2283
418
944
445
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
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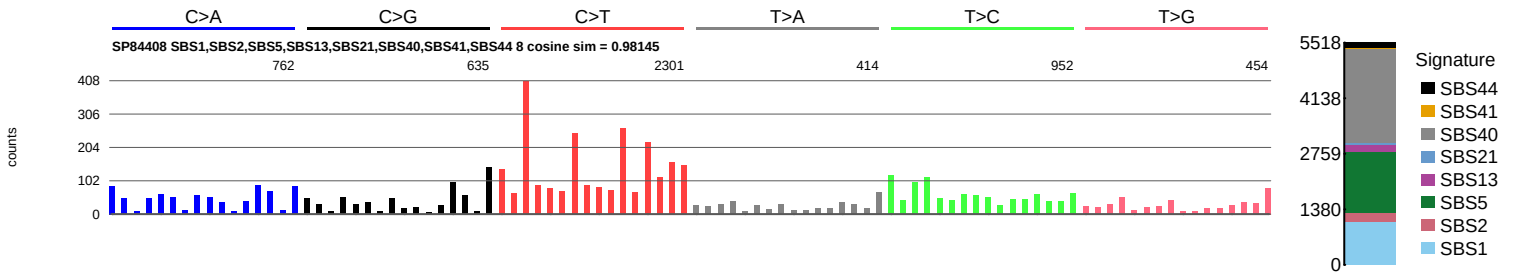
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98145
762
635
2301
414
952
454
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
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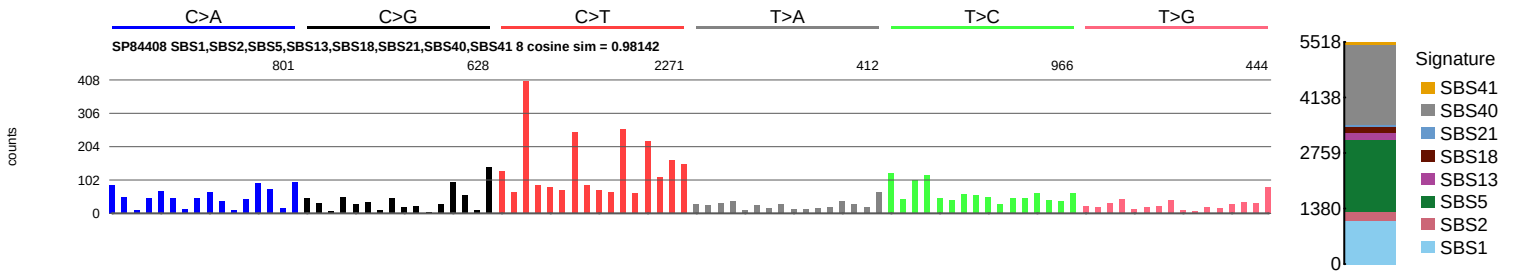
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98142
801
628
2271
412
966
444
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
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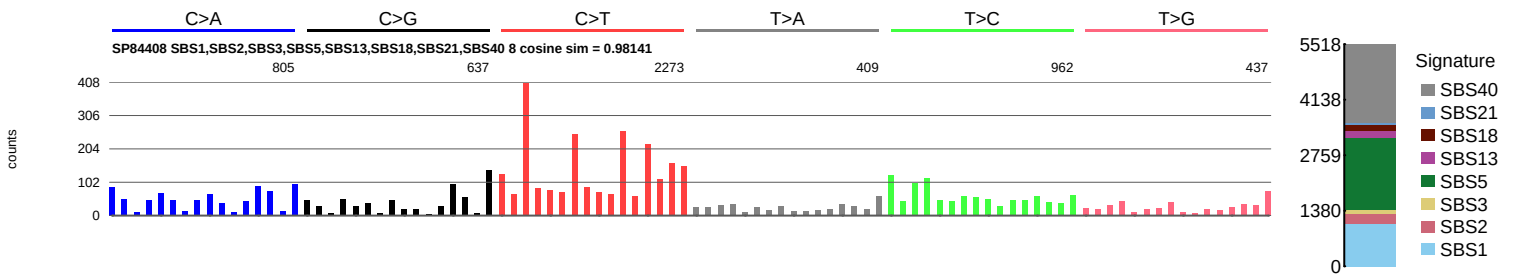
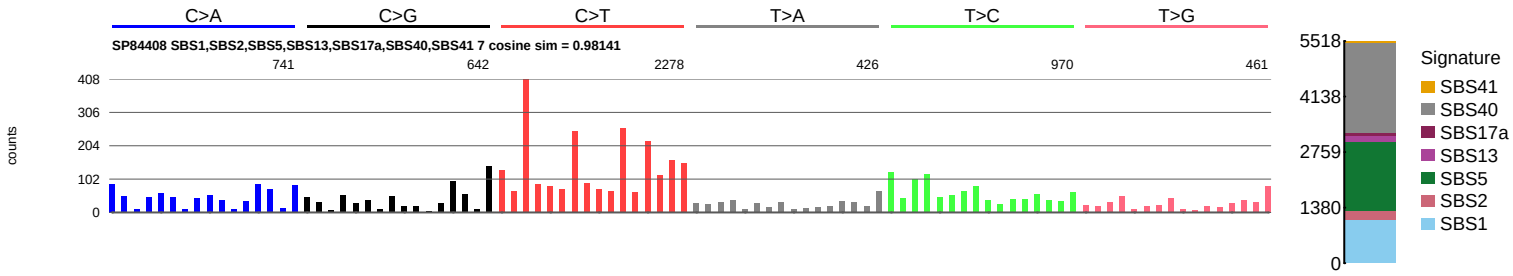
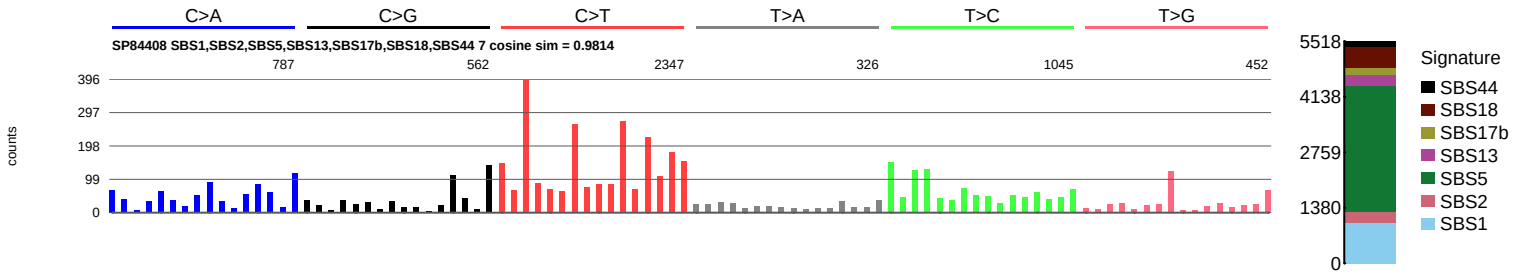
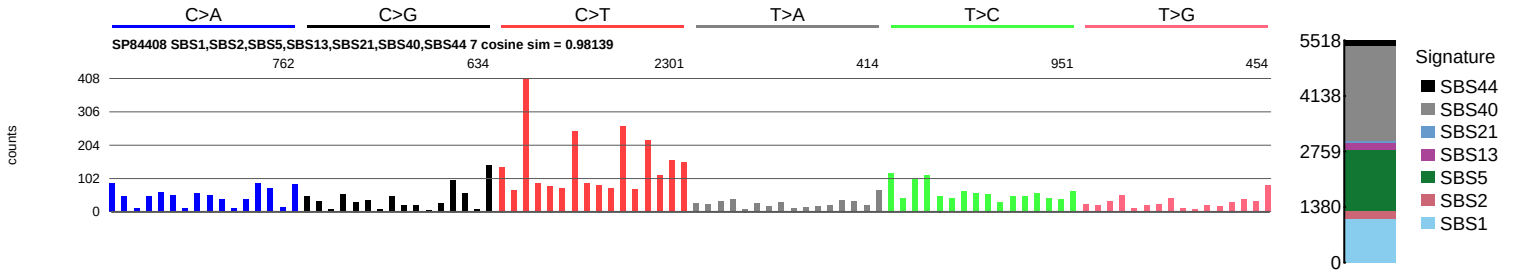
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40 8 cosine sim = 0.98141
805
637
2273
409
962
437
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40,SBS41 7 cosine sim = 0.98141
741
642
2278
426
970
461
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS44 7 cosine sim = 0.9814
787
562
2347
326
1045
452
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40,SBS44 7 cosine sim = 0.98139
762
634
2301
414
951
454
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
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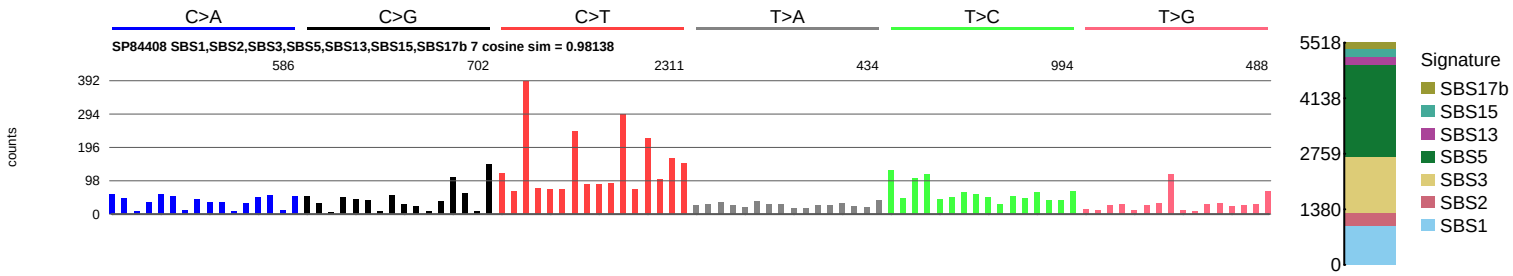
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b 7 cosine sim = 0.98138
586
702
2311
434
994
488
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS17b
SBS15
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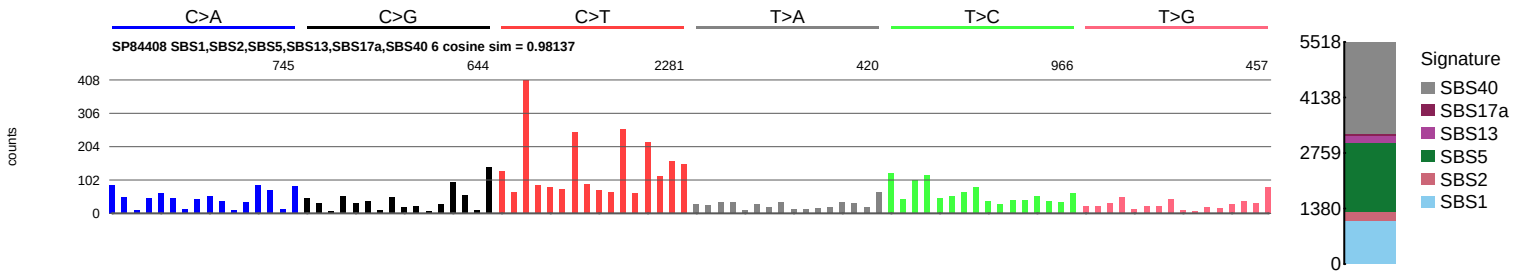
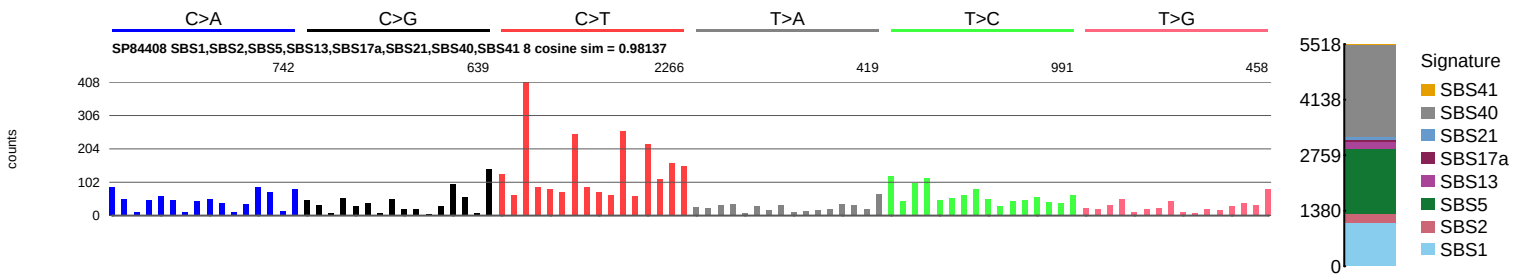
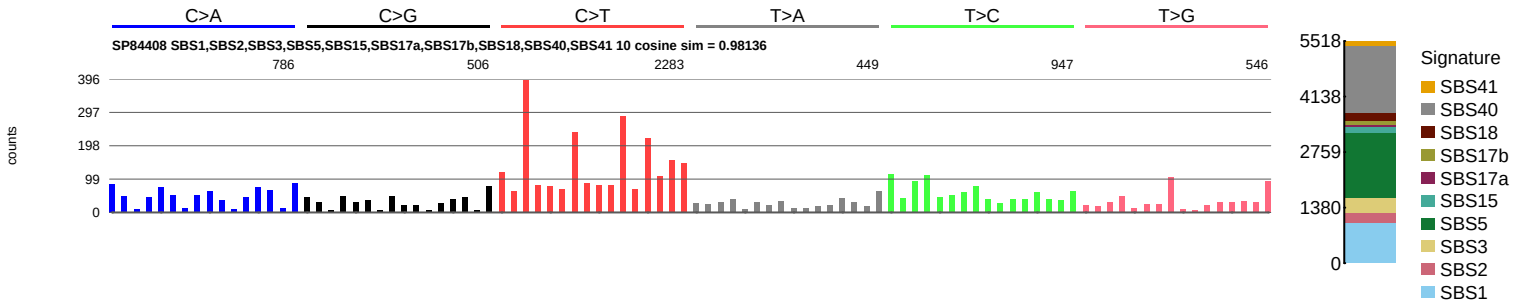
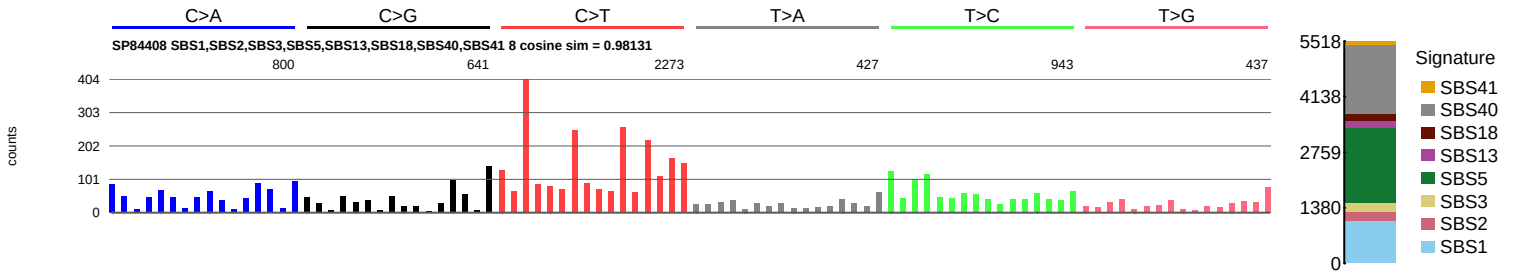
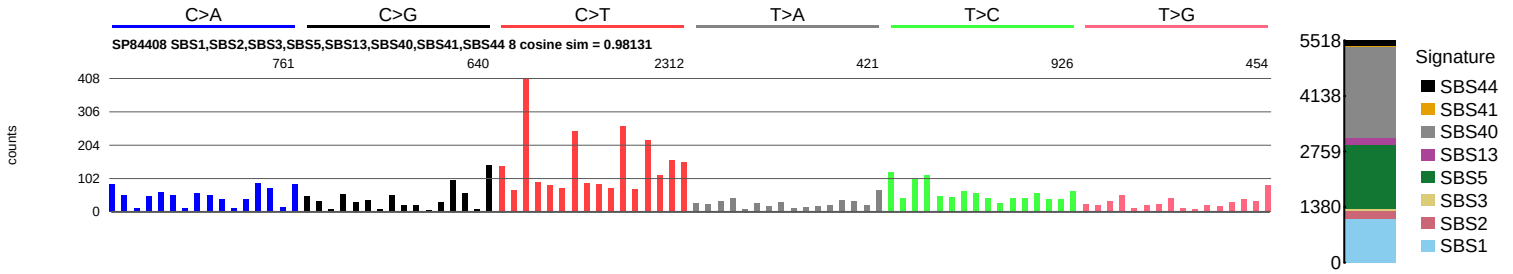
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40 6 cosine sim = 0.98137
745
644
2281
420
966
457
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS41 8 cosine sim = 0.98137
742
639
2266
419
991
458
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98136
786
506
2283
449
947
546
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41 8 cosine sim = 0.98131
800
641
2273
427
943
437
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40,SBS41,SBS44 8 cosine sim = 0.98131
761
640
2312
421
926
454
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS13
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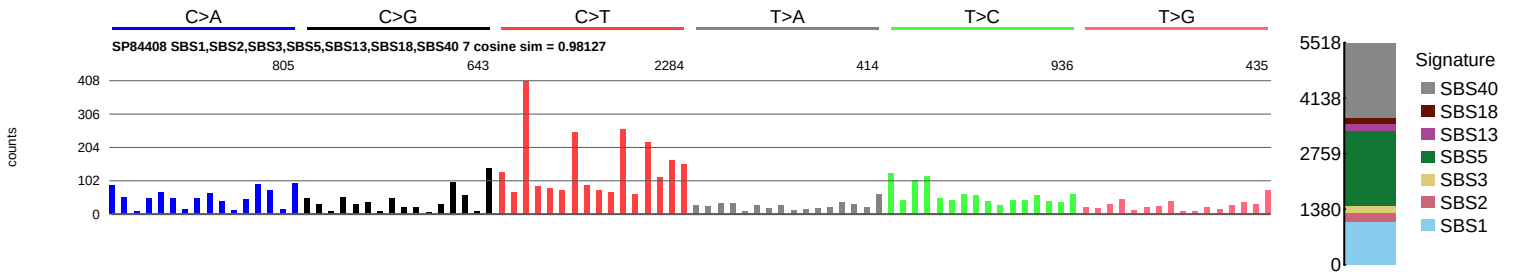
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40 7 cosine sim = 0.98127
805
643
2284
414
936
435
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
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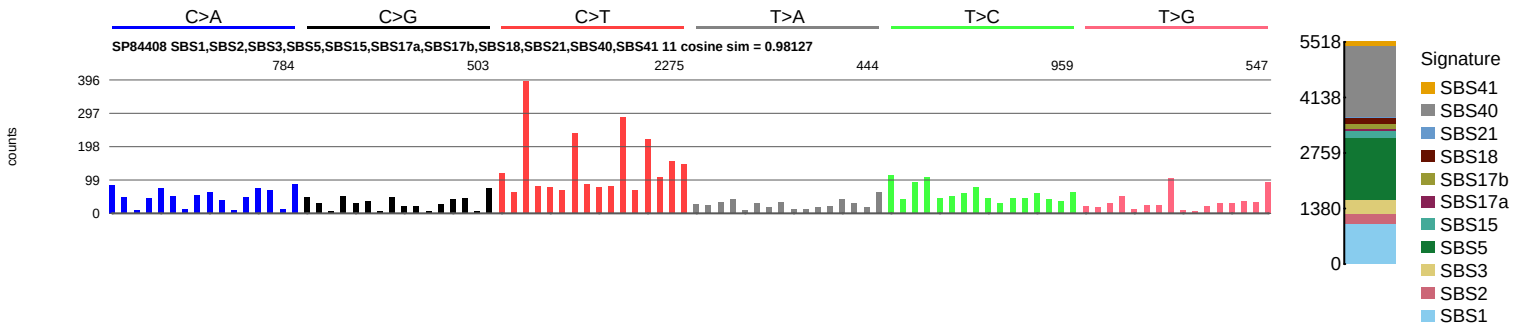
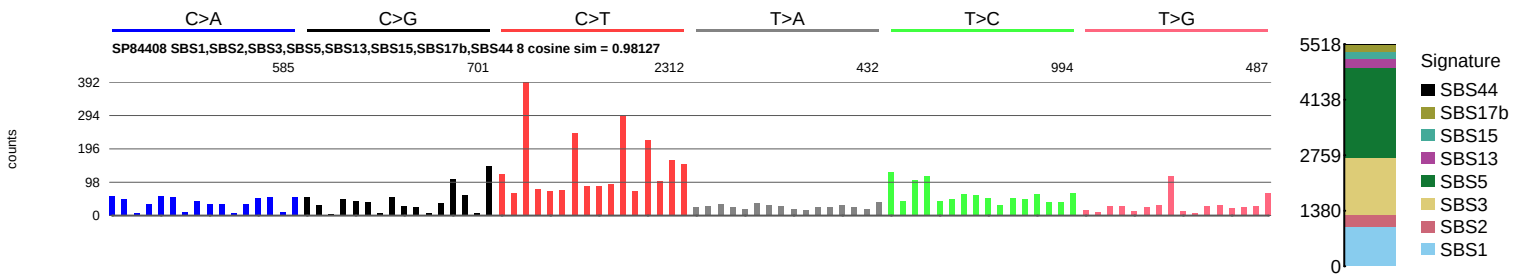
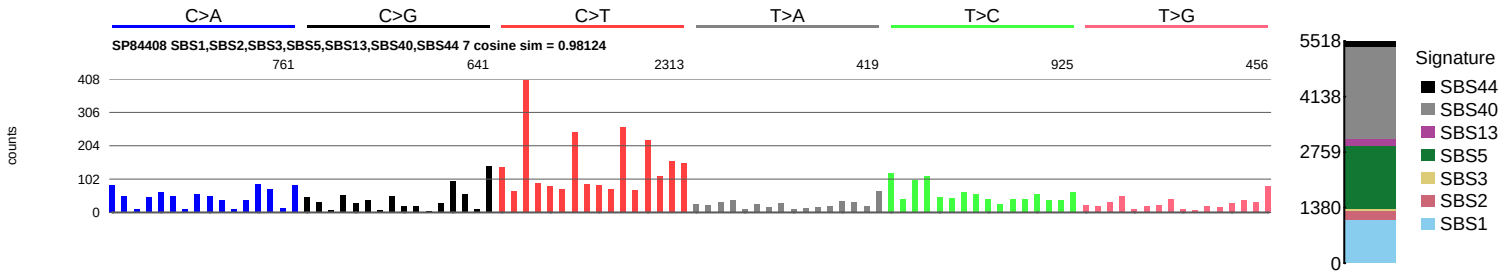
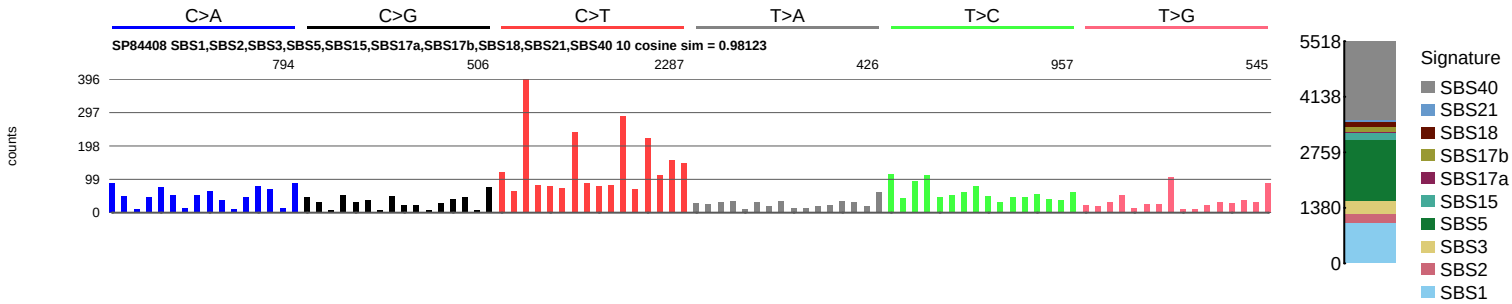
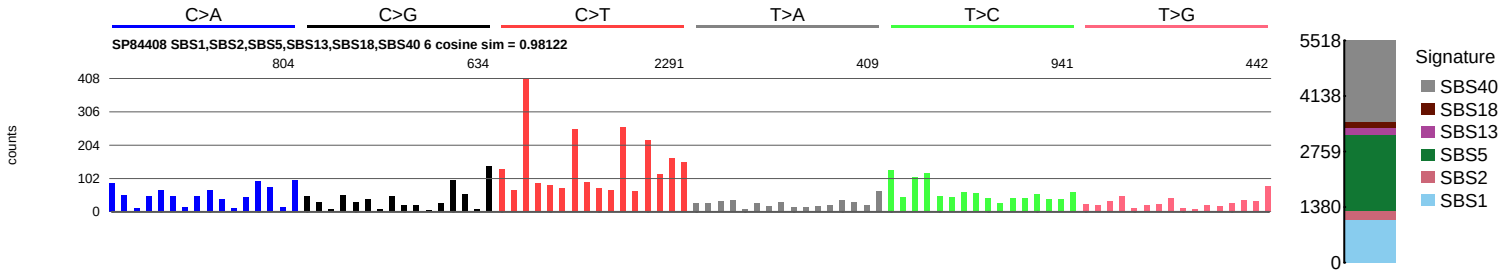
SBS5
SBS3
SBS2
SBS1

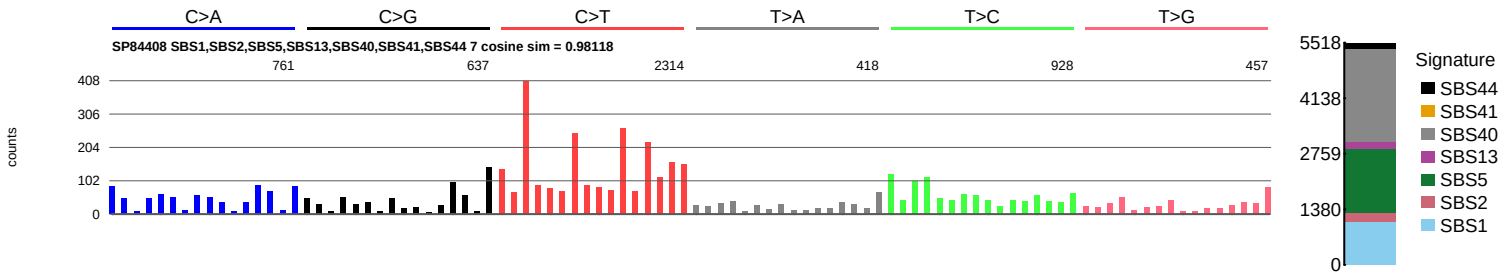

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98127
784
503
2275
444
959
547
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

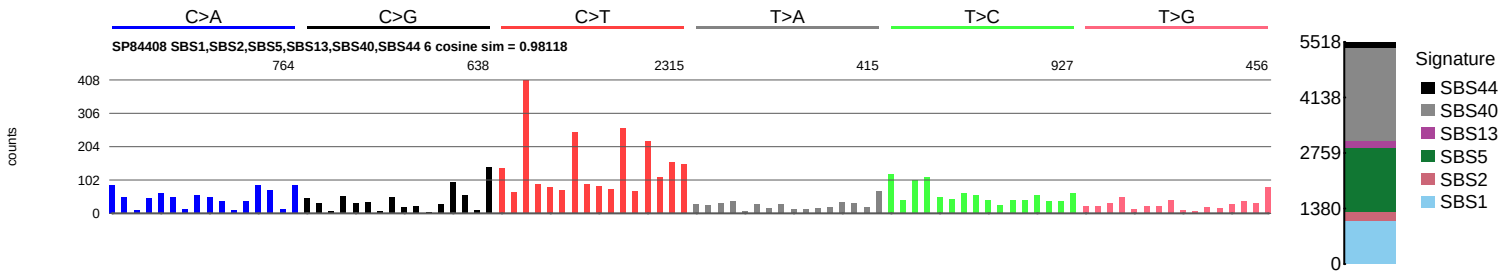

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS44 8 cosine sim = 0.98127
585
701
2312
432
994
487
counts
392
294
196
98
0
5518
4138
2759
1380
0
Signature
SBS44
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

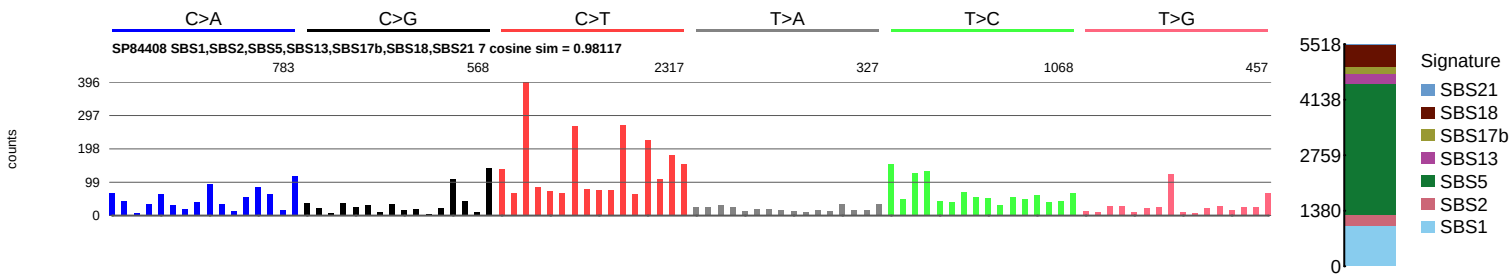

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40,SBS44 7 cosine sim = 0.98124
761
641
2313
419
925
456
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS13
SBS5
SBS3
SBS2
SBS1

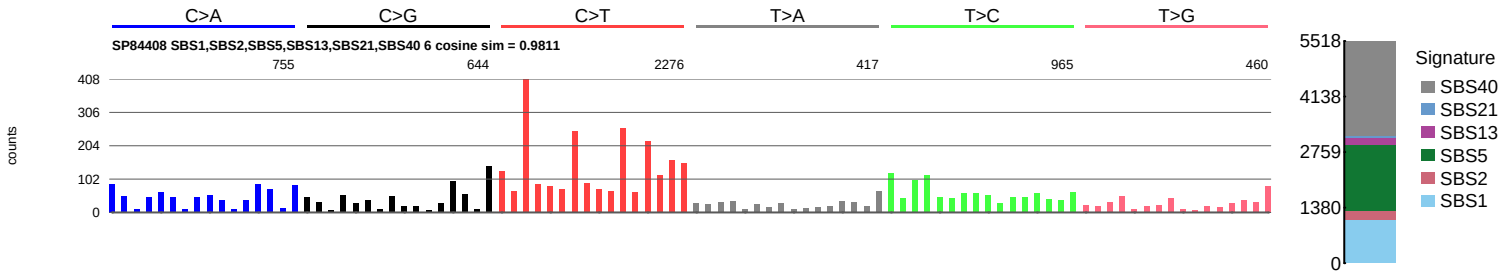

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98123
794
506
2287
426
957
545
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

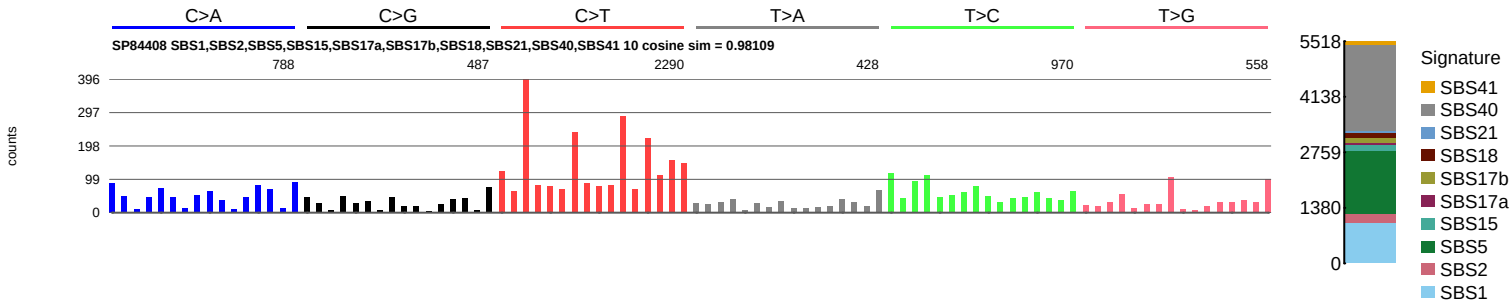

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40 6 cosine sim = 0.98122
804
634
2291
409
941
442
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1

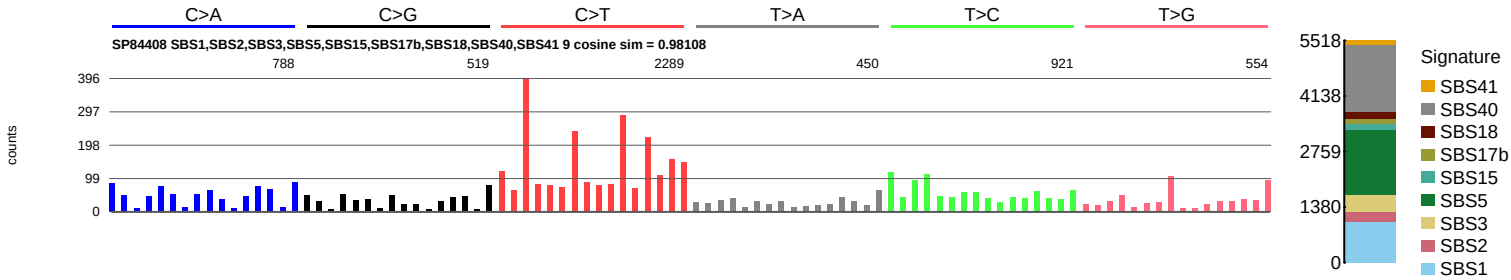

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS40,SBS41,SBS44 7 cosine sim = 0.98118
761
637
2314
418
928
457
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS13
SBS5
SBS2
SBS1

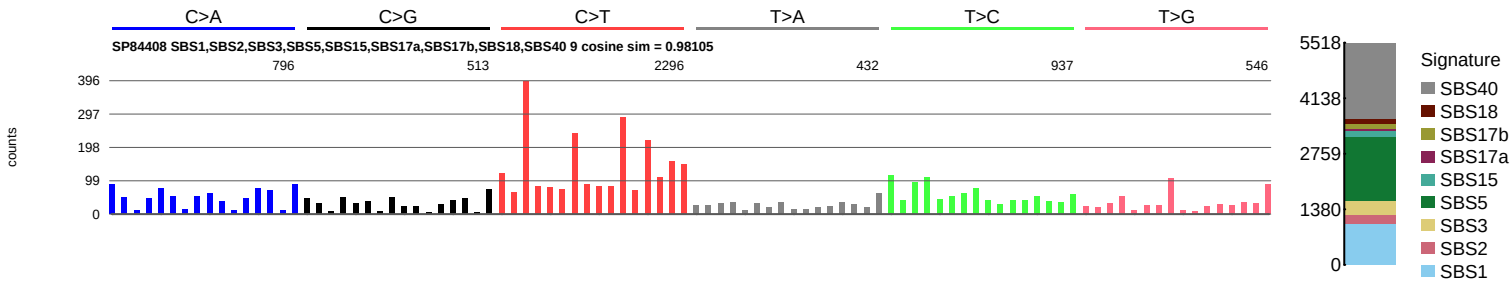

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS40,SBS44 6 cosine sim = 0.98118
764
638
2315
415
927
456
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS13
SBS5
SBS2
SBS1

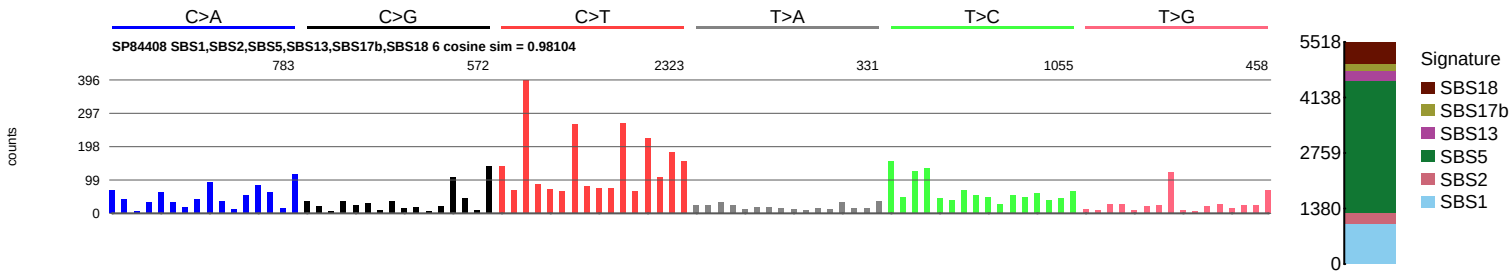

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21 7 cosine sim = 0.98117
783
568
2317
327
1068
457
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

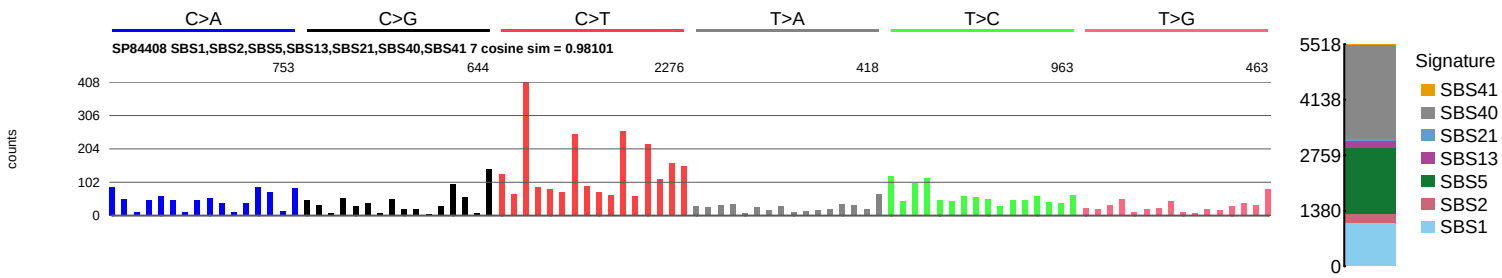

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40 6 cosine sim = 0.9811
755
644
2276
417
965
460
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS13
SBS5
SBS2
SBS1

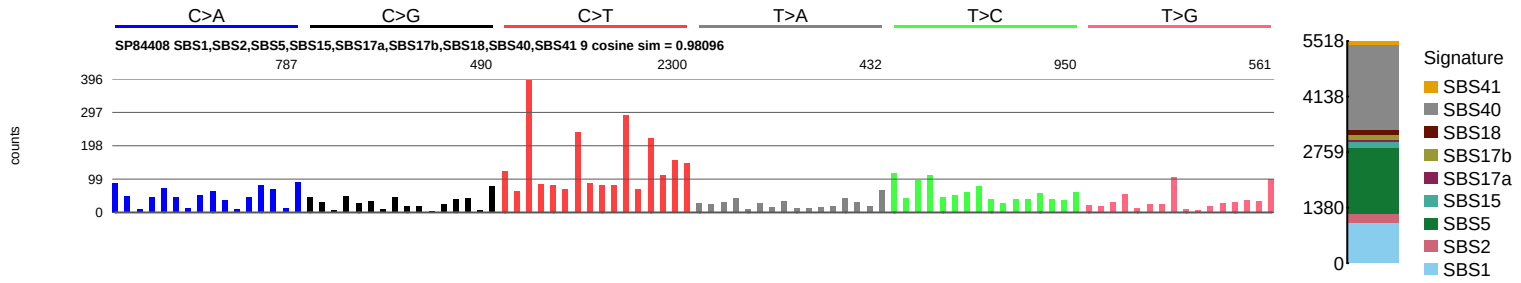

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98109
788
487
2290
428
970
558
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

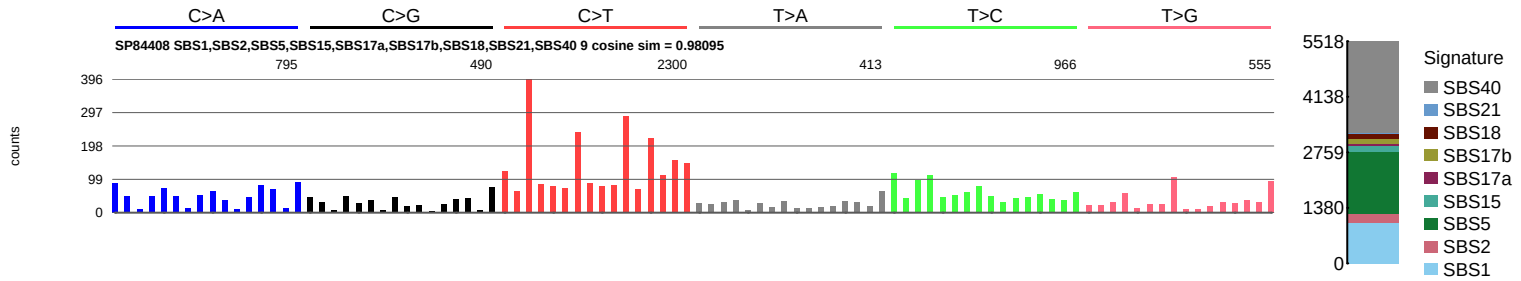

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98108
788
519
2289
450
921
554
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

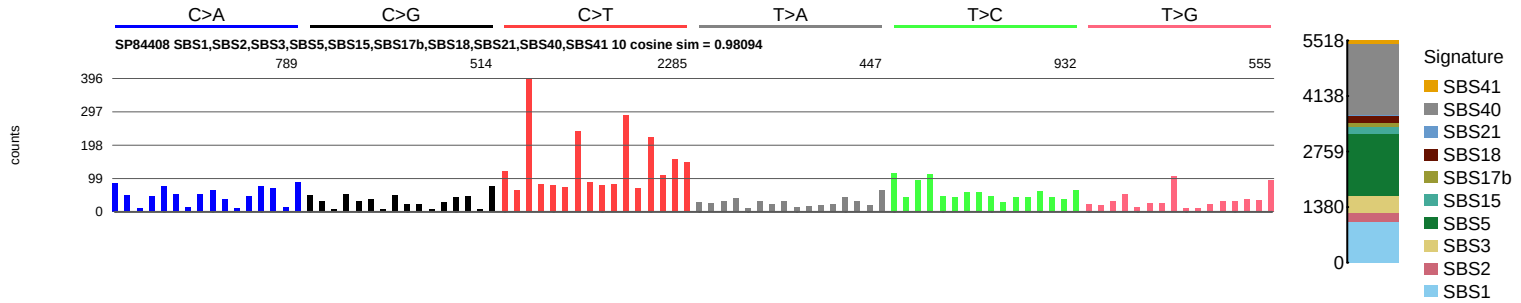

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98105
796
513
2296
432
937
546
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

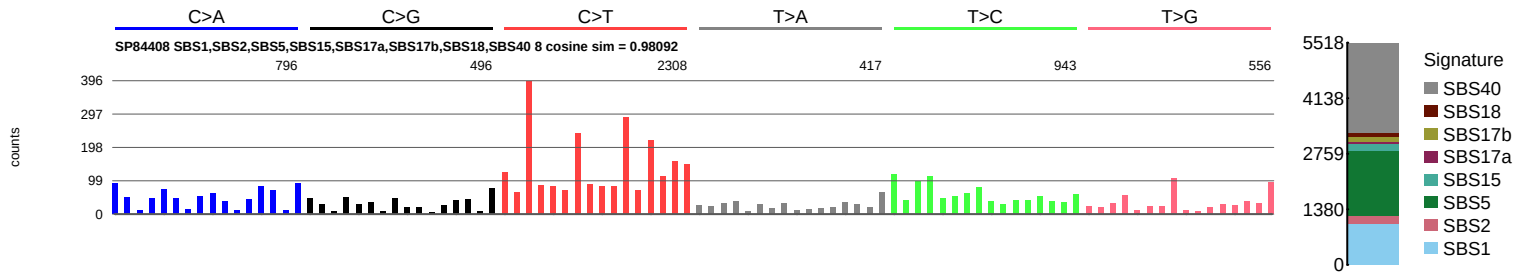

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18 6 cosine sim = 0.98104
783
572
2323
331
1055
458
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

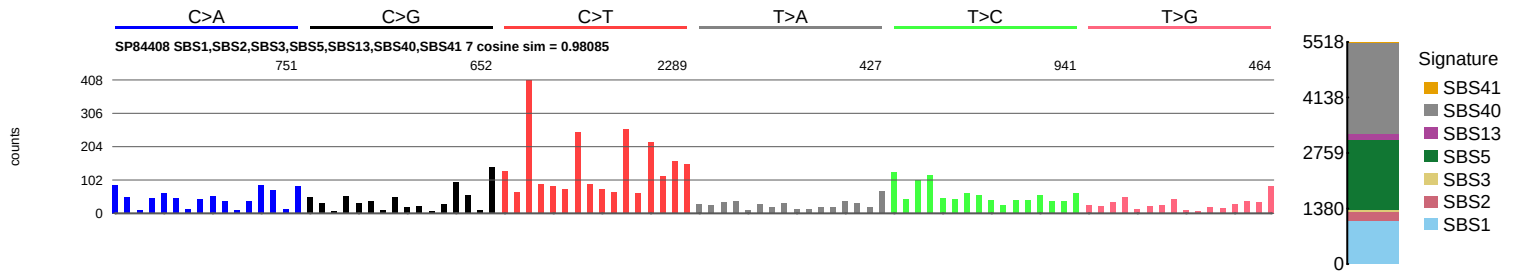

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40,SBS41 7 cosine sim = 0.98101
753
644
2276
418
963
463
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS13
SBS5
SBS2
SBS1

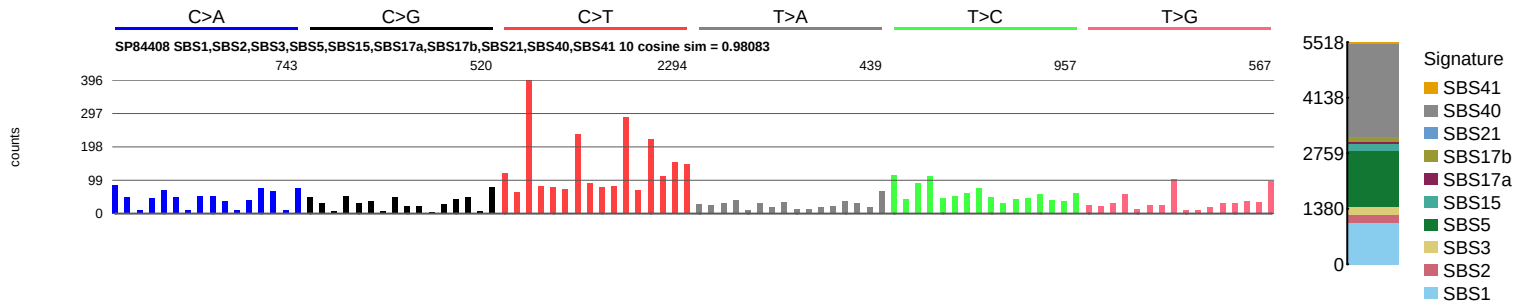

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98096
787
490
2300
432
950
561
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

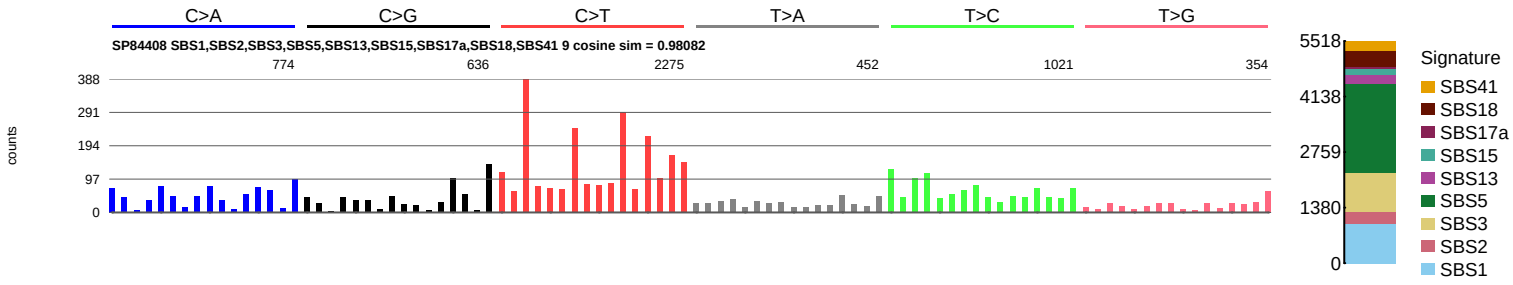

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98095
795
490
2300
413
966
555
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

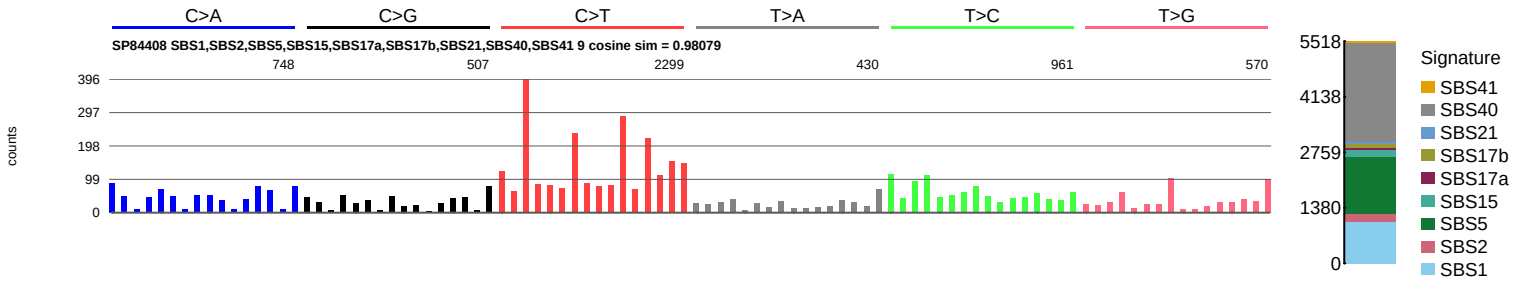

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98094
789
514
2285
447
932
555
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

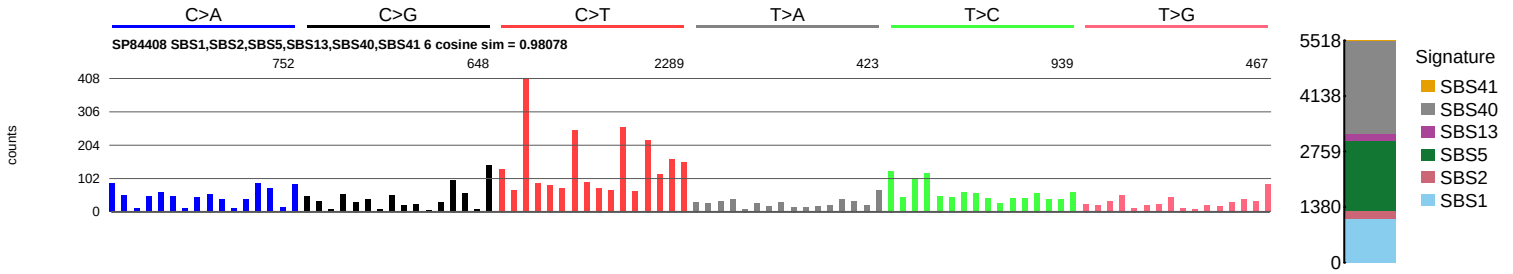

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98092
796
496
2308
417
943
556
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

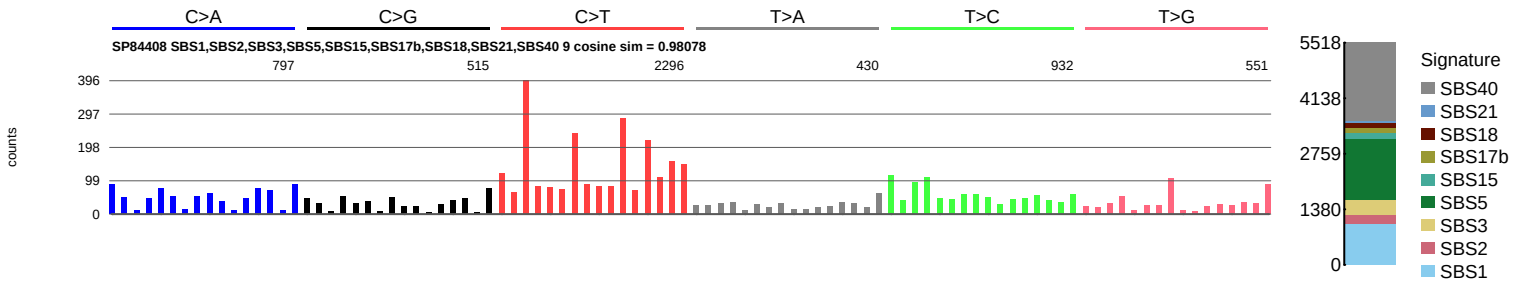

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40,SBS41 7 cosine sim = 0.98085
751
652
2289
427
941
464
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS13
SBS5
SBS3
SBS2
SBS1

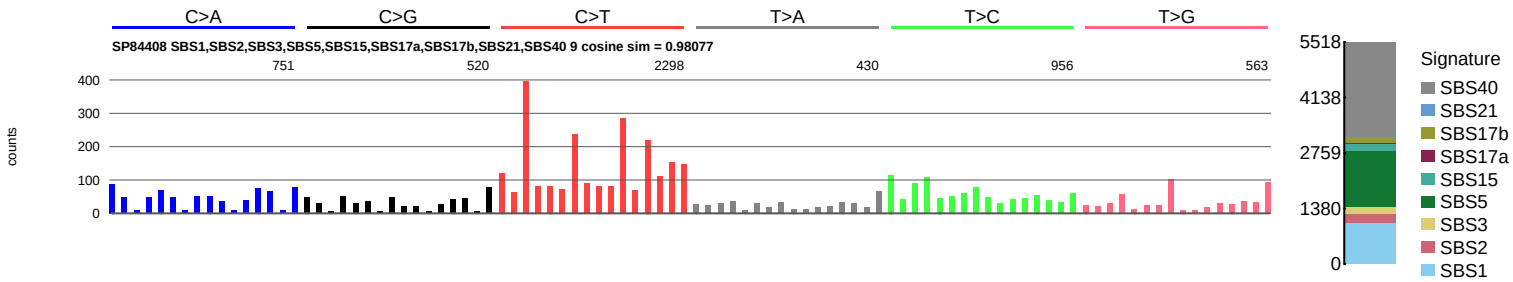

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98083
743
520
2294
439
957
567
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

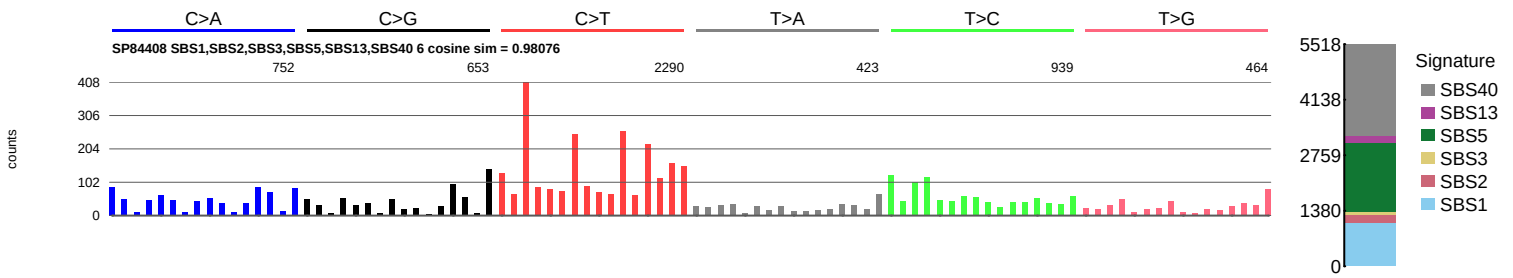

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 9 cosine sim = 0.98082
774
636
2275
452
1021
354
counts
388
291
194
97
0
5518
4138
2759
1380
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98079
748
507
2299
430
961
570
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS40,SBS41 6 cosine sim = 0.98078
752
648
2289
423
939
467
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98078
797
515
2296
430
932
551
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98077
751
520
2298
430
956
563
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40 6 cosine sim = 0.98076
752
653
2290
423
939
464
counts
408
306
204
102
0
5518
4138
2759
1380
0
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SBS40
SBS13
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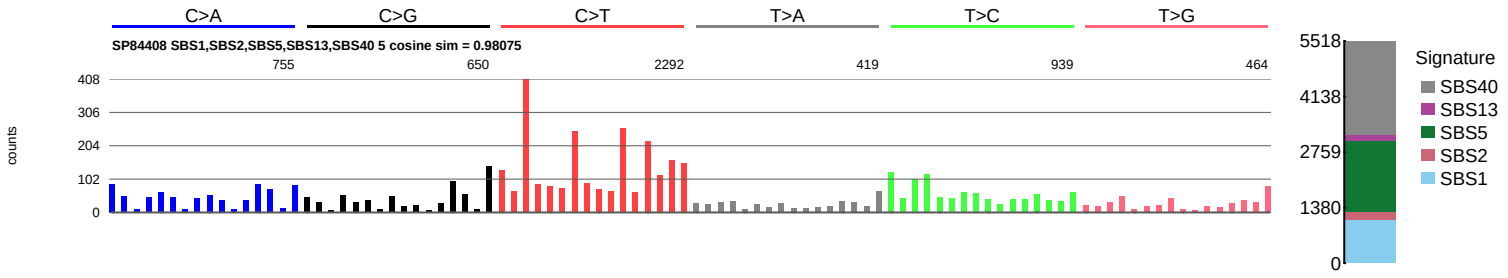
SBS3
SBS2
SBS1


C>A
C>G
C>T
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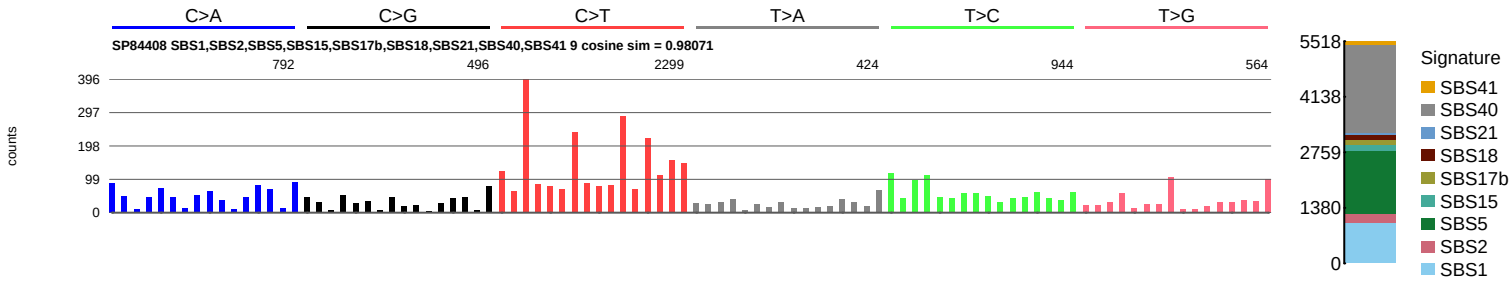
T>G
SP84408 SBS1,SBS2,SBS5,SBS13,SBS40 5 cosine sim = 0.98075
755
650
2292
419
939
464
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS40
SBS13
SBS5
SBS2
SBS1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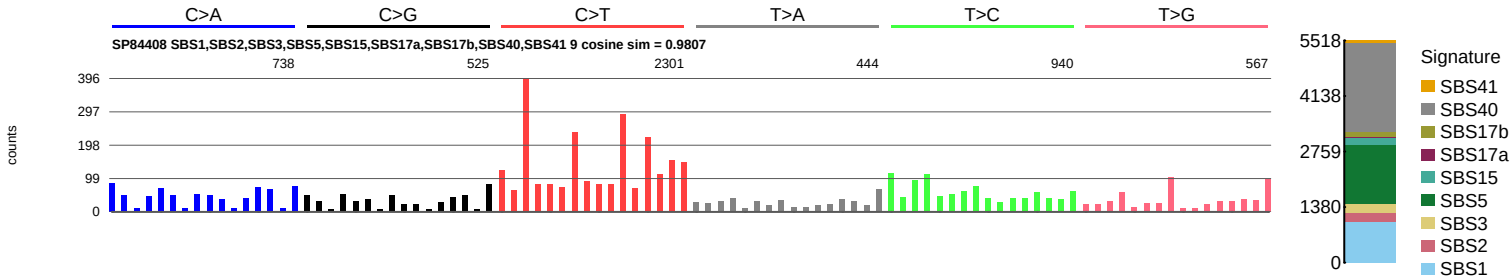
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98071
792
496
2299
424
944
564
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.9807
738
525
2301
444
940
567
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
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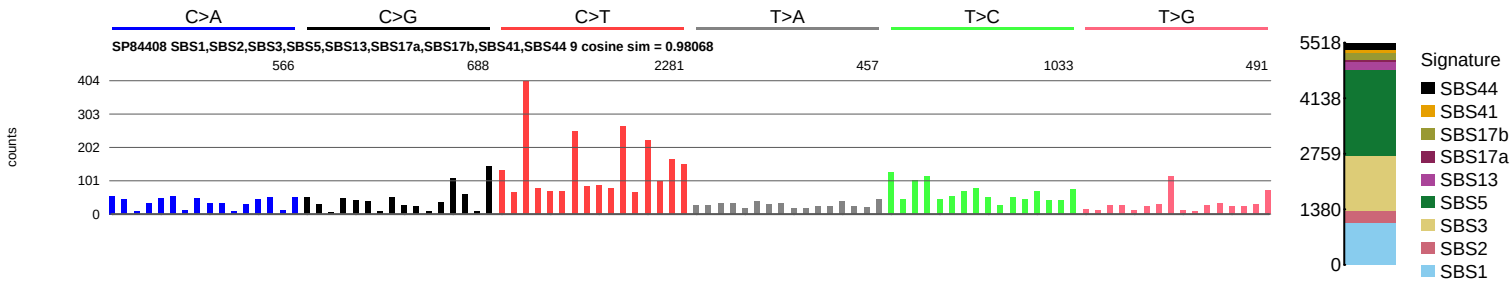
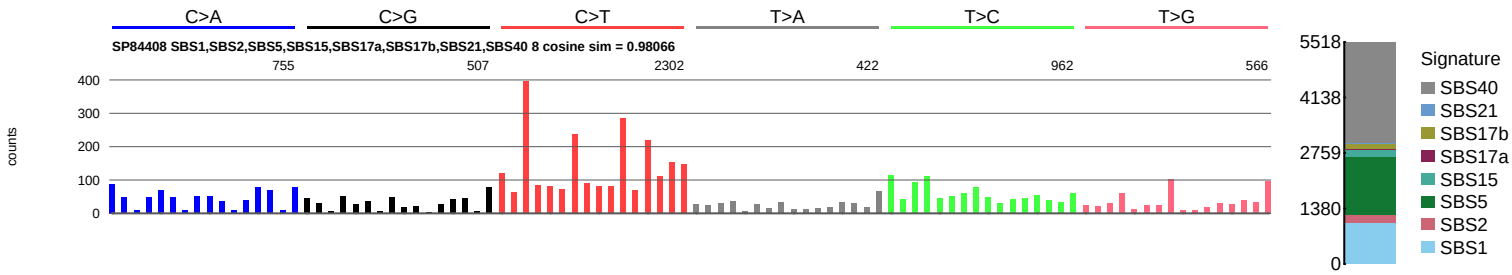
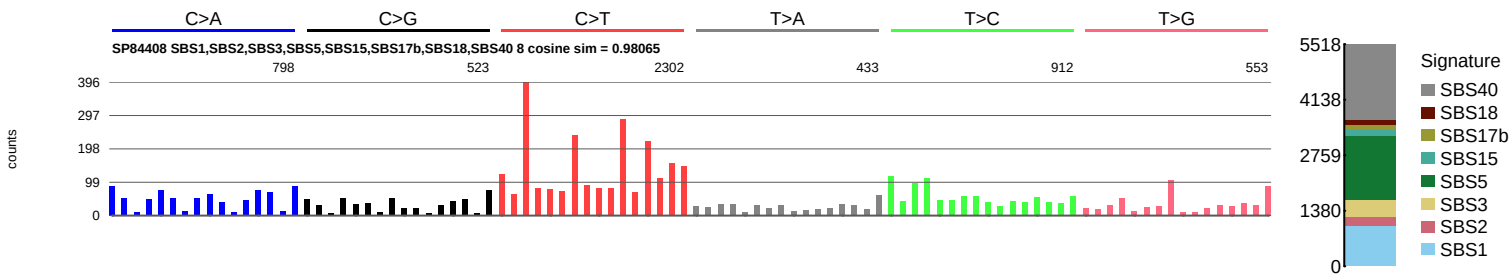
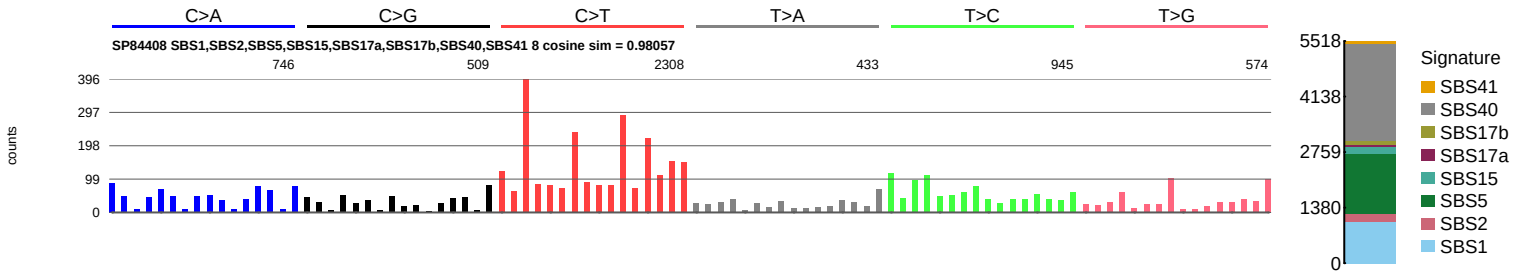
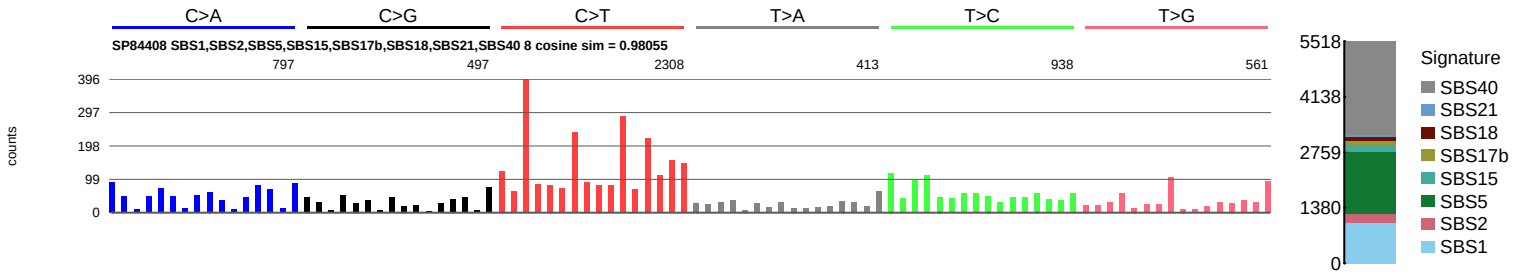
SBS2
SBS1

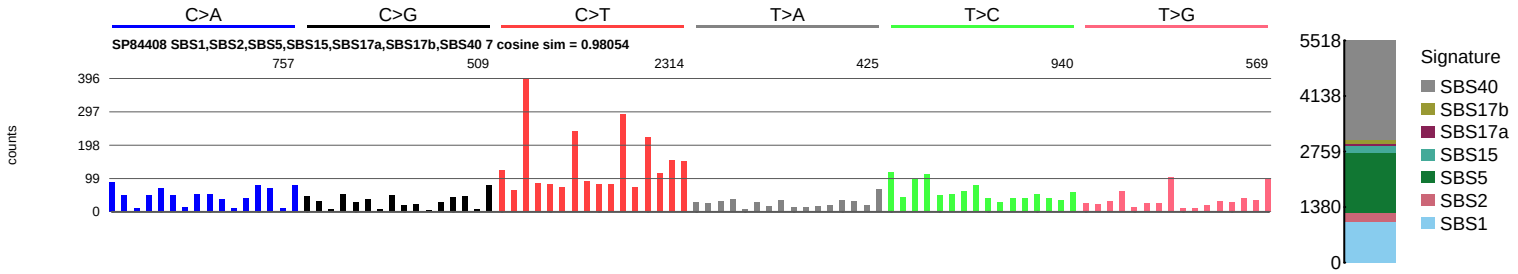

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98068
566
688
2281
457
1033
491
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

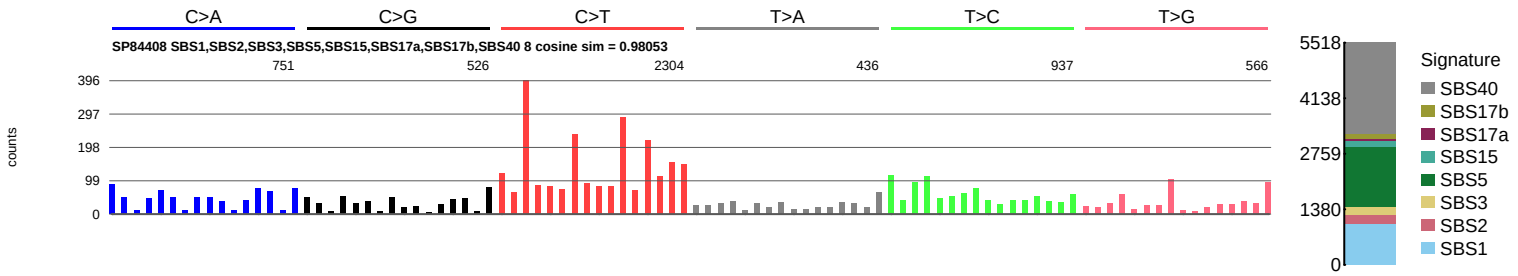

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98066
755
507
2302
422
962
566
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

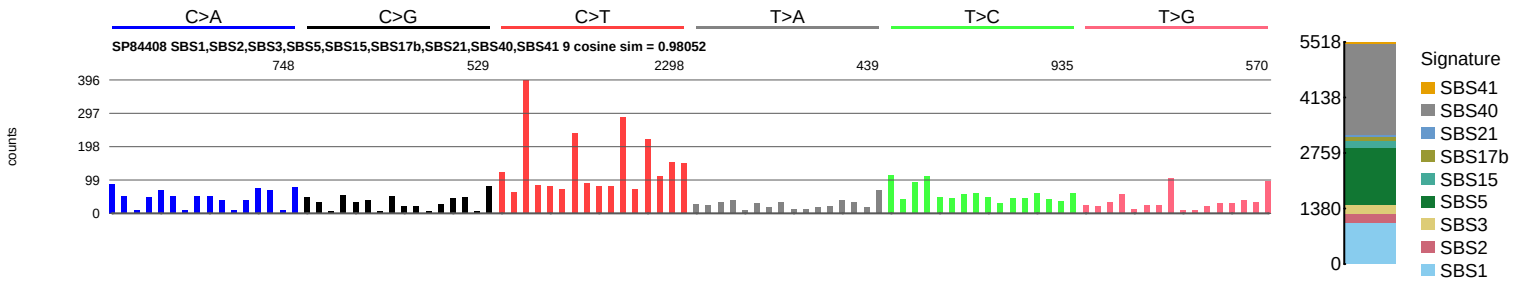

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98065
798
523
2302
433
912
553
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98057
746
509
2308
433
945
574
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98055
797
497
2308
413
938
561
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98054
757
509
2314
425
940
569
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98053
751
526
2304
436
937
566
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98052
748
529
2298
439
935
570
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
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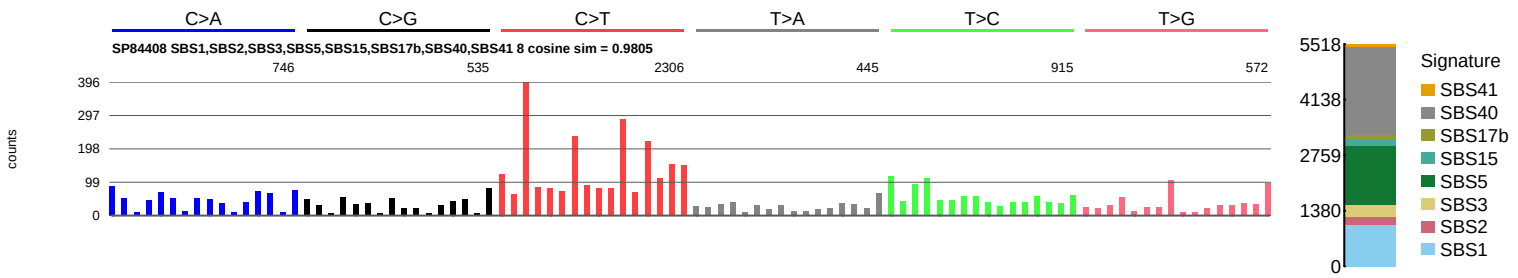
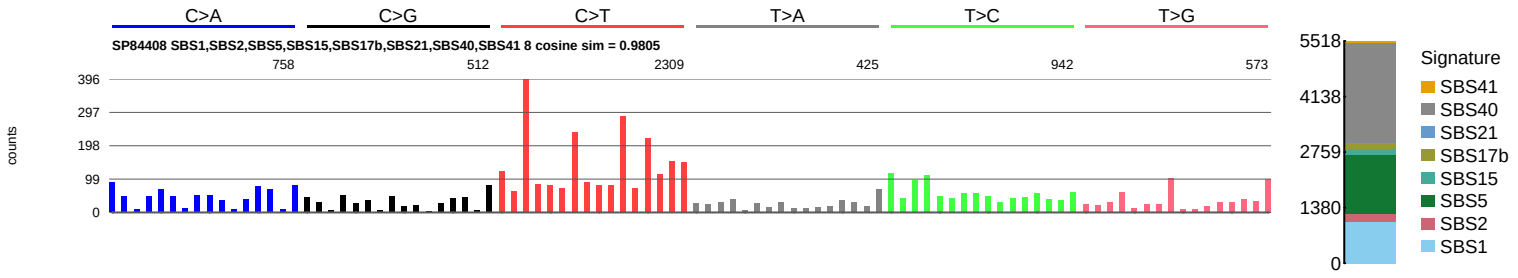
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS40,SBS41 8 cosine sim = 0.9805
746
535
2306
445
915
572
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.9805
758
512
2309
425
942
573
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS21
SBS17b
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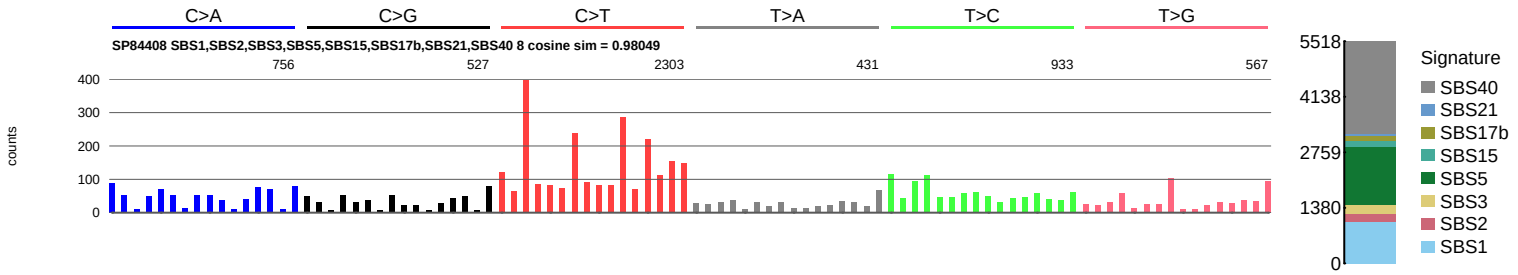
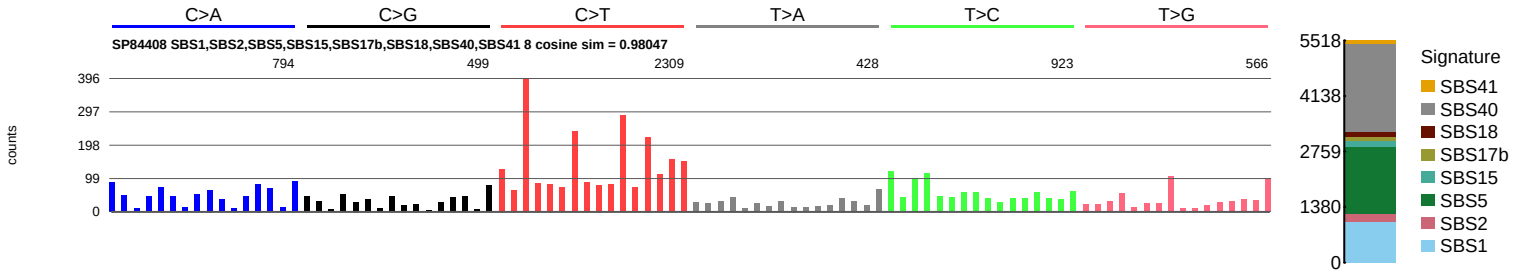
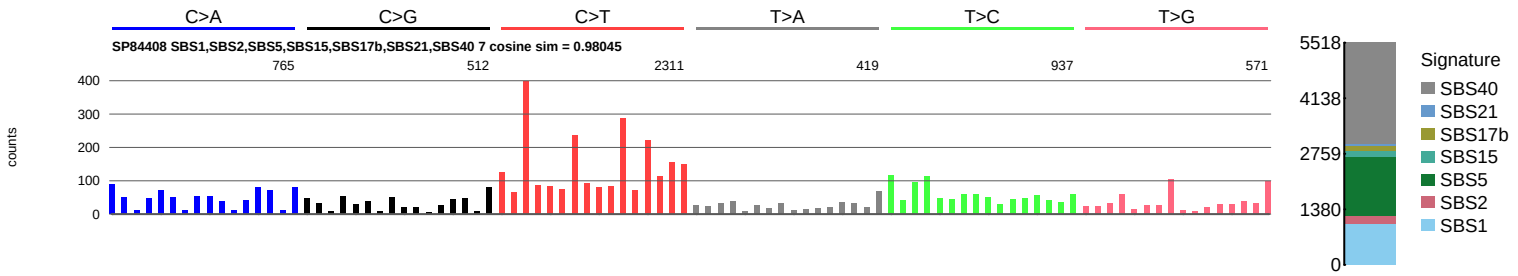
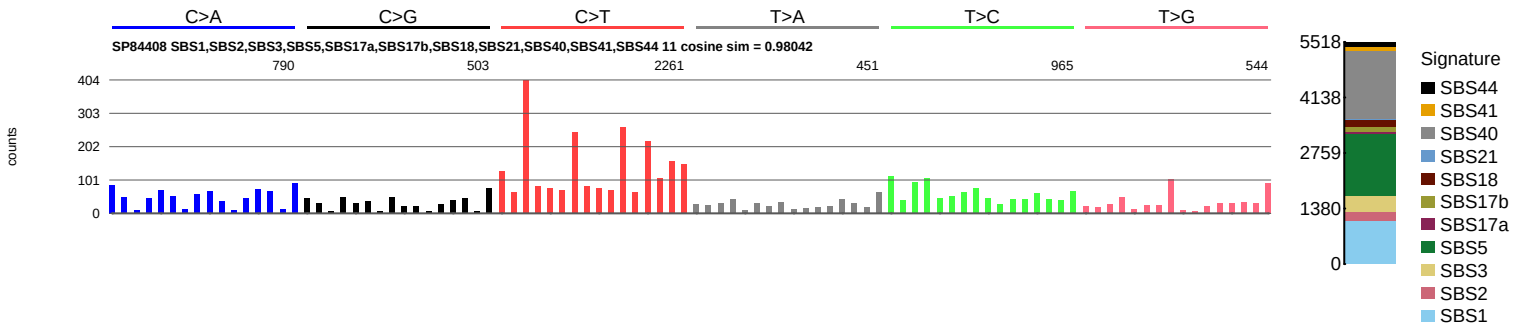
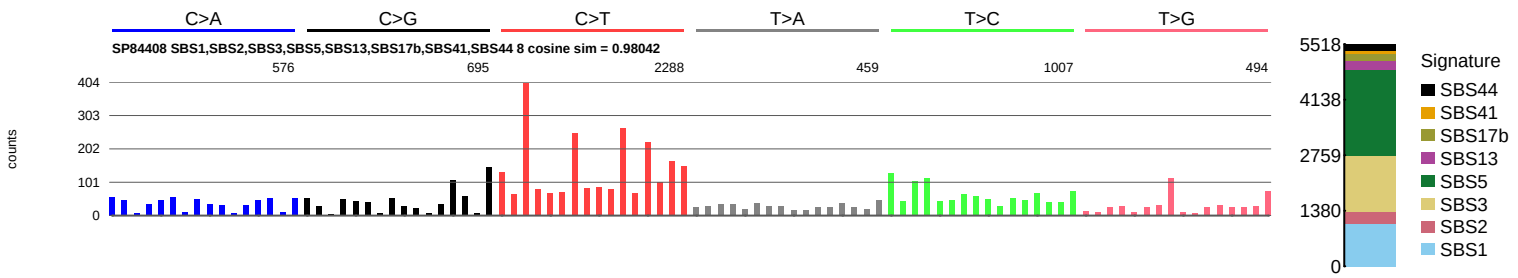
SBS5
SBS2
SBS1

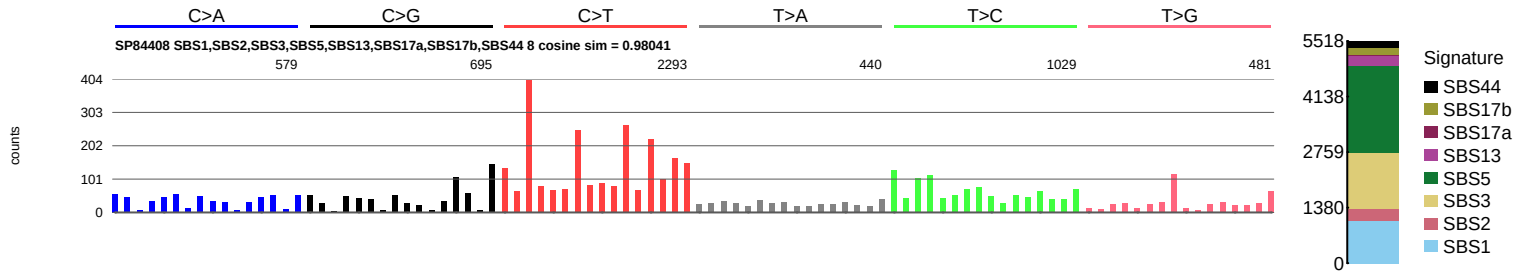

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS21,SBS40 8 cosine sim = 0.98049
756
527
2303
431
933
567
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

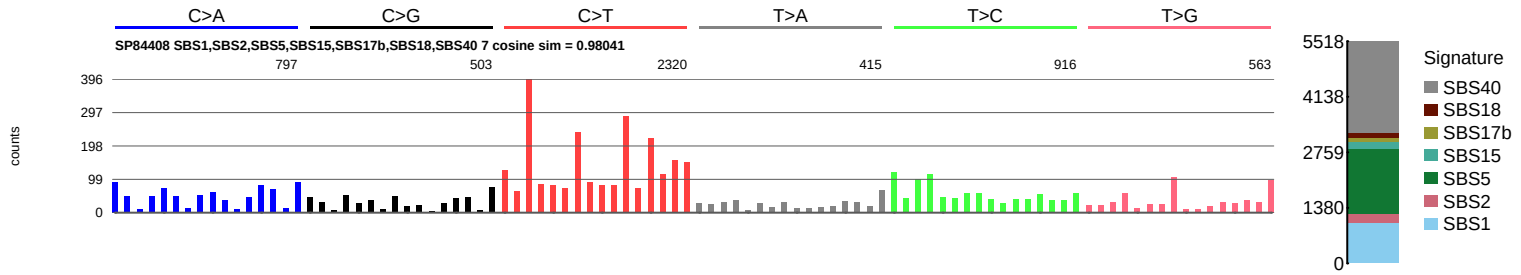

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98047
794
499
2309
428
923
566
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

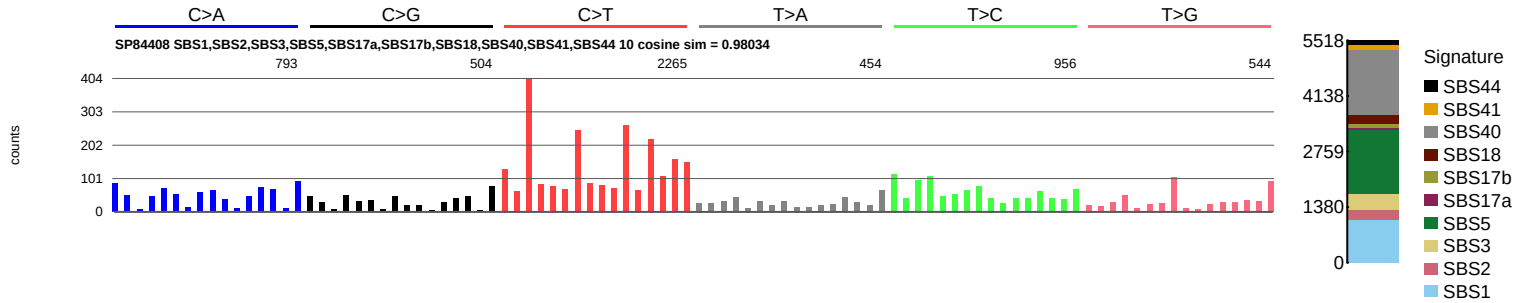

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS40 7 cosine sim = 0.98045
765
512
2311
419
937
571
counts
400
300
200
100
0
5518
4138
2759
1380
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS5
SBS2
SBS1

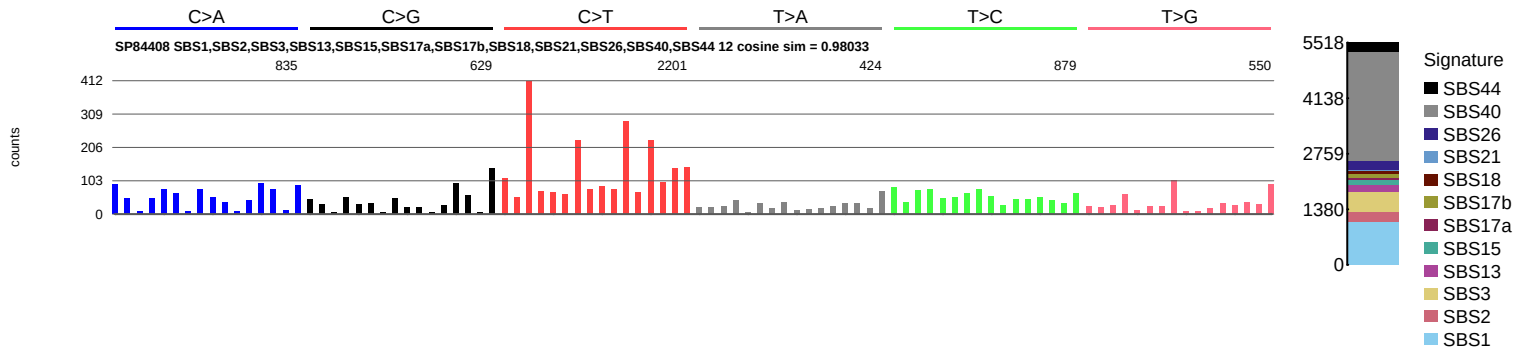

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98042
790
503
2261
451
965
544
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

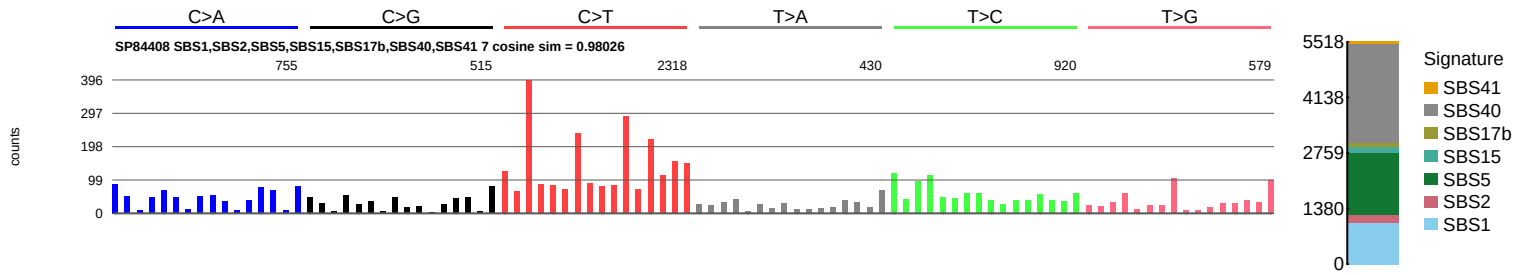

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS41,SBS44 8 cosine sim = 0.98042
576
695
2288
459
1007
494
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

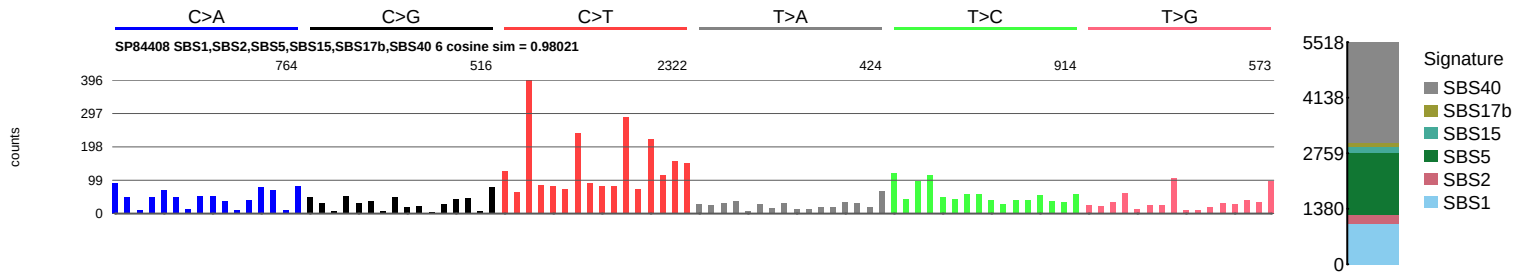

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98041
579
695
2293
440
1029
481
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98041
797
503
2320
415
916
563
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98034
793
504
2265
454
956
544
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98033
835
629
2201
424
879
550
counts
412
309
206
103
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS40,SBS41 7 cosine sim = 0.98026
755
515
2318
430
920
579
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS40 6 cosine sim = 0.98021
764
516
2322
424
914
573
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
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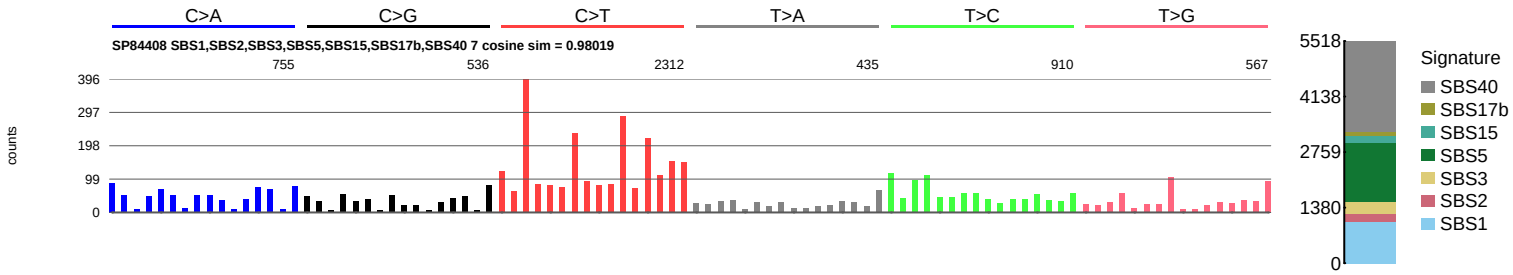
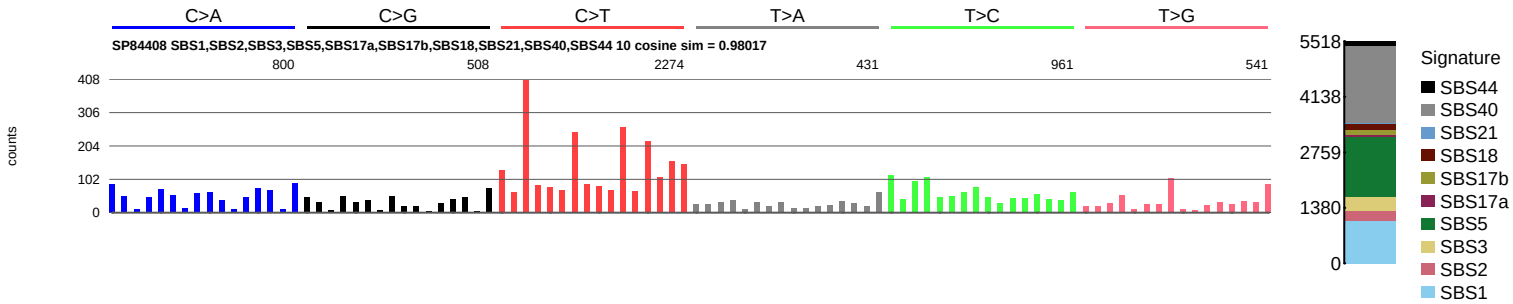
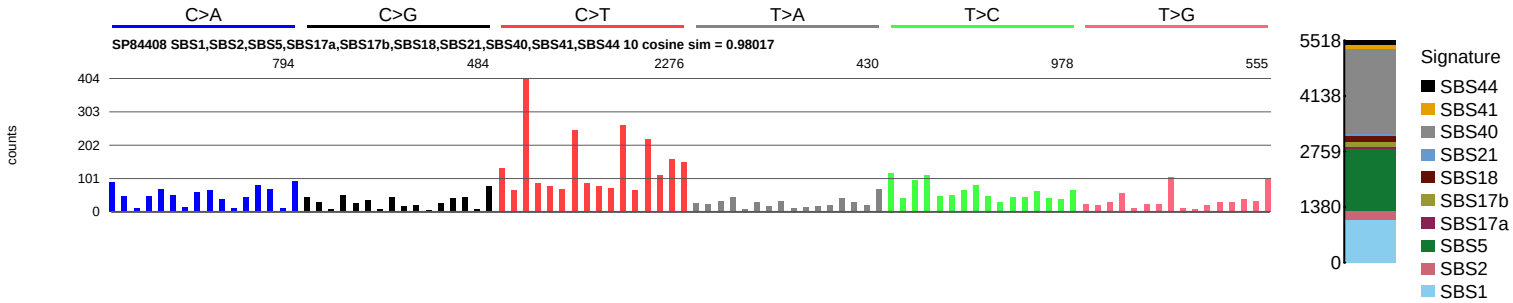
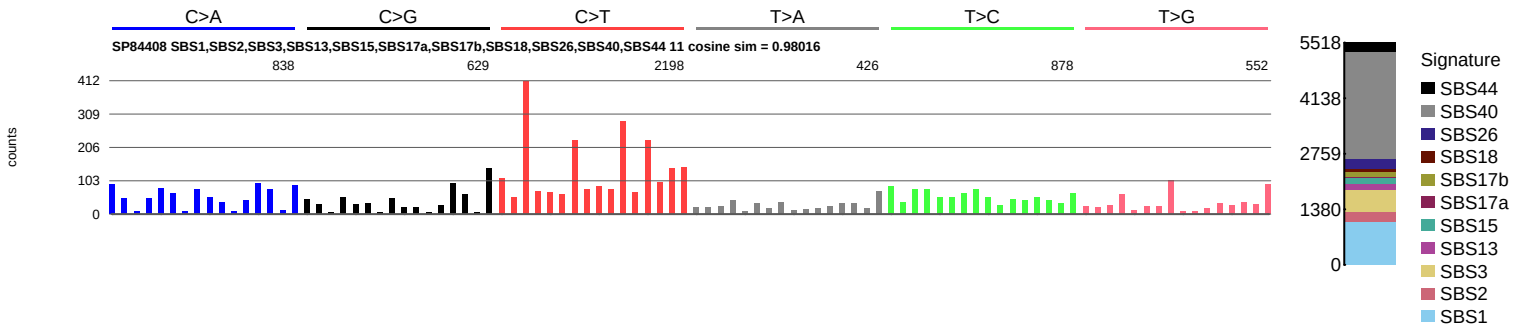
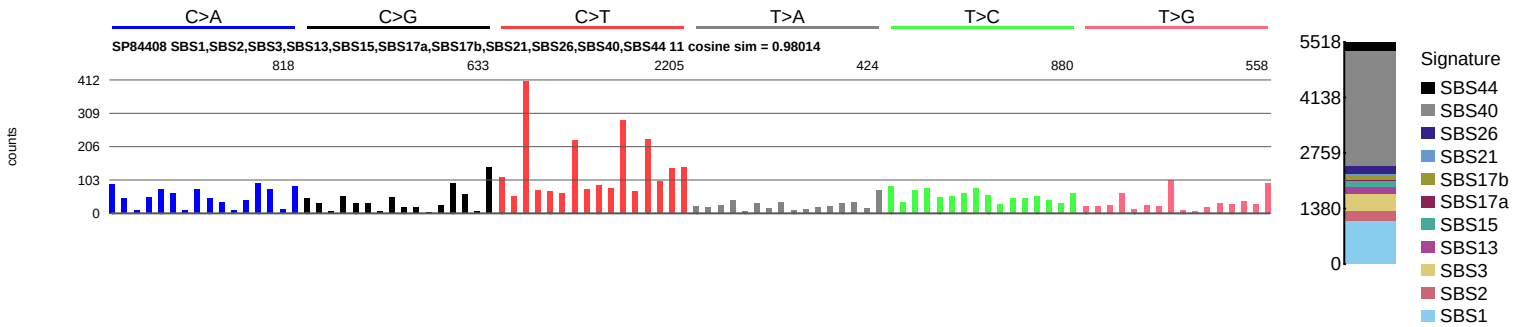
SBS5
SBS2
SBS1

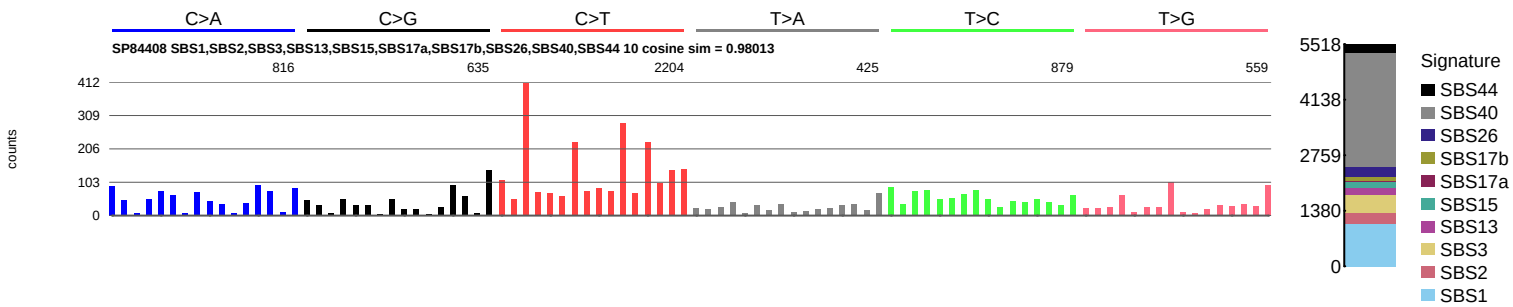

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS40 7 cosine sim = 0.98019
755
536
2312
435
910
567
counts
396
297
198
99
0
5518
4138
2759
1380
0
Signature
SBS40
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

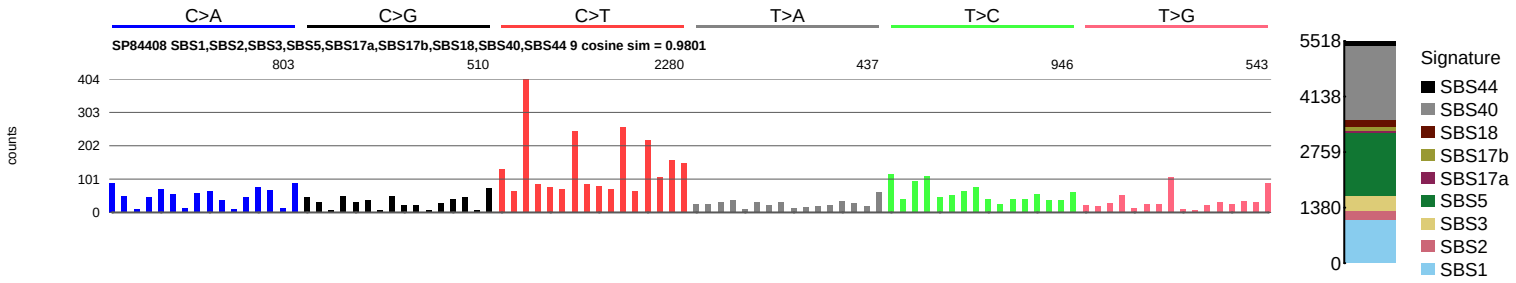

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98017
800
508
2274
431
961
541
counts
408
306
204
102
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

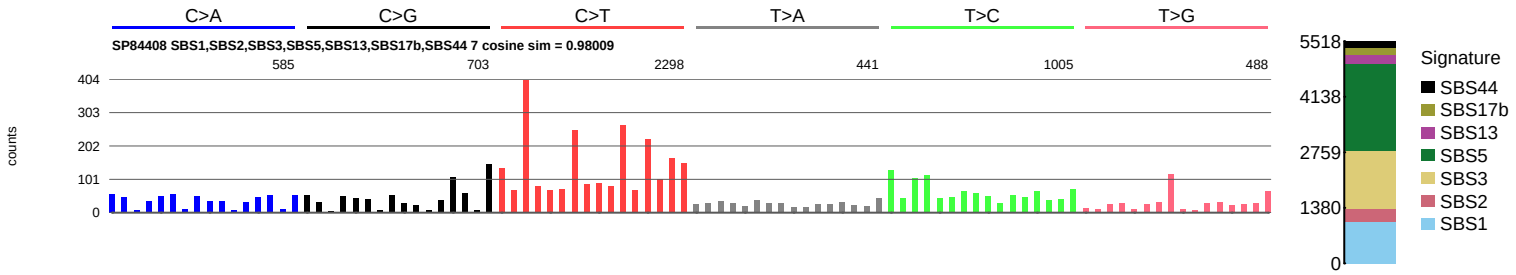

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98017
794
484
2276
430
978
555
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

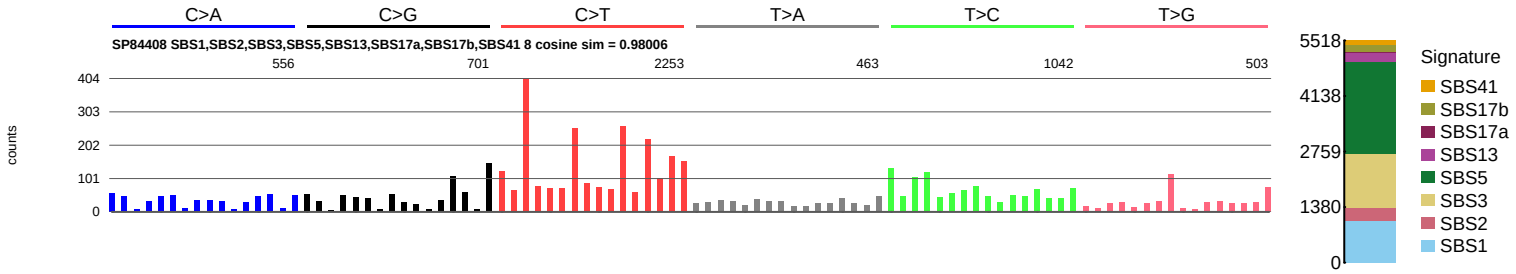

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98016
838
629
2198
426
878
552
counts
412
309
206
103
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

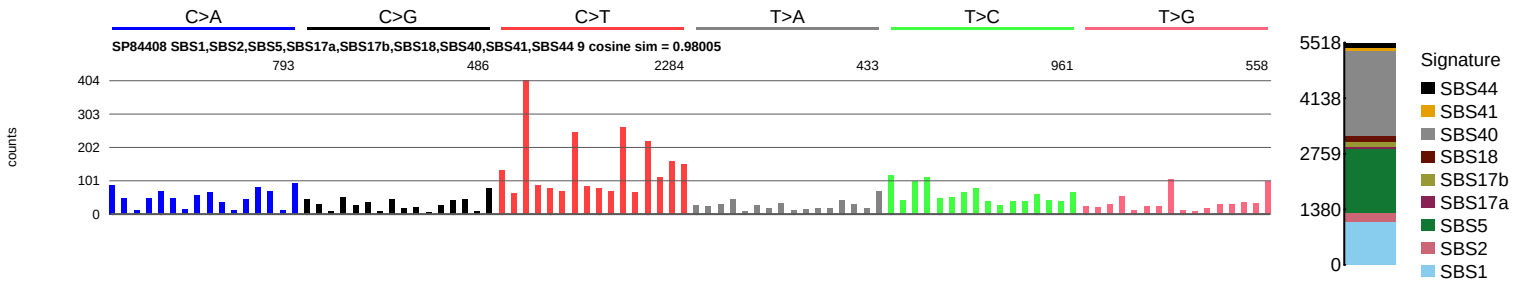

C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98014
818
633
2205
424
880
558
counts
412
309
206
103
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98013
816
635
2204
425
879
559
counts
412
309
206
103
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.9801
803
510
2280
437
946
543
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS44 7 cosine sim = 0.98009
585
703
2298
441
1005
488
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98006
556
701
2253
463
1042
503
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84408 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98005
793
486
2284
433
961
558
counts
404
303
202
101
0
5518
4138
2759
1380
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1